



STRATEGIC PLAN
2020/21 - 2024/25

ANNUAL PERFORMANCE PLAN
2024/25 - 2026/27

**AMENDED
STRATEGIC PLAN
2020/21 – 2024/25**



PART A



PART B



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Statement by the Chairperson

When the National Energy Regulator of South Africa (NERSA) initially developed the Strategic Plan 2020 to 2025, we were energised by the fact that the period covered in this Strategic Plan coincides with the start of a new decade. It ignited a sense of new beginnings, new ideas, new challenges, and new opportunities. In the development of this Plan, the Energy Regulator took cognisance of where our mandate fits into the bigger picture of energy.

In regulating the energy industry, we acknowledge the pivotal role it plays in economic growth. In modern economies, economic growth is closely associated with increasing energy consumption. The availability of secure, reliable, and affordable energy supply is essential for industrial processes and the provision of public services such as lighting, heating, cooking, information and communication technology, and mobility.

The global energy system is undergoing unprecedented change, driven by forces such as technological innovation, changes in consumption patterns, supply dynamics and policy shifts. These forces offer opportunities to resolve the challenges that the global energy system faces today, namely:

- providing energy access to the more than one billion people who lack it;
- meeting demand for an additional two billion people by 2050 while delivering that energy at an affordable cost; and
- ensuring that the carbon footprint declines.

In addition, the geopolitical landscape of energy is quickly shifting, and environmental concerns pose a serious challenge. At the same time, the economics of competing energy sources have changed, and the advent of Fourth Industrial Revolution technologies has enabled new business models, while making others obsolete. The latter has created significant uncertainty about the pace and destination of the transformation, making a strong case for a systemic, multi-stakeholder approach that increases the transparency of the enablers and reforms needed for countries to achieve an effective energy transition.

It is important to note that despite the continued rapid growth in renewable energy last year, renewable energy provided only a third of the required increase in energy generation, with coal providing a broadly similar contribution. The increasing use of coal within the energy sector is estimated to have more than accounted for the entire growth of global coal consumption last year. Overall, the electricity sector is estimated to have absorbed approximately 50% growth in primary energy

in 2018 and accounted for approximately 50% of the increase in carbon emissions.

Over the next 20 years, the global energy system will face a critical challenge in respect of decarbonising the power sector while at the same time endeavouring to meet the rapid increase in the demand for power, especially in developing countries. Renewable energy has a vital role to play in meeting that challenge, but it is unlikely to be able to do so on its own. A variety of different technologies and fuels are likely to be required, including extensive coal-to-gas switching and the widespread deployment of carbon capture, use and storage.

Regarding continental developments, sub-Saharan Africa accounts for 4.5% of global primary energy demand. Energy demand is very low. However, there are several factors pointing towards potentially rapid and prolonged growth in demand: strong economic expansion; increasing urbanisation; industrialisation and modernisation; a burgeoning middle class in



many countries; as well as a legacy of unmet energy demand. Bioenergy demand will increase by 40% in absolute terms by 2040, exacerbating stress on the forestry stock. The sub-Saharan Africa power system is expanding rapidly, with generation capacity quadrupling to 385 GW. The power mix becomes more diverse, with coal (mainly South Africa) and hydropower (all regions), being joined by greater use of gas (Nigeria, Mozambique, Tanzania), solar (South Africa and Nigeria) and geothermal (East Africa).

The share of renewables in total capacity more than doubled to 44%. Natural gas resource holders can power domestic economic development and boost export revenues, but only if the right regulation, prices, and infrastructure are in place. It is predicted that natural gas will nearly triple its share of the energy mix in Africa to 11% by 2040.

The Southern African region is relatively well endowed with energy resources. It has vast energy potential from solar, wind, nuclear, hydro, thermal, gas and petroleum sources in several countries. Biomass is by far the largest source of energy in most regional countries. Electricity, as the dominant source of energy in the region, is generated mainly through thermal or hydroelectric resources. The coal industry is the backbone of power generation in the region and a significant share of the resource is allocated for export. The region has a large reserve of low-cost

hydroelectricity in the north (especially Inga Reservoir in the Democratic Republic of Congo [DRC]) and Kariba Dam on the Zambia/Zimbabwe border in the middle of the regional system, as well as large reserves of cheap coal in Botswana, Mozambique, South Africa, and Zimbabwe.

Natural gas is becoming more significant to the region's energy sector, as Mozambique, Namibia, South Africa and Tanzania are developing the natural gas fields in their respective countries. New natural gas discoveries by international oil companies in Mozambique and Tanzania during the past decade, have ignited investor interest in this previously under-explored region.

Furthermore, the region has some of the most significant known reserves of uranium. The mineral is being mined in Namibia and South Africa for use as fuel for nuclear power plants while exploration is underway in Botswana and Zimbabwe. Nuclear technology is included in the electricity subsector, but it must be demonstrated that nuclear power can be a safe electricity generation option and the confidence of the population and governments must be won to endorse nuclear energy deployment in the SADC region. Only South Africa has nuclear capacity, with tentative plans for a new nuclear programme.

As we are planning to go into the new decade, we need to acknowledge the developments that took

place in South Africa since we published our previous Strategic Plan. The country has been able to commit to a total of 18 000 MW of new generation capacity. Coal will remain a key factor in electricity generation in South Africa in the near future. Government decided to extend Koeberg's design life and then expand the nuclear power programme into the future to ensure that nuclear power remains a factor in the energy mix. Gas-to-power technologies provide the flexibility required to complement renewable energy. Exploration to assess the magnitude of local recoverable shale and coastal gas is being pursued. Cooperation and partnerships-with neighbouring countries are critical for South Africa.

The Government, through the National Development Plan, envisages that by 2030 South Africa will have an energy sector that provides reliable and efficient energy service at competitive rates; that is socially equitable through expanded access to energy at affordable tariffs; and that is environmentally sustainable through reduced emissions and pollution.

Our Strategic Plan was geared to regulate the energy sector considering the consequences or impact relating to the aforementioned developments and challenges. Then the unimaginable happened. As with so many others, the roll-out of our Strategic Plan commenced in parallel with the nation-wide lockdown, which was instituted as a Government's critical measure to deal



with threat of the COVID-19 pandemic to the country and its people. At that time, the effect of this virus was evident through the myriad of the reports from all over the world. These reports highlighted the adverse effect of the virus on countries' economies, the resultant social crises of job losses and financial hardships, and the rapid infection rate of the virus for which there is currently no cure nor a vaccine.

The impact of this pandemic on the energy sector emerged slowly since the pandemic became a global health crisis. Projections were made that the impact of COVID-19 on energy demand in 2020 would be more than seven times larger than the impact of the 2008 financial crisis on global energy demand. The COVID-19 crisis and measures taken to slow its spread have had a profound impact on energy demand, the likes of which have not been seen for the past 60 to 70 years. It is also said that the energy sector that will emerge from the COVID-19 crisis may look significantly different from what came before. Low prices and low demand in all subsectors will leave energy companies with weakened financial positions and often strained balance sheets. Business lines that are insulated to a degree from market signals, including those with renewable electricity projects, will emerge in the best financial position. Private firms that are the most exposed to market prices will experience the most severe financial impact. Market concentration and consolidations are likely.

In addition, it is predicted that the COVID-19 crisis will have a significant impact on investment in the energy sector. This could raise concerns about energy security because investment is necessary even if global energy demand takes a long time to return to the pre-crisis trajectory. A considerable proportion of global energy investment is devoted to just sustaining existing levels of energy supply: maintaining oil and gas production at current levels, replacing aging power generation capacity - often with a capital-intensive combination of renewables and flexibility sources - and reinvesting in aging electricity networks. Investment in these activities will have to remain robust even with a subdued recovery.

Importantly, governments around the world will play a major role in shaping the energy sector's recovery from the COVID-19 crisis, just as they have long been in the driving seat in orienting energy investment. In particular, the design of economic stimulus packages presents a major opportunity for governments to link economic recovery efforts with clean energy transitions - and steer the energy system onto a more sustainable path. While the clean energy transitions and stimulus discussions are gathering momentum, a coordinated policy effort will be needed to harvest its opportunities and lead to a more modern, cleaner and more resilient energy sector for all.

The impact of COVID-19 on South Africa's economy was severe, prior to the global crisis pandemic, South

Africa's domestic outlook for 2020 was already very weak. South Africa is currently in its longest downward business cycle since records started in 1945, with rising unemployment, declining real gross domestic product (GDP) per capita, and declining business confidence. Forecasts suggest that this could be worse because of the COVID-19 effects. It is projected that this pandemic would increase poverty, inequality, and unemployment in South Africa.

NERSA does not function in isolation and needs to take cognisance of the developments, trends, and challenges within the global energy environment. This will assist in deciding on appropriate response by incorporating any relevant trends and energy-related developments into its strategy. This was the basis for the Energy Regulator's decision to review the initial Strategic Plan. This was done to ensure that we not only address the impact of COVID-19 in the short to medium term, but that we look at the long term as well, which will go beyond the period covered by this Strategic Plan. Since the outbreak of this pandemic, it has become clear that we will be faced with this health threat longer than expected and that we will have to adapt to the 'new normal'. This refers to the new way we will have to carry on with our lives, work and interactions with other people following this crisis. Our challenge would be to ensure that we align our move towards the new normal for NERSA with our endeavours to facilitate the availability of reliable, affordable, and clean energy,



which will lead to sustainable economic and social development. The pivotal role that NERSA plays in the energy sector is underpinned by its mandate that is enshrined in its founding legislation and is aligned to the objectives of our government. The focus on this role of NERSA will be intensified in the aftermath of the COVID-19 pandemic.

NERSA will continue to align its regulatory mechanisms with the transformation of the energy sector by ensuring the development of a sustainable energy mix that comprises coal, solar, wind, hydro, gas, and nuclear energy. NERSA will also continue to execute its mandate in such a manner that the country's energy constraints are addressed to create a conducive environment for growth and to endeavour to strike a fair balance between the interests of consumers on the one hand and regulated entities on the other hand.

Another priority for NERSA is the availability of secure, adequate, and reliable energy supply. The challenges South Africa experienced in the last few years with load shedding and unplanned power outages accentuated the importance of the reliable supply of energy, because it severely affected all sectors of society. NERSA is committed to collaborate with Government and all stakeholders to address this challenge, within the parameters of its mandate. NERSA will also investigate what the most appropriate

regulatory framework would be for the licensing of the restructured electricity supply industry following the unbundling of Eskom.

We are aware that during this planning period NERSA, as the rest of the country, will face challenges not necessarily known at this point in time. The current level of uncertainty about future developments over the next four years will require innovative approaches. A high level of agility to adapt to changing circumstances in the most effective manner to ensure the most effective and relevant regulation of South Africa's energy sector will also be required. I believe that NERSA is up to this task.

I would like to take this opportunity to acknowledge the important work that the Members of the Energy Regulator, the management team, and staff are executing, and would like to encourage a collective and innovative spirit in implementing the legislative mandate of NERSA and future strategic programmes.

MS MALEHO M D NKOMO

Deputy Chairperson

National Energy Regulator of South Africa



Chief Executive Officer's Statement

The National Energy Regulator of South Africa (NERSA) was established on 1 October 2005 in terms of the National Energy Regulator Act, 2004 (Act No. 40 of 2004). Its mandate is to regulate the electricity industry in terms of the Electricity Regulation Act, 2006 (Act No. 4 of 2006), the piped-gas industry in terms of the Gas Act, 2001 (Act No. 48 of 2001), and the petroleum pipelines industry in terms of the Petroleum Pipelines Act, 2003 (Act No. 60 of 2003).

NERSA's mandate, as contained in the relevant legislation, is summarised as follows:

- Issuing of licences and setting pertinent conditions.
- Setting and/or approving tariffs and prices.
- Monitoring and enforcing compliance with licence conditions.
- Dispute resolution including mediation, arbitration and the handling of complaints.
- Gathering, storing and disseminating industry information.
- Setting of rules, guidelines and codes for the regulation of the three industries.
- Determination of conditions of supply and applicable standards.
- Registration of import and production activities.

In carrying out its mandate, NERSA endeavours to achieve its vision to be a recognised world-class leader in energy regulation. NERSA is expected to implement its mandate and to take the necessary regulatory decisions proactively in anticipation of and in response

to the changing circumstances in the energy industry. The role of NERSA is to ensure the development and sustainability of the electricity, piped-gas and petroleum pipelines industries, while facilitating the affordability of and accessibility to these industries to balance the economic interests of all stakeholders to ensure sustainable socio-economic development of South Africa and a better life for all.

During the previous planning period, the Regulator upheld its regulatory principles of transparency, neutrality, consistency and predictability, independence, accountability and integrity in regulating the electricity, piped-gas and petroleum pipelines industries.

The highlights of NERSA's achievements range from putting in place structures and systems that ensure sound corporate governance and capacity building, to the provision of current and user-friendly information, as well as the development, fine tuning and implementation of regulatory methodologies, processes, procedures and systems to ensure that NERSA delivers on its

mandate. NERSA's focus was the continued alignment of its regulatory mechanisms with the transformation of the energy sector by ensuring the development of a sustainable energy mix.

The Regulator also continued to ensure the orderly development in the energy sector, mainly through licensing, setting and approving of prices and tariffs, compliance monitoring and enforcement, and dispute resolution in the electricity, piped-gas and petroleum pipelines industries. In addition, NERSA commenced with a process to contribute towards the transformation of the energy industry, within the ambit of our mandate.

We continuously endeavoured to strike the balance between the interests of producers and investors, and those of South African citizens across all our primary activities, namely licensing, setting and approving of prices and tariffs, compliance monitoring and enforcement, and dispute resolution in the electricity, piped gas and petroleum pipelines industries.



Furthermore, the Regulator continued to engage with other regulatory bodies to align its regulatory processes with international best practices.

In developing this Strategic Plan, the mandate of NERSA as well as key policy priorities were taken into account. The strategic focus stated in this Strategic Plan is in line with and in support of the following key priorities of Government, as espoused in the Medium-Term Strategic Framework:

- **Priority 2:** Economic transformation and job creation
- **Priority 3:** Education, skills and health
- **Priority 7:** A better Africa and world.

It also supports the National Development Plan (NDP), which is a plan for the country to eliminate poverty and reduce inequality by 2030 through uniting South Africans, unleashing the energies of its citizens, growing an inclusive economy, building capabilities, enhancing the capability of the state and leaders working together to solve complex problems.

However, due to the changing circumstances as a result of the global COVID-19 pandemic, the Regulator decided to review this Strategic Plan. The reason being that we needed to ensure that NERSA is geared to effectively regulate an energy sector in a country that is adversely affected by this pandemic.

In view of the far-reaching impact of the COVID-19 pandemic, NERSA will put measures in place to ensure we continuously identify areas affected by this pandemic to be able to timeously put mitigating strategies in place. With the realisation that this pandemic will be prevalent for the foreseeable future, NERSA will have to be more innovative and agile in ensuring that we continue to ensure the orderly development in the energy sector, mainly through licensing, setting and approving of prices and tariffs, compliance monitoring and enforcement, and dispute resolution in the electricity, piped-gas and petroleum pipelines industries.

During the review of this Strategic Plan, it became clear that NERSA needs to re-assess how it operates and put an operating model in place that will improve our efficacy. This is informed by our commitment to achieve our vision through our mission to regulate the energy industry in accordance with government laws and policies, standards and international best practices in support of sustainable socio-economic development.

The severe impact of COVID-19 on the economy of South Africa, when our economy was already severely strained, will require NERSA to focus more on its role in advancing economic growth and social development within South Africa. Another priority for NERSA will be to contribute towards the transformation of the energy industry, within the ambit of our mandate.

In view of the aforementioned, NERSA remains committed to increasing delivery on its mandate as well as evaluating the impact of our actions.

Specific outcomes were identified that will guide the Regulator's programmes for the next five years in respect of each of the regulated industries:

Electricity Industry

1. Accessible and cost-reflective electricity that is equitably distributed for consumption
2. Diverse energy supply that is certain and secure for current and future user needs
3. Conducive regulatory environment that results in regulatory certainty and increased investment in the electricity industry.

Petroleum Pipelines Industry

1. Equitable access to affordable petroleum products, services and infrastructure at competitive prices
2. Efficient, sustainable, equitable and orderly development of a transformed, competitive and accessible petroleum pipelines industry
3. A conducive regulatory environment that results in regulatory certainty and increased investment in the petroleum industry.



Piped-Gas Industry

1. Equitable access to affordable gas services at competitive prices
2. Efficient, sustainable, equitable and orderly development of a transformed, competitive and accessible piped-gas industry
3. A conducive regulatory environment that results in regulatory certainty and increased investment in the piped-gas industry.

The **organisational** outcomes are as follows:

1. Creation of an enabling environment for internal and external stakeholders through proactive, dynamic and data-driven advisory, advocacy and decision-making.
2. An effective operating model that enables the organisation to fulfil its role effectively.

The achievement of the above outcomes will be enabled through, among others, revised regulatory methodologies and rules; continued monitoring of licensees' performance; contributing towards the restructuring of the energy industry; periodic assessment of adequacy of competition; decreasing regulatory burden; improved critical business and regulatory processes.

We believe the aforementioned will enable NERSA to contribute towards and address the challenges identified by the Department of Mineral Resources and Energy, namely the impact of high electricity prices and the security of energy supply.

NERSA will continue to place emphasis on facilitating the entry of new players into the energy sector, particularly in the light of the generally monopolistic nature of the electricity, piped-gas and petroleum pipelines industries.

Our overall aim remains that the impact of implementing this Strategic Plan is to facilitate a secure, reliable, affordable, sustainable, competitive and transformed energy industry, which contributes to the economic growth of South Africa. In order to achieve this, NERSA places a high premium on capacity building of its staff complement. In addition to the training and development for staff members, NERSA has been and will continue to run successful internship and learnership programmes.

NERSA is fully committed to the implementation of this Strategic Plan, with the strategic guidance and support of the Energy Regulator. I would like to take this opportunity to acknowledge the important work that the staff are executing and would like to encourage an innovative and collaborative spirit in implementing the legislative mandate of NERSA and future strategic programmes.

NOMALANGA P SITHOLE

*Chief Executive Officer of the National Energy
Regulator of South Africa*



Official Sign-Off

It is hereby certified that this Strategic Plan.

- was developed by the Executive Management of NERSA under the guidance of the Energy Regulator;
- takes into account all the relevant policies, legislation and other mandates for which the Energy regulator is responsible; and
- accurately reflects the impact, outcome and outputs that the Energy Regulator will endeavour to achieve over the period 2020/21 – 2024/25.

Gerda Gräbe

Senior Manager: Strategic Planning and Monitoring

MS. Bulelwa Pono

Chief Financial Officer

Adv. Nomalanga Sithole

Acting Chief Executive Officer (Accounting Officer)

Approved by:

Mrs. Maleho M D Nkomo

Deputy Chairperson (on behalf of the Accounting Authority)



Acronyms and Abbreviations

AFDB	African Development Bank
AFUR	African Forum for Utility Regulators
APP	Annual Performance Plan
B-BBEE	Broad-Based Black Economic Empowerment
CAGR	Compound Annual Growth Rate
CBM	Coal Bed Methane
CCGT	Closed Cycle Gas Turbine
CNG	Compressed Natural Gas
CPI	Consumer Price Index
CTL	Coal-to-Liquid
DJP	Durban-to-Johannesburg Pipeline
DoE	Department of Energy
EEDSM	Energy Efficiency and Demand Side Management
ELR	Electricity Regulation
ELS	Electricity Subcommittee
EPP	Electricity Pricing Policy
ER	Energy Regulator
ESI	Electricity Supply Industry
FBE	Free Basic Electricity

FID	Final Investment Decision
FLNG	Floating Liquefied Natural Gas
GAR	Piped-Gas Regulation
GDP	Gross Domestic Product
GHG	Greenhouse Gases
GJ	Gigajoule
GSA	Gas Supply Agreement
GTL	Gas-to-Liquid
GUMP	Gas Utilisation Master Plan
HDI/ HDSA	Historically Disadvantaged Individuals/ South Africans
IBT	Inclining Block Tariff
ICT	Information and Communication Technologies
IDM	Integrated Demand Management
IEA	International Energy Agency
IEP	Integrated Energy Plan
IGU	International Gas Union
IPAP	Industrial Policy Action Plan
IPP	Independent Power Producer
IRP	Integrated Resource Plan

Ke	Cost of Equity
LNG	Liquefied Natural Gas
MCEP	Manufacturing Competitive Enhancement Programme
MOA	Memorandum of Agreement
MOU	Memorandum of Understanding
MPP	Multi-Product Pipeline
MTEF	Medium-term Expenditure Framework
Mtoe	Million Tonnes of Oil Equivalent
MTPA	Metric Tons Per Annum
MTSF	Medium-Term Strategic Framework
MW	Megawatt
NDP	National Development Plan
NERSA	National Energy Regulator of South Africa
NIPF	National Industrial Policy Framework
NMPP	New Multi-Product Pipeline
NFI	Non-Financial Information
OCGT	Open Cycle Gas Turbine
OECD	Organisation for Economic Co-operation and Development



PASA	Petroleum Association of South Africa
PE(R) STEL	Political, Economic, Regulatory, Social, Technological, Environmental and Legal
PICC	Presidential Infrastructure Coordinating Committee
PFMA	Public Finance Management Act, 1999 (Act No. 1 of 1999)
PGS	Piped-Gas Subcommittee
PPA	Power Purchase Agreement
PPR	Petroleum Pipelines Regulation
PPS	Petroleum Pipelines Subcommittee
PV	Photovoltaic
REC	Regulator Executive Committee
REIPP	Renewable Energy Independent Power Producer
REIPPPP	Renewable Energy Independent Power Producer Procurement Programme
RERA	Regional Electricity Regulatory Association
RESAP	Renewable Energy Strategy and Action Plan
RIA	Regulatory Impact Assessment

ROMPCO	Republic of Mozambique Pipeline Investment Company
SACREEE	SADC Centre for Renewable Energy, Energy and Efficiency
SADC	Southern African Development Community
SAPIA	South Africa Petroleum Industry Association
SAPP	Southern African Power Pool
SCOA	Standard Chart of Accounts
SFF	Strategic Fuel Fund
SIP	Strategic Integrated Project
SQAM	Standards, Quality Assurance, Accreditation and Metrology

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PART A: OUR MANDATE

1. Constitutional Mandate

- 1.1. The National Energy Regulator of South Africa is listed as a public entity in terms of Schedule 3A of the Public Finance Management Act, 1999 (Act No. 1 of 1999).
- 1.2. The Constitution of South Africa is applicable to NERSA in conduct of its business, with specific reference to the Bill of Rights.
- 1.3. NERSA's responsibility is carried out through licensing, setting or approving of prices and tariffs, compliance monitoring and enforcement, and dispute resolution in the electricity, piped-gas and petroleum pipelines industries. It facilitates, through its regulatory functions, the construction of power stations, pipelines and storage facilities to ensure continued access to energy and security of supply in the country. NERSA's commitment to the protection of the environment and the growth of cleaner, more resource-efficient production of energy is built into its regulatory functions.

2. Legislative and Policy Mandates

2.1. Relevant Legislation

- 2.1.1. NERSA is the regulatory authority established in terms of the National Energy Regulator Act, 2004 (Act No. 40 of 2004) with the mandate to 'undertake the functions of the National Electricity Regulator as set out in the Electricity Regulation Act, 2006 (Act No. 4 of 2006), undertake the functions of the Gas Regulator as set out in the Gas Act, 2001 (Act No. 48 of 2001), undertake the functions of the Petroleum Pipelines Regulatory Authority as set out in the

Petroleum Pipelines Act, 2003 (Act No. 60 of 2003) and to perform such other functions as may be assigned to it by or under these Acts'.

- 2.1.2. NERSA's mandate is anchored in the following four primary Acts:

- the National Energy Regulator Act, 2004 (Act No. 40 of 2004);
- the Electricity Regulation Act, 2006 (Act No. 4 of 2006) (ERA);
- the Gas Act, 2001 (Act No. 48 of 2001); and
- the Petroleum Pipelines Act, 2003 (Act No. 60 of 2003).

- 2.1.3. The regulatory functions of NERSA, as contained in the legislation relevant for the regulation of the energy industry, are summarised as follows:

- issuing of licences with conditions;
- setting and/or approving tariffs and prices;
- monitoring and enforcing compliance with licence conditions; dispute resolution including mediation, arbitration and the handling of complaints;
- gathering, storing and disseminating industry information;
- setting of rules, guidelines and codes for the regulation of the three industries;
- determining of conditions of supply and applicable standards;
- consulting with government departments and other bodies with regard to industry development and regarding any matter contemplated in the three industry Acts;
- expropriating land as necessary to meet the objectives of the relevant legislation;
- registration of import and production facilities; and
- performing any activity incidental to the execution of its duties

2.1.4. Each one of the industry-specific Acts that NERSA is deriving its mandate from, has certain objects that should be achieved if NERSA carries out its functions as defined in these Acts.

a) The objects of the **Electricity Regulation Act** as stipulated in section 2 of the Act, are to:

- achieve the efficient, effective, sustainable and orderly development and operation of electricity supply infrastructure in South Africa;
- ensure that the interests and needs of present and future electricity customers and end users are safeguarded and met, having regard to the governance, efficiency, effectiveness and long-term sustainability of the electricity supply industry within the broader context of economic energy regulation in the Republic;
- facilitate investment in the electricity supply industry;
- facilitate universal access to electricity;
- promote the use of diverse energy sources and energy efficiency;
- promote competitiveness and customer and end user choice; and
- facilitate a fair balance between the interests of customers and end users, licensees, investors in the electricity supply industry and the public.

b) The objects of the **Gas Act** as stipulated in section 2 of the Act, are to:

- promote the efficient, effective, sustainable and orderly development and operation of gas transmission, storage, distribution, liquefaction and re-gasification facilities and the provision of efficient, effective and sustainable gas transmission, storage, distribution, liquefaction, re-gasification and trading services;
- facilitate investment in the gas industry;
- ensure the safe, efficient, economic and environmentally responsible transmission, distribution, storage, liquefaction and re-gasification of gas;
- promote companies in the gas industry that are owned or controlled by historically disadvantaged South Africans by means of licence conditions so as to enable them to become competitive;

- ensure that gas transmission, storage, distribution, trading, liquefaction and re-gasification services are provided on an equitable basis and that the interests and needs of all parties concerned are taken into consideration;
- promote skills development among employees in the gas industry;
- promote employment equity in the gas industry;
- promote the development of competitive markets for gas and gas services;
- facilitate gas trade between the Republic and other countries; and
- promote access to gas in an affordable and safe manner.

c) The objects of the **Petroleum Pipelines Act** as stipulated in section 2 of the Act, are to:

- promote competition in the construction and operation of petroleum pipelines, loading facilities and storage facilities;
- promote the efficient, effective, sustainable and orderly development, operation and use of petroleum pipelines, loading facilities and storage facilities;
- ensure the safe, efficient, economic and environmentally responsible transport, loading and storage of petroleum;
- promote equitable access to petroleum pipelines, loading facilities and storage facilities;
- facilitate investment in the petroleum pipeline industry;
- provide for the security of petroleum pipelines and related infrastructure;
- promote companies in the petroleum pipeline industry that are owned or controlled by historically disadvantaged South Africans, by means of licence conditions to enable them to become competitive;
- promote the development of competitive markets for petroleum products;
- promote access to affordable petroleum products; and
- ensure an appropriate supply of petroleum to meet market requirements.

d) The object of the **National Energy Regulator Act** as stipulated in section 1 of the Act, is to:

- establish a National Energy Regulator for the regulation of the electricity, piped-gas and petroleum pipelines industries

2.1.5. The Electricity Regulation Act, the Gas Act and the Petroleum Pipelines Act gives the Minister of Mineral Resources and Energy (the Minister) the power to make Regulations in terms of which NERSA must discharge its mandate.

a) The Minister has published the following **Electricity Industry Regulations**:

- the Electricity Regulations for Expropriation on behalf of a licensee;
- the Electricity Regulations for compulsory norms and standard for reticulation services;
- the Electricity Regulations on deviation from set or approved tariffs; and
- the Revised New Generation Regulations were issued on 4 May 2011.

b) The Minister has published the following **Piped-Gas Industry Regulations** on 20 April 2007, which deal with, amongst others:

- third-party access to transmission and storage facilities;
- expropriation procedures and timelines;
- mechanisms to promote historically disadvantaged South Africans;
- mediation and arbitration procedures; and
- price regulation principles and procedures.

c) The Minister has published the following **Petroleum Pipelines Industry Regulations** on 4 April 2008. The Regulations deal with, amongst others:

- third-party access to storage facilities;
- setting of tariffs for petroleum pipelines and approval of tariffs for petroleum loading and storage facilities;
- expropriation procedures and timelines;
- mechanisms to promote historically disadvantaged South Africans; and
- mediation and arbitration procedures.

2.1.6. NERSA derives its revenue by, among others, imposing prescribed levies on the regulated industries following a prescribed transparent procedure. In this regard, the following Acts govern the imposition of such levies:

- the Gas Regulator Levies Act, 2002 (Act No. 75 of 2002);
- the Petroleum Pipelines Levies Act, 2004 (Act No. 28 of 2004); and
- section 5B of the Electricity Act, 1987 (Act No. 41 of 1987).

2.1.7. Apart from the afore-mentioned industry specific legislation that anchors NERSA's mandate and the imposition of levies, the following facilitating and foundational legislation are also applicable to NERSA's conduct of its business:

- the Public Finance Management Act, 1999 (Act No. 1 of 1999) (PFMA), which specifies the accounting of NERSA as a Section 3A Public Entity;
- the Promotion of Access to Information Act, 2000 (Act No. 2 of 2000) (PAIA), which determines the way that NERSA has to treat access to information;
- the Promotion of Administrative Justice Act, 2000 (Act No. 3 of 2000) (PAJA), which determines just administrative action of NERSA;
- the Protection of Personal Information, 2013 (Act No 4 of 2013), which determines the way that NERSA has to treat personal information; and
- all other applicable laws of the Republic of South Africa.

2.2. Relevant Policies

NERSA's mandate is further derived from published government policies and regulations developed by the Minister in terms of the Electricity Regulation Act, Gas Act and Petroleum Pipelines Act. As outlined in these legislative prescripts, NERSA must make decisions that are not at variance with published government policy. The relevant applicable policies are:

- White Paper on Energy Policy for South Africa of 1998;
- Electricity Pricing Policy (EPP) of the South African Electricity Supply Industry;
- Free Basic Electricity Policy;
- White Paper on Renewable Energy Policy for South Africa of 2003;
- Energy Security Master Plan: Liquid Fuels published by the Department of Energy in 1998 and 2007;
- National Development Plan;
- Industrial Policy Action Plan (IPAP); and
- Integrated Resource Plan (IRP) 2019.

3. Institutional Policies and Strategies over the Five-Year Planning Period

- 3.1. Although policy formulation is outside of NERSA's realm of authority, specific policy gaps are continuously identified that require ongoing dialogue and strategic engagement with the Department of Mineral Resources and Energy in order to ensure that there is alignment between NERSA's strategic direction and the Department's policy thrusts.
- 3.2. In the previous five-year planning period, NERSA has seen that there are developments in the three industries that are not covered by the current industry-specific Acts. This requires a review of the regulatory legislation.

- 3.3. In addition to its mandate as per the legislation mentioned in the previous section, the Energy Regulator's decisions are informed by published policies of government. Within the parameters of NERSA's mandate and the resultant functions, NERSA contributes towards critical government priorities and programmes. Below is a summary of NERSA's contributions towards the –

- enabling milestones in the National Development Plan (NDP);
- strategic integrated projects in the National Infrastructure Plan; and
- seven priorities announced by the Honourable President, Mr Cyril Ramaphosa during the State of the Nation Address (SONA) in Parliament on 20 June 2019

3.3.1. NERSA's contribution to the National Development Plan

The National Development Plan (NDP) is a plan for the country to eliminate poverty and reduce inequality by 2030 through uniting South Africans, unleashing the energies of its citizens, growing an inclusive economy, building capabilities, enhancing the capability of the state and leaders working together to solve complex problems. The high-level objectives of the NDP are to:

- reduce the number of people who live in households with a monthly income below r419 per person (in 2009 prices) from 39% to zero; and
- reduce inequality, as measured by the Gini Coefficient, from 0.69 to 0.6.

Chapter 4 of the NDP deals with *Economic infrastructure – the foundation of social and economic development*. This chapter places emphasis on the need for South Africa to maintain and expand, among others, its electricity infrastructure in order to support economic growth and social development goals. In respect of the regulation of the energy sector, NERSA noted that the NDP calls for more emphasis on stimulating market competition and promoting affordable access to quality services when issuing licences and setting tariffs.

In order to achieve the NDP goals by 2030, 19 enabling milestones were identified. Even though NERSA contributes indirectly to most of the enabling milestones, NERSA contributes specifically to 4 pertinent enabling milestones. Table 1 below summarises NERSA's contribution to the relevant enabling milestones.

Table 1: NERSA's contribution to the NDP

Relevant enabling milestones	NERSA's contribution
1: Increase employment from 13 million in 2010 to 24 million in 2030	• Implementation of the Youth Employment Accord; • Implementation of a Learnership Programme as well as an Internship Programme; • Training and development of staff and stakeholders; • Techno Girls programme where ten girls from grade 9 to grade 12 are exposed to NERSA's activities through visits to the organisation during school holidays.
4: Establish a competitive base of infrastructure, human resources and regulatory frameworks	• Publication of rules, codes and guides for the regulation of the electricity, piped-gas and petroleum pipelines industries; • Setting rules and frameworks that facilitate the building of new infrastructure; • Setting and/or approving cost reflective tariffs and market related prices that encourage investment; • Facilitating and enforcing third-party access to facilities through licence conditions; • Monitoring compliance through undertaking technical audits leading to regular maintenance and refurbishment of infrastructure and thus contributing to an increase in quality of supply;
5: Ensure that skilled, technical, professional and managerial posts better reflect the country's racial, gender and disability makeup	• NERSA ensures continued compliance with the Skills Development Act. No. 97 of 1998; • Implementation of an Employment Equity Plan; • When recruiting new staff members, NERSA ensures as far as possible that the representation within the relevant department and division reflects the country's racial, gender and disability makeup.
6: Broaden ownership of assets to historically disadvantaged groups	• Licensing and the setting and/or approving of tariffs and prices, as in this manner NERSA creates pre-conditions towards the achievement of this milestone; • Issuing licences to eligible applicants to facilitate the meeting of stated socio-economic development targets; • Facilitating and enforcing third-party access to facilities; • Promoting companies that are owned and controlled by Historically Disadvantaged Individuals (HDIs) to become competitive; and • Regulatory advocacy for strengthening the powers of the Regulator.

Relevant enabling milestones	NERSA's contribution
10: Produce sufficient energy to support industry at competitive prices, ensuring access for poor households, while reducing carbon emissions per unit of power by about one-third	• Regulating in a manner that facilitates security of supply; • Taking affordability into consideration when setting and/or approving tariffs and prices; • Determining inclining block tariffs and free basic electricity tariffs to protect the low income electricity consumers; • Facilitating the conclusion of Power Purchase Agreements between the buyer and the renewable energy Independent Power Producers; • Facilitation of the implementation of the Integrated Resource Plan (IRP) through considering concurring with determinations made by the Minister in line with section 34 of the Electricity Regulation Act, 2006 (Act No. 4 of 2006); • Development and implementation of the Grid Code for renewable energy to facilitate the introduction of renewable energy power producers; • Registration of gas importation and production facilities; • Monitor the implementation of the Gas Utilisation Master Plan (once promulgated).Facilitating access to electricity in setting aside some funds for the Electrification Cross-subsidy as part of determining electricity prices; • Incorporating compliance with the National Environmental Management Act, 1998 (Act No. 107 of 1998) into licence conditions; • Promoting energy efficiency in general in South Africa and in particular in the NERSA building; • Facilitating the transition to a low carbon economy; and • Regulatory advocacy with regard to cleaner fuels policy.

3.3.2. Nersa's Contribution to the Medium Term Strategic Framework 2019-2024

- a) The Medium Term Strategic Framework (MTSF) is a five-year plan of government that is intended to implement the electoral mandate and the National Development Plan Vision (NDP) 2030.
- b) It aims to address the challenges of unemployment, inequality and poverty through three pillars of the NDP:
 - Achieving a more capable State
 - Driving a strong and inclusive economy;
 - Building and strengthening the capabilities of South Africans
- c) The seven priorities, which will be achieved through more focused implementation, coordination and integration by the various levels of government including state owned enterprises, the private sector and civil society, are as follows:
 - **Priority 1:** A capable, ethical and developmental state
 - **Priority 2:** Economic transformation and job creation
 - **Priority 3:** Education, skills and health
 - **Priority 4:** Consolidating the social wage through reliable and quality basic services
 - **Priority 5:** Spatial integration, human settlements and local government
 - **Priority 6:** Social cohesion and safe communities
 - **Priority 7:** A better Africa and world
- d) NERSA identified the following government priorities to which it can contribute – as part of implementing its mandate:
 - **Priority 2:** Economic transformation and job creation
 - **Priority 3:** Education, skills and health
 - **Priority 7:** A better Africa and world

Table 2: NERSA's contribution to government's priorities

Relevant Priorities	NERSA's contribution
2: Economic Transformation and Job Creation	By facilitating investment in the energy industry and thereby contributing to economic growth, leading to job creation, NERSA contributes through: • licensing and the setting and/or approving of tariffs and prices, as in this manner NERSA creates pre-conditions towards the achievement of this priority; • approving renewable energy licenses to ensure that the socio-economic development commitments specified in the bidding process are met; • promoting companies that are owned and controlled by Historically Disadvantaged Individuals (HDIs) to become competitive; and • regulating in a manner that facilitates security of supply. Contributing to a competitive and responsive economic infrastructure network through: • Setting rules and frameworks that facilitate the building of new infrastructure; • Setting and/or approving cost reflective tariffs and prices that encourage efficient investment; • Facilitating and enforcing third-party access to facilities; • Monitoring compliance and undertaking technical audits leading to regular maintenance and refurbishment of the infrastructure and therefor to the improvement in quality of supply; and • Promoting competition and competitiveness in the energy industry.
3: Education, skills and health	• Implementation of the Learnership and Internship Programmes; • Implementation of the bursary programme for qualifying external applicants; • Coordinating the design of a regulatory course at an accredited institution of higher learning; and • Coordinating the development of a technical regulatory training and development programme.
6: A Capable, Ethical and Developmental State	• Transparent regulatory processes; • All decisions and reasons thereof are made public through being published on the website; • The public is invited to make comments prior to decisions being made (written or in public hearing); • Customer education programmes and awareness campaigns; • Training and development of staff and stakeholders, including training to electricity distributors on the completion of the forms requesting information from them; and • Techno Girls programme - where ten girls from grade 9 to grade 12 are exposed to NERSA's activities through visits to the organisation during school holidays.

3.3.3. Nersa's Contribution to the National Infrastructure Plan

The South African Government adopted a National Infrastructure Plan (NIP) in 2012 that intends to strengthen the delivery of basic services and transform South Africa's economic landscape, while simultaneously creating significant numbers of new jobs. The plan also supports the integration of African economies. The New Growth Path sets a goal of five million new jobs by 2020, identifies structural problems in the economy to be overcome and points to opportunities in specific sectors and markets or 'jobs drivers'.

In order to address these challenges and goals, a total of 18 strategic integrated projects (SIPs) have been developed. The following three SIPs were identified for energy:

1. SIP 8: Green energy in support of the South African economy

- Support sustainable green energy initiatives on a national scale through a diverse range of clean energy options as envisaged in the Integrated Resource Plan (IRP2010).
- Support bio-fuel production facilities.

2. SIP 9: Electricity generation to support socio-economic development

- Accelerate the construction of new electricity generation capacity in accordance with the IRP2010 to meet the needs of the economy and address historical imbalances.
- Monitor implementation of major projects such as new power stations: Medupi, Kusile and Ingula.

3. SIP 10: Electricity transmission and distribution for all

- Expand the transmission and distribution network to address historical imbalances, provide access to electricity for all and support economic development.
- Align the 10-year transmission plan, the services backlog, the national broadband roll-out and the freight rail line development to leverage off regulatory approvals, supply chain and project development capacity.

Table 3 below summarises NERSA's contribution to the relevant strategic integrated projects.

Table 3: NERSA's contribution to the NIP

Relevant Priorities	NERSA's contribution
8: Green energy in support of the South African economy	- Facilitating the conclusion of Power Purchase Agreements between the buyer and the renewable energy Independent Power Producers; - Incorporating compliance with the National Environmental Management Act, 1998 (Act No. 107 of 1998) into licence conditions; - Facilitation of the implementation of the Integrated Resource Plan (IRP) through considering concurring with determinations made by the Minister in line with section 34 of the Electricity Regulation Act, 2006 (Act No. 4 of 2006); - Facilitating the transition to a low carbon economy; and - Regulatory advocacy with regard to cleaner fuels policy.
9: Electricity generation to support socio-economic development	- Regulating in a manner which facilitates security of supply and investment; - Facilitating the conclusion of Power Purchase Agreements between the buyer and the renewable energy Independent Power Producers; - Setting rules and frameworks that facilitate the building of new infrastructure; - Setting and/or approving cost reflective tariffs and prices that encourage investment; - Monitoring compliance through undertaking technical audits leading to regular maintenance and refurbishment of infrastructure and thus contributing to an improvement in quality of supply.
10: Electricity transmission and distribution for all	- Facilitating access to electricity in setting aside some funds for the Electrification Cross-subsidy as part of determining electricity prices; - Taking affordability into consideration when setting and/or approving tariffs and prices, while allowing a provision for expansion of current operations; - Determining inclining block tariffs and free basic electricity tariffs to protect the low income electricity consumers; - Facilitating reliability of supply; - Determining benchmarks and monitoring maintenance of infrastructure; - Auditing of the implementation of the Transmission Development Plan; - Monitoring compliance with licence conditions; and - Dispute resolution, including mediation, arbitration and handling of complaints

4. Relevant Court Rulings

The ruling by the courts in the following two cases have a significant impact on the operations or service delivery obligations:

4.1. Interruption of supply of electricity to Emfuleni, which includes supply to Cape Gate Pty (Ltd).

4.1.1. **Applicant:** Cape Gate Pty (Ltd) and Others

4.1.2. **Defendant / Respondent:** Eskom, Emfuleni, NERSA and other

4.1.3. **Synopsis:** The Applicant sought an:

- interdict against Eskom to prevent it from implementing its power supply interruption decision;
- order that the decision to implement interruptions in the electricity supply be reviewed and set aside; and
- order that Eskom supply electricity on an uninterrupted basis to the Municipality on the basis that direct payment will be made to Eskom.

4.1.4. **Court ruling:** The following orders were issued:

- The dispute regarding non-payment by Emfuleni to Eskom was referred to the respondents for resolution in terms of section 41(3) of the Constitution.
- Eskom was interdicted from interrupting electricity supply to Emfuleni, pending resolution of the dispute within six months of this order or pending the outcome of the final determination of Part B of the application, whichever is earlier.

- The applicants were authorized, subject to appropriate oversight by NERSA, performing its statutory functions, to discharge their debt to Emfuleni by:

- Making payment directly to Eskom for electricity they consume at the rate of Eskom, and submitting proof thereof to Emfuleni.
- Continuing to pay the difference between the municipal tariff and Eskom tariff (the municipal portion) to Emfuleni.
- The respondents, including NERSA, were directed to do all things necessary to give effect to the temporary order.
- Emfuleni's obligations and duties to the Applicants will not be affected by this order.

4.1.5. **Ongoing impact on operations or service delivery obligations:**

- The order authorising end users to make direct payments to Eskom for electricity they consume is not in line with the current legal framework. It was made as a just and equitable relief.
- It has serious implications for municipalities and the work that NERSA does.
- There is no timeline that the court has set for the operation of this disruptive process.

4.1.6. **What has been done to remedy the disruptive effect of the judgement?**

- Gauteng Provincial Government (COGTA Department) has kick started the intergovernmental process to deal with the effect of the judgement and soliciting means to remedy it.
- Eskom and NERSA is part of the IGR process.

4.2. Issues related to the approved maximum prices of gas and approved transmission tariffs for Sasol Gas

4.2.1. **Applicant:** NERSA and Sasol Gas

4.2.2. **Defendant / Respondent:** PG Group and Others

4.2.3. **Synopsis:**

- PG Group & Others, together called the Gas Users Group (GUG), were unhappy about the maximum prices of gas and transmission tariffs approved for Sasol Gas by NERSA, which came into operation on 26 March 2014. GUG submitted that the prices are excessive and therefore sought an order to:
 - review and set aside the abovementioned approvals by NERSA; and
 - review and set aside the methodology used by NERSA to consider the abovementioned maximum price application, or declaring such methodology to be invalid for purposes of such consideration. NERSA contests the action by the applicants.
- After the Court granted judgement in favour of NERSA and SASOL, the GUG appealed.

4.2.4. **Court ruling:**

- Both the Supreme Court of Appeal and the Constitutional Court granted a judgement against NERSA and Sasol. The Constitutional Court effectively criticised the price indicator method used to determine maximum prices.

4.2.5. **Ongoing impact on operations or service delivery obligations:**

- Following the ruling of the Constitutional Court, the Energy Regulator has been unable to process any maximum price applications using the price indicator approach. Work is in progress to develop an interim mechanism, while attending to the review of the Maximum Price Methodology, in line with the Constitutional Court ruling.

4.2.6. **What has been done post the judgement?**

- NERSA has reviewed the Methodology to align it with the ConCourt judgement.

PART B: OUR STRATEGIC FOCUS

1. Vision

NERSA strives to regulate the South African electricity, piped-gas and petroleum pipelines industries by ensuring that the most efficient and effective industries are in place to exceed the requirements of existing and future energy customers. This is encapsulated in our **our vision statement**, which is:

‘To be a recognised world-class leader in energy regulation’

In this context, being ‘world-class’ means that NERSA:

- Is recognised as a leader within the league of Regulators.
- Regulates the energy industry within its mandate without losing sight of its shared vision and values.
- Creates an environment that has low regulatory risk as viewed by all stakeholders.
- Promotes competition and competitiveness and continues to provide sound, objective and professional regulation of monopolies given the existing socio-economic conditions.
- Subscribes to the best regulatory practices and standards, including corporate governance principles.
- Continually evaluates its performance and benchmark itself against the “best-in-class energy regulators in the world”.
- Is passionate and sensitive to the needs of its stakeholders, especially employees, consumers, energy suppliers and government, to ensure equity.
- Is considered as an efficient and effective regulator.
- Encourages new ideas, innovation, processes and systems that engender economic efficiency, effectiveness and continuous improvement to meet its aim to be a learning organisation.
- Maintains synergy between input, work processes and results through its capable, diverse, highly motivated and dedicated teams.

2. Mission

By regulating the energy industry in accordance with government laws and policies, NERSA makes a valuable contribution to the socio-economic development and prosperity of the people of South Africa. Our **mission statement**, commits NERSA:

‘To regulate the energy industry in accordance with government laws and policies, standards and international best practices in support of sustainable socio-economic development’

3. Values

Values are the expression of what we stand for and how we will conduct ourselves. In this context and in addition to our commitment to comply with the requirements of section 9 (11) of the National Energy Regulator Act, 2004 (Act No. 40 of 2004) and its Code of Conduct, we have adopted the following values

Passion	We conduct our business with a sense of urgency and commitment and are proud to be part of NERSA.
Spirit of Partnership	In working with all our stakeholders, we deliver on our promises for the purpose of sustainable development.
Excellence	In striving for the best results, we promote growth/development of our staff, and benchmark ourselves against the ‘best-in-class’ energy regulators across the globe.
Inclusivity	We embrace, value and treat all our stakeholders fairly and equally.

Innovation	As a learning organisation, we strive to set trends and promote creativity by challenging the norm in order to continuously improve.
Integrity	Being honest, fair and sincere with all stakeholders and among ourselves.
Responsibility	We practice responsibility and take ownership of our actions and decisions.
Professionalism	We encourage maintenance of high standards of professional competence, responsiveness, respect, trust and collaboration between our teams
Pride	We take pride in what we do.

4. Regulatory Principles

In regulating the three industries, NERSA must adhere to sound principles and approaches to be able to deliver on its mandate and achieve its objectives. NERSA has given consideration both to international best practice and the key principles stated in the African Forum for Utility Regulators (AFUR) Framework for Utility Regulation in Africa¹. Following the completion of the *Benchmarking the National Energy Regulator of South Africa against international good practice*, NERSA reviewed the literature on good regulatory principles and identified those principles that emerge strongly and consistently as international good practice. Supported by its legal mandate, NERSA adopted the following internationally accepted regulatory principles to underpin its regulatory approach:

Transparency	We are required to explain its decisions and processes to regulated entities and other interested parties, which implies that the data or information on which the decision is based is readily available and the reasoning behind it is readily explained. This covers public consultation and accessibility.
Neutrality	We should be neutral to all market players without favouring any one group (non-discriminatory).
Consistency and Predictability	Our decisions must be consistent and should have a reasonable degree of predictability based on previous rulings in similar cases.
Independence	The independence of NERSA from the regulated companies is a prerequisite for any sound regulatory system. Independence from political influence is also desirable to ensure the long-term stability of regulatory practices. Avoidance of regulatory capture by some customer groups is also necessary for successful regulation.
Accountability	We should be accountable for its actions and decisions. Independence must not be confused with the lack of accountability.
Integrity	We should exercise professionalism, honesty and objectivity in the management of the Energy Regulator's affairs and in all its dealings with stakeholders.

¹ This Framework was adopted by AFUR in November 2003.

5. Situational Analysis

5.1. External Environment Analysis

The performance environment of NERSA is impacted upon by energy demand and supply trends and developments in the global, continental, regional and national environments.

5.1.1. Global Trends

According to the World Economic Forum insight report², the following key issues of the energy system and energy transition are worth noting, as summarised below:

- a) Energy is a key element of the modern economy. The availability of secure and reliable energy supply is essential for industrial processes and the provision of public services such as lighting, heating, cooking, information and communication technology, and mobility.
- b) The energy system is undergoing unprecedented change, driven by forces such as technological innovation, changes in consumption patterns, supply dynamics and policy shifts. These forces offer opportunities to resolve the challenges that the global energy system faces today, namely:
 - providing energy access to the more than one billion people who lack it;
 - meeting demand for an additional two billion people by 2050 while delivering that energy at an affordable cost; and
 - ensuring that the carbon and emissions footprint decline.
- c) The geopolitical landscape of energy is quickly shifting and environmental concerns have shaken the system's foundations. At the same time, the economics of competing energy sources have changed, and the advent of Fourth Industrial Revolution technologies have enabled new business models, while making others obsolete. The latter has created significant uncertainty about the pace and destination of the transformation, making a strong case for a systemic, multi-stakeholder approach that increases the transparency of the enablers and reforms needed for countries to achieve an effective energy transition.
- d) Energy systems are complex and are at the heart of every country's economy. These systems aim to support society in the three dimensions of the energy triangle, namely:
 - inclusive economic development;
 - environmental sustainability; and
 - secure and reliable access to energy.
- e) The boundaries of energy systems have recently started shifting. The stakeholders are diverse, including:
 - end users and industrial consumers;
 - energy companies;
 - financial sector entities;
 - policy-makers;
 - cities;
 - international energy organizations; and
 - civil society.

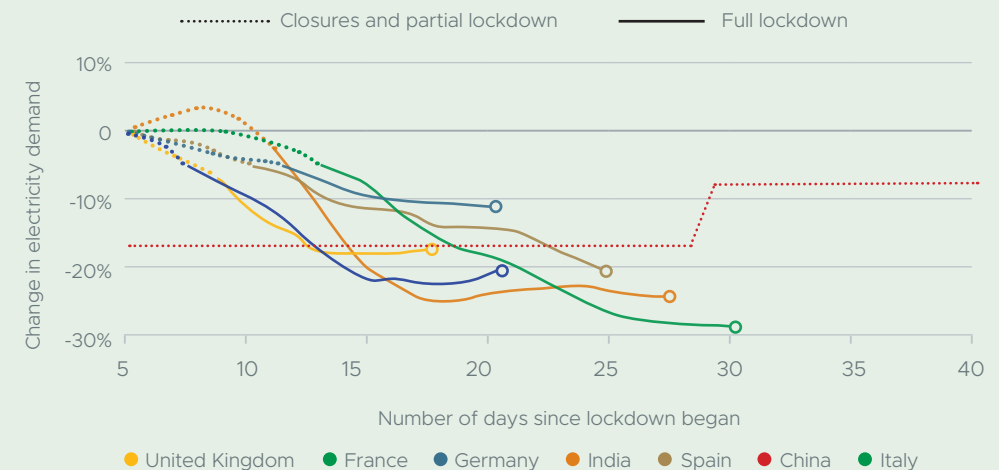
² World Energy Forum report (2018) on Fostering Effective Energy Transition: A Fact-Based Framework to Support Decision-Making

- f) In the last decade the following trends have emerged:
- Technological progress has allowed new forms of producing, storing, transforming and consuming energy, altering the nature of the energy system.
 - Energy consumption patterns have fundamentally shifted, resulting in new demand dynamics.
 - Policy-makers have started to adapt energy policies, and new coalitions have been formed to address challenges and harness opportunities associated with these developments.
- g) Countries can use these game-changing trends to enhance their energy systems and improve the wellbeing of their populations.
- h) The COVID-19 pandemic is unprecedented in its scale and speed in recent times, and it has the potential to redefine economic, political and social aspects relevant to the energy transition. It has forced countries around the world to change and relinquish valuable commodities and freedoms to collectively address this COVID-19 outbreak. An effect of similar magnitude is required for a successful energy transition. Beyond the uncertainty over its long-term consequences, it has unleashed gushing effects in real time. Compounded disruptions from the erosion of almost a third of global energy demand, delayed or stalled investments and projects, uncertainties over the employment prospects of millions of energy-sector employees, in addition to unprecedented oil price volatilities and subsequent geopolitical implications have created a perfect storm for energy markets. The 'new Earth' that will emerge after COVID-19 will be a 'new normal', but many fundamental challenges will still exist. Chief among them is the imperative to collectively work towards an effective and inclusive energy transition (World Economic Forum insight report).

5.1.2. The impact of the COVID-19 crisis on global energy demand and CO² emissions

- a) International Energy Agency (IEA) (2020) daily data shows that through mid-April countries in full lockdown experienced an average 25% decline in energy demand per week and countries in partial lockdown an average 18% decline. Daily data collected for 30 countries until 14 April, representing over two-thirds of global energy demand, show that demand depression depends on duration and stringency of lockdowns (see Figure 1 below).
- b) Global energy demand declined by 3.8% in the first quarter of 2020, with most of the impact felt in March as lockdown measures were enforced in North America, Asia, Europe and elsewhere.

Figure 1: Evolution of electricity demand following lockdown implementation³

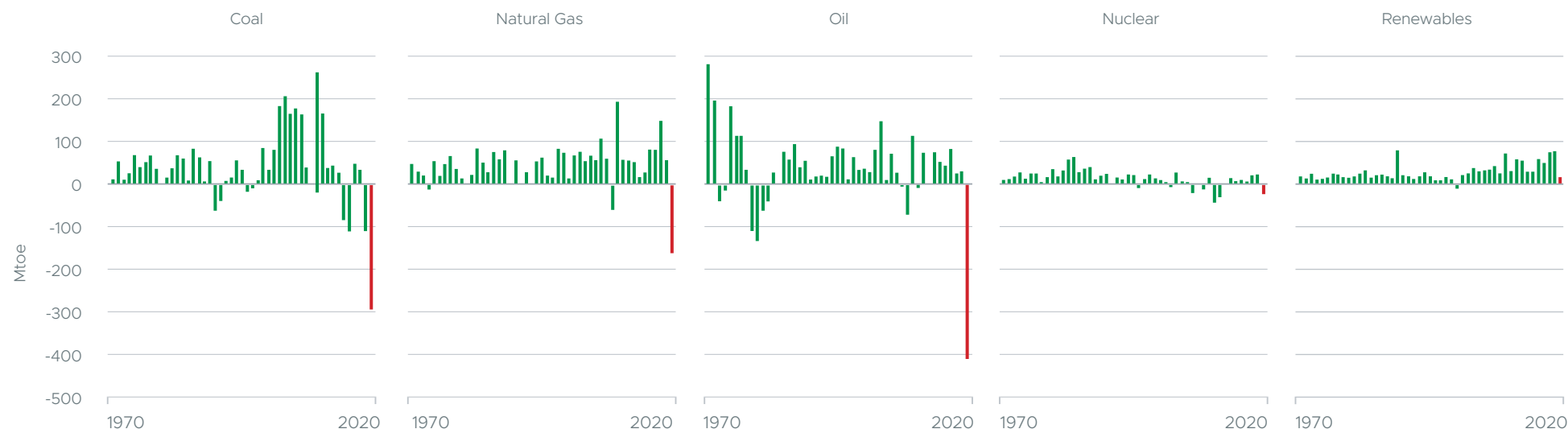


Source: International Energy Agency (2020)

³ Electricity demand drops to Sunday levels under lockdown, with dramatic reductions in services and industry only partially offset by higher residential use. Service-based economies suffer the most.

- c) **Global coal demand** was hit the hardest, falling by almost 8% compared with the first quarter of 2019 (see Figure 2 below). There are three reasons that have advanced to explain falling global coal demand, First, China – a coal-based economy – was the country the hardest hit by COVID-19 pandemic in the first quarter of 2020; Second, cheap gas and continued growth in renewables elsewhere challenged the usage of coal; and third, mild weather also capped coal use.
- d) **Oil⁴ demand** was also hit strongly, down nearly 5% in the first quarter, mostly by curtailment in mobility and aviation, which account for nearly 60% of global oil demand. By the end of March, global road transport activity was almost 50% below the 2019 average and aviation 60% below.
- e) The impact of the pandemic on **gas demand** was more moderate, at around 2%, as gas-based economies were not strongly affected in the first quarter of 2020.
- f) **Renewables** were the only source that posted a growth in demand, driven by larger installed capacity and priority dispatch.
- g) **Electricity demand** has been significantly reduced as a result of lockdown measures, with knock-on effects on the power mix (see Figure 1 on the previous page). Electricity demand has been depressed by 20% or more during periods of full lockdown in several countries, as upticks for residential demand are far outweighed by reductions in commercial and industrial operations. For weeks,

Figure 2: Change in global energy demand by fuel, 1970–2020⁵



Source: International Energy Agency (2020)

4 Oil means crude oil and the refined product produced from oil.

5 Coal is set for the largest decline since World War II, alongside sharp reductions for gas and oil. Nuclear power is less affected by lockdown measures, while renewables are the only energy source on the rise in 2020.

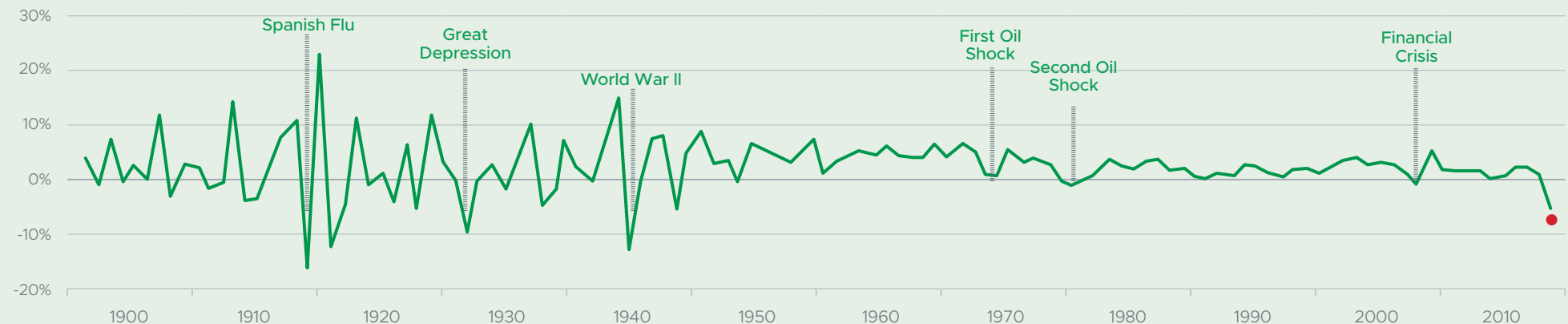
the shape of demand resembled that of a prolonged Sunday. Demand reductions have lifted the share of renewables in the electricity supply, as their output is largely unaffected by demand. Demand fell for all other sources of electricity, including coal, gas and nuclear power. Global electricity demand falls by 5%, with 10% reductions in some regions. Low-carbon sources would far outstrip coal-fired generation globally, extending the lead established in 2019.

- h) Looking at the full year ahead (2020), the International Energy Agency (2020) explored a scenario that quantifies the energy impacts of a widespread global recession caused by months-long restrictions on mobility and social

and economic activity. Within this scenario (forecast), the recovery from the depths of the lockdown recession is only gradual and is accompanied by a substantial permanent loss in economic activity, despite macroeconomic policy efforts.

- i) The IEA (2020) forecasts that energy demand contracts by 6%, the largest in 70 years in percentage terms and the largest ever in absolute terms. The impact of COVID-19 on energy demand in 2020 would be more than seven times larger than the impact of the 2008 financial crisis on global energy demand (see Figure 3 below).

Figure 3: Change in global energy demand, 1990-2020



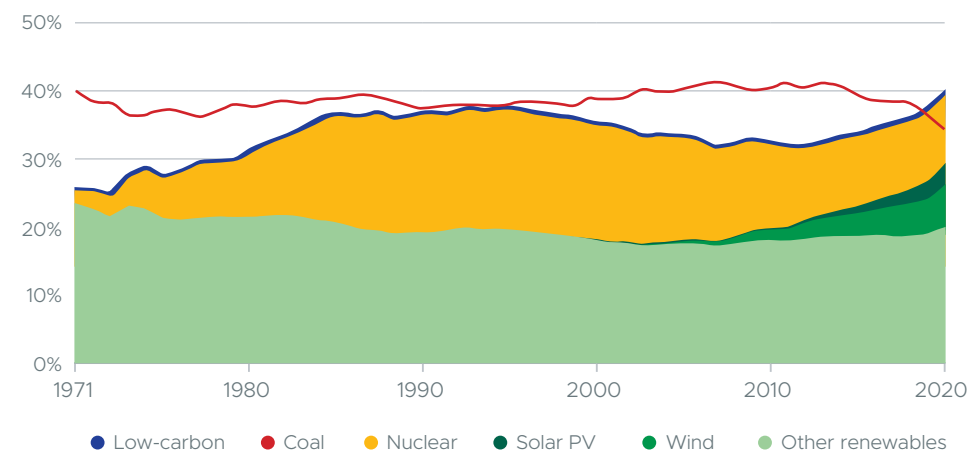
j) All fuels will be affected as follows:

- i) **Oil demand** could drop by 9%, or 9 mb/d on average across the year, returning oil consumption to 2012 levels.
- ii) **Coal demand** could decline by 8%, in large part because electricity demand will be nearly 5% lower over the course of the year. The recovery of coal demand for industry and electricity generation in China could offset larger declines elsewhere.
- iii) **Gas demand** could fall much further across the full year than in the first quarter, with reduced demand in power and industry applications.
- iv) **Nuclear power demand** would also fall in response to lower electricity demand.
- v) **Renewables demand** is expected to increase because of low operating costs and preferential access to many power systems. Recent growth in capacity, some new projects coming online in 2020, would also boost output.

k) Global CO₂ emissions are expected to decline by 8%, or almost 2.6 gigatonnes (Gt), to levels of 10 years ago (see Figure 4). Such a year-on-year reduction would be the largest ever, six times larger than the previous record reduction of 0.4 Gt in 2009 – caused by the global financial crisis – and twice as large as the combined total of all previous reductions since the end of World War II. As after previous crises, however, the rebound in emissions may be larger than the decline, unless the wave of investment to restart the economy is dedicated to cleaner and more resilient energy infrastructure.

l) CO₂ emissions drop the most due to the COVID-19 crisis (see Figure 5 on the next page). Global energy-related CO₂ emissions are set to fall nearly 8% in 2020 to their lowest level in a decade. Reduced coal use contributes the most. However, evidence suggests that a large rebound is likely post the COVID-19 crisis.

Figure 4: Global generation shares from coal and low-carbon sources, 1971-2020⁶

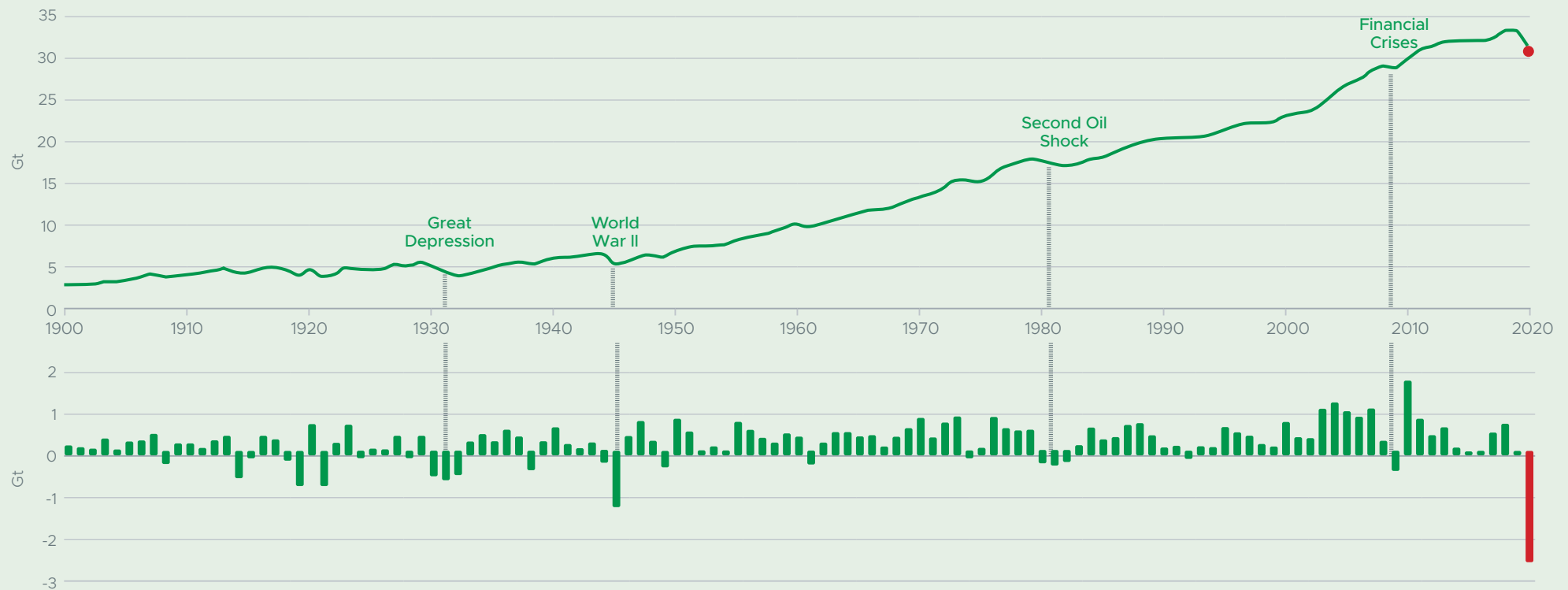


m) The COVID-19 crisis and measures taken to slow its spread have had a profound impact on energy demand, the likes of which have not been seen for 60-70 years. The full impact of the current situation, is yet unknown, will be determined by the duration of lockdown measures and the recovery paths taken around the world. This unprecedented situation and the stimulus packages that governments around the world are putting in place will shape the energy sector for years to come, with significant consequences for the energy industry at large, energy security and clean energy transitions.

n) The industry is feeling the financial impact throughout value chains, with most energy companies losing substantial revenues. In effect, they are being hit twice, first by lower demand for their products – including oil, gas, coal and electricity – and again by lower prices for these products. Average oil prices fell sharply, with West Texas Intermediate hitting negative prices for the first time in history as excess storage became scarce.

⁶ For the first time in 50 years, low carbon technologies overtook coal as the leading source of electricity in 2019, and they are moving further ahead in 2020

Figure 5: Global energy-related CO2 emissions and annual change 1900-2020



- o) LNG prices have declined to all-time lows in European and Asian markets, which were abundantly supplied even before the COVID-19 crisis depressed demand. Natural gas prices have gone negative in parts of the United States, where storage is full. The smallest impact is on coal: as the supply chain is less affected by logistical constraints than oil and natural gas. A combination of cheap gas and weakening demand have also led to power prices declining by one-third to one-half in liberalised wholesale markets. Market prices for electricity have dipped below zero in the United States and a number of countries in Europe, including Germany, Denmark, France, Belgium, Sweden, Finland and Switzerland.
- p) The energy sector that emerges from the COVID-19 crisis may look significantly different from what came before. Low prices and low demand in all subsectors will leave energy companies with weakened financial positions and often strained balance sheets. Business lines that are insulated to a degree from market signals, including those with renewable electricity projects, will emerge in the best financial position. Private firms that are the most exposed to market prices will experience the most severe financial impacts. Market concentration and consolidations are likely.

- q) Across the energy sector, the COVID-19 crisis will have a significant impact on investment. This could raise concerns about energy security because investment is necessary even if global energy demand takes a long time to return to the pre-crisis trajectory. A considerable proportion of global energy investment is devoted to just sustaining existing levels of energy supply: maintaining oil and gas production at current levels, replacing aging power generation capacity – often with a capital-intensive combination of renewables and flexibility sources – and reinvesting in aging electricity networks. Investment in these activities will have to remain robust even with a subdued recovery.
- r) Energy security has been put to the test in new ways by the crisis, including in oil and gas markets. Simultaneous supply and demand shocks have sent oil markets into turmoil. Oil plays a central role in global macro finance, both as a share of international trade and as a critical source of government revenues for several major producers. National lockdown measures have caused unprecedented demand declines, whose speed and magnitude greatly exceed the normal market flexibility of supply. As a result, even with attempts at coordinated management, a disorderly production shutdowns resulted some places. The consequent macroeconomic and financial disruptions could undermine the industry's ability to ramp up production as the world economy and oil demand recover. The supply situation has stabilised as countries globally relaxed lock down restrictions.
- s) The supply of **natural gas** is critical to operations in all sectors, including industry, residential and services heating, and electricity supply. Due to large investments in recent years and the slump in demand because of COVID-19, global gas markets are abundantly supplied and storage levels are very high. At the same time, intense financial strain is hurting the industry, including companies who own and operate critical infrastructure facilities. Policymakers and regulators need to ensure that operational, maintenance and safety expenditures are prioritised and appropriately maintained. US liquefied natural gas (LNG) has played a major role in improving energy security and market efficiency in several regions, but the ongoing challenging market conditions risk significant shut-in of US LNG facilities.
- t) **Electricity** security's place at the heart of modern economies has been underscored by the COVID-19 crisis. A robust, uninterrupted electricity supply is a key precondition of both the functioning of the health care system and the maintenance of social welfare and online economic activity. Robust power systems have enabled adaptations to the ongoing crisis, including a huge expansion of teleworking activities, particularly in advanced economies. In some parts of the world, however, a reliable supply cannot be taken for granted. In Africa, several thousand hospitals and health care facilities have no access to electricity. In both Africa and South Asia, electricity reliability problems limit social distancing.
- u) Electricity security has remained robust as the COVID-19 crisis has accelerated the shift to renewable energy in the power mix. The share of renewables has jumped several years ahead of pre-pandemic expectations, including the shares of wind and solar, curbing CO2 emissions and air pollution. The rise of renewables has posed some problems for electricity security. However, in advanced economies, the main cause of blackouts is the inability of the system to manage sudden changes in power flows and various network problems. Lower electricity demand paired with continued growth of wind and solar PV has stepped up the share of variable renewables, calling for more flexibility to keep the lights on. At the same time, available flexibility has been limited by the shutdown of industrial facilities that provide demand response and because dispatchable power plants are idle because power prices are extremely low. As the energy industry's financial challenges grow, the cost of restarting dispatchable power capacity that had been mothballed could emerge as a significant energy security concern as economies and electricity demand recover. To date, electricity systems in major economies have maintained robust reliability, but continuous vigilance will be needed from system operators, regulators and governments.
- v) The COVID-19 crisis is also influencing the path for **clean energy transitions**. Global CO2 emissions are set for the largest year-to-year reduction on record, but a sustainable energy pathway calls for continuous efforts and commitment. The unprecedented decline in emissions in 2020 may only be temporary

without structural changes. Recoveries from past crises have caused immediate rebounds in CO₂ emissions, including the highest year-on-year increase on record in 2010.

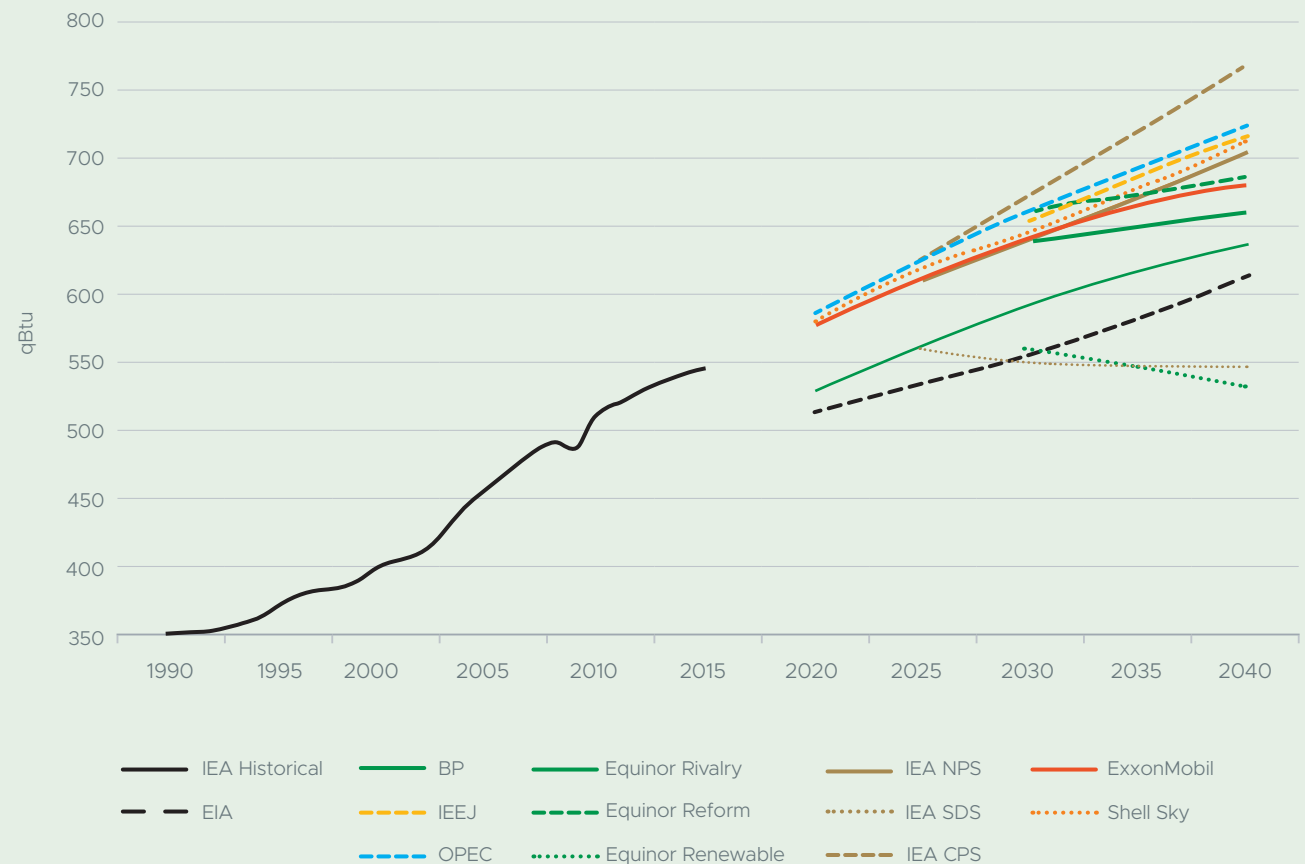
- w) Importantly, governments around the world will play a major role in shaping the energy sector's recovery from the COVID-19 crisis, just as they have long been in the driving seat in orienting energy investment. In particular, the design of economic stimulus packages presents a major opportunity for governments to link economic recovery efforts with clean energy transitions – and steer the energy system onto a more sustainable path. While the clean energy transitions and stimulus discussions are gathering momentum, a co-ordinated policy effort will be needed to harvest its opportunities and lead to a more modern, cleaner and more resilient energy sector for all.

5.1.3. Global Energy Consumption and Demand Trends

- a) The global energy sector has changed dramatically over the last 25 years, with larger changes possible over the next 25. The magnitude and direction of these changes, however, is highly uncertain. According to the Global Energy Outlook (2019), global primary energy consumption has grown

rapidly over the past 25 years, reaching 546 quadrillion Btu (qBtu) in 2015, more than 190 qBtu higher than 1990 levels. Over the next 25 years, growth is projected to slow down, increasing by roughly 110 to 160 qBtu in Evolving Policies scenarios, and declining by as much as 4 qBtu under Ambitious Climate scenarios (see Figure 6 below).

Figure 6: Global primary energy consumption⁷

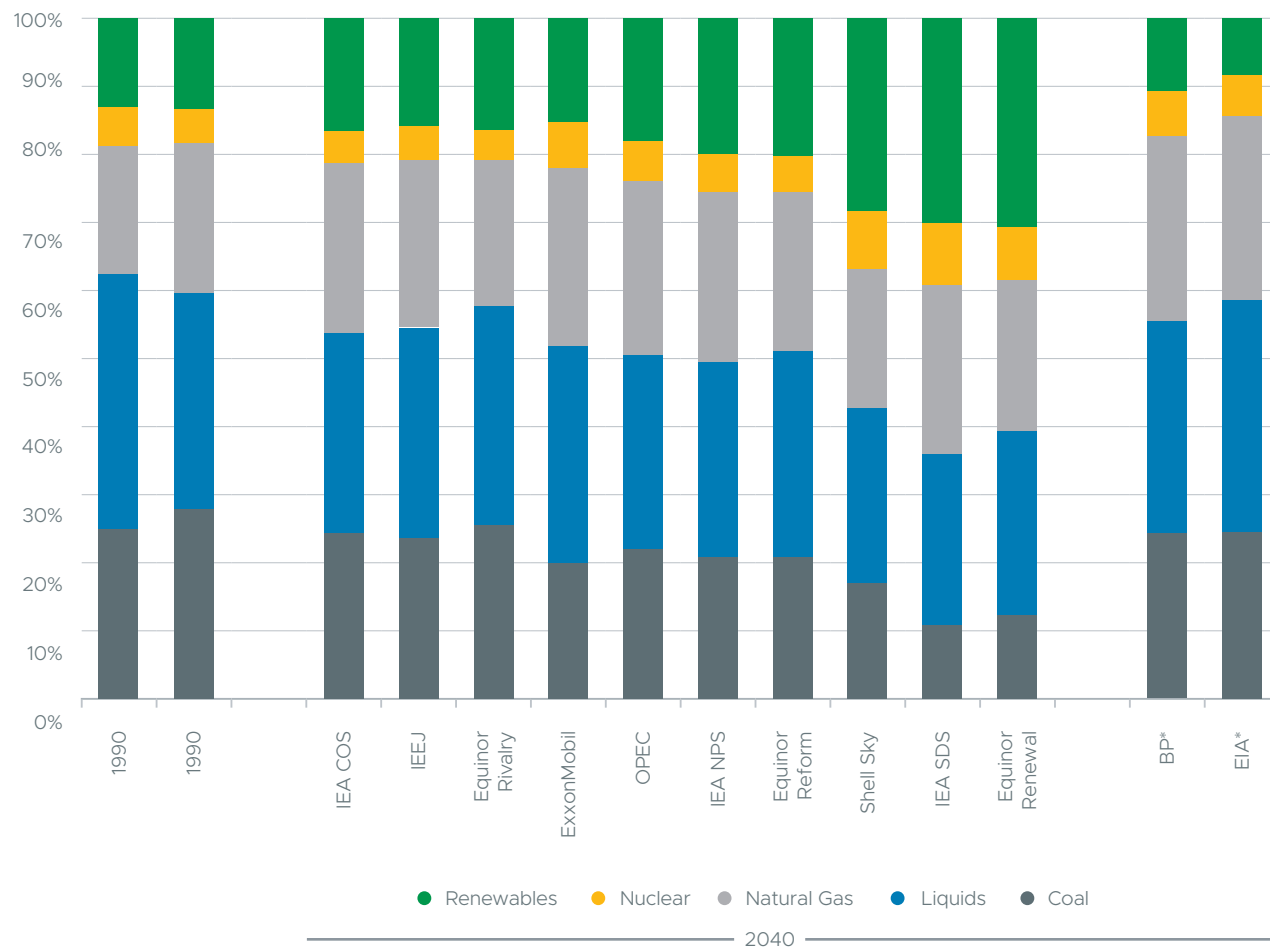


7 Global Energy Outlook (2019)

- b) Global energy consumption is marked by a series of deep disparities with more than 1 billion people with no access to electricity. In addition, the gap between expectations of fast, renewables-driven energy transitions and the reality of today's fossil fuel-dependent energy remains significant.
- c) The International Energy Outlook current policies scenario (IEA CPS) shows the highest consumption in 2040 at 767 qBtu, an increase of 41% over 2015. OPEC and the Institute of Energy Economics Japan (IEEJ) project consumption of roughly 720 qBtu in 2040, similar to the absolute levels of growth from the previous 25 years. Evolving Policies scenarios project moderately slower growth, led by the IEA new policy scenario (NPS) (703 qBtu), ExxonMobil (681 qBtu), and Equinor's Reform Scenario (659 qBtu). Under two of three Ambitious Climate scenarios (IEA sustainable development scenario (SDS) and Shell Sky), global energy consumption is roughly flat to 2040. In the IEA SDS, demand is 544 qBtu in 2040, while Equinor Renewal projects consumption falling to 534 qBtu in 2040. On the other hand, under Shell's Sky, demand grows to 711 qBtu by 2040, higher than any Evolving Policies scenarios.
- d) With regard to the shares of global primary energy consumption by fuel projections, the Global Energy Outlook (2019) report states that fossil fuels, which made up 82% of global primary

energy in 2015, dominate across Reference and Evolving Policies scenarios, ranging from 74% to 79% in 2040 (see Figure 7 below). Under Ambitious Climate scenarios, fossil fuels decline from 60% to 62%.

Figure 7: Shares of global primary energy consumption by fuel⁸



- e) Liquid fuels (primary oil) will continue to be the single largest fuel source in the energy mix across most outlooks, though its share shifts from 32% in 2015 to between 28% and 32% in the Evolving Policies scenarios. Under ambitious climate policies, liquids still account for 26% to 27% by 2040, but of a smaller aggregate energy base in the case of IEA SDS and Equinor Renewal.
- f) On the other hand, natural gas becomes the second largest source in most outlooks, rising from 21% in 2015 to between 21% and 27% by 2040.
- g) According to forecasts, coal loses market share across all projections. Under Ambitious Climate scenarios, coal declines from 28% of the mix in 2015 to between 12% and 17% by 2040. Under Evolving Policies, it falls to 20% to 22%.
- h) Renewables, led by wind and solar, will grow under all projections, though the rate of growth varies widely. Under Reference scenarios, renewables increase from 14% of the mix in 2015 to between 16% and 17%. Under Ambitious Climate scenarios, they become the largest source of global primary energy, overtaking petroleum to reach as high as 31% in 2040.
- i) Projections for nuclear's share of the mix also vary substantially, and is highest under Ambitious Climate scenarios, where it provides 8% to 9% of global primary energy, up from 5% in 2015. For other scenarios, nuclear accounts for 4% to 7% of the mix.
- j) According to the BP Report (2019)⁹, world energy demand is projected to grow by 1.3% per annum from 2016 to 2040 with all the growth coming from emerging economies. China and India will account for over a quarter of this increase. Global energy intensity [the ratio of energy demand to Gross Domestic Product (GDP)] is projected to decline by 1.9% per annum over this period. Renewables are the fastest growing fuel source, however oil and gas are still expected to account for more than half of global energy in 2040. Coal demand peaks, with its share of primary energy expected to fall to 21% by 2040. Natural gas is expected to replace coal as the second largest source of energy, after oil.
- k) The World Energy Outlook (2018) report notes that as economies continue to grow, energy demand grows as well. Average GDP in the non-Organisation for Economic Co-operation and Development (OECD). Over the past 25 years, world economic growth has been led by the non-OECD countries, accompanied by strong growth in energy demand in those countries. From 1990 to 2015, real GDP grew by 4.9% per year in the non-OECD, compared with 2.1% per year in the OECD. In the future, the difference in economic growth rates between OECD and non-OECD countries is expected to narrow somewhat, as economic growth in non-OECD countries moderates, and as their industrial sectors move from reliance mainly on production in energy-intensive industries to more service-oriented industries.
- l) The emerging trends are as follows¹⁰:
- Renewables are the world's fastest-growing energy source over the projection period. Renewable energy consumption is expected to increase by an average of 2.6% per year between 2012 and 2040.
 - Nuclear power is the world's second fastest growing energy source, with consumption increasing by 2.3% per year over that period.
 - Even though the consumption of non-fossil fuels is expected to grow faster than the consumption of fossil fuels, it is projected that fossil fuels will still account for 78% of energy use in 2040.
 - Natural gas is expected to grow faster than other fossil fuels in the next two decades. Abundant natural gas resources and robust production, including rising supplies of tight gas, shale gas, and coalbed methane, will contribute to the strong competitive

9 BP Statistical Review of World Energy, 2019

10 World Energy Outlook (2018) report: The gold standard of energy analysis

position of natural gas. Shell has warned in its annual report released in March 2018 that there could be a shortage in the Liquefied Natural Gas (LNG) market in the next decade unless new investment is undertaken soon. Investment decisions on new LNG supply have come to a near standstill over the last two years. In 2017, only one large-scale LNG project reached Final Investment Decision, namely the 3.4 MTPA Coral South FLNG in Mozambique, marking the lowest volume of sanctioned LNG in nearly twenty years (IGU, 2018)¹¹. According to the IGU (2018), the total volume and number of LNG contracts signed has declined consistently over the past three years.

- Although liquid fuels (mostly petroleum based) will remain the largest source of world energy consumption, the liquids share of world market energy consumption falls from 33% in 2012 to 30% in 2040. Contributing to the decline are rising oil prices in the long term, which lead many energy users to adopt more energy efficient technologies and to switch away from liquid fuels.
- Coal, the world's slowest growing energy source, will rise by 0.6% per annum and will be surpassed by natural gas by 2030.

Oil/Petroleum

- m) World consumption of liquid fuels rises from 95 million barrels per day (b/d) in 2015 to 113 million b/d in 2040 (International Energy Organisation, 2017). Most of this growth is seen in the transportation and industrial sectors with an average increase of 0.7% per year from 2015 to 2040. Non-OECD nations account for most of the increase, with demand rising by 1.3% per year compared with a slight decrease in the OECD. Most of the growth (80% of the total increase) in world liquid fuels consumption from 2015 to 2040 comes from non-OECD countries, where strong economic and population growth increase the demand for liquid fuels by 39%.
- n) The use of petroleum and other liquids in the industrial sector to power equipment and serve as chemical feedstocks will increase slowly between 2015 and 2040. Furthermore, the use of petroleum and other liquids to generate electricity declines over the projection period as a result of increasing oil prices and relatively less costly natural gas, encouraging producers to switch to alternative energy sources.
- o) Energy security remains paramount, and oil remains in the spotlight, as a result, a broader and dynamic approach is required to ensure energy security.

Natural gas

- p) Global natural gas consumption is expected to grow in both the OECD and non-OECD countries from 2015 to 2040. However, the growth is higher with an expected average of 1.9% per year in non-OECD countries that have expanding industrial sectors and electricity demand, in contrast to 0.9% per year in OECD countries. The share of world natural gas consumption in non-OECD countries increases from 53% in 2015 to 59% in 2040. Natural gas continues to be an attractive fuel for the electric power and industrial sectors in many countries, accounting for nearly 75% of the projected increase in total consumption between 2015 and 2040. Natural gas-fired generation is attractive for new power plants because of low capital costs, favourable heat rates, and relatively low fuel cost. Natural gas-intensive industries, such as chemicals, refining, and primary metals, are expected to expand over the period of 2015 to 2040 – particularly in non-OECD countries – driving industrial demand higher. The largest increases in natural gas production from 2015 to 2040 occur in the Middle East (11.8 Tcf), China (9.5 Tcf), the United States (10.7 Tcf), and Russia (4.8 Tcf).
- q) Demand for natural gas is expected to grow by more than half, the fastest rate among the fossil fuels, and increasingly flexible global trade in LNG offers some protection against the risk of supply

11 International Gas Union (IGU) World Gas LNG Report – 2018 Edition, 27th World Gas Conference Edition

disruptions. The growth in LNG increased by 29 million tonnes to 293 million tonnes in 2017. The main regions that push global gas demand higher are China and the Middle East, but gas is also expected to become the leading fuel in the OECD energy mix by around 2030. China has overtaken South Korea to become the second-largest importer of LNG as a result of switching its policies from coal to gas to reduce air pollution¹². Japan is still the largest LNG importer, but according to Capital Markets Outlook 2018, a structural change in its energy policy could see it lose the top slot by as early as 2020. The key uncertainty is whether gas can be made available at prices that are attractive to consumers while still offering incentives for the necessary large capital-intensive investments in gas supply.

- r) The global interest in LNG power generation is increasing. There is also a growing demand for LNG as a bunkering fuel worldwide, albeit slowly. From 2018, ships operating in European waters will be required to report their annual greenhouse gas emissions. At the moment, ships operating in Europe must comply with a 0.5% sulphur limit, but there are options aside from using LNG as a fuel, such as installing scrubbers. The 0.5% limit will apply globally from 2020, down from the current 3.5%¹³. LNG exports from the Americas are set to rise from 2018 as supplies ratchet up from the United States, Trinidad and Tobago and Peru.

The three countries exported a combined 19.74 mt of LNG during the first nine months of 2017, a year-on-year increase of 51.5%. The LNG market is expected to grow going forward (2017 – 2021)¹⁴. The contributing factors for this rapid increase is the start-up of several new projects in Australia and Indonesia, rapid economic growth especially of emerging economies, and rising demand for environmentally cleaner fuels. Global Gas Analytics (GGA) forecasts that LNG exports from the Americas to increase by 10% year-on-year in 2018, to 28.4 mt.

Coal

- s) According to the *Coal transitions in South Africa Report, 2018*, South Africa's Nationally Determined Contribution (NDC) is based on the long-term benchmark emissions trajectory range, which is contained in the National Climate Change Response Strategy White Paper (DEA, 2011). The NDC commits to limiting emissions to a range between 398 and 614 Mt CO₂-eq, between 2025 and 2030. Known as the Peak, Plateau, and Decline trajectory (PPD), the goal is to peak emissions between 2020 and 2025, plateau for approximately a decade and decline in absolute terms thereafter (RSA, 2016). The National Climate Change Policy Framework thus extends the NDC commitment to 2050, with a goal to reduce emissions to between 212

and 428 Mt CO₂-eq in 2050 (DEA, 2011). The key finding of the NDC scenario is that South Africa can meet its NDC and mid-PPD primarily through decarbonising the electricity sector. The scenario results in 71% of electricity generated from wind and solar photovoltaic (PV) by 2050. There is substantial investment in gas capacity because of a conservative assumption that renewable energy cannot be considered firm capacity during the peak, though the gas plants contribute relatively less to electricity generated (14%).

- t) Considering the role of coal in South Africa's economy, it should be noted that coal is an important foreign exchange earner. It accounts for approximately 12% of the total merchandise exports from South Africa over the period 1993 to 2015 (CoM, 2016). The State benefits via taxes and royalties associated with coal mining. Coal royalties are approximately 18% of total mining royalties. The coal-mining sector employed around 77,000 workers in 2015. In comparison, the entire mining sector employed approximately 457,000 workers in 2016 (Chamber of Mines, 2016), out of a total employed workforce of 15.8 million people (StatsSA, 2017). Coal jobs therefore account for nearly 0.5% of the national workforce¹⁵.
- u) According to the International Energy Outlook, forecast worldwide coal consumption remains roughly the same between 2015 and 2040 (about

¹² 4th Quarter report on the development of new gas sources in South Africa and neighbouring countries

¹³ 4th Quarter report on the development of new gas sources in South Africa and neighbouring countries

¹⁴ http://www.researchandmarkets.com/research/s9wds5/global_liquefied

¹⁵ Coal transitions in South Africa Report, 2018

160 quadrillion Btu), with decreasing consumption in China and the United States offsetting growth in India. China remains the largest single consumer of coal in 2040 (about 73 quadrillion Btu), despite a steady decline in the country's consumption over time. A slowing economy and plans to implement policies to address air pollution and climate change emphasises the decline over the projection period. India's coal consumption continues to grow by an average of 2.6% per year from 2015 to 2040, with the country surpassing the United States as the second-largest coal consumer before 2020.

- v) In OECD countries, coal consumption declines by an average 0.6% per year over the period of 2015 to 2040 because of increasing competition from natural gas and renewables and only moderate increases in electricity demand. Africa, the Middle East and other non-OECD countries, are projected to gradually expand coal capacity and generation through 2040, but their use of this resource starts from a low base. Despite significant increases in coal consumption, coal's share in overall energy consumption in India is projected to decrease from 49% in 2015 to 43% by 2040, due in part to policies promoting renewable and nuclear-based generation.

Electricity

- w) According to the World Energy Outlook (2018) report, electricity is the fastest-growing source of final energy demand, and over the next 25 years, it continues to outpace energy consumption as a whole. The power sector now attracts more

investment than oil and gas combined – necessary investments as the generation mix changes and ageing infrastructure is upgraded.

- x) According to forecasts, net electricity generation in OECD Europe is expected to increase slowly, by an average of 1.1% per year from 2015 to 2040, compared to the world average increase of 1.5% per year (International Energy Outlook, 2018). India's net electricity generation increases by an average of 3.2% per year over the same period, driven by strong industrial growth and policies to increase the availability of electricity in rural areas.
- y) The generation mix in OECD Europe changes considerably by 2040, with renewables and natural gas growing, coal remaining flat, and nuclear power and liquid fuels declining. Nuclear generation's share is expected to decline from around 25% in 2015 to less than 15% by 2040. This is a result of stated policies to either cap or phase out nuclear power, including those adopted in France, Germany, and Sweden. The use of natural gas electricity generation in OECD Europe does not expand until 2030, mostly because of the large increases in projected renewables generation. In OECD Europe, when natural gas begins to gain market share in 2030, it displaces nuclear power, coal, and renewable generation.
- z) The number of people without access to electricity declined from 1.7 billion in 2000 to 1.1 billion in 2016 and is forecast at 650 million by 2030 (World Energy Outlook, 2018). The remaining population without access becomes increasingly

concentrated in sub-Saharan Africa as developing countries in Asia reach a 99% electrification rate, with universal access achieved by the mid-2020s in India and Indonesia (see Figure 3 below).

- aa) The number of people without access to clean cooking falls, but only to 2.2 billion by 2030. According to the World Energy Outlook (2018) report, the greatest challenge in achieving universal access to electricity is providing access to people living in the most remote areas in sub-Saharan Africa. Although most of the access is done through generation from renewables, the grid expansion also has an important part to play.
- bb) Universal access strategies should be diverse. Local conditions and practices need to be underpinned by firm political commitments with supportive and enabling regulatory frameworks; engagement with the private sector; appropriate financing options and investment; capacity building and close consultation from the outset with local communities, especially women (World Energy Outlook, 2017).
- cc) Globally the dependence on electricity is growing and society is becoming more and dependent on the use of electricity for the sustainability of life as they know it. Cities would not survive without electricity. Yet as this is taking place, there are growing concerns about the security of supply. Apart from all the normal reasons for this, there is a new

threat that is attracting attention globally – cybersecurity and the vulnerability of the power system to cyber-attacks. This is a global problem and South Africa is not excluded. However, this is an area of regulation that has not yet been addressed.

dd) Globally, the trend in renewable energy that is receiving the most attention from regulators is the installation of rooftop solar PV from a domestic customer point of view. This is putting a big dent in utility revenues and there are implications for regulators as well, among others:

- the sustainability of licensees;
- restructuring of tariffs by licensees in response to Small-Scale Energy Generation;
- tariff structures for feeding power onto the grid;
- the network impact of these installations; and
- control of quality of supply for other customers.

The most difficult implication to deal with is the sustainability of the licensees. NERSA has addressed these issues in varying degrees, but it requires ongoing attention.

5.1.4. Continental Developments

a) Sub-Saharan Africa accounts for almost 14% of the world's population, but only 4.5% of global primary energy demand [619 million tonnes of oil equivalent (Mtoe)]. According to latest statistics from the World Energy Outlook (2017) report, the number of people without access to electricity in sub-Saharan Africa continues to decline, albeit slowly. Over 200 million people have gained access since 2000, less than the overall population increase. As a result, there remain more than 600 million people without access, despite an increase in the access rate of 20 percentage points to 43%.

b) Average energy consumption per person in most African countries is well below the world average of around 2 tonnes of oil equivalent (toe) per capita and is broadly comparable to India's average of 0.7 toe/capita.

African energy demand has been driven by the growing needs of North Africa, Nigeria and South Africa. In 2018, primary energy demand in Africa was more than 830 million tonnes of oil equivalent (Mtoe): North Africa (24%), Nigeria (19%), and South Africa (16%) together accounted for almost 60% of this despite making up only 35% of the population.

c) Furthermore, recent efforts have been uneven, with around 60% of the progress seen since 2011 concentrated in just four countries (Kenya, Ethiopia, Tanzania and Nigeria), which together account for only 31% of the population without

electricity access in sub-Saharan Africa. In Kenya, the access rate has increased by over 65 percentage points in 2000, to 73% today, and the Last Mile Connectivity Project aims to deliver universal access by 2022. In Ethiopia, electricity now reaches 45% of the population compared with 5% in 2000. The National Electrification Programme, launched in 2017, outlines a plan to reach universal access by 2025, aiming to reach 35% of the population with off-grid solutions.

d) In South Africa, while the current electrification rate is relatively high (84%) it has been declining since 2014, in large part because electrification in urban areas has not kept pace with migration from rural areas.

e) Energy demand in sub-Saharan Africa is very low. However, there are several factors pointing towards potentially rapid and prolonged growth in demand: strong economic expansion; increasing urbanisation; industrialisation and modernisation; a burgeoning middle class in many countries; as well as a legacy of unmet energy demand. The sub-Saharan Africa energy system is expected to expand rapidly by 2040 and so do the demands placed upon it. According to the World Energy Outlook Report (2018), the sub-Saharan Africa economy will quadruple in size, the population will nearly double (to 1.75 billion) and energy demand grows by around 80% by 2040. The capacity and efficiency of the system improve, and access to modern energy services grows, but many of the existing energy challenges are only partly overcome.

- f) Bioenergy demand will increase by 40% in absolute terms by 2040, exacerbating stress on the forestry stock. However, the share of bioenergy in the energy mix declines from above 60% to below half and the share of modern fuels edges higher. Oil demand will more than double to 4 million barrels/day (Mb/d) in 2040 [over 0.5 Mb/d is the residential use of Liquid Petroleum Gas (LPG) and kerosene] and becomes the second-largest fuel in the mix, overtaking coal. Natural gas use grows by nearly 6% per year, to reach 135 bcm.
- g) According to BP, Africa will have the highest Compound Annual Growth Rate ('CAGR') for oil and gas consumption over the next 20 years while having the lowest existing energy consumption base¹⁶. There is an urgency to address the current and future power supply, transmission and distribution needs. Therefore, the proven nature of Open Cycle Gas Turbines (OCGT) and Combined Cycle Gas Turbines (CCGT) technology coupled with the increased global volumes of LNG and potential for subdued future prices appear to offer an opportunity for African gas to power to grow. Africa has significant natural gas reserves, with increasing numbers of countries joining the list of countries that have discovered resources. For countries lacking domestic gas today, importing LNG for gas to power projects has become feasible due to the reason that there is an increase in countries that have discovered natural gas.
- h) The sub-Saharan Africa power system is expanding rapidly, with generation capacity quadrupling to 385 GW. The power mix becomes more diverse, with coal (mainly South Africa) and hydropower (all regions), being joined by greater use of gas (Nigeria, Mozambique, Tanzania), solar (notably in South Africa and Nigeria) and geothermal (East Africa). The share of renewables in total capacity more than doubled to 44%. The total power sector investment averages around \$46 billion per year, with just over half of it in transmission and distribution.
- i) Oil production will rise above 6 Mb/d by 2020, but will then taper off to 5.3 mb/d in 2040. Nigeria and Angola remain the dominant producers, although Uganda and Kenya are expected to ramp up oil output in the 2020s. Gas production will rise to 230 bcm in 2040, led by Nigeria, and there will be an expansion of the output from Mozambique (60 bcm in 2040), as well as Angola and Tanzania (each 20 bcm). Coal supply is expected to grow by 50% to reach 325 Mtoe, still concentrated in South Africa, but joined increasingly by Mozambique and others. Sub-Saharan energy exports are drawn increasingly towards Asian markets. Crude oil net exports will decline to just over 3.8 Mb/d by 2040, partly due to a greater share being refined and consumed domestically. Rising gas output from Mozambique and Tanzania will bring sub-Saharan LNG export towards 100 bcm by 2040 (around 17% of inter-regional LNG trade), and Mozambique joins South Africa as a key coal exporter.
- j) Furthermore, sub-Saharan Africa makes only a small contribution to the global energy-related CO2 emissions. It is envisaged that it will account for a mere 3% of the total in 2040, but is on the front line when it comes to the potential impacts of a changing climate. In particular, hydropower prospects can be affected by changing patterns of rainfall. The fuelwood and charcoal sectors operate largely outside the formal economy, meaning that policymakers have few levers to promote more sustainable forestry.
- k) Sub-Saharan Africa is rich in energy resources and holds approximately 50% of the continent's oil and gas resources and nearly all of the coal resources, but very poor in energy supply (International Energy Agency, 2017). The political instability in Sub-Saharan Africa limits the realisation of future gas infrastructure. A clear and comprehensive plan is needed to attract Foreign Direct Investment (FDI) into a country's gas sector.
- l) Natural gas resource-holders can power domestic economic development and boost export revenues, but only if the right regulation, prices and infrastructure are in place. The incentives to use gas within sub-Saharan Africa are expected to grow as power sector reforms and gas infrastructure projects move ahead. International

16 PwC (2018). Staking on tomorrow: Africa oil and Gas review report

Energy Agency, (2017) predicts that natural gas will nearly triple its share of the energy mix in Africa to 11% by 2040.

m) Sub-Saharan Africa has 221.6 trillion cubic feet of proved natural gas reserves. The Middle East has almost 13 times that amount and Eurasia has almost 10 times that amount. Sub-Saharan Africa produced 1.69 trillion cubic feet of natural gas in 2011, accounting for 1% of total global natural gas production. Natural gas production in Sub-Saharan Africa grew by an annual average of 10% over the past ten years. The growth was led by Nigeria, Equatorial Guinea, and Mozambique. Nigeria produces around two-thirds of the region's natural gas. The largest gas discovery was made in Egypt in the Zohr field with more than 30tcf of gas, which is located within the offshore Shorouk Block. Over the next year or two, Egypt plans to bring online all four trains of the first phase of Zohr, as well as expanding operations at the Abu Qir acreage and starting up the Atoll project and Phase 9B of the West Delta Deep Marine project¹⁷.

n) Sub-Saharan Africa exports about 1.22 tcf of natural gas and LNG via pipeline. Nigeria, Equatorial Guinea, and Mozambique are the only sizable natural gas exporters in the region. Angola joined the group in 2013 when it began exporting LNG. According to the IGU (2019) report¹⁸, several new gas projects came online in Algeria, leading to an increase of

0.8 MT to reach 12.4 MT of exports, which is the country's highest since 2014.

- o) The African Energy market has a required energy investment of US\$65 to US\$90 billion, with actual current investment at US\$23 billion. This translates to a funding gap of between US\$40 and US\$60 billion. To address this issue, the African Development Bank (AFDB) has since established a new fund for energy that is aimed at achieving universal access to energy by 2025. It envisages 200 million connections and doubling the continent's generation capacity by 2025. The AFDB fund aims to increase new off-grid connections by 130 million, new generation capacity by 160 GW and new clean cooking solutions by a further 150 million.
- p) With all these developments and growth in energy demand and supply, there is a need to harmonise regulations with regulatory authorities across the continent to ensure efficient energy landscape.

5.1.5. Regional Developments

a) Energy is vital to development in the Southern African Development Community (SADC). Beyond its use in daily life, fuel and electricity catalyse infrastructure projects that drive both regional integration and economic growth. As the SADC region industrialises on its path to improved human development, energy production and distribution

will only increase in importance. Recognising the fundamental role of energy in accomplishing its goals, the SADC passed the Protocol on Energy in 1996, which provides a framework for cooperation on energy policy among SADC Member States.

- b) Since the adoption of the Protocol on Energy, the SADC has enacted several strategic plans for energy development in the region. Although implementation of these strategies has been slow, the region has made significant strides, particularly in electricity. At present, nine Member States of the SADC have merged their electricity grids into the Southern African Power Pool (SAPP), reducing costs and creating a competitive common market for electricity in the region. Similarly, the SADC has established the Regional Electricity Regulatory Association (RERA), which has helped in harmonising the region's regulatory policies on energy and its subsectors.
- c) While the SADC is enacting a number of initiatives to address these issues, it has identified two chief points of focus, as follows:

- Electricity Generation – Southern Africa has ample resources for electricity generation, though it occasionally lacks the capacity for development.
- Hydropower and Renewable Energy – Renewable energy has grown in importance for both regional and global energy markets.

¹⁷ "1st phase of Zohr gas field about to be finished: Min." Egypt Today. January 2018

¹⁸ IGU World Gas LNG Report – 2018 Edition, 27th World Gas Conference Edition

- d) In 2015, the SADC also launched the Industrialisation Strategy and Road Map for 2015–2063. Based on the Strategy and Roadmap, a SADC Industrialisation Action Plan had been drafted which covers how industrialisation should unfold; competitiveness; regional integration; crosscutting issues; institutional arrangements; and the monitoring and evaluation process. The successful implementation of this roadmap is essential for socio-economic development in the region and will have a bearing on the activities undertaken by regulators – the energy requirements for meeting the regional growth targets of 4–7% per annum as part of the industrialisation process are expected to be enormous. There has also been cooperation by SADC Member States on the establishment of the SADC Centre for Renewable Energy, Energy and Efficiency (SACREEE) in Namibia and the Southern Africa Research and Documentation Centre, which will function as platforms for capacity building, distribution of energy-related information, and energy-related projects.
- e) The region is relatively well endowed with energy resources. The SADC region has vast energy potential from solar, wind, nuclear, hydro, thermal, gas and petroleum sources in several countries. However, biomass is by far the largest source of energy in most regional countries.
- f) Electricity, as the dominant source of energy in the region, is generated mainly through thermal or hydroelectric resources. The coal industry is the backbone of power generation in the region,

supplying almost 62% in Southern Africa and a significant share of the resource is allocated for export. The region has a large reserve of low-cost hydroelectricity in the north [especially Inga Reservoir in the Democratic Republic of Congo (DRC)] and Kariba Dam on the Zambia/Zimbabwe border in the middle of the regional system, as well as large reserves of cheap coal in Botswana, Mozambique, South Africa and Zimbabwe.

- g) Natural gas is becoming more significant to the region's energy sector, as Mozambique, Namibia, South Africa and Tanzania are developing the natural gas fields in their respective countries. New natural gas discoveries by international oil companies in Mozambique and Tanzania during the past decade have ignited investor interest in this previously under-explored region. The nascent petroleum and gas sub-sector is however plagued by volatile prices. Although the region is endowed with some petroleum and gas resources, these are not directly available to the region due to either foreign commitments or the lack of the necessary infrastructure to exploit, process, store and distribute throughout the region.
- h) Furthermore, the region has some of the most significant known reserves of uranium. The mineral is being mined in Namibia and South Africa for use as fuel for nuclear power plants while exploration is underway in Botswana and Zimbabwe. Nuclear technology is included in the electricity sub-sector, but it must be demonstrated that nuclear power can be a safe electricity generation option and the

confidence of the population and governments must be won to endorse nuclear energy deployment in the SADC region. Only South Africa has nuclear capacity, with tentative plans for a new nuclear programme.

- i) The region has great potential for renewable energy, including hydropower, which is already being utilised on a commercial scale. However, the necessary infrastructure for grid connection is poor. The prices for most renewable energy technologies are decreasing, but more must be done in the form of innovative financing. A key factor of the SADC energy sector is the fact that the region has faced an electricity deficit since 2007 due to a combination of factors that have contributed to a diminishing generation surplus capacity against an increasing growth in demand. In recent years, the sub-region has experienced a power deficit situation due to a number of reasons, including growing demand against limited expansion in generation capacity.

Electricity

- j) Only 32% of rural areas in the region have access to electricity, as a result the SADC region falls behind in Africa regarding access to electricity.
- k) Although plans have been put in place to address the supply shortage by 2020, projects intended to address the shortage lag behind the deadline due to failure to package projects for funding, below-cost tariffs, poor project preparation, issues with Power

Purchase Agreements (PPAs), and the absence of regulatory frameworks, among other constraints. Massive investment in generation, transmission and distribution infrastructure will be required to sustain the projected increase in power demand in the region. Between US\$93 billion and US\$212 billion is required for short and long-term projects to boost power supply by 2027.

- l) One of the most pressing constraints is the need to improve the transmission line capacity and strengthen the regional grid. Approximately 60–70% of the matched bids in the Southern African Power Pool cannot take place due to transmission capacity constraints. Eskom, for example, would be able to sell all of its ‘excess’ capacity to its northern neighbours if the transmission capacity existed.
- m) More than 24,000 MW of new generation capacity was commissioned between 2014 and 2017. A number of rehabilitation and new generation projects are being undertaken to address the generation supply gap. The SAPP Plan indicates that 57,000 MW would need to be commissioned in the next 20 years. The generation mix is expected to change in the future with more emphasis on renewable energy particularly hydropower development. Currently, hydropower constitutes 21% of the generation mix, which will increase to at least 26% in the next 20 years. However, there is a need to diversify the energy source base in view of the experiences of Zambia and Zimbabwe, particularly in 2015, when hydropower generation dropped by nearly 40% due to low water levels

in the Zambezi river and the Kariba Dam as a result of poor rainfall. This, therefore, calls for the prioritisation of solar and other renewable energy projects in line with the climate change efforts being pursued internationally.

- n) Nearly all the SAPP Member States have high solar penetration levels, which provide great potential and a meaningful contribution of solar energy to the current power deficit. The total renewable energy contribution is expected to rise to at least 35% of the regional energy mix by 2030.
- o) Renewable energy targets in the SADC region are provided in Table 4 on the next page.

Table 4: Renewable Energy Targets in the SADC Member States¹⁹

Country	Sector/Technology	Target
Angola	Electricity access Renewable energy (small-scale) Hydropower Biofuels	Increase in renewable energy capacity of the following amounts by 2025: • Small hydro: 100 MW, with 60 MW for municipalities • Solar: 100 MW, with 10 MW off-grid • Wind: 100 MW • Biomass: 500 MW
Botswana	Energy access Renewable electricity Renewable energy	• 82 per cent access to modern energy services by 2016; 100 per cent access by 2030 • Capacity increases expected from REFIT programme (delayed) • 15 per cent renewable share in final energy consumption by 2036, but may increase to 20 per cent in 2017 Renewable Energy Strategy once approved
DRC	Energy access (non-renewable energy-specific)	• 60 per cent overall energy access (not renewable-specific) by 2025 (from 9 per cent currently)
Lesotho	Grid extension (non-renewable energy-specific)	• Targets pending completion of Sustainable Energy Strategy 2018
Madagascar	Renewable electricity	• 85 per cent renewable share in electricity generation by 2030
Malawi	Electricity access Electricity efficient device Renewable energy Biofuels	By 2025/2030: • 30 per cent access to electricity (up from 9 per cent since 2010) • 100 per cent use of efficient cook stoves in off-grid households • 6 per cent renewable share in energy mix (up from 1 per cent) • Biofuels mandate of 20 per cent ethanol and 30 per cent biodiesel
Mauritius	Renewable electricity	• 35 per cent of electricity from renewables by 2025; generation shares of 17 per cent bagasse, 8 per cent wind, 4 per cent waste, 2 per cent solar, 2 per cent geothermal by 2025 (under review)

19 SADC Renewable Energy And Energy Efficiency Status Report, 2018

Country	Sector/Technology	Target
Mozambique	Renewable electricity	400 MW increase in installed renewable energy capacity by 2024, including: • Wind: 150 MW • Hydro: 100 MW large-scale, 100 MW small-scale • Solar: 30 MW • Biomass: 30 MW
Namibia	Renewable electricity	• 70 per cent renewable share in electricity generation by 2030
Seychelles	Renewable electricity	• 5 per cent renewable share in electricity generation by 2020; 20 per cent by 2030
South Africa	Renewable electricity Transport	• 21 per cent renewable share in electricity generation by 2030 • 17.6 GW solar capacity, 37.4 GW wind capacity by 2050 (IRP 2016)
Eswatini	Renewable electricity	• 60 MW of intermittent resources such as solar PV by 2030 • 50 per cent renewable share in energy consumption by 2030
Tanzania	Renewable electricity	• 5 per cent renewable share in electricity generation by 2030 (up from less than 1 per cent)
Zambia	Electricity access Biofuel	• 200 MW increase in renewable energy capacity by 2020
Zimbabwe	Electricity access Renewable energy Hydropower (small-scale) Biofuel	• 1,100 MW increase in renewable energy capacity by 2025; 2,100 MW increase by 2030 (16.5 per cent increase overall) • 2,400 GWh increase in renewable energy generation by 2025; 4,600 GWh increase by 2030 (26.5 per cent increase overall) • Note: targets are conditional on final approval by government

p) In its bid to meet the rising demand of electricity, the SADC region is implementing several Generation and Transmission projects across the region. Some of the projects include the following:

- Zambia–Tanzania–Kenya Interconnector
- Mozambique–Malawi Interconnector and Namibia–Angola Interconnector
- Zimbabwe–Zambia–Botswana–Namibia Interconnector
- Mozambique–Zimbabwe–South Africa Interconnector

Petroleum and Gas

- q) The SADC region is endowed with significant deposits of coal (and associated coal bed methane gas), crude oil, shale gas and natural gas. This optimal exploitation could potentially prove to be the 'missing ingredient' in terms of diversifying the region's energy mix, reducing the cost of energy and improving its accessibility to the citizens of the region. It could also reduce carbon dioxide emissions, which are associated with advancing global warming and climate change. Natural gas is becoming more significant to the region's energy sector as Angola, DRC, Madagascar, Mozambique, Namibia, South Africa and Tanzania develop natural-gas fields in their respective countries. Parallel to these developments, countries endowed with coal resources, particularly Botswana, Mozambique, South Africa and Zimbabwe, are redoubling efforts to extract coal-bed methane gas on a commercial scale.
- r) Investments in the oil and gas sector are rising, particularly in Angola, Mozambique and Tanzania due to the vast resources found in those countries. However, the sector is plagued by volatile prices, which have been uncharacteristically low in the past two years, thus generally discouraging investment.
- s) The petroleum and gas industries in the region only exist in the national context with isolated underdeveloped regulatory systems where they do exist. The SADC region has no developed common frameworks aimed at facilitating the development of regional markets and integration of the petroleum and gas sectors within the region.
- t) Presently, most Member States in the petroleum sector have no domestic fuel production capability, but import fuel from other Member States, at different standards. The fuel standards should be harmonised to allow the ease of movement of blended fuels as well as biofuels as blending feedstock within the region. Furthermore, the issues around refinery and storage capacity in the region must be addressed to encourage intra-regional trade especially between the landlocked and coastal Member States.
- u) The projected demand for petroleum products/ liquid fuels in the SADC region is expected to grow significantly in the period up to 2027. The projected growth in demand will have to be matched by the expansion of the necessary infrastructure for production, refinery, storage and pipeline/ transport that goes with uninterrupted supply to the region.
- v) In 2009, SADC adopted a Framework on Sustainable Biofuels, which provides guidelines for production and development of biofuels. Some Member States are already blending bioethanol with petrol/gasoline, and producing biodiesel to optimise the utilisation of their natural resources while reducing the importation of fuel products. However, the success of this programme will also depend on the harmonisation of fuel specifications and standards in the region. Since 2015, the SADC has been advocating for the migration of the region towards low Sulphur fuels and the introduction of cleaner vehicles, since the use of high Sulphur fuel diesel is still common in the region.
- w) There are several ports to import product to South Africa, but the Port of Durban is deemed the port of entry. From there, the inland areas as well a number of adjacent SADC countries are supplied. Matola in Mozambique is also an alternative supply route to the Mpumalanga and Gauteng provinces.
- x) Only six countries have proven gas reserves, with Namibia being the only one with no gas production. The remaining SADC countries Lesotho, Madagascar, Malawi, Mauritius, Seychelles, Swaziland, and Zambia have no known reserves.
- y) The main producers of gas in the SADC region are Angola, Tanzania, DRC and Mozambique. Angola leads the region in deposits of gas and petroleum, while South Africa is rich in shale gas and coal-bed methane gas. Tanzania is emerging as a force in this sector as new discoveries of natural gas continue to be made along its Indian Ocean coast. Mozambique has also seen a rapid expansion of its gas industry since the commissioning of the

865 km-long gas pipeline from the Pande and Temane gas fields in south-central Mozambique to Secunda in South Africa by the multinational Republic of Mozambique Pipeline Investment Company (ROMPCO), headquartered in South Africa.

- z) The Rovuma area, in the far north of Mozambique near the Tanzanian border, has seen positive results of natural gas exploration. Between 150 to 200 trillion cubic feet of gas has been found offshore Mozambique's Cabo Delgado province and final investment decisions have already been made for two liquefied natural gas (LNG) projects, the most recent being Anadarko's Rovuma Area 1 Mozambique LNG project. The final capital estimate has not been made, but Anadarko has indicated that the project will involve two LNG trains with total yearly nameplate capacity of 12.88-million tons.
 - aa) Separately, the Italian Energy Group, Eni, is building the \$4.7-billion Coral South floating LNG facility, while Eni and ExxonMobil are making progress on an LNG project based on the Rovuma Area 4 block offshore, which will share infrastructure with Anadarko's project. Over the coming two decades, it is estimated that more than \$100-billion will be invested in the territory because of the gas projects and several countries, including Portugal, Brazil and France that are actively mobilizing their business communities around the opportunities associated with Mozambican LNG projects.
 - bb) State-owned freight logistics firm, Transnet, plans to launch a tender next year for South Africa's first terminal to import liquefied natural gas (LNG) at Richards Bay port, with first gas expected to land in 2024. The target source of gas for this project is LNG from Mozambique. For this project to be successful, it is of vital importance for South Africa to secure the new gas supplies. Angola and Mozambique are potential LNG suppliers due to their shorter shipping distances, which would give South Africa advantage in securing relatively favourable delivered ex-ship prices.
 - cc) In addition, there is also an opportunity for South African companies to explore other business opportunities that will arise from the development of the three multibillion-dollar gas projects in Mozambique. The region in which these megaprojects are to be developed is both rural and remote, which means just about everything that is needed to support the projects, from ports and roads, to housing and retail developments, still has to be built. In other words, this is not only a game changing prospect for Mozambique, but also a significant business opportunity for South African companies, especially those willing to collaborate with local companies in line with Mozambique's localization requirements.

5.1.6. Economic Outlook

- a) According to the South African Reserve Bank, South Africa's current domestic economic performance can be summarised as follows:
 - South Africa's potential is significant, yet growth over the past ten years has not benefitted from the global recovery.
 - The economy is globally positioned, sophisticated, and diversified.
 - The following were identified as binding constraints to growth:
 - policy uncertainty;
 - the regulatory environment not being conducive to investment; and
 - there is no sustained long-term partnership/cooperation between government, business and labour (Social Compact).
 - SA credit ratings downgrade: South Africa's rapidly worsening fiscal metrics during the course of 2019 alerted all three major ratings agencies to place the country on a negative outlook during the second half of 2019. This has had an impact on the inflow of the FDI
 - A recent World Bank Study (2018)²⁰ on South Africa reveals that it is one of the most unequal economies in the world, with consumption inequality having increased since 1994. Wealth inequality is high and has been rising over time.

- Currently, more than 50% of the population lives in poverty and the economy is not generating sufficient jobs, with 29% of the labour force being unemployed.
 - According to statistics from StatsSA, investment as a percentage of GDP has been declining since 2014. The total investment is now at 19.4% of the GDP, down from 23.5% in 2009.
- b) South Africa's per capita growth rate is currently just above 1%, alongside Colombia with 1.8%, Chile with 1.5%, Brazil with 1.1% and Venezuela with -3.9 (2014 data). Among the highest per capita growth rate in 2018 is China with 6.9%, Malaysia with 5.9% and Indonesia with 5.1%.
- c) Real GDP growth in South Africa is expected to remain below 2% through 2019. However, this is not sufficient to make a meaningful dent in unemployment, poverty, and inequality. Global events, including the Eurozone debt crisis (2010–2012) and weak commodity prices (2014–2015) have contributed to the poor domestic growth performance since 2010. However, at least since 2012, a worsening domestic political, policy and socioeconomic climate ensured that SA was unable to benefit fully from the more recent broad-based improvement in global growth and rebound in key commodity prices. Year-on-year,
- economic growth improved slightly from 0.6% in 2016 to 1.3% in 2017. However, there was a slight dip in 2018 with 0.7% growth recorded.
- d) The average annual consumer price inflation (CPI) was 4.7% in 2018, down from 6.4% in 2016 and 5.3% in 2017. CPI has averaged 5.4% over the past five years, which is in line with the South African Reserve Bank inflation target range. After averaging below 5% in 2018, headline CPI inflation is projected to average of 5.06% during the period of 2019–2023 (BER 2019). This implies that CPI will remain stuck at the lower end of the South African Reserve Bank's (SARB) inflation target band.
- e) The petrol price has increased considerably over the years, with a percentage change of 13.00% in 2018 up from 8.1% in 2017 and 1.4% in 2016. This petrol price is still expected to increase over the coming years, starting with a decrease of approximately 1.2% in 2019. Cumulatively, the 2018 petrol price has increased with 106.2% since 2007 and is expected to continue increasing to 118.2% in 2023.
- f) Impact of COVID-19 on the South African economic outlook can be summarised as follows:
- Real GDP decreased by a record 51% in the second quarter of 2020 owing to the impact of the COVID-19 lockdown restrictions since the end of March 2020. This follows a 2.1% decline in the first quarter of 2020.
 - The biggest negative contributors to GDP growth were the manufacturing (-10.8%), trade (-10.5%) transport (-6.6%) mining (-6%) and finance (-5.4%) sectors.
 - National Treasury and the SARB bank forecasts GDP to decline by -7% and -8.2% in 2020 respectively.
- g) In response to the COVID 19-pandemic, the Department of Mineral Resources and Energy (DMRE) and its entities presented economic interventions in response to the economic impact induced by the Covid 19 pandemic and the economic downgrade of the country. The interventions focused on DMRE's operational
- The Covid-19 pandemic is projected to increase poverty, inequality and unemployment.
 - Estimates suggest that an additional 3.5 million people have been falling into poverty since the beginning of national lockdown restrictions in March 2020.
 - Unemployment increased from 29.1% (6.6million) in December 2019 to 30.1% (7million) in March 2020. Latest statistics show the unemployment rate at 23.3% representing 4.3 million people.

readiness, SAMI Health & Safety readiness, interventions and responses within the broader mining and energy sectors. For example;

- Additional procurement of electricity capacity from existing IPPS (approximately 128 MW and Eskom to procure short-term power (approximately 128 MW).
- Acceleration of the nuclear-built programme
- Ensuring of fuel price benefits being passed to end consumers
- Energy security: shifting of power stations (open cycle turbines) demand from diesel to natural gas within the next 5 years.
- Energy security: conversion of PetroSA from being a gas-2-liquid to be a liquid refining facility.

5.1.7. Impact of BRICS on the Energy Sector

- a) The establishment of the New Development Bank (BRICS Bank) has highlighted its main funding areas as sustainable development and sustainable infrastructure among BRICS countries (Brazil, Russia, India, China and South Africa) and other strategic developing countries (especially in Africa). One of the focus areas of the Bank is to scale up low carbon and climate-resilient investments for sustainable infrastructure, including in particular speeding up the energy transition consistent with the Paris Agreement. The envisaged approach to this is aligning their financial flows with the countries' pathways to low carbon and climate

resilient development, increasing the predictability and ease of access to concessional resources, such as the Green Climate Fund, and leveraging private finance for climate investments.

- b) Most of the Bank's projects involve green energy or infrastructure. According to the Bank, between 1 and 1.5 trillion US dollar is needed to fully harness renewable energy among the trading bloc. The bank approved two infrastructure projects with a loan value of US\$413.8 million during the 12th Board of Directors meeting in Shanghai on November 2017. Non-resident portfolio flows into BRICS nations rose to \$166.5 billion in May 2017, up from \$28.3 billion in outflows 12 months prior, according to data compiled by the Institute of International Finance and EPFR Global. The bank sold its first 3 billion (\$437 million) yuan-denominated bonds in China in July 2017, to fund clean energy projects in member states.
- c) The BRICS Bank has 25 projects at various stages of preparation for 2018 to 2019, with a total lending amount of \$6 billion. Three of these projects are in South Africa and include the Greenhouse Gas Emissions Reduction and Energy Sector Development Project (US\$300m), Durban Container Terminal Berth Reconstruction Project (US\$200m) and Eskom Renewable Energy (Transmission) project (US\$180m).
- d) Between 2003 and 2017, BRICS has invested about US\$3383m in 11 South African Energy projects. This investment translated to 809 jobs created

(Deloitte, 2018). However, in January 2018, an agreement was signed between the Russian state energy company Rosatom and the South African government to construct small hydropower plants in Mpumalanga to power rural regions of the country. This is a key component of South Africa's energy security strategy. Each mini hydropower plant is expected to power 250 to 400 houses. This project could be the first of several small hydro projects aimed at generating innovative and affordable energy in South Africa and across the continent.

5.1.8. SA credit ratings downgraded

- a) South Africa's rapidly worsening fiscal metrics during the course of 2019 alerted all three major ratings agencies to put the country on a negative outlook during the second half of 2019. These led to credit rating downgrades in March and April 2020, taking the country to general sub-investment grade and SA exiting the Financial Times Stock Exchange (FTSE) World Government Bond Index on 30 April following the final downgrade from investment-grade status by Moody's at the end of March. The impact of the COVID-19 pandemic added to reasons for the downgrade and has since overshadowed the economic fight. Different projections of sustained contractions in real GDP for the full year range between 5% and 10%.
- b) In April 2020, both Fitch and Standards & Poor's (S&P) Global Ratings downgraded SA's sovereign credit rating by another notch to push it deeper

into sub-investment grade (sub-IG) territory. Fitch rating agency highlighted that the downgrade was due to “the lack of a clear path towards government debt stabilisation”, with a further shock to government finances and growth due to COVID-19. Fitch had South Africa’s foreign currency rating two notches below investment grade, while S&P Global Ratings is at three notches below IG.

- c) On 20 November 2020, South Africa sunk deeper into junk territory after Moody’s Investors Service joined Fitch Ratings in lowering the country’s credit ratings. Moody’s cut the nation’s foreign and local-currency ratings to Ba2, two levels below investment grade, from Ba1. The outlook remains negative (Bloomberg). Fitch cut the nation’s foreign and local-currency ratings to BB-, three levels below investment grade, from BB, also with a negative outlook. S&P kept its assessment of South Africa’s foreign-currency debt three levels below investment grade, with a stable outlook²¹.

5.1.9. National Environment

Electricity

- a) There is currently no annual growth in electricity demand – there has not been for the last 10 years and there is no sign of that changing. Eskom has 51 943MW of licenced capacity and

the renewable licenced capacity is 6 592.7MW. In April 2018, the then Minister of Energy announced the signing of the agreements for the 27 projects procured under the Renewable Energy Independent Power Producer Procurement Programme (REIPPPP) Bid Windows 3.5 and 4. This is by far the biggest Independent Power Producer (IPP) procurement by the Department of Energy to date, representing a total of R56 billion of investment and about 2300MW of generation capacity to be added to the grid over the next 5 years. This investment is injected into the economy by the private sector, with no contribution from Government other than support to Eskom in the event of a default by the buyer. The new projects are as detailed below:

- 15 new wind, solar PV and concentrated solar power (CSP) projects, Northern Cape;
- 4 new wind projects, Eastern Cape;
- 4 new solar PV projects, North West;
- 2 wind projects, Western Cape;
- 1 a biomass project, Mpumalanga; and
- 1 small hydro project, Free State.

- b) The Integrated Resource Plan (IRP) 2010–2030 estimated that South Africa would require over 40,000 MW of new generation capacity by 2025. The IRP 2018 was released in August 2018 and should provide clarity on the way forward as well as a predicted price path.

- c) The percentage of South African households that were connected to the main electricity supply increased from 76.7% in 2002 to 84.7% in 2018.

- Mains electricity was most common in Limpopo (92.7%), Northern Cape (91.7%), Free State (91.2%) and Mpumalanga (90.7%), and least common in Gauteng (77.7%), KwaZulu-Natal (83.5%) and North West (83.7%).
- The largest increases between 2002 and 2018 were observed in Eastern Cape (36.7%), and Limpopo (21.6%).
- The percentage of households with access to mains electricity actually declined in Gauteng (12.2%) and Western Cape (0.68%). These declines can be associated with the rapid in-migration experienced by these provinces.

- d) Industrial and commercial demand is higher than residential consumption across different forms of energy.

Petroleum and Gas Sector

- e) Inputs of petroleum products play an important role in transport and production activities of various other sectors of the South African economy. However, South Africa does not have its own economically extractable natural crude oil resources, therefore, South Africa relies on

21 Source: <https://businesstech.co.za/news/finance/450475/what-the-latest-rating-downgrades-mean-for-the-average-south-african/>

imports of crude oil and refined fuels to meet its liquid fuels needs.

- f) Approximately 11 142 million litres of petrol and 12 539 million litres of diesel were consumed in South Africa in 2018 representing a decrease of 0.28 per cent and an increase of 3.12 per cent respectively compared to 2017 (DoE, 2018). More illuminating and power paraffin was consumed in 2018 than in 2017, with 702 million litres and 648 million litres consumed respectively. This represents a 7.69 per cent increase in paraffin consumption. Approximately 552 million litres of furnace oil were consumed in 2018, representing a 5.25 per cent increase from consumption in 2017. Furthermore, there was a decrease of 9.32 per cent in the consumption of LPG, with 504 million litres and 551 million litres being consumed in 2018 and 2017 respectively.
- g) The majority of South Africa's refinery output is transported via pipeline, but product is also uplifted directly using road, or transported by rail, to other distribution facilities. The Transnet Pipelines Division operates the main liquid petroleum pipeline system running between Durban and the inland region, comprising the Multi-Product Pipeline (MPP) and the crude oil pipeline to Sasolburg servicing the NATREF refinery. It then extends into the northern network with delivery depots in Gauteng (Alrode, Langlaagte, Waltloo, OR Tambo International

Airport, Tarlton), North West (Klerksdorp, Rustenburg) and Mpumalanga (Witbank) as well as Free State (Kroonstad). The MPP has a coastal accumulation facility as well as an Inland Accumulation facility. At each of these, as well as at the aforementioned delivery depots, the various NERSA licensees have storage facilities interconnecting to the pipeline system. In the eight national ports, there are also marine loading facilities interconnecting to the coastal refineries and/or storage facilities located within or adjacent to the ports. In the inland areas, the storage facilities are mainly replenished by road or rail. In total, NERSA has issued licences to operate 194 storage facilities, 23 marine loading facilities and 19 pipelines to 59 licensees. As of 31 March 2018, TPL has stopped injecting petroleum products into the Durban-to-Johannesburg Pipeline (DJP) and this pipeline will be decommissioned. In an effort to alleviate the supply burden resulting from demand growth, there were plans to build a 300 000 boe/d refinery located in the Eastern Cape Province called 'Project Mthombo'. However, the Government recently announced new plans for the refinery to be located in Richards Bay. Current refinery operators are reluctant to expand present capacity due to the high investment cost involved in meeting cleaner fuel standards while there is a surplus of liquid petroleum products available in the international market. Nonetheless, South Africa's refineries are well placed on a cash operating basis within

its regional peer group (European and African countries that have more than one refinery), indicating their current competitive situation relative to these other manufacturers.

- h) It is expected that small-scale importation and trading of LNG will precede the establishment of LNG storage and gasification terminals in South Africa. In this regard, the Energy Regulator has recently licenced the operations of Volco (Pty) Ltd (Volco) and Volco Alfa (Pty) Ltd (Volco Alfa), which will import the small-scale LNG into South Africa in the Western Cape Province. The LNG will then be transported to customers' sites via trucks using 40' ISO containers, where it will be stored, regasified and traded to the customers in gaseous form.
- i) Renergen is the first company in South Africa to build a small-scale onshore LNG plant. It intends to monetise its LNG by developing between 12 - 18 LNG filling stations across South Africa by 2023. Renergen has signed a deal with Total under which the French major will rebrand two of its filling stations on the N3 national route between Johannesburg and Durban as LNG outlets. The LNG sold at these filling stations would be exclusively for the use of trucks and buses. The first phase of the project is planned to supply 400 trucks from 2021, with the second phase supplying approximately 3,000 - 5,000 trucks from 2023.

- j) South Africa's intentions to expand the role of LNG in its energy mix is reflected in the country's Integrated Resource Plan (IRP), which was published in October 2019. The IRP envisages the creation of an additional 8 100MW of gas and diesel-fired generation capacity by 20230 in order to support energy security.
- k) Sasol has confirmed the much speculated intention to sell its equity interests in the ROMPCO. Speculations emerged as early as April this year that the petrochemicals producer was seeking to sell off some of its African assets as part of its restructuring. It was said that the company had appointed advisers to sell its stakes in a power plant in Mozambique and a gas pipeline running from the country into South Africa. Sasol said that the sale is part of the its drive to raise as much as \$5 billion through asset sales amid cost overruns and lower oil prices by end of its 2021 financial year.
- l) Sasol's gas supply from Mozambique is scheduled to start falling from 2023 onwards, with a forecasted yearly shortfall of 98 million gigajoules from 2025 onwards.
- m) The gas sector looks forward to a potential increased investment due to a boost in investor confidence affirmed by the second gas-condensate discovery

by Total in the Western Cape, 175 kilometers off the southern coast of South Africa.

- n) Total made a significant gas condensate discovery after drilling its Brulpadda prospects on Block 11B/12B in the Outeniqua Basin, offshore South Africa. The area is 175km off the southern coast of South Africa. The estimated gas reserves are in the range of 56 million cubic meters, of which around 450 million cubic meters can be recovered²².

Gas-to-Power procurement programme

- o) In order to support the implementation of the Integrated Energy Plan, the DMRE is currently finalising the Gas Utilisation Master Plan (GUMP) for South Africa. The GUMP would act as a roadmap for the development of the gas industry in the South African economy. It analyses the potential and opportunity for the development of South Africa's gas economy and sets out a path of how this could be achieved. One of the main objectives is to enable the development of indigenous gas resources and to create the opportunity to stimulate the introduction of a portfolio of gas supply options.
- p) The key challenges in the sector are to bring gas demand and supply on stream at the same time and spread geographically to stimulate broader localised demand. Without local demand, it would

be difficult to develop distributed gas supply and without such distributed gas supply, it would be difficult to develop local gas demand. One way of overcoming this challenge is to develop a Gas-to-Power Programme. This would potentially anchor gas demand while creating a long-term sustainable gas demand. The intention of the Gas-to-Power Programme is not only supplying power, but also supplying a limited amount of gas, marketed in the form of a Gas Supply Agreement (GSA), for use by industrial and other users.

- q) The Gas-to-Power Programme has stalled until the completion and publishing of the Integrated Energy Plan (IEP) and the updated Integrated Resource Plan.

Regulated Energy Industry

- r) Energy is at the core of current and future industrial and technological development. The National Development Plan envisages that the country will have an energy sector that promotes economic growth and development through adequate investment in energy infrastructure by 2030. Furthermore, the plan envisages that South Africa will have an adequate supply of electricity and liquid fuels to ensure that economic activity and welfare are not disrupted and that 95% of the population will have access to some form of energy.

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s) NERSA has commenced with a process to determine the size of the NERSA-regulated activities within the energy sector (Electricity, Piped-Gas and Petroleum Pipelines).

- The Energy Regulator has seen a rapid increase in the number of operational licensees over the 2014 to 2018 period with the exception of 2017 to 2018, where a significant decrease occurred in the electricity distribution space. Currently, there are 367 licensees operating under the regulation of the Energy Regulator. The bulk of these licensees are in the Electricity sector, followed by the Petroleum Pipelines and Piped-Gas sectors respectively.
- In particular, electricity generation has seen a rapid increase in licences issued since the implementation of the DMRE's REIPPPP that was officially launched in 2011. Between 2014 and 2018, an additional 31 licensees were licensed (13% increase).
- In 2017, a decrease of 11 licensees (-5.8%) occurred due to mergers of 26 distribution licensees into 12. Of the 26 merged licensees, four (Indaka, Imbabazane, Ezingoleni and Khara Hais) were under Eskom Distribution.
- The Petroleum Pipelines industry's regulated facilities had a regulated capacity of 16,764,237 m3 transported by pipelines, 12,014,534 m3 in storage facilities and 16,173,861 MT in loading

facilities in 2017. Of particular interest is the storage sub-sector, which saw a 12% increase in regulated facilities from 2014 to 2015. There was a slight decline from 2015 to 2016, due to the implementation of the bulk determination by the Regulator.

- The Petroleum Pipelines industry's regulated facilities had a regulated capacity of 22,127,097 m3 transported by pipelines in 2018, 12,329,854 m3 in storage facilities and 16,177,014 MT in loading facilities. Of particular interest is the storage sub-sector, which saw a 12% increase in regulated facilities from 2014 to 2015. There was a slight decline from 2015 to 2016, due to the implementation of the bulk determination by the Regulator.
- With regard to the Electricity sector, there are 131 regulated facilities, of which 30 are owned by Eskom, 16 by general IPPs, 78 by renewable IPPs and 7 by municipalities. This jointly represents 61 074.90MW of electricity generation in the country. There is a 15.48% decrease in the number of regulated facilities from 2017. Interestingly, IPPs combined represent 94 facilities with a capacity of 8 593MW in 2018. This represents an increase of 2.76 per cent of electricity added to the national grid since 2017.
- In addition, as per the Gas Act, the Energy Regulator is mandated to register certain gas activities in order to keep abreast of key

developments in the gas industry. As of 2018, 118 biogas facilities and 35 biogas registrants are registered with the Energy Regulator.

- t) There is a significant amount of energy assets in operation under the ambit of the Energy Regulator. As of 2018, there are R830.020 billion worth of operational assets under regulation, with the Electricity industry being the dominant player representing 94.32%, and 4.56% and 1.11% for Petroleum Pipelines and Piped-Gas respectively.
- u) The energy sector is undergoing major reforms with the construction of a number of projects that will add significant amounts of capacity in the short term. As of 2018, there are R430 180 billion assets under construction, of which R146 896 billion assets are in the Electricity sector, R265million in Petroleum Pipelines and R18 283 million in Piped-Gas. The electricity sector's construction projects include the approved DoE REIPPPP projects and Eskom power projects. IPPs, in particular, have investment projects worth R66.478billion (45.2%) and Eskom, through its new build programme, accounts for R80 418billion (54.7%), with projects such as Medupi and Kusile power stations still under construction. It should be noted that some of these projects are nearing completion and will be adding significant amounts of electricity to the South African power grid.

5.1.10. PE(R)STEL Factors Analysis

The specific factors considered in the environmental scan are shown in the tables below.

Table 5: Political factors

Political factors	Impact if factor is not addressed	NERSA response to the factor
Electricity Industry Regulation		
1. Municipalities' executive authority for funding of municipal infrastructure	- Some municipalities are unable to fund, build, operate and maintain adequate electricity infrastructure – which has a negative impact on security of supply - Ring-fencing of municipal electricity revenues - Unsustainable cross subsidising of municipal services	- Engage with relevant ministries regarding municipal funding more broadly - Base municipal tariffs within the broader municipal funding model
2. Role of SOEs in economic recovery	- Regulatory mandates that promote a just energy transition undermined - Reputational damage to NERSA	- Eskom Political Task Team (PTT) involvement - Establish and execute Eskom Engagement Task Team under steer of the PTT - Develop collaborative relationships with key delivery ministries, such as, inter alia, National Treasury (Operation Vulindlela), Department of Public Enterprises, COGTA etc.
Piped-Gas Industry Regulation		
1. Delays in finalisation of legislative amendments and developments (with specific reference to the Gas IPP and the Gas Utilisation Master Plan)	- Cost of gas may be too high - It may deter / delay entry into the gas market	Develop a report on regulatory advocacy and engagements with relevant policy makers
2. Lack of policy on gas infrastructure investment	- Uncertainty for investment - Lost opportunity to encourage competition in piped-gas industry - Impedes growth of the gas market in SA - It may deter / delay entry into the gas market	- Continued regulatory advocacy and engagements with relevant policy makers - Advocate the development of the Gas Utilisation Master Plan, Gas IP, Gas Infrastructure Plan

Political factors	Impact if factor is not addressed	NERSA response to the factor
Piped-Gas Industry Regulation (continued)		
3. Emerging gas policy in Mozambique	Security of gas supply – Supply diversification	- Monitor ability of SASOL to supply - Undertake regulatory and intergovernmental engagements - Monitor utilisation of excess capacity in ROMPCO Pipeline - Approve tariffs for SA side of cross border assets to facilitate investment and additional gas supply
4. Regulating the gas market – bundled and unbundled approach to LNG projects	- May deter infrastructure investments - Regulatory uncertainty	- Develop a NERSA position paper on regulating the gas market – bundled and unbundled - Continued regulatory advocacy and engagements with relevant policy makers - Revisit the Gas Rules
5. Alignment of Gas Infrastructure Plan, the IRP and IEP	- Possible duplication or contradictions - Regulatory uncertainty	Continued regulatory advocacy and engagements with relevant policy makers
Petroleum Pipelines Industry Regulation		
1. Geo-political upheavals impacting on petroleum producing transient countries	- Higher and volatile fuel prices - Rand/dollar exchange rate volatility	- Regulatory advocacy on price regulation by the DMRE - Participate in fuel price policy and regulatory framework reviews - Participating in regional structures dealing with petroleum matters
2. Neighbouring countries finding alternative sources of fuel	- Low tariffs through the NMPP and concomitant high tariffs - Threats to security of supply	- Monitor interventions by Transnet to increase the volumes - Regulate in a manner that promotes immigration from pipelines to other modes of transport - Participate in supply managers forums and other security of supply committees - Continued regulatory advocacy

Table 5: Political factors (continued)

Political factors	Impact if factor is not addressed	NERSA response to the factor
Petroleum Pipelines Industry Regulation (continued)		
3. Decline in investment friendliness of South Africa	- Further large-scale investments in petroleum infrastructure (and demand sectors) slows down. - Petroleum Infrastructure may not be sufficient to meet future demand - Decline in fuel demand which can lead to higher tariffs and/or stranded assets	- Adjust regulatory framework to attract investments - Continued regulatory advocacy and engagements with relevant policy makers to ensure efficiencies - Identify and implement key measures to improve regulatory certainty through consistent and defensible decisions, based on world-class regulatory frameworks, methodologies and mechanisms - Regulate in a manner that promotes competition
Transversal Regulatory and Organisational		
1. Developmental State	Decisions of NERSA could be in conflict with policy	Continued regulatory advocacy and engagements with relevant policy makers
2. Manage interface between different policy thrusts of Government (new growth path, IPAP2)	Decisions of NERSA could be in conflict with policy	- Make decisions that are not in conflict with the Acts - Develop and implement a strategic engagement framework on developing legislation/policy changes
3. Policy gaps and inconsistencies	- Regulatory uncertainty - Lack of credibility of regulatory system	- Review impact on NERSA's mandate - Continued regulatory advocacy and engagements with relevant policy makers - Develop a report on the cost of projects, the impact and implications thereof e.g. Integrated Resource Plan
4. Discussion/debate around nationalisation	Uncertainty for investment	Identify and implement key measures to improve regulatory certainty through consistent and defensible decisions, based on world-class regulatory standards, procedures and processes
5. Review of Sustainable Development Goals	NERSA may not assist the country in achieving its goals	Regulate in such a manner that accessibility and affordability is enhanced

Table 6: Economic factors

Economic factors	Impact if factor is not addressed	NERSA response to the factor
Electricity Industry Regulation		
1. Lack of competition in electricity supply industry	- Impact on the ability of the Independent Power Producers to access the industry - High electricity prices to industrial consumers	- Enforce Third-Party Access through regulatory decisions - Amend the dispatch rules to include balancing rules - Continued regulatory advocacy and engagements with relevant policy makers
2. Subsidies in Industry	Subsidies cause wrong investment decisions	- Continued regulatory advocacy and engagements, also focusing on the following: - approval of municipal tariffs that rationalise application of subsidies; and - limiting surpluses that municipalities can accumulate for cross-subsidisation.
3. Electricity Price to commercial entities in the municipalities has reached a critical level	Commerce and industry closing down	- Develop a paper on tariffs in municipalities, focusing on, among others: - Influencing tariff structures - Determining whether the actual application of tariffs yields expected result.
4. Impact of poverty	Lack of affordability and accessibility	Focus on pro-poor regulation
5. Increased consumption of coal by China and India	Security of supply	- Regulate the stock piles - Develop a report on the introduction of renewable energy in the energy mix (taking into account its limitations)
6. Inter-dependency of SADC on SA economy	SADC countries' power plans not realised	- Contribute through regional structures such as RERA towards the realisation of SADC countries' power plans - Review NERSA's role in international trade
7. Economic decline and low credit rating	- Depressed economy leading to less disposable income, which in turn would result in an increase in bad debt and an ESI that is not economically viable. - Low credit rating Limits investment attraction, - Reduction in economic growth affects affordability	- Ensure that electricity price increases are kept to the minimum by enforcing efficient licensee operations and ensure that pro-poor regulation is strengthened - Infrastructure investments and development implementation has been affected and delayed. - Requires regulation review to align the economy and investment attraction

Table 6: Economic factors (continued)

Economic factors	Impact if factor is not addressed	NERSA response to the factor
Electricity Industry Regulation (continued)		
8. Credit worthiness of State-Owned Entities (SOEs)	- Impact on infrastructure investment due to higher cost of debt and inability to issue bonds - Higher tariffs	- Regulate in a manner that drives efficiency - Set credit rating criteria in the MYPD methodology
9. Drought – water infrastructure	- Development of shale gas prospects to encourage gas-to-power projects in the country - Security of supply	Review the efficient management of water resources in generation of electricity
10. Decline in electricity demand due to COVID-19 pandemic	Low demand has led to low income and profit sustainability. In addition this has threatened energy security and investment attraction as delays in manufacturing have halted mega projects	There is a need to review tariffs and price methodologies to determine whether it is responsive to the long, medium and short term economic impact of COVID-19 and develop appropriate responses
Piped-Gas Industry Regulation		
1. Lack of competition in gas industry	- Barrier to competitive outcomes (key barriers including lack of gas supplies and infrastructure to enable such supplies) - Likely perpetuation of current monopoly in the industry	- Continued regulatory advocacy and engagements with relevant policy makers to facilitate entry - Enforce Third-Party Access through regulatory decisions - Review and implement Maximum Prices Methodology and Tariff Guidelines
2. Lack of infrastructure investment	- No/limited growth in the gas market - Lack of gas import infrastructure - Lack of entry of new gas suppliers	- Develop a regulatory advocacy report to the DMRE and IPPs regarding gas-to-power procurement programme - Continued advocacy with policy makers to expedite finalisation of Gas Masterplan and alignment of IEP, IRP and Gas Infrastructure Plan

Economic factors	Impact if factor is not addressed	NERSA response to the factor
Piped-Gas Industry Regulation (continued)		
3. Economic growth stagnation	May deter investments and present barriers to entry	Continued advocacy with policy makers
4. Lack of indigenous gas sources	- Impact growth of gas industry - Discourage investment - Lack of competition in gas industry	- Continued research and monitoring of developments in new gas sources - Develop and maintain gas trade relations with neighbouring countries. - Explore prospects for LNG imports
5. Gas industrialisation campaign	Ineffective regulation of the gas market	- Continued regulatory advocacy - Undertake intergovernmental engagements
6. Gas supply certainty – Sasol Gas indicated in FY19 that it expects its gas fields to start declining in 2023	- Sasol Gas may not be able to meet supply obligations going forward - May jeopardise existence and growth of the gas industry.	- Engagements with relevant stakeholders, including <i>inter alia</i> Sasol Gas, the Industrial Gas Users Association –Southern Africa regarding the viability of potential new sources of supply - Gather data from Sasol Gas in terms of S28 and Regulation 9 of the Gas Act, in terms of which Sasol is expected to provide information on its gas reserves - Continued regulatory advocacy and engagements with relevant policy makers to facilitate the entry of new gas suppliers, and the development of infrastructure to enable such supplies
Petroleum Pipelines Industry Regulation		
1. Low economic growth in South Africa	- Reduced demand for liquid fuel - Further large-scale investments in petroleum infrastructure will stop. - Petroleum Infrastructure may not be sufficient to meet future demand	Identify and implement key measures to improve regulatory certainty through consistent and defensible decisions, based on world-class regulatory standards, procedures and processes
2. HDSA and B-BBEE participation	- No third-party access to storage facilities - Non-transformed petroleum pipelines industry - Social upheavals	- Participate in Charter Counsel - Develop and implement a strategic engagement framework on transformation
3. Importation of fuels via trucks through other ports of entry into South Africa	- Lower volumes through pipelines leading to higher tariffs. - Disruption of regulatory framework	- Monitor developments in this regard - Continued regulatory advocacy

Table 6: Economic factors (continued)

Economic factors	Impact if factor is not addressed	NERSA response to the factor
Transversal Regulatory and Organisational		
1. Impact of environmental levies and the Carbon Tax Act on prices	Impossible to facilitate achievement of affordable energy services	Develop a position paper on the impact of environmental levies to policy makers
2. Manage interface between different policy thrusts of Government	Decisions of NERSA could be in conflict with policy	- Make decisions that are not in conflict with the Acts - Develop and implement a strategic engagement framework on developing legislation/policy changes
3. Downgrade of South Africa's credit status	Capital flight (foreign and local)	Identify and implement key measures to improve regulatory certainty through consistent and defensible decisions, based on world-class regulatory standards, procedures and processes.
4. Persistently low economic growth rate	Cost of energy – impact on consumers	Review tariffs to encourage manufacturing

Table 7: Regulatory factors

Regulatory factors	Impact if factor is not addressed	NERSA response to the factor
Electricity Industry Regulation		
1. Regulatory reform in the electricity sector	- Electricity supply and demand misaligned with weak market signals to curb inefficient electricity use - Electricity market reforms poorly managed with avoidable unintended consequences - Information asymmetry - Poor quality of evidence used to base decisions - Unsubstantiated decisions taken due to lack of all relevant information available - Contraction in energy intensive usage sectors - Loss of value from natural resource endowments - Economic recovery constrained	- Establish regulatory reform department with capability to assess: - Technical aspects - Economic aspects - Legal aspects - Technoeconomic evaluation of a regulated ESI that promotes choices that encourages: - Productive (technical) efficiency (least cost of supply); - Allocative efficiency (provide the greatest benefit relative to costs). - Dynamic efficiency (timely responses to changes that enhance economic efficiency) - Acquisition of global, regional and national data to support decision making and advocacy - Establishment of an Integrated Energy Modelling capability and associated Integrated Energy Modelling System (IEMS) - Review of licencing/registration regulations/rules - Promoting collaboration and information sharing with stakeholders whose activities are affected by Energy Regulator decisions and advice - Policy, legislative and regulatory advice to relevant ministries, - Research and implement programmes to progress electricity sector reforms with specific focus on, inter alia: - Tariff setting methodology reviews – cost reflective tariffs driven by efficiency - Capacity investments in a high reserve margin environment – underutilised/ stranded assets - Transition to ‘smart’ tariffs – to reflect how and when electricity is consumed.

Table 7: Regulatory factors (continued)

Regulatory factors	Impact if factor is not addressed	NERSA response to the factor
Electricity Industry Regulation (continued)		
2. Rationally regulated electricity supply industry	- Weakly coordinated and poorly managed unbundling of Eskom - Unpredictable and uncertain electricity price path - Inefficient use of electricity resulting from weak regulatory signals - Inefficient investment decisions resulting in stranded assets - NERSA reputational risks	- Implementation of the Regulatory Reporting System for financial data and a Regulatory Reporting System for non-financial data: - Revision of ERTSA - Establish 'municipal' ERTSA - Development of energy database that integrates energy production and consumption data as evidence for: - Developing and regularly updating a benchmarked and trusted electricity price path - Making sound and substantiated decisions, including inter alia, review of the tariff setting methodology and all other tariffs setting/approval processes - The transition to efficient cost reflective tariffs; - The integrated Type of Use and Time of Use tariffs, - The development of regulatory instruments that promote equitable access to electricity, including, inter alia, a review of the Inclining Block Tariffs, the efficiency of the Free Basic Electricity subsidy etc. - Conclusion of Eskom matters – regulatory, legal or otherwise, including, inter alia: - MYPD applications (consolidated or otherwise) - RCA reviews - Supplementary applications, - Review and revision of MYPD methodology - Development and finalisation of MYPD5 methodology - implement mechanisms to address EAF and reserve margin to address the 'fallacy of capacity constraints'.

Regulatory factors	Impact if factor is not addressed	NERSA response to the factor
Electricity Industry Regulation (continued)		
3. Compliance of municipalities with electricity licence conditions	- Undermine reliability of municipal distribution of electricity - Security and quality of supply - Undermine affordability of, and accessibility to, electricity - Continued tariff misalignment between Eskom, IPPs and municipalities - Key national programmes will be undermined - Undermine service delivery	- Benchmarking of municipal electricity supply metrics - Base approval of municipal tariffs on cost of supply studies - Increased compliance monitoring and robust enforcement of licence conditions – penalties, tribunals etc. - Continued regulatory advocacy and engagements, also focusing on the following: - Interdepartmental engagement to locate evidence-based electricity tariffs within the broader municipal funding model; - limiting surpluses that municipalities can accumulate for cross-subsidisation. - approval of municipal tariffs based on cost of supply studies
4. Coordinated regulation of gas and electricity industries	- Inconsistent policy messages deterring investment - Incorrect signals sent to the market resulting in inefficient investment decisions and stranded assets	- Strengthen internal coordination and strategic interactions with government structures - Collaboration with other regulators to address regulatory asymmetry
5. Management of concurrent jurisdiction with other regulators or institutions	- Regulatory overlap - No clear roles and responsibilities - Lack of cooperation may lead to delay in decision-making.	- Continued regulatory advocacy and engagements with relevant policy makers - Develop and implement Memorandums of Understanding (MOUs) and Memorandums of Agreement (MOAs) with appropriate regulators or institutions
Piped-Gas Industry Regulation		
1. Light-handed approach of current regulatory framework and weak enforcement powers	Difficult to effectively enforce regulatory mandate	- Continued regulatory advocacy and engagements with relevant policy makers, with specific reference to the review of the Gas Act and the National Energy Regulator Act - Develop and implement MOUs with the appropriate regulators or institutions, focusing among others on reducing confusion and unnecessary regulatory burden and cost

Table 7: Regulatory factors (continued)

Regulatory factors	Impact if factor is not addressed	NERSA response to the factor
Piped-Gas Industry Regulation (continued)		
2. Regulatory gaps, limited discretion and fragmentation of legislation (gas) (not regulating entire value chain)	- Unnecessary regulatory burden - Unintended consequences (e.g. High distribution tariffs) - Ineffective regulation of industry - Difficulty in approving vs setting gas prices and tariffs	- Report on regulatory advocacy and engagements regarding provisions/ measures to be included in the Gas Amendment Bill - Amendments to the Gas Act by the DMRE
3. Lack of experience in regulating new activities (e.g. LNG, Shale gas, FSRU, regasification)	Inappropriate regulation of new activities	- Develop the rules, norms and standards for the regulation of the new activities - Develop and implement a skills gap analysis and appropriate training for staff in regulating new activities
4. Information asymmetry	Possible incorrect decisions taken due to lack of accurate/ adequate information for decision making	- Develop and implement an appropriate method of ensuring the collection of accurate data - Implement the Regulatory Reporting Manuals to overcome information asymmetry
5. Concurrent jurisdiction regarding the regulation of gas	Lack of cooperation may lead to delay in decision making	Development and implementation of MOUs and MOAs with regulators with concurrent jurisdiction
6. Gaps and inconsistencies between regulations and the Act	- Regulatory uncertainty - Leads to confusion among stakeholders - Legal challenges	- Continued regulatory advocacy and engagements with relevant policy makers - Advocate for the gas amendment process by DMRE
7. Cross-border regulation and harmonisation of processes, methodologies and procedures	Regulatory uncertainty	- Continued engagement with INP to harmonise regulatory processes. - Finalise and implement MOU with Mozambique regarding sharing of information and mutual co-operation on regulatory matters

Regulatory factors	Impact if factor is not addressed	NERSA response to the factor
Piped-Gas Industry Regulation (continued)		
8. Complementary jurisdiction misalignment in application of policy objectives	Regulatory and investment uncertainty	- Continued regulatory advocacy and engagement in with relevant policy makers - Develop appropriate MOUs
Petroleum Pipelines Industry Regulation		
1. Concurrent and complementary jurisdiction	Regulatory uncertainty	- Harmonise regulatory methodologies (internally and externally) - Continued regulatory advocacy and engagements with relevant policy makers and other regulators
2. Cross-border regulation and harmonisation of processes, methodologies and procedures	- Regulatory uncertainty - Reduce intra-regional and/or intercontinental trade	- Participation in RERA's Petroleum and Gas Regulatory Subcommittee - Participation in regional and continental regulatory structures
3. Policies lagging behind	Impacting NERSA's ability to effectively regulate the industry	Continued alignment and revisions between DMRE mandate and associated policies.
4. Possible market interventions by Government: - biofuels - strategic stocks - security of supply - cleaner fuels - New refinery LNG importation	Inadequate regulatory framework	- Continued regulatory advocacy and engagements with relevant policy makers - Continued participation in SADC structures (e.g. Oil and Gas Subcommittee) - Identify potential regulatory process amendments - Provide inputs on suggested policy and regulatory amendments - Pro-actively engage on possible market interventions and adjust framework accordingly
5. Inconsistency in storage and loading tariff and storage methodology	Undue over-compensation	Revise the methodology

Table 7: Regulatory factors (continued)

Regulatory factors	Impact if factor is not addressed	NERSA response to the factor
Transversal Regulatory and Organisational		
1. Management of concurrent jurisdiction	- Regulatory overlap - No clear roles and responsibilities	- Continued regulatory advocacy and engagements with relevant policy makers - Develop and implement MOUs and MOAs with regulators with concurrent jurisdiction
2. Perception of independence of the Regulator	Uncertainty for investment	- Develop and execute a Stakeholder Engagement Strategy to inform a Stakeholder relations management system - Communication strategy, including, inter alia, attention to NERSA's activities, information dissemination, approach to Records of Decision etc.
3. Review of the Energy Regulator Act	Negative impact on regulatory ability if identified gaps are not addressed in the Act	Continued regulatory advocacy and engagements with relevant policy makers
4. Implementation of regulatory programmes and projects approved at continental and regional level	NERSA may not be in a position to contribute to continental and regional matters that may have an impact on the energy industry, and the country as a whole	NERSA needs to incorporate continental and regional programmes in its regulatory activities (<i>since RSA is a member and an important role player in regional and continental structures, e.g. RERA & AUC</i>)

Table 8: Social factors

Social factors	Impact if factor is not addressed	NERSA response to the factor
Electricity Industry Regulation		
1. Regulatory instruments to reduce poverty	- Increased poverty - Boycotting of payments of electricity - Social unrest and ongoing service delivery protests - Destruction of electricity supply infrastructure	- Public consultations to understand community grievances and extent to which regulatory instruments can influence outcomes - Develop regulatory approaches and instruments that promote equitable and appropriate access to electricity - Continued regulatory advocacy and engagements with relevant policy makers – with specific reference to poverty reduction measures - Review Free Basic Electricity and other proactive poverty reduction subsidies to reduce social wealth gaps
2. Social unrest and ongoing service delivery protests	Destruction of electricity supply infrastructure	There is a need to regulate in a manner that promotes equitable distribution
Piped-Gas Industry Regulation		
1. Implementation of HDSA/B-BBEE participation policy	- Limited participation in market by HDSA/B-BBEE and industry transformation - Access to gas and infrastructure	- Ensure third-party access - Continued regulatory advocacy and engagements with relevant policy makers – with specific reference to the development of a charter - Enforce transformation provisions in BBBEE legislation
2. Uncontrolled building on pipeline servitudes	May result in damage to pipelines, posing a threat to security of supply	Increase pressure on licensees to consult with municipalities by monitoring and enforcing compliance with licence conditions and Regulations
3. Skills development	Inadequate skills to match new technically inclined developments upstream	- Monitor construction plans - Ensure skills transfer in interactions with specialist service providers (e.g. skills transfer clauses in service level agreements with consultants) - Ensure continued training on new developments in the industry

Table 8: Social factors (continued)

Social factors	Impact if factor is not addressed	NERSA response to the factor
Petroleum Pipelines Industry Regulation		
1. Lack of awareness of positioning of pipelines by other relevant authorities	Health, safety and environmental risks – bad publicity or reputational risk for NERSA	- Public awareness campaigns to explain NERSA's role and responsibilities - Monitor and enforce compliance with licence conditions and Regulations for licensees to liaise with municipalities
2. Increase of attempted theft on the pipelines	- Security of supply compromised - Health and safety risk	- Monitor and enforce compliance with licence conditions - Promote improved coordination and cooperation with other regulatory authorities, municipalities and law enforcement agencies
Transversal Regulatory and Organisational		
1. High level of unemployment	Political instability that can affect delivery of infrastructure to the poor	- Ensure that NERSA's Internship and Learnership programmes are current and effective - Investigate how NERSA can use tariffs to allow licensees to employ young people as apprentices
2. Service delivery protests (consumer activism)	- Alienated and marginalised communities - Potential increase in tariffs	- Conduct customer education and public consultation initiatives - Develop a position paper on the most appropriate funding mechanisms - Develop a position paper on tariff reducing instruments in order to obtain policy clarity
3. Perception of independence of the Regulator	Uncertainty for investment	- Develop a strategic engagement framework with all role players - Develop a proactive communication strategy on NERSA's activities – particularly on how decisions are reached
4. Resistance to energy infrastructure close to settlements	Security of supply	Ensure that the sector is ready for expropriation proceedings in terms of the Electricity Regulation Act

Table 9: Technological factors

Technological factors	Impact if factor is not addressed	NERSA response to the factor
Electricity Industry Regulation		
1. Technological innovation e.g. Smart Grid	- Security of supply - Stranded assets	- Develop appropriate rules to cater for technological innovation in the sector - Monitor compliance with robust enforcement - Develop measures in order to protect user information - As the grid becomes more sophisticated, NERSA may need new regulations to protect the grid over the long-term - Customer education - Engagement with smart technology providers and platforms (especially SOEs, where potential leverage and social benefit – such as Telkom, SITA etc.) to develop smart tariff applications towards real-time monitoring of the electricity systems.
2. Renewable Generation	- Security of supply - SA not meeting environmental targets	- Amend the Grid Code to include dispatch rules - Create market and balancing rules
3. Gas as primary energy source	Security of supply	Continued regulatory advocacy and engagements with relevant policy makers
4. Nuclear Generation	- Security of supply - Higher tariffs	- Develop an evidence-based report on the expansion of nuclear energy in the energy mix - Conduct customer education - Conduct a skills analysis and develop a strategy to upgrade NERSA skills
5. Energy efficiency	Revenue shortfall for municipalities/ distributors/ Eskom	- Continued regulatory advocacy and engagements with relevant policy makers with specific reference to a different funding model for municipalities so that they do not have to depend mainly on electricity revenues - Continued monitoring of the implementation and the impact of energy efficient measures
6. Storage technologies	- Could impact prices and security of supply - Will not harness the benefits of e.g. renewable energy, mini grids, etc.	- Create a regulatory environment to include this technology and capacity building of NERSA staff to improve understanding - Develop rules codes to define how these technologies connect with the electricity grid

Table 9: Technological factors (continued)

Technological factors	Impact if factor is not addressed	NERSA response to the factor
Electricity Industry Regulation (continued)		
7. Embedded and self-generation	Eskom and Municipal sustainability at risk	- Engage with stakeholders - Develop a framework to address sustainability issues - Develop rules for registration - Develop systems to ensure monitoring to form inputs into planning processes.
Piped-Gas Industry Regulation		
1. Regulatory framework lags technological innovation	- Unregulated gas activities (risk) - Deters entry and investment - Regulatory uncertainty - NERSA could be exposed to possible legal action - Ineffective regulation	- Continued regulatory advocacy - Incentivise through tariffs, prices and licensing - Monitor developments in the industry - Ensure that a regulatory framework is developed in order to be ready for the regulation of the industry with technological innovation
2. Lack of piped-gas infrastructure for new technology (Liquefied Natural Gas, regasification, Compressed Natural Gas, Floating Liquefied Natural Gas, Liquefied Natural Gas tanks etc.)	Deters investment and growth of downstream industry	Continued regulatory advocacy and engagements with relevant policy makers
3. Resistance to new gas technology (e.g. Shale Gas hydraulic fracturing)	SA misses out on opportunity to replace crude imports with domestic GTL	- Conduct research on new gas technology and the impact on regulation - Continuously monitor developments of gas technologies - Review adequacy of current regulatory regime and rules - Continued regulatory advocacy and engagements with relevant policy makers - Conduct a skills analysis and develop a strategy to upgrade NERSA skills on regulation of new gas technologies
4. Lack of gas storage infrastructure	Security of supply could be compromised	Continued regulatory advocacy and engagements with relevant policy makers

Technological factors	Impact if factor is not addressed	NERSA response to the factor
Petroleum Pipelines Industry Regulation		
1. Alternative forms of energy and technological improvements that reduce demand for petrol	• Risk of stranded assets • Risk of bankrupting new entrants • Lower pipeline volumes will lead to higher tariffs, which may result in incentives to use alternative modes of transport	• Forward looking regulatory framework • Monitor trends and potential alignment of tariff methodologies • Create an environment to regulate within changing landscape • Monitor supply and demand
2. Fragmentation of the different product grades of fuel – losing economies of scale	• Lower volumes will lead to higher tariffs. Higher Transnet Pipeline costs due to higher interface volumes. • It will reduce available storage capacity for individual products • It will reduce availability of storage capacity per product grade and may consequently further reduce third-party access	• Licence tanks to store more than one type of product
Transversal Regulatory and Organisational		
1. Rapid development in ICT sector	• Lost efficiencies and limited communication impact and reach	• Harness technologies to speed up processes and improve efficiency • Implement cyber security controls
2. Technological Developments	• There are several advancements that affect NERSAs ability to deliver and respond	• NERSA needs to assess how to take advantage of technological advancement in their operations beyond COVID-19

Table 10: Environmental factors

Environmental factors	Impact if factor is not addressed	NERSA response to the factor
Electricity Industry Regulation		
1. Climate change imperatives	- Can impact the security of supply because renewable energy generators cannot contribute to meeting peak demand and are unreliable in delivery of energy. - The current high cost of renewable energy generators will impact on the accessibility to all end users.	- Evidence based regulatory advocacy and engagements with, inter alia: - Relevant policy makers; - Civil Society; and - Consumers
2. Environmental activism	Security of supply	Continued regulatory advocacy and engagements with relevant policy makers
3. Growing awareness of environmental factors	SA not meeting its reduction in greenhouse gas emission targets	Utilise the Multi-Year Price Determination to facilitate contributing towards the reduction of greenhouse gas emissions
4. Carbon tax (off sets and carbon trading)	Higher prices of all non-renewable energy	- Continued regulatory advocacy and engagements with relevant policy makers - Monitor developments and decisions taken by the G20
5. Minimum Emission Standard	- Shutting down of power stations that do not comply - Security of supply	Sensitise stakeholders on the impact of the standard
6. Reduction in emission due to low activities	This has presented an opportunity for use of alternative energy sources	NERSA needs to adjust its policies and processes to address procurement of large renewable energy projects. The regulatory methodologies to deal with these need to be revised

Environmental factors	Impact if factor is not addressed	NERSA response to the factor
Piped-Gas Industry Regulation		
1. Environmental activism, global warming, carbon taxes and emissions reduction	Gas market cannot grow	- Continued regulatory advocacy and engagements with relevant policy makers – specifically to promote gas as a more attractive option and environmentally friendly energy source - Monitor developments and decisions taken by the G20 and climate change agreements
2. Shale Gas hydraulic fracturing perceived as an environmental threat	- SA misses out on shale gas potential - SA misses out on an opportunity to become energy self-sufficient	- Conduct research on shale gas and the environment - Continued regulatory advocacy and engagements with relevant policy makers - Participate in national debate on shale gas and task teams where possible
Petroleum Pipelines Industry Regulation		
1. Reduction of carbon emissions	- Additional cost to the economy with no alternative fuel source of any scale - Taxes applied by the economy cannot respond to the signal	Develop a report on the impact of the introduction of the Carbon Tax Act
2. Automotive industry is globally moving towards cleaner fuels and the market demand for cleaner fuels is increasing.		
Transversal Regulatory and Organisational		
1. Environmental levies and Carbon tax policy	- SA not meeting its environmental targets - Lack of affordability - Policy uncertainty	- Continued regulatory advocacy and engagements with relevant policy makers - Monitor developments and decisions taken by the G20
2. Delays in issuing environmental Impact Assessments	Security of supply	Continued regulatory advocacy and engagements with relevant policy makers
3. Health and Safety	Possible environmental disasters such as petroleum/gas leaks from pipelines, wind turbine blades coming loose etc.	NERSA to ensure that it discharges its responsibility regarding health and safety

Table 11: Legal factors

Legal factors	Impact if factor is not addressed	NERSA response to the factor
Electricity Industry Regulation		
1. Electricity Regulation Act under review	It will compromise the regulation of electricity supply industry	Continued regulatory advocacy and engagements with relevant policy makers, with specific reference to the need for effective regulation of electricity supply industry
2. Regulatory Principles compromise	- Loss of credibility - Listed as Regulatory Risk - NERSA subject to liability claims	Make sure all decisions are made in accordance with sound regulatory principles.
Piped-Gas Industry Regulation		
1. Delays in legislative amendments and developments	- May deter entry into the gas market - Weak mandate on regulation of piped-gas - Uncertainty in terms of the separation of the oil and gas provision in the Bill	Continued regulatory advocacy and engagements with relevant policy makers
Petroleum Pipelines Industry Regulation		
1. Fragmentation of legislations – possible consolidation of downstream petroleum legislation	- Regulatory burden to licensees - Duplication of resources	- Continued regulatory advocacy and engagements with relevant policy makers - Prepare for defragmentation
2. Possible legal / legislative intervention: - Petroleum Liquid Fuels Sector Codes - Petroleum Pipelines Act and Regulations - Mineral and Petroleum Resources Act	- Regulatory uncertainty - Non-compliance with the BBBEE Act in issuing licenses	- Continued regulatory advocacy and engagements with relevant policy makers - Continued efficient regulation - Amend licensing rules to include BBBEE requirements

Legal factors	Impact if factor is not addressed	NERSA response to the factor
Transversal Regulatory and Organisational		
1. National Energy Regulator Amendment Bill	- NERSA's views not taken into consideration - NERSA not ready when the National Energy Regulator Amendment Bill becomes operational	- Continued regulatory advocacy and engagements with relevant policy makers Regulatory Advocacy - Proactively start preparing for a change in mandate
2. Ability to influence supplementary legislation	- NERSA's views not included - NERSA's powers weakened	- Develop a strategic engagement framework on developing legislation/policy changes - Continued regulatory advocacy and engagements with relevant policy makers
3. Compliance with regulatory requirements (Public Finance Management Act and others)		
4. Fragmentation of legislations		
5. Infrastructure Development Act	Expectation to fund out of tariff and tax instead of by investment.	Develop a position paper on what the funding model should be
6. Pending legal cases	Uncertainty on regulatory decisions and regulatory tools	Implement decisions of the court as soon as the judgement is given

5.2. Internal Environment Analysis

5.2.1. Organisational Capacity

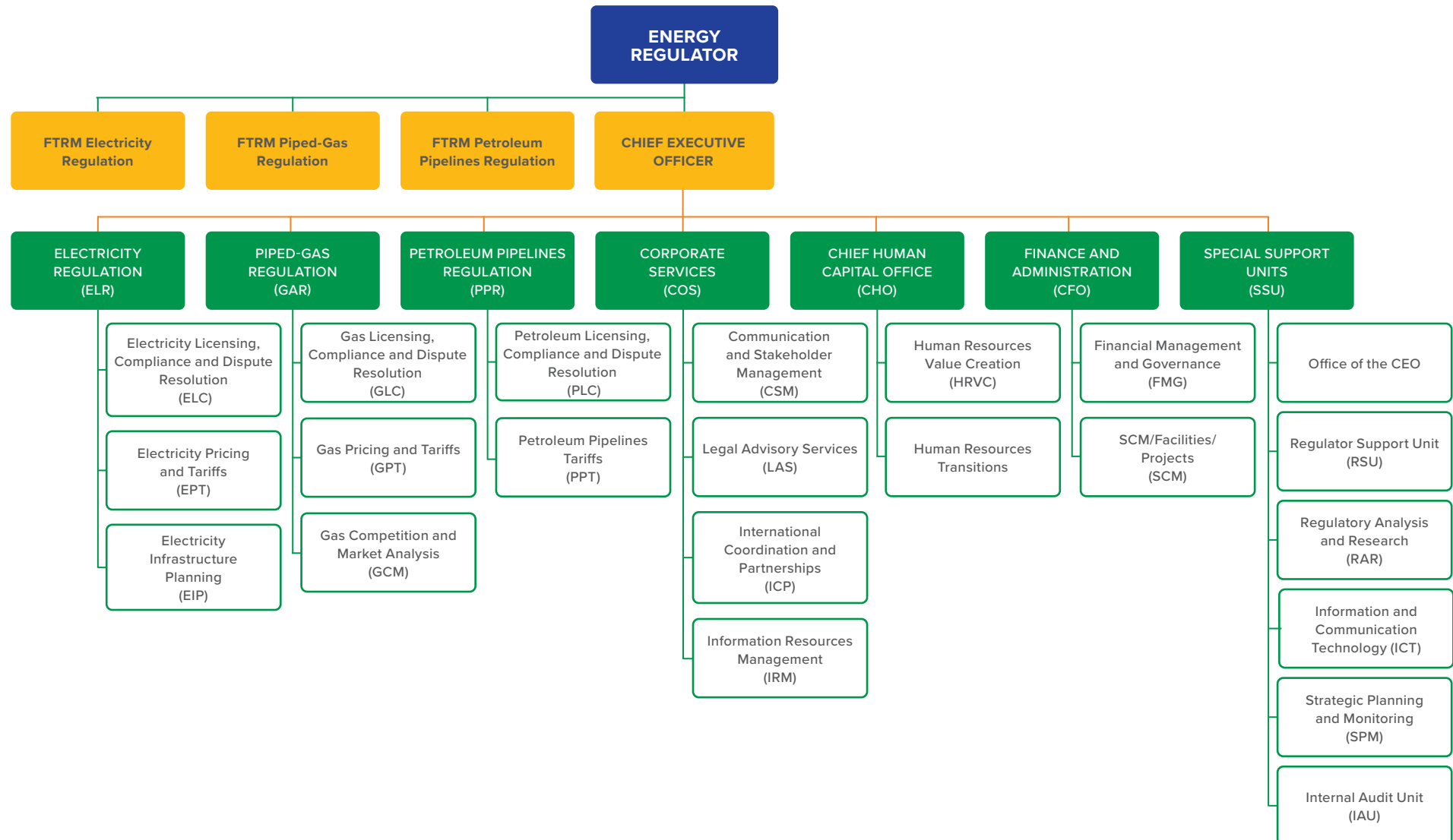
- a) NERSA has an approved structure of 253 staff members. The staff strength as at 31 January 2020 is 250. This includes the 224 permanent employees, three Full-Time Regulator Members (FTRMs), 4 fixed-term contract employees and 19 interns.
- b) Table 12 below summarises the staff complement of NERSA.

Table 12: NERSA Staff complement

Division	Department	Complement
Electricity Regulation (ELR)	FTRM	3
	Executive	3
	Electricity Pricing and Tariffs (EPT)	35
	Electricity Licensing, Compliance and Dispute Resolution (ELC)	34
	Electricity Infrastructure Planning (EIP)	13
Total		88
Piped-Gas Regulation (GAR)	FTRM	3
	Executive	5
	Gas Pricing and Tariffs (GPT)	8
	Gas Licensing, Compliance and Dispute Resolution (GLC)	11
	Gas, Competition and Market Analysis (GCM)	4
Total		31
Petroleum Pipelines Regulation (PPR)	FTRM	3
	Executive	6
	Petroleum Pipelines Tariffs (PPT)	9
	Petroleum Licensing, Compliance and Dispute Resolution (PLC)	9
Total		27
Finance and Administration (CFO)	Executive	3
	Financial Management and Governance (FMG)	7
	Supply Chain Management	13

Division	Department	Complement
Total		23
Human Resources (CHO)	Executive	2
	Human Resources – Value Creation	8
	Human Resources -Transactions	3
Total		13
Corporate Services (COS)	Executive	3
	Legal Advisory Services (LAS)	6
	Communication and Stakeholder Management (CSM)	9
	International Co-ordination and Partnerships (ICP)	3
	Information Resources Management (IRM)	7
Total		28
Specialised Support Units (SSU)	Internal Audit (IAU)	7
	Strategic Planning and Monitoring (SPM)	4
	Regulator Support (RSU)	11
	CEO's Office Operations (COO)	5
	Regulatory Analysis and Research (RAR)	6
	Information and Communication Technology (ICT)	10
Total		43
Grand Total NERSA Staff Complement		253

c) Below is the approved NERSA Organisational Structure:



5.2.2. Status regarding compliance with the BBEE Act

In 2017 /2018, NERSA embarked on its first B-BBEE accreditation and was awarded a Level eight (8) B-BBEE contribution status level. According to the B-BBEE report, NERSA was accredited a Level seven (7) B-BBEE contribution Status. However, because NERSA's skills development and enterprise development did not meet the minimum threshold, NERSA was discounted to a Level eight (8) contribution level.

Plans have been developed and implemented to improve the skills development and enterprise development requirements. In March 2019 the Energy Regulator approved the Enterprise Development Strategy and implementation commenced from April 2019.

5.2.3. Status regarding women and people with disabilities

- As at the end of 31 December 2020, NERSA's staff strength is 236 and comprises 100 (42%) males and 136 (58%) females.
- As at the end of 31 December 2020, there are 13 (54%) females and 11 (46%) males in management positions.
- As at the end of 31 December 2020, the percentage of persons with disabilities is 2%.

5.2.4. Strengths, Weaknesses, Opportunities and Threats facing NERSA

A Strengths, Weakness, Opportunities and Threats (SWOT) framework was used to analyse the internal situation at NERSA. Each element of the SWOT analysis was further categorized into key themes and documented in the table below.

Strengths	
Theme	Factors
Financial Outlook	- Year-on-year annual budget increase (pre COVID-19) - Stable revenue stream as actual revenue reported was 0.9% greater than what was budgeted for before the pandemic
Skilled Workforce	- Skilled personnel with extensive knowledge and understanding on how licensees work and how to support/ respond to solve problems - There is a high degree of transparency - Intimate knowledge of regulated industries - Staff complement constitutes 57% females, which positively contributes to the is developmental agenda
COVID-19 Response	- Adequate management of NERSA's operational, maintenance and safety expenditures in order to effectively respond to the COVID-19 pandemic and ensure investments stabilise/ improve over time - New ways of working in response to COVID-19 pandemic have enhanced staff wellbeing and ensured business continuity
Corporate Governance	NERSA has been able to achieve consecutive clean audit reports during the past five years
Environmentally Sustainable	NERSA is continuing their journey towards becoming a green organisation by reducing paper usage and carbon footprint. The organisation is also busy with the refurbishment of its building where after it will receive a Green Building certificate on completion of the project

Weaknesses	
Theme	Factors
B-BBEE Threshold Compliance	• Inability to meet the minimum threshold targets on spend for skills development and enterprise development resulting in NERSA's overall B-BBEE level being discounted from a Level 7 to a Level 8
Cultural Issues	• Perceptions of bureaucracy exist widely throughout the organisation • High staff turnover, especially at the top management band • Slow decision making and delegation of associated activities • Sense of “bullied” / “unhappy” employees resulting in the increased legal matters/ issues and the rise of grievances • Span of control in the divisional organisational structure is not optimal • Trust issues persists and result in challenges in obtaining external support in the development of information systems • Ambiguous roles and responsibilities between NERSA governance and management/ executive committees • Lack of clear direction from members as instructions continuously change • Lack of exposure of employees to the industry • Lack of training • Lack of innovation/ creation of new ideas • Misalignment in terms of actual vs expected remuneration • Decisions are not aligned to the developments within the global and regional energy industry • Skills of individuals are under-utilised • Several team members feel as though they are working in silos • Lack of knowledge sharing, cooperation, and collective decision making/ contributions from various departments • Gap between organisational structure and skills required in specific areas (e.g. relevant and needed skillsets among board composition)

Weaknesses	
Theme	Factors
External Relationships	There is a need to focus on improving relationships between NERSA and industry players and ensure information is shared, collected and used in a timely manner that benefits the collective
	Negative perceptions of stakeholders exists
	There is a need to clarify the roles between NERSA and the Environmental department
Processes and Procedures	Lack of effective and efficient documentation management (process to receive and archive/store documents in soft and hard copy formats)
	Government policies make it difficult to automate and streamline processes
	Internal processes and systems need to be updated, digitised and improved in order to support the “new normal” ways of working (e.g. home working)
	Decentralised and fragmented data – as information is not stored on a single platform and cannot be easily accessed for informed decision-making (licensee data and information)
	NERSA is mainly responsive to industry developments (need to become more proactive)
	Improvements in retention processes and procedures is required
	There is a need for formalised procedures to manage time (e.g. meeting) and ensure productivity
Organisational Strategy	Lack of alignment of structure to strategy
	Lack of common understanding of the strategy

Opportunities	
Theme	Factors
Implementation of Technological Innovations	Technological progress has allowed for new forms of producing, storing, transforming, and consuming energy, altering the nature of the energy system (need to keep up with the pace of technological change)
	There is a need to integrate new technologies and business models into existing structures
	There is a need to establish a process to collect information from industries
Response to changing customer Needs	NERSA should consider the option of utilising electrification funds that are collected through the tariff to support vulnerable customers who are unable to afford their energy bills
	Use of analytics (energy modelling, investment, and economic driver analytics) to inform a demand-led strategy
	Ensure energy security through; reduction of the regulatory burden on new electricity applications, ensure sector regulatory certainty, fast-tracking of application processing and consider proposals on the reduction of energy prices
Collaboration and Relationship Development	Continue interactions with DMRE in order to legislate and establish a structure to implement mandate
	Implement DMRE regulations to unlock significant local production and importation (when there is a shortfall) of LPG
	Implement SADC's established Regional Electricity Regulatory Association (RERA) that will assist in harmonising the region's cross border policies and regulations (once finalised)
Departments/ Divisions	Opportunity exists for NERSA to develop/ employ individuals responsible for data modelling in order to accelerate the decision-making processes
	Improve the alignment of internal characteristics to the external environment
	There is potential to invest in additional digital infrastructure / innovations across the value chain in order to stay up to date/ ahead of the market
	Prepare NERSA for different outcomes/ responses to disaster using scenario based responses (e.g. mild, harsh, severe)
Employees	Balance diverse expectations of employees in order to build trust within NERSA
	Capitalise on the ability to learn from new colleagues, to generate new ideas and remain relevant

Threats	
Theme	Factors
COVID-19 Pandemic	- The COVID-19 crisis may have a significant impact on investments, sustainability of energy supply, ability to invest in aging electricity networks, infrastructure and revenues due to changes in industry volumes - Adjust to new ways of working, upskilling staff and continued virtual activities/ operations
Energy Finance Sector	There has been a slow migration to cost reflective tariffs, inadequate project preparation, issues with Power Purchase Agreements, and absent regulatory frameworks which stunt investment and financing in the energy sector
Regulatory Landscape	- Lack of region-wide regulatory framework that addresses renewable energy - Limited relevance of regulation within the emerging distributed energy landscape - Regulatory control within the entire supply chain of the regulated industries is limited - Projects intended to address the supply shortage are delayed due to absent regulatory frameworks and below-cost tariffs which indirectly impacts the ability for energy operators/ suppliers to sustain demand - Decisions have been legally challenged - Encroachment of various departments in running NERSA affairs
Technological Advancements	- Fast changing energy landscape due to emerging innovative energy generation technologies - NERSA will have to move fast to keep up with the pace of technological change and the rising need for flexible operation of power systems - Regulatory frameworks need to balance the need for providing certainty while being flexible enough to effectively integrate new technologies and business models
Economic Outlook	There is an unknown long-term impact on the economy and industry as a result of the recession, pandemic, credit downgrade, poverty, and inequality
Industry Changes	- There has been changes to the various operating industries (Sasol Gas' intentions to divest in some of its infrastructure assets) - There have been amendments to the competition act which need to be accounted for from a regulatory perspective - Industry development creates challenges in terms of legislation
Legislation Issues	Several instances of legislative shortcomings persist

PART C: MEASURING OUR PERFORMANCE

1. Institutional Performance Information

NERSA's mandate is to regulate the electricity, piped-gas and petroleum pipelines industries in line with each industry's specific legislation and regulations. Therefore, this part of the Strategic Plan will be divided into sections for each of the regulated industries as well as a section dealing with transversal regulatory and organisational matters.

2. Impact Statement

In line with Government's priorities, NERSA's overall impact statement is as follows:

Secure, reliable, affordable, sustainable, competitive and transformed energy industry, which contributes to the economic growth of South Africa.

3. Measuring our Outcomes

The attainment of the above impact statement will be driven by the industry specific and organisational impact statements and accompanying outcomes, as described in the sections below.

3.1. Electricity Industry Regulation

IMPACT STATEMENT		<i>Sustainable, cost reflective, reliable, accessible, affordable, transparent, and demand-led electricity supply from diverse energy sources in a manner that maximises economic growth and transformation</i>		
Outcome	Outcome Indicator	Baseline	Five Year Target	
1. Accessible and cost reflective electricity that is equitably distributed for consumption	1.1. Energy Regulator decision on the review of Eskom's revenue application for year 1 of MYPD 5 considered by the relevant committee or the Energy Regulator within the stated timeframe of receiving all the required information for the application	Approved MYPD 4	Energy Regulator decision on the review of Eskom's revenue application for year 1 of MYPD 5 considered by the ER within 6 months after receipt of complete application	
	1.2. Percentage of complete tariff applications of licensed distributors considered by the relevant committee or the Energy Regulator within the stated timeframe	100% of complete tariff applications of licensed distributors considered by the REC/ELS within 60 working days of receipt of complete application	100% of complete tariff applications of licensed distributors considered by the REC/ELS within 60 working days of receipt of complete application	
2. Diverse energy supply that is certain and secure for current and future user needs	2.1. Percentage of complete applications for a licence or for the registration of electricity generation activities are considered by the relevant committee or the Energy Regulator within the stated timeframes of receiving all the required information for the application	100% of complete licence applications considered by the ER within 120 working days after the period of objections expired and no objections were received or after objections are addressed 100% of complete applications for registration of electricity generation facilities considered by the ELS within 60 days from receipt of all required information	100% of complete licence applications considered by the ER within 120 working days after the period of objections expired and no objections were received or after objections are addressed 100% of complete applications for registration of electricity generation facilities considered by the ELS within 60 days from receipt of all required information 120 days for licence applications	
	2.2. Approved regulatory framework for the licensing of the restructured electricity supply industry following the unbundling of Eskom	Approved regulatory framework for the licensing of the current electricity generation activities	Approved regulatory framework for the licensing of the restructured electricity supply industry following the unbundling of Eskom by 31 March 2025	

IMPACT STATEMENT

Sustainable, cost reflective, reliable, accessible, affordable, transparent, and demand-led electricity supply from diverse energy sources in a manner that maximises economic growth and transformation

Outcome	Outcome Indicator	Baseline	Five Year Target
2. Diverse energy supply that is certain and secure for current and future user needs (continued)	2.3. Number of audit reports on the state of compliance with licence conditions considered by the relevant committee or the Energy Regulator within the stated timeframe	3 audit reports by 31 March (one each for generation, transmission and distribution) on the state of compliance with licence conditions considered by the ELS/REC by 31 March	3 audit reports (one each for generation, transmission and distribution) on the state of compliance with licence conditions considered by the ELS/REC by 31 March
	2.4. Number of reports on the System Adequacy considered by the relevant committee or the Energy Regulator within the stated timeframe	1 report considered annually by the ELS/REC by 31 March	1 report considered annually by the ELS/REC by 31 March
	2.5. Number of reports on the performance of renewable technologies considered by the relevant committee or the Energy Regulator within the stated timeframe	1 report considered annually by the ELS/REC by 31 March	1 report considered annually by the ELS/REC by 31 March
	2.6. Number of reports on the analysis of Eskom's actual performance against the regulatory framework considered by the relevant committee or the Energy Regulator within the stated timeframe	1 report considered annually by the ELS/REC by 31 March	1 report considered annually by the ELS/REC by 31 March

IMPACT STATEMENT		<i>Sustainable, cost reflective, reliable, accessible, affordable, transparent, and demand-led electricity supply from diverse energy sources in a manner that maximises economic growth and transformation</i>	
Outcome	Outcome Indicator	Baseline	Five Year Target
3. Conducive regulatory environment that results in regulatory certainty and increased investment in the electricity industry	3.1. Number of reports on new entrants into the electricity supply industry considered by the relevant committee or the Energy Regulator within the stated timeframe	1 report considered annually by the ELS by 31 March	1 report considered annually by the ELS by 31 March
	3.2. Approved revenues and tariffs to facilitate the sustainability of electricity supply considered by the relevant committee or the Energy Regulator within the stated timeframe	Eskom's Retail Tariff Structural Adjustments (ERTSA) in line with statutory requirements considered by the relevant committee of the Energy Regulator annually for tabling in Cabinet before 15 March	Report on analysis of Eskom's ERTSA for the coming financial year considered by the ELS/ER within 6 months after receipt of complete application
		Approved tariff methodology for current electricity supply industry	Approved tariff methodology for the restructured electricity supply industry following the unbundling of Eskom considered by the ER by 31 March 2025
	3.3. Percentage of applications from the ESI requiring amendment to or exemption from the South African grid code, considered by the relevant committee or the Energy Regulator within stated timeframe from receipt of complete information	100% applications from the ESI requiring amendment to or exemption from the South African grid code considered by the ELS/REC within 3 months from receipt of complete information	100% applications from the ESI requiring amendment to or exemption from the South African grid code considered by the ELS/REC within 3 months from receipt of complete information
	3.4. Number of reports on regulatory advocacy aimed at improvement of the regulatory framework provided through legislation, regulation and government policies considered by the relevant committee or the Energy Regulator	1 report considered annually by the ELS by 31 March	1 report considered annually by the ELS by 31 March

3.1.1. Explanation of Planned Performance over the Five Year Planning Period

- a) The rationale for the choice of the outcome indicators relevant to the respective outcomes is aligned with legislative requirements.
- To approve municipal tariffs that ensure the financial viability and sustainability of all licensed municipal distributors while also protecting the poor from rapidly increasing electricity prices;
 - To approve Eskom's revenue requirements and prices/tariffs that allows for the sustainability of Eskom and therefore overall viability of the electricity supply industry.
 - Ensure certainty for new licensees, in making sure they know all the applicable conditions in order to be connected to the grid.
 - Ensure oversight of non-compliance to Grid Code to ensure speedy compliance
 - Ensure risk mitigating measures are implemented in time to support security of supply
 - Medium to long term infrastructure development planning is implemented according to the set license conditions
- b) The following enablers were identified to achieve the five-year targets:
- Revised MYPD methodology;
 - Monitoring of licensed distributor's performance;
 - Tariff methodology;
 - Wheeling methodology;
 - Automated assistance to the licensing application and evaluation process;
 - Grid Governance Code;
 - Restructuring of the electricity supply industry; and
 - Increasing resources within NERSA.
- c) The identified outcomes should contribute as follows to the achievement of the impact statement:
- Sustainability of the electricity supply industry;
 - Protection of the poor from rapidly increasing electricity prices;
 - Make available grid code requirement for each technology;
 - Audits will highlight areas of need and tariff decisions will provide funds to perform refurbishment;
 - An up-to-date data base containing all submitted information in a format that can be easily interrogated;
 - Reporting requirements are regularized by inclusion in the Grid Code;
 - Encourage entry of new players;
 - The licensing of operators ensures orderly development and the license conditions ensure that the licensees comply with proper standards;
 - Regulatory certainty through appropriate pricing and tariffs methodologies.

3.2. Piped-Gas Industry Regulation

IMPACT STATEMENT		<i>Sustainable, cost reflective, reliable, accessible, affordable, transparent, and demand-led electricity supply from diverse energy sources in a manner that maximises economic growth and transformation</i>		
Outcome	Outcome Indicator	Baseline	Five Year Target	
1. Equitable access to affordable gas services at competitive prices	1.1. Percentage of complete maximum price applications considered by the relevant committee or the Energy Regulator within the stated timeframe	100% of complete maximum price applications considered by the ER within 120 working days after date of publication of the preliminary assessment of the maximum price applications	100% of complete maximum price applications considered by the ER within 120 working days after date of publication of the preliminary assessment of the maximum price applications	
	1.2. Percentage of complete transmission tariff applications considered by the relevant committee or the Energy Regulator within the stated timeframe	100% of complete transmission tariff applications considered by ER within 120 working days after date of publication of preliminary assessment of tariff applications	100% of complete transmission tariff applications considered by the ER within 120 working days after date of publication of preliminary assessment of tariff applications	
	1.3. Number of reports on the review of the definition of the piped-gas market considered by the relevant committee or the Energy Regulator within the stated timeframe	No report available	1 report considered annually by the PGS by 31 March	
	1.4. Number of reports on the impact of developments on competition in the gas industry considered by the relevant committee or the Energy Regulator within the stated timeframe	No report available	One report on the impact of developments on competition in the gas industry considered annually by the PGS by 31 March	

3.2. PIPED-GAS INDUSTRY REGULATION (continued)

IMPACT STATEMENT	<i>Sustainable, cost reflective, reliable, accessible, affordable, transparent, and demand-led electricity supply from diverse energy sources in a manner that maximises economic growth and transformation</i>		
2. Efficient, sustainable, equitable and orderly development of a transformed, competitive and accessible piped gas industry	2.1. Percentage of complete licence applications considered by the relevant committee or the Energy Regulator within the stated timeframe	100% of complete applications considered by the PGS/REC within 60 working days from date of close of public comment period or period of applicant's response to objections received	100% of complete applications considered by the PGS/REC within 60 working days from date of close of public comment period or period of applicant's response to objections received
	2.2. Percentage of complete registration applications of gas activities considered by the relevant committee or the Energy Regulator within the stated timeframe	100% of complete applications for the registration of gas activities are processed and considered by the PGS within 60 working days from date of close of public comment period	1 audit report on the compliance of ROMPCO pipeline considered annually by the PGS by 31 March
	2.3. Number of audit reports on compliance of the ROMPCO pipeline according to the compliance frameworks considered annually by the relevant committee or the Energy Regulator within the stated timeframe	1 audit report on the compliance of ROMPCO pipeline considered annually by the PGS by 31 March	1 audit report on the compliance of ROMPCO pipeline considered annually by the PGS by 31 March
	2.4. Number of monthly volume balance reports assessed and analysis reports to monitor the supply of 120m GJ p.a. from Mozambique to South Africa considered by the relevant committee or the Energy Regulator within the stated timeframe	12 monthly volume balance reports considered annually by the PGS by 31 March	12 monthly volume balance reports considered annually by the PGS by 31 March
	2.5. Number of reports on licensees' compliance with licence conditions considered by the relevant committee or the Energy Regulator within the stated timeframe	One annual report considered by PGS by 31 March regarding compliance inspections conducted on licensed facilities	One annual report considered by PGS by 31 March regarding compliance inspections conducted on licensed facilities

IMPACT STATEMENT	<i>Sustainable, cost reflective, reliable, accessible, affordable, transparent, and demand-led electricity supply from diverse energy sources in a manner that maximises economic growth and transformation</i>		
3. A conducive regulatory environment that results in regulatory certainty and increased investment in the piped-gas industry	3.1. Number of reports on regulatory mechanisms required for the review of licensing rules considered by the relevant committee or the Energy Regulator within the stated timeframe	No report available	1 report on regulatory mechanisms required for the review of licensing rules considered by the PGS by 31 March 2022
	3.2. Revised tariff methodology considered by the relevant committee or the Energy Regulator within the stated timeframe	Current tariff methodology	Revised tariff methodology considered by the ER by March 2025
	3.3. Refined framework for conducting adequacy of competition in the gas industry considered by the relevant committee or the Energy Regulator within the stated timeframe	Approved framework for the determination of the adequacy of competition in the gas sector	Refined framework for conducting adequacy of competition in the gas industry by 31 March 2021
	3.4. Number of reports on regulatory advocacy aimed at improvement of the regulatory framework provided through legislation, regulation and government policies considered annually by the relevant committee or the Energy Regulator within the stated timeframe	1 report on regulatory advocacy considered annually by the PGS by 31 March	1 report on regulatory advocacy considered annually by the PGS by 31 March

3.2.1. Explanation of Planned Performance over the Five Year Planning Period

- a) The rationale for the choice of the outcome indicators relevant to the respective outcomes are the following:
- To allow customers to have a choice on the source of supply which will improve customer countervailing power, as well as the quality, cost and efficiency of supply of gas;
 - To promote enhanced entry into the gas supply market;
 - To improve access to gas supply and services;
 - To facilitate the growth of the gas sector in support of industrialization;
 - To increase access to and utilisation of gas in the market;
 - To promote compliance with licence conditions;
 - To regulate maximum prices and tariffs so as to mimic competitive outcomes in the gas market; and
 - To facilitate effective regulation of cross border assets.
- b) The following enablers to achieve the five-year targets were identified:
- Revised methodology for gas prices and tariffs to attract investment;
 - Efficient licensing of gas infrastructure;
 - Facilitation of 3rd party access to uncommitted capacity;
 - Effective framework for regulation of the gas industry;
 - Periodic assessment of adequacy of competition;
 - Compliance investigations;
 - Effective compliance monitoring and enforcement;
 - Adequate supply of gas to meet demand; and
 - Effective collaboration with other regulatory bodies such as TNPA and Competition Commission on matters of common interest
- c) The identified outcomes will contribute to the achievement of the impact as follows:
- Improved competition, leading to more competitive pricing and wider choice for customers;
 - Improved security of supply;
 - Effective regulation of licensed activities, maximum prices and tariffs;
 - Promote import competition;
 - Growth in gas imports and production;
 - Switching to gas as an alternative energy source;
 - Review of Methodologies and the tariff guidelines; and
 - Enforcement of third party access.
- d) The following challenges have been identified:
- Current gaps in the Gas Act present a challenge on effective regulation of the gas industry e.g. Nersa has no mandate to regulate distribution tariffs, no third party access to gas distribution infrastructure etc. This increases barriers to entry and expansion at all levels of the gas supply chain, and also leads to other unintended consequences such as inefficient tariffs, and eligible customers migrating from distribution to transmission infrastructure, which may inhibit the orderly development of gas infrastructure.
 - Vertically integrated sole/dominant supplier with monopoly position;
 - Lack of adequate gas supply - SA does not have its own indigenous gas sources and currently relies on supply from Mozambique. This presents a challenge for security of supply, especially given the noted potential decline of gas supply from Mozambique from 2024;
 - No mandatory third party access to gas distribution pipelines;
 - Impact of exclusivity on distribution licenses with potential market foreclosure. This may inter alia affect new investments and entry of new players into the market as it would be the prerogative of the incumbent distribution network owner to allow entry into exclusive distribution areas;

- Impact of COVID-19 has weakened enforcement abilities of the Energy Regulator on on-site inspections;
- Weak enforcement model in the current Gas Act;
- Inadequate competition; and
- Dated gas infrastructure in some areas - results in increased maintenance costs with impact on tariffs.

3.3. PETROLEUM PIPELINES INDUSTRY REGULATION

IMPACT STATEMENT		Efficient, safe, effective, sustainable, competitive and transformed petroleum pipelines industry		
Outcome	Outcome Indicator	Baseline	Five Year Target	
1. Equitable access to affordable petroleum products, services and infrastructure at competitive prices	1.1. Number of reports on the percentage utilisation of pipelines, storage facilities and loading facilities and third party access considered annually by the relevant committee or the Energy Regulator	2 reports on the percentage utilisation of pipelines, storage facilities and loading facilities and third party access considered annually by the PPS by 31 March	1 report on the percentage utilisation of pipelines, storage facilities and loading facilities and third party access considered annually by the PPS by 31 March	
	1.2. Percentage of complete pipeline, storage and loading facility tariff applications considered by the relevant committee or the Energy Regulator within the stated timeframe	75% of complete pipeline, storage and loading facility tariff applications considered by the PPS/ER within 6 months from receipt of application	75% of complete pipeline, storage and loading facility tariff applications considered by the PPS/ER within 6 months from receipt of application	
	1.3. Approved efficiency adjustment factor for inclusion into the tariff methodology	None	Approved efficiency adjustment factor for inclusion into the tariff methodology by 31 March 2024	

3.3. PETROLEUM PIPELINES INDUSTRY REGULATION (continued)

IMPACT STATEMENT	<i>Efficient, safe, effective, sustainable, competitive and transformed petroleum pipelines industry</i>			
2. Efficient, sustainable, equitable and orderly development of a transformed, competitive and accessible petroleum pipelines industry	2.1. Percentage of complete licence applications considered by the relevant committee or the Energy Regulator within the stated timeframe	100% of complete licence applications considered by the PPS/REC/ER within 60 working days under the conditions as prescribed in Section 19(1) of the Petroleum Pipelines Act	100% of complete licence applications considered by the PPS/REC/ER within 60 working days under the conditions as prescribed in Section 19(1) of the Petroleum Pipelines Act	
	2.2. Percentage of complete applications for licence amendments / revocations considered by the relevant committee or the Energy Regulator within the stated timeframe	100% of complete applications for licence amendments / revocations considered by the PPS/REC/ER within 60 working days under the conditions as prescribed in Sections 23 or 24 of the Petroleum Pipelines Act	100% of complete applications for licence amendments / revocations considered by the PPS/REC/ER within 60 working days under the conditions as prescribed in Sections 23 or 24 of the Petroleum Pipelines Act	
	2.3. Number of reports on the geographic spread of licences issued for petroleum infrastructure and new entrants considered annually by the relevant committee or the Energy Regulator	No report available	1 report on the geographic spread of licences issued for petroleum infrastructure and new entrants considered annually by the PPS by 31 March	
3. A conducive regulatory environment that results in regulatory certainty and increased investment in the petroleum industry	3.1. Reviewed tariff methodology for storage, loading facilities and petroleum pipelines considered by the relevant committee or the Energy Regulator within the stated timeframe	Commenced with the review of the tariff methodology for storage and loading facilities to provide regulatory certainty and facilitate investment in the industry.	Approved reviewed tariff methodology for storage and loading facilities; and petroleum pipelines by 31 March 2025	
		Approved tariff methodology for petroleum pipelines		
	3.2. Number of reports on regulatory advocacy aimed at improvement of the regulatory framework provided through legislation, regulation and government policies for the petroleum pipelines industry considered annually by the relevant committee or the Energy Regulator	1 report on regulatory advocacy considered annually by the PPS by 31 March	1 report on regulatory advocacy considered annually by the PPS by 31 March	

3.3.1. Explanation of Planned Performance over the Five Year Planning Period

- a) The rationale for the choice of the outcome indicators relevant to the respective outcomes are the following:
 - To promote competition in the construction.
 - To facilitate access to affordable petroleum products.
- b) The following enablers to achieve the five-year targets were identified:
 - Enabling legislation to be amended;
 - Revised tariff methodology;
 - Benchmark study to be able to assess prudency;
 - Efficient processing of applications;
 - Review of licensing rules;
- c) The identified outcomes will contribute to the achievement of the impact as follows:
 - Enforcement of compliance by the Tribunal; and
 - Audits for compliance monitoring.
- Lower the barriers to entry;
 - More transformed Industry;
 - Affordable tariffs;
 - Promote import competition;
 - Sufficient capacity to meet market demand;
 - Revised enabling legislation;
 - Reduced regulatory burden;
 - Improved third party access; and
 - Harmonized regulatory framework.

3.4. TRANSVERSAL REGULATORY AND ORGANISATIONAL

IMPACT STATEMENT		<i>NERSA established and perceived as an efficient, effective and credible regulator</i>		
Outcome	Outcome Indicator	Baseline	Five Year Target	
1. Creation of an enabling environment for internal and external stakeholders through proactive, dynamic and data-driven advisory, advocacy and decision making	1.1. Percentage of regulatory processes is based on appropriate Research within stated timeframe	None	100% of regulatory processes is based on appropriate research by 31 March 2025	
	1.2. Number of reports on partnership creation to position NERSA as a recognised regulator nationally, regionally and internationally considered annually by the relevant committee or the Energy Regulator by 30 September and 31 March	2 reports on partnership creation to position NERSA as a recognised regulator nationally, regionally and internationally considered annually by the REC by 31 March	2 reports on partnership creation to position NERSA as a recognised regulator nationally, regionally and internationally considered annually by the REC by 31 March	
	1.3. Number of reports on the implementation of the Learnership and Internship Programmes considered annually by the relevant committee or the Energy Regulator by 31 March	1 report on the implementation of the Learnership and Internship Programmes considered annually by the HRRC by 31 March	1 report on the implementation of the Learnership and Internship Programmes considered annually by the HRRC by 31 March	
	1.4. Improved level of contribution towards BBBEE	BBBEE Level 8	BBBEE Level 3 by 31 March 2025	

Outcome	Outcome Indicator	Baseline	Five Year Target
2. An effective operating model that enables the organisation to fulfil its role effectively	2.1. Percentage of business processes are automated and efficient within the stated timeframe	Majority of business processes are manual	70% of processes are automated and efficient by 31 March 2025
	2.2. Improved efficacy of the NERSA based on an organisational knowledge management approach	Knowledge management framework, Strategy and implementation plan 2010	Reviewed Knowledge management framework and Strategy with an implementation plan considered by the ER by 31 March 2022 Report on the improved of the efficacy of the regulator based considered by the REC by 31 March 2024 Reviewed Knowledge management framework and Strategy with an implementation plan considered by the ER by 31 March 2025
	2.3. Good governance demonstrated in audit outcomes	Unqualified audit	Unqualified with no findings in the management report

3.4.1. Explanation of Planned Performance over the Five Year Planning Period

- a) The rationale for the choice of the outcome indicators relevant to the respective outcomes is to focus on the key requirement for the effective operations of the Energy Regulator.
- b) The following enablers to achieve the five-year targets were identified:
 - Improved data analysis
 - Trends analysis (market study)
 - Speedy processing of applications
 - GIS
 - Reviewed PPA
 - Online application system
- c) The identified outcomes will contribute to the achievement of the impact as follows:
 - Proactively improving critical business and regulatory processes

4. KEY RISKS

The key risks which may affect the achievement of the identified outcomes are those that need to be mitigated in the medium to long-term, as the critical mitigating strategies relates to amendment of legislations and regulation which falls within the ambit.

NERSA's Integrated Enterprise Risk Management Framework provides for processes to manage the mitigating identified risks with quarterly reporting on progress made with the mitigating strategies. These strategies are solely within NERSA's control to mitigate the identified risks.

The risks indicated below are those that will be prevailing in the long-term due to the length of time it will take to mitigate those risks.

OUTCOMES	KEY RISK	RISK MITIGATION
- Conducive regulatory environment that results in regulatory certainty and increased investment in the electricity industry - A conducive regulatory environment that results in regulatory certainty and increased investment in the piped-gas industry - A conducive regulatory environment that results in regulatory certainty and increased investment in the petroleum industry	Unresponsive regulatory framework to landscape changes in the sector	- Improve approach to Regulatory Advocacy (continuously align as the market evolve) - Conduct environmental scan and develop draft studies (methodologies, guidelines and rules) in anticipation of industry changes - Review Methodology and Guidelines - Issue Rules for implementation of IRP
- Diverse energy supply that is certain and secure for current and future user needs - Efficient, sustainable, equitable and orderly development of a transformed, competitive and accessible piped gas industry - Equitable access to affordable petroleum products, services and infrastructure at competitive prices - Efficient, sustainable, equitable and orderly development of a transformed, competitive and accessible petroleum pipelines industry	Regulatory uncertainty	- Emphasise compliance with NERSA's Methodologies in decisions - Review Methodologies to align to current circumstances - Improve approach to Regulatory Advocacy (continuously align as the market evolve) - Regular review of Rules to provide regulatory certainty - Improve MOAs with relevant Authorities
- Accessible and cost reflective electricity that is equitably distributed for consumption - Equitable access to affordable gas services at competitive prices	Rising energy costs - (High energy prices and tariffs)	- Full implementation of Prudency Guidelines - Enforce compliance to ring-fencing of electricity business costs in municipalities to ensure infrastructure maintenance using the allocated revenue - Facilitate development of projects to deliver domestic supply of primary fuels and electricity - Assist DMRE in the implementation National Free Basic Electricity Policy of Government¹.

OUTCOMES	KEY RISK	RISK MITIGATION
- Diverse energy supply that is certain and secure for current and future user needs - Efficient, sustainable, equitable and orderly development of a transformed, competitive and accessible piped gas industry - Equitable access to affordable petroleum products, services and infrastructure at competitive prices - Efficient, sustainable, equitable and orderly development of a transformed, competitive and accessible petroleum pipelines industry	Reputational damage	- Ensure accuracy and high quality of RFDs - Regular review of Regulatory Tools to align with dynamics of the industry (Methodologies/ Rules/ Procedures) - Independent Peer review of NERSA's Regulatory Tools (Methodologies/ Rules/ Procedures) - Enhance compliance with Methodologies - Stakeholder Engagement - Streamline the decision making - Improve quality of Reasons for Decision - Improve regulatory advocacy Develop Standard Operation Procedures - Increase Stakeholder engagement (Public Hearing) - Effective implementation of Stakeholder Engagement Strategy and Plan - Conduct Stakeholder Surveys - Improve NERSA's PR systems
- Diverse energy supply that is certain and secure for current and future user needs - Efficient, sustainable, equitable and orderly development of a transformed, competitive and accessible piped gas industry	Constraints of gas supply	- Coordinated engagement with relevant stakeholders including Sasol Gas, DMRE and PASA /Regulatory Advocacy to develop a coordinated policy to incentivize investment - Prioritize licensing of new projects for LNG imports - Cooperation with authorities in Mozambique for continued gas supply opportunities - Fast-track current activities for local exploration of natural gas resources - Develop Gas-to-Power projects to accelerate development of GAS industry
A conducive regulatory environment that results in regulatory certainty and increased investment in the piped-gas industry	Disruptions of regulatory regime	Harmonisation of regulatory processes between SA and Mozambique

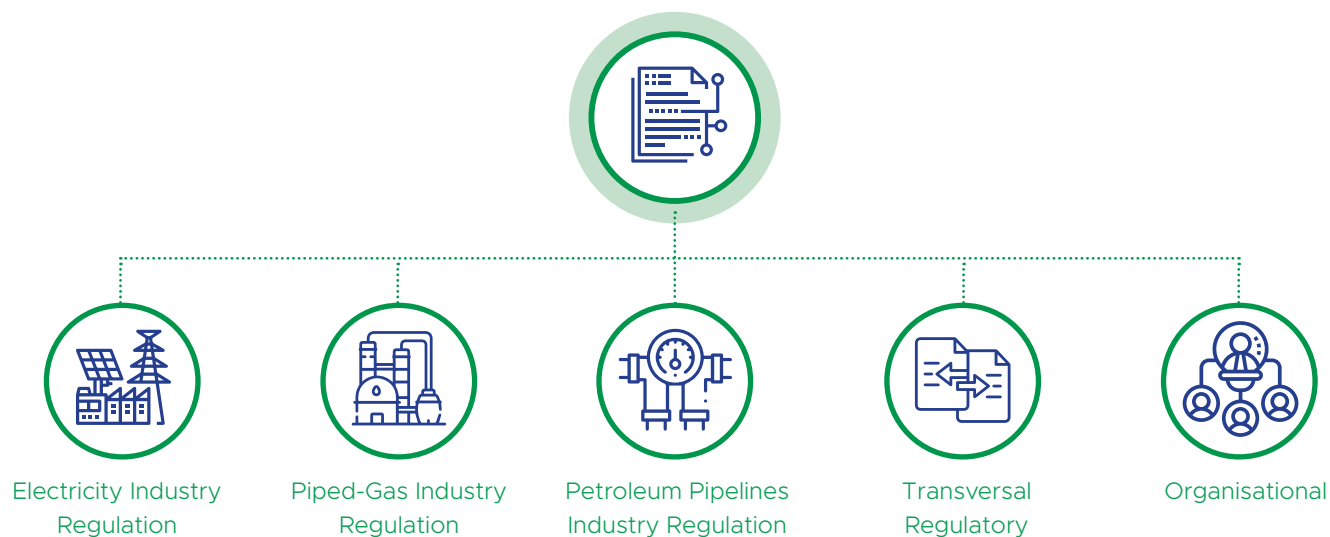
OUTCOMES	KEY RISK	RISK MITIGATION
- An effective operating model that enables the department to fulfil its role effectively - Creation of an enabling environment for internal and external stakeholders through proactive, dynamic and data-driven advisory, advocacy and decision making	Business continuity/ disruptions	- Review financing Model of NERSA as Legislated - Review and implement Business Continuity Plan - Review and implement Business Continuity Management Policy - Conduct Disaster Management Training - Review IT Disaster Recovery Plan - Fund and implement ICT Strategy - Improve current IT support processes - Develop Financial sustainability Strategy - Review and monitoring of Going-concern assessment models - Review current NERSA strategy to make NERSA more relevant and effective
- Efficient, sustainable, equitable and orderly development of a transformed, competitive and accessible piped gas industry - Equitable access to affordable petroleum products, services an infrastructure at competitive prices - Efficient, sustainable, equitable and orderly development of a transformed, competitive and accessible petroleum pipelines industry	Uncompetitive outcomes	- Advocacy with DMRE – to strengthen NERSA’s regulatory powers / address regulatory gaps by amending the Gas and electricity Acts and addressing fragmentation of regulation in the Petroleum Pipelines industry. - Advocacy with DMRE and other regulatory bodies to develop a coordinated policy to incentivize investment - Finalisation of MoA between NERSA and other relevant regulatory authorities to set out processes for collaboration on competition matters in the energy sector - Review Methodologies - Lower barriers to entry

PART D: TECHNICAL INDICATOR DESCRIPTIONS

The technical indicator descriptions (TID) below is a description of the outcome indicators stated in this Plan, which defines the data collection processes and gathering of portfolios of evidence.

These indicators are divided in the following functional areas:

- Electricity Industry Regulation
- Piped-Gas Industry Regulation;
- Petroleum Pipelines Industry Regulation;
- Transversal Regulatory; and
- Organisational.



1. ELECTRICITY INDUSTRY REGULATION

Indicator title	Energy Regulator decision on the review of Eskom's revenue application for year 1 of MYPD 5 within the stated timeframe of receiving all the required information for the application	Percentage of tariff applications of licensed distributors considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	This is the decision of the Energy Regulator on review of Eskom's revenue application, based on the Multi Year Price Determination (MYPD) – which incorporates some of the Rate of Return (RoR) and incentive based principles through the introduction of the transmission and distribution service incentive schemes and the energy efficiency demand side management (EEDSM) schemes.	This is the percentage of complete tariff applications from licensed distributors for increases within the guideline and benchmark that are considered within a specified time frame.
Source of data	Eskom's revenue application; Information supplied by Eskom	Tariff Applications and D Forms; Tariff analysis schedules
Method of calculation / assessment	Decision of the Energy Regulator	(number of tariff applications approved within 60 days of receipt of complete application) / (number of received tariff applications)*100
Assumptions	Eskom submits complete application	Complete applications received from licensees
Disaggregation of Beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired Performance	The Energy Regulator decided upon the recommendations following the review Eskom's revenue application for year 1 of MYPD 5 within 120 working days after receipt of complete application	100% of tariff applications of licensed distributors were considered by the REC/ELS (depending on delegation) within 60 working days of receipt of complete application
Indicator Responsibility	EM (ELR) and HOD (EPT)	EM (ELR) and HOD (EPT)

1. ELECTRICITY INDUSTRY REGULATION (continued)

Indicator title	Percentage of complete applications for a licence or for the registration of electricity generation activities are considered by the relevant committee or the Energy Regulator within the stated timeframes of receiving all the required information for the application	
Definition	This is the percentage of complete licence applications that are considered in compliance within the legislated timeframes.	This is the percentage of complete applications for registration of electricity generation activities that are considered within the legislated timeframes.
Source of data	Licence application	Registration applications
Method of calculation / assessment	(number of processed licence applications within 120 days / number of received licence applications)*100	(number of processed licence applications within 60 days / number of received licence applications)*100
Assumptions	Applicants provide all required information to accept application for analysis	All required information is received from applicants
Disaggregation of Beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired Performance	100% of complete licence applications are considered by the REC/ELS (depending on delegation) and submitted to the Energy Regulator for a final decision, within 120 working days of receiving all the required information for the application	100% of complete applications for the registration of electricity generation activities are considered by the REC/ELS (depending on delegation) and submitted to the Energy Regulator for a final decision within 120 working days of receiving all the required information for the application
Indicator Responsibility	EM (ELR) and HOD (ELC)	EM (ELR) and HOD (ELC)

Indicator title	Number of audit reports on the state of compliance with licence conditions considered by the relevant committee or the Energy Regulator within the stated timeframe	Number of reports on the System Adequacy annually considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	These are annual reports on all the audits NERSA conducted on the state of distribution, generation and transmission licensees' compliance with licence conditions, including audit findings.	These are annual reports in which NERSA indicates the adequacy and performance of the generation system, as well as a capacity outlook for the near future.
Source of data	Compliance audit reports	Reports from Eskom
Method of calculation / assessment	Number of reports per year	Number of reports per year
Assumptions	Audits completed as planned	Information from Eskom received timeously
Disaggregation of Beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired Performance	3 audit reports - one each for distribution, generation and transmission licensees - on the state of compliance of licensees with licence conditions considered annually by the REC/ELS (depending on delegation) by 31 March	1 System Adequacy Report considered annually by the REC/ELS (depending on delegation) by 31 March
Indicator Responsibility	EM (ELR) and HOD (ELC)	EM (ELR)

1. ELECTRICITY INDUSTRY REGULATION (continued)

Indicator title	Number of reports on the performance of renewable technologies considered by the relevant committee or the Energy Regulator within the stated timeframe	Number of reports on the analysis of Eskom's actual performance against the regulatory framework considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	These are annual reports on the performance of renewable technologies aimed at informing all stakeholders and decision makers on the status.	These are annual reports on the analysis of Eskom's actual performance against the regulatory framework in order to determine the level of compliance.
Source of data	Reports on the performance and progress of Renewable Energy	Reports from Eskom; compliance audit reports
Method of calculation / assessment	Number of reports per year	Number of reports
Assumptions	Analysis completed	Analysis completed
Disaggregation of Beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired Performance	2 monitoring reports on the performance and progress of Renewable Energy projects for 2020/21 considered annually by the REC/ELS (depending on delegation) by 30 September and 31 March respectively	1 report on the outcome of the Eskom's actual performance against the regulatory framework considered annually by the REC/ELS (depending on delegation) by 31 March
Indicator Responsibility	EM (ELR) and HOD (EIP)	EM (ELR) and HOD (EIP)

Indicator title	Number of reports on new entrants into the electricity supply industry considered by the relevant committee or the Energy Regulator within the stated timeframe	Number of reports on refurbishments and upgrades of electricity infrastructure considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	These are annual reports on new entrants (licensees in generation, transmission and distribution) into the electricity supply industry in order to determine the growth of the industry.	These are annual reports on the analysis of refurbishments and upgrades of electricity infrastructure by licensees – in order to monitor the level of security of electricity supply.
Source of data	License and registration applications	Reports from Eskom; compliance audit reports
Method of calculation / assessment	Number of reports	Number of reports
Assumptions	Analysis completed	Analysis completed
Disaggregation of Beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired Performance	1 report on the assessment of new entrants into the electricity supply industry considered annually by the REC/ELS (depending on delegation) by 31 March	1 report on refurbishments and upgrades of electricity infrastructure considered annually by the ELS/REC by 31 March
Indicator Responsibility	EM (ELR) and HOD (EIP)	EM (ELR) and HOD (EIP)

1. ELECTRICITY INDUSTRY REGULATION (continued)

Indicator title	Approved revenues and tariffs to facilitate the sustainability of electricity supply considered by the relevant committee or the Energy Regulator within the stated timeframe	Percentage of applications from the ESI requiring amendment to or exemption from the South African grid code, considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	To ensure that Eskom's revenue requirements for the next Multi-Year Price Determination period is approved annually to allow Eskom enough revenue to be a going concern but also to protect the consumers against inflated prices.	This is the percentage of applications for amendment of or exemption from the Grid Code considered within specified timeframe
Source of data	Applications from Eskom	Applications for amendment of or exemption from the grid code
Method of calculation / assessment	Application of tariff model	(number of applications requiring amendments or exemptions completed within 60 days / number of applications for exemptions received)*100
Assumptions	Eskom submits complete application	Recommendations from Grid Code Advisory Committee submitted with all required supporting documents
Disaggregation of Beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired Performance	Energy Regulator approved revenues and tariffs for Eskom within 6 months after receipt of the complete application	100% of applications from the ESI requiring amendment to or exemption from the distribution and transmission grid code, considered by the REC/ELS (depending on delegation) within 60 days from receipt of complete information
Indicator Responsibility	EM (ELR) and HOD (EPT)	EM (ELR) and HOD (EIP)

Indicator title	Number of reports on regulatory advocacy aimed at improvement of the regulatory framework provided through legislation, regulation and government policies considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	These are annual reports on regulatory advocacy engagements with decision-makers on identified legislative and policy matters.
Source of data	Reports on each engagement indicating the reason for and outcome of the engagement
Method of calculation / assessment	Number of reports
Assumptions	Reports on each engagement compiled
Disaggregation of Beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired Performance	1 report on regulatory advocacy aimed at improvement of the regulatory framework provided through legislation, regulation and government policies s considered annually by the REC/ELS (depending on delegation) by 31 March
Indicator Responsibility	EM (ELR), HOD (ELC), HOD (EPT) and HOD (EIP)

2. PIPED-GAS INDUSTRY REGULATION

Indicator title	Percentage of complete maximum price applications considered by the relevant committee or the Energy Regulator within the stated timeframe	Percentage of complete tariff applications considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	This is the percentage of complete applications for maximum prices of piped-gas considered within a set timeframe, subject to a finding that there is inadequate competition.	This is the percentage of complete tariff applications considered within a set timeframe, subject to a finding that there is inadequate competition.
Source of data	Applications for maximum prices of gas	Applications for tariffs
Method of calculation / assessment	(number of applications for maximum prices completed within 120 days / number of applications for maximum prices received)*100	(number of tariff applications completed within 120 days / number of applications for tariff applications received)*100
Assumptions	Complete applications received from licensees	Complete applications received from licensees
Disaggregation of beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired performance	100% of complete maximum price applications considered by the Energy Regulator within 120 working days after date of publication of preliminary assessment of the maximum price applications	100% of complete tariff applications considered by Energy Regulator within 120 working days after the date of the publication of preliminary assessment of the applications
Indicator Responsibility	EM (GAR) and HOD (GPT)	EM (GAR) and HOD (GPT)

Indicator title	Percentage of complete trading margin applications considered by the relevant committee or the Energy Regulator within the stated timeframe	Number of reports on the review of the definition of the piped-gas market considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	This is the percentage of trading margin applications considered within a set timeframe, aimed at enabling the licensee to: a) Recover all efficient and prudently incurred investment and operational costs, and b) Make a profit commensurate with risk.	These are annual reports on the review of the definition of the piped-gas market.
Source of data	Applications for trading margin	Research reports
Method of calculation / assessment	(number of trading margin applications completed within 120 days / number of applications for maximum prices received)*100	Number of reports
Assumptions	Complete applications received from licensees	Analysis completed
Disaggregation of beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired performance	100% of complete trading margin applications considered by the Energy Regulator within 120 working days after the date of the publication of preliminary assessment of the applications	1 report on the review of the definition of the piped-gas market considered annually by the PGS by 31 March
Indicator Responsibility	EM (GAR) and HOD (GPT)	EM (GAR) and HOD (GLC)

2. PIPED-GAS INDUSTRY REGULATION (continued)

Indicator title	Percentage of complete licence applications considered by the relevant committee or the Energy Regulator within the stated timeframe from date of close of public comment period or period of applicant's response to objections received	Percentage of complete registration applications of gas activities considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	This is the percentage of complete licence applications considered within a set timeframe and conditions.	This is the percentage of the registration applications for operations or activities related to the production and importation of gas, considered within a set timeframe and conditions.
Source of data	Licence applications	Registration applications
Method of calculation / assessment	(Number of licence applications considered within 60 days after the end of the objection period or period of applicant's response to objections received) / (total number of applications received) * 100	(Number of registration applications considered within 120 days from receipt of complete application) / (total number of applications received) * 100
Assumptions	Complete applications submitted	Complete applications submitted
Disaggregation of beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired performance	100% of complete licence applications considered by the REC or PGS (depending on the delegation) within 60 working days from date of close of public comment period or period of applicant's response to objections received	100% of complete registration applications of gas activities considered by the PGS within 60 working days from date of close of public comment period
Indicator Responsibility	EM (GAR) and HOD (GLC)	EM (GAR) and HOD (GLC)

Indicator title	Number of audits conducted on the ROMPCO pipeline according to the compliance frameworks and audit reports considered by the relevant committee or the Energy Regulator within the stated timeframe	Number of monthly volume balance reports to monitor the supply of 120m GJ p.a. from Mozambique to South Africa assessed and analysis reports considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	These are annual audits conducted on the ROMPCO pipeline according to the compliance framework, non-compliance notices issued (where necessary) and audit reports compiled.	These are reports on the assessment and analysis of Sasol's monthly volume balance reports considered within a stated timeframe, in order for NERSA to have regular, systematic, consistent, and sufficient non-financial information relevant to economic regulation, to enhance the efficiency and transparency of the regulatory process.
Source of data	Audit reports	Volume balance report assessment reports from Sasol
Method of calculation / assessment	Number of audits	Number of reports
Assumptions	Approved received to travel to Mozambique to conduct audit	Information received timeously from Sasol
Disaggregation of beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired performance	1 audit conducted annually on the ROMPCO pipeline according to the compliance frameworks and audit reports considered by the PGS by 31 March	12 monthly volume balance reports assessed and analysis reports considered by the PGS to monitor the supply of 120m GJ p.a. from Mozambique to South Africa
Indicator Responsibility	EM (GAR) and HOD (GLC)	EM (GAR) and HOD (GLC)

2. PIPED-GAS INDUSTRY REGULATION (continued)

Indicator title	Number of inspections conducted, non-compliance notices issued (where necessary) and quarterly inspection reports considered by the relevant committee or the Energy Regulator within the stated timeframe	Number of reports on the assessment of criteria for the allocation of uncommitted capacity considered annually by the relevant committee or the Energy Regulator within the stated timeframe
Definition	These are planned inspections conducted during the reporting year, aimed at enforcing monitoring and compliance of licensed entities with licence conditions. Notices of non-compliance are issued when necessary, and quarterly inspection reports compiled.	These are annual reports on the assessment of criteria for the allocation of uncommitted capacity
Source of data	Approved plan to annual inspections, Inspection reports	Reports from licensees
Method of calculation / assessment	Number of inspections	Number of reports
Assumptions	Inspections completed	Inputs received
Disaggregation of beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired performance	Inspections conducted (as per the annual plan for inspections), non-compliance notices issued (where necessary) and quarterly inspection reports considered by the PGS	1 report on the assessment of criteria for the allocation of uncommitted capacity considered annually by the PGS by 31 March
Indicator Responsibility	EM (GAR) and HOD (GLC)	EM (GAR) and HOD (GLC)

Indicator title	Number of reports on the implementation of the reviewed mechanism for enforcement of 3rd party access considered by the relevant committee or the Energy Regulator within the stated timeframe	Number of reports on the engagements on harmonised requirements/ processes for regulation of cross border assets between SA and Mozambique considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	This is a report on the implementation of the reviewed mechanism for enforcement of 3rd party access	This is a report on continued engagements with the Mozambique regulator to facilitate harmonization of regulatory frameworks and policies required for the effective regulation of cross border assets between SA and Mozambique considered by 31 March 2025.
Source of data	Reports from licensees; compliance reports	Existing regulatory frameworks, MOUs
Method of calculation / assessment	Number of reports	Number of reports
Assumptions	Completed reports received	Engagements completed
Disaggregation of beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired performance	1 report on the implementation of the reviewed mechanism for enforcement of 3rd party access considered by the Energy Regulator by 31 March 2025	1 report on the engagements on harmonised requirements/ processes for regulation of cross border assets between SA and Mozambique considered by the PGS by 31 March 2025
Indicator Responsibility	EM (GAR) and HOD (GLC)	EM (GAR)

2. PIPED-GAS INDUSTRY REGULATION (continued)

Indicator title	Number of reports on the assessment of adequacy of competition in the piped-gas industry considered by the relevant committee or the Energy Regulator within the stated timeframe	Number of reports on regulatory mechanisms required for the review of licensing rules considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	This is a report on the assessment conducted to determine the adequacy of competition in the piped-gas industry – which is an important component for tariff and pricing methodologies considered by 31 March 2025.	This is a report on regulatory advocacy engagements with key decision-makers on the review of licensing rules for the piped-gas industry considered by 31 March 2022.
Source of data	Research reports; analysis report	Reports on each engagement indicating the reason for and outcome of the engagement
Method of calculation / assessment	Number of reports	Number of reports
Assumptions	Research and analysis completed	Reports on each engagement compiled
Disaggregation of beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired performance	1 report on the assessment of adequacy of competition in the piped-gas industry considered by the Energy Regulator by 31 March 2025	1 report on regulatory mechanisms required for the review of licensing rules considered by the PGS by 31 March 2022
Indicator Responsibility	EM (GAR) and HOC (GCM)	EM (GAR) and HOD (GLC)

Indicator title	Revised tariff methodology considered by the relevant committee or the Energy Regulator within the stated timeframe	Refined framework for conducting adequacy of competition in the gas industry considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	The price and tariff methodology for the piped-gas industry will be reviewed to ensure to ensure correct decision-making in respect of price and tariff applications and considered by 31 March 2025.	The framework for conducting adequacy of competition in the piped-gas industry will be reviewed to guide the process to determine adequacy of competition in the piped-gas industry considered by 31 March 2021.
Source of data	Revised price and tariff methodology	Revised framework for conducting adequacy of competition
Method of calculation / assessment	Approved revised price and tariff methodology	Approved revised framework for conducting adequacy of competition
Assumptions	Participation by key stakeholders	Participation by key stakeholders
Disaggregation of beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired performance	Revised tariff methodology is considered by the Energy Regulator by 31 March 2025	Reviewed framework for conducting adequacy of competition in the piped-gas industry is considered by the Energy Regulator by 31 March 2021
Indicator Responsibility	EM (GAR) and HOD (GPT)	EM (GAR) and HOD (GCM)

2. PIPED-GAS INDUSTRY REGULATION (continued)

Indicator title	Number of reports on regulatory advocacy aimed at improvement of the regulatory framework provided through legislation, regulation and government policies considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	These are annual reports on gas regulatory advocacy engagements with decision-makers on identified legislative and policy matters considered by 31 March annually.
Source of data	Reports on each engagement indicating the reason for and outcome of the engagement
Method of calculation / assessment	Number of reports considered per annum
Assumptions	Reports on each engagement compiled
Disaggregation of beneficiaries (where applicable)	• Target for women: N/A • Target for youth: N/A • Target for people with disabilities: N/A
Spatial transformation (where applicable)	• Contribution to spatial transformation priorities: N/A • Description of Spatial Impact: N/A
Desired performance	1 report on gas regulatory advocacy engagements with decision-makers on identified legislative and policy matters considered annually by the PGS by 31 March
Indicator Responsibility	EM (GAR), HOD (GLC) and HOD (GPT)

3. PETROLEUM PIPELINES INDUSTRY REGULATION

Indicator title	Number of reports on the percentage utilisation for pipelines, storage facilities and loading facilities and third party access considered by the relevant committee or the Energy Regulator within the stated timeframe	Percentage of complete pipeline, storage and loading facility tariff applications considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	These are annual reports on the percentage utilisation for pipelines, storage facilities and loading facilities and third party access aimed at promoting competition in the industry, considered annually by the PPS by 31 March.	This is the percentage of all complete pipeline, storage and loading facility tariff applications considered by the PPS/ER (depending on the delegation) within 8 months of receipt of complete application
Source of data	Analysis reports	Applications for tariffs
Method of calculation / assessment	Number of reports	(Number of tariff applications considered by the relevant Subcommittee within 8 months of receipt of complete application) / (Total number of tariff applications received)*100
Assumptions	Analysis of trends completed	Complete applications received
Disaggregation of beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired performance	1 report on the percentage utilisation for pipelines, storage facilities and loading facilities and third party access considered annually by PPS by 31 March	100% complete pipeline, storage and loading facility tariff applications considered by the considered by the PPS/ER (depending on the delegation) within 6 months from receipt of complete application
Indicator Responsibility	EM (PPR) and HOD (PLC)	EM (PPR) and HOD (PPT)

3. PETROLEUM PIPELINES INDUSTRY REGULATION (continued)

Indicator title	Percentage of complete licence applications considered by the relevant committee or the Energy Regulator within the stated timeframe	Percentage of complete applications for licence amendments / revocations considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	This is the percentage of complete licence applications that will be decided upon by the PPS/REC/ER (depending on the delegation) within the timelines as prescribed in Section 19(1) of the Petroleum Pipelines Act.	This is the percentage of complete applications for licence amendments that will be decided upon by the by the PPS/REC/ER (depending on the delegation)within the timelines as prescribed in Section 19(1) of the Petroleum Pipelines Act.
Source of data	Licence applications	Licence amendment applications
Method of calculation / assessment	(number of applications decided upon within statutory deadlines / number of received licence applications) * 100	(number of applications decided upon within statutory deadlines / number of received licence applications) * 100
Assumptions	Complete applications	Complete applications
Disaggregation of beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired performance	100% of complete licence applications considered by the PPS/REC/ER (depending on the delegation) within 60 working days under the conditions as prescribed in Section 19(1) of the Petroleum Pipelines Act	100% of percentage of complete applications for licence amendments that will be decided upon by the by the PPS/REC/ER (depending on the delegation)within the timelines as prescribed in Section 19(1) of the Petroleum Pipelines Act
Indicator Responsibility	EM (PPR) and HOD (PLC)	EM (PPR) and HOD (PLC)

Indicator title	Reviewed tariff methodology for storage and loading facilities considered by the relevant committee or the Energy Regulator within the stated timeframe	Number of reports on regulatory advocacy aimed at improvement of the regulatory framework provided through legislation, regulation and government policies for the petroleum pipelines industry considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	The tariff methodology for storage and loading facilities to provide regulatory certainty and facilitate investment in the petroleum pipelines industry considered by the Energy Regulator by 31 March 2015.	These are annual reports on regulatory advocacy aimed at the improvement of the regulatory framework provided through legislation, regulation and government policies for the petroleum pipelines industry considered annually by the PPS by 31 March.
Source of data	Reviewed tariff methodology for storage and loading facilities	Reports on each engagement indicating the reason for and outcome of the engagement
Method of calculation / assessment	Reviewed tariff methodology for storage and loading facilities	Number of reports
Assumptions	Participation of key stakeholders	Reports on each engagement compiled
Disaggregation of beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired performance	The reviewed tariff methodology for storage and loading facilities approved by the Energy Regulator by 31 March 2015	1 report on regulatory advocacy aimed at the improvement of the regulatory framework provided through legislation, regulation and government policies for the petroleum pipelines industry considered annually by the PPS by 31 March
Indicator Responsibility	EM (PPR) and HOD (PPT)	EM (PPR), HOD (PLC) and HOD (PPT)

3. PETROLEUM PIPELINES INDUSTRY REGULATION (continued)

Indicator title	Number of reports on new entrants into the petroleum pipelines industry considered by the relevant committee or the Energy Regulator within the stated timeframe	Number of reports on the geographic spread of petroleum pipelines infrastructure considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	These are annual reports on new entrants into the petroleum pipelines industry in order to determine the growth of the industry, considered by the PPS by 31 March annually.	These are annual reports indicating the geographic spread of petroleum pipelines infrastructure in order to provide information on the level of access to petroleum pipelines services across South Africa, considered by the PPS by 31 March annually.
Source of data	License and registration applications	GIS reports; data base of licensees
Method of calculation / assessment	Number of reports	Number of reports
Assumptions	Analysis completed	Analysis completed
Disaggregation of beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired performance	1 report on new entrants into the petroleum pipelines industry considered annually by the PPS by 31 March	1 report on the geographic spread of petroleum pipelines infrastructure considered annually by the PPS by 31 March
Indicator Responsibility	EM (PPR) and HOD (PLC)	EM (PPR) and HOD (PLC)

Indicator title	Number of reports on the pipelines, storage and loading licenses issued considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	These are annual reports on the pipelines, storage and loading licenses issued in a particular financial year, considered annually by the PPS by 31 March
Source of data	Data base of licence applications
Method of calculation / assessment	Number of reports
Assumptions	Analysis completed
Disaggregation of beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired performance	1 report on the pipelines, storage and loading licenses issued considered annually by the PPS by 31 March
Indicator Responsibility	EM (PPR) and HOD (PLC)

4. TRANSVERSAL REGULATORY AND ORGANISATIONAL

Indicator title	Percentage of business processes are automated and efficient within the stated timeframe	Percentage of regulatory processes is based on appropriate Research within stated timeframe
Definition	Business processes and internal control measures are digitized to improve efficacy	All processes applied for the regulation of the energy sector is based on relevant research
Source of data	Business process analysis	Research reports
Method of calculation / assessment	IT design	Approved regulatory processes
Assumptions	Business process analysis complete	Research completed
Disaggregation of beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired performance	70% of processes are automated and efficient by 31 March 2025	100% of regulatory processes is based on appropriate research by 31 March 2025
Indicator Responsibility	CIO	SM (RAR)

Indicator title	Number of reports on partnership creation to position NERSA as a recognised regulator nationally, regionally and internationally considered by the relevant committee or the Energy Regulator within the stated timeframe	Number of reports on the implementation of the Learnership and Internship Programmes considered by the relevant committee or the Energy Regulator within the stated timeframe
Definition	This is the number of reports on partnership creation, which include engagements with other regulators; participation in regulatory associations, events and conferences; and partnerships with other institutions for capacity building purposes – aimed at positioning NERSA as a recognised regulator nationally, regionally and internationally considered by the relevant subcommittee.	This is reports on the implementation of the learnership and Internship programmes
Source of data	Reports on an overview of international engagements and partnerships activities	Learnership and Internship programmes
Method of calculation / assessment	Number of reports	Number of reports
Assumptions	Analysis completed	Approved Learnership and Internship programmes
Disaggregation of beneficiaries (where applicable)	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A	- Target for women: N/A - Target for youth: N/A - Target for people with disabilities: N/A
Spatial transformation (where applicable)	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A	- Contribution to spatial transformation priorities: N/A - Description of Spatial Impact: N/A
Desired performance	2 reports on partnership creation to position NERSA as a recognised regulator nationally, regionally and internationally considered annually by the REC by 30 September and 31 March	1 report the implementation of the Learnership and Internship Programmes considered by HRRC annually by 31 March
Indicator Responsibility	EM (COS) and HOD (ICP)	CHCO

**ANNUAL
PERFORMANCE PLAN
2024/25 – 2026/27**



PART A



PART B



PART C



PART D



PART E



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STATEMENT BY THE CHAIRPERSON

The National Energy Regulator (NERSA) was established on 1 October 2005 in terms of the National Energy Regulator Act of 2004, and is mandated to regulate South Africa's electricity, piped-gas and petroleum pipelines industries.

Within the parameters of its regulatory mandate, NERSA continuously endeavours to ensure security of supply; a level playing field and contribute to improved competition in the energy sector. In regulating the electricity, piped-gas and petroleum pipelines industries, NERSA adheres to the regulatory principles of transparency; neutrality; consistency and predictability; independence; accountability; integrity; efficiency; and public interest.

In developing this Annual Performance Plan, the Energy Regulator's premise was that the performance environment of NERSA is impacted upon by the energy demand and supply trends and developments in the global, continental, regional and national environments. These trends and developments were identified and informed the development of this Plan. In addition, NERSA acknowledged the fact that although our mandate, in terms of our functions, is clear, the manner in which we implement that mandate cannot be done in isolation. In order to be an effective regulator, we take a broader view of other factors that could impact on our performance environment

Increases in oil prices in response to the conflict, triggered by the attack by Hamas in Israel on October 7, 2023, have so far been relatively mild. Israel is not an oil producer and no major international oil infrastructure runs close to the Gaza Strip or southern Israel (October 2023). Brent, the European benchmark, gained around 10 percent, while its American equivalent has gained around 9 percent. Prices rose to around \$90 per barrel in October 2023, still far from their historical highs before declining significantly in December 2023 and January 2024 respectively. Major oil companies have suspended production from the Tamar natural gas field off the Israel's northern coast. Moreover, some cargo was diverted from the Ashdod port to Haifa, the heart of the potential India-Saudi-Arabia-Israel corridor, causing a moderate backlog and calls for force majeure on carriers. Looking ahead, expectations are that the conflict will remain concentrated in the territories around Israel in 2024, notably the northern part of the Gaza Strip.

A strike in the western region of Australia disrupted exports and raise prices of LNG, which is used for

electricity generation. European gas prices have been volatile since August 2023 over the labour unrest in Australia and spiked by about 28% in what has been famously referred to as the "butterfly effect". Thus this phenomenon has demonstrated that the global LNG and electricity markets remain vulnerable to disruptions and volatility due to localised geopolitical developments.

Qatar Energy has also halted LNG vessels that were in the Suez Canal on their way to the Asian Pacific market through the Red Sea. Houthi rebels in Yemen have significantly stepped up attacks on commercial shipping vessels, including those transporting crude oil and LNG travelling through the lower Red Sea since mid-November in response to Israel's bombardment of Gaza. The Houthis have declared war on Israel since October 2023 to the present.

In 2023, global gas demand rose by just 0.5%, as growth in China, North America and gas-rich countries in Africa and the Middle East was partially offset by declines in other regions. As the pandemic restrictions were relaxed and economic activity returned, China regained



its position as the world's largest LNG importer, even though still below the 2021 levels as natural gas demand grew by 7%. In contrast, natural gas consumption in Europe fell by 7%, reaching its lowest level since 1995. This decline was compounded by the rapid expansion of renewables and an increased availability of nuclear power weighing on natural gas demand in both Europe and mature markets in Asia, driving prices lower.

On the supply side, gas availability remained relatively tight in 2023, as the increase in global LNG production fell short of expectations. As such, production growth was not sufficient to offset the continued decline of Russian piped gas deliveries to Europe. The growth in supply was also highly concentrated geographically, with the United States becoming the world's largest LNG exporter, accounting for 80% of additional LNG supply in 2023.

In South Africa, natural gas has become the natural alternative to coal and the country is currently planning LNG import terminals on its coasts to switch several gigawatts of coal based power generation capacity. Discoveries onshore and offshore are also providing the foundation to grow the gas value chain and serve already well-established manufacturing and transport industries. The country became a small LNG producer in 2022 and will seek to further develop its domestic reserves to supply industrial facilities and decarbonise its transport sector.

Geopolitical uncertainties are the biggest risk factor for global gas markets in 2024, according to the report.

Russia's invasion of Ukraine, heightened tensions in the Middle East and concerns over deliberate interference with critical infrastructure such as pipelines all have the potential to generate further volatility. Russia is the world's second-largest oil producer and a key supplier of crude oil to European refineries. It has also been historically the largest supplier of natural gas to Europe, providing about 35 per cent of its supply. In 2023, policy measures and new regulations in key import markets were introduced with a focus on affordability and security of supply. The European Union launched its Joint Gas Purchasing mechanism; Japan initiated its Strategic Buffer LNG ahead of the 2023/24 winter season; and China is formulating its Natural Gas Utilization Policy, setting out the guiding principles for an "orderly growth in natural gas demand" in the coming years.

The Rand has weakened significantly against major world currencies. Interest rates have gone up and affected economic growth. South Africa's value-added exports have not increased significantly. Political instability and hostility in Central Eastern Europe (CEE) is affecting emerging economies.

The Minister of Mineral Resources and Energy has approved the implementation of a revised zone differentials into the price structures of petrol, diesel

and Illuminating Paraffin (IP) with effect from the 5th of April 2023. The annual adjustments to road transport tariffs applicable in petrol, diesel and IP price structures will range from an increase of 40.8 c/l in Gordonia Central Magisterial District Pricing Zone to an increase of 7.8 c/l for petrol and diesel as well as 14.1 c/l for IP (Zone 9C-Gauteng).

The Standard Offer Programme enables Eskom to acquire electricity at a predetermined price calculated at the avoided cost of own generation (including long-term energy purchases from independent power producers). Emergency Generator programme enables independent generators to consistently supply energy so they can compete with ESKOM generators in the domestic market. The Bilateral Power Import Programme is to ensure that neighbouring countries imports are secure.

During the State of the Nation Address (SONA) in March 2023, President Ramaphosa first announced the creation of the portfolio for a Minister of Electricity in the Presidency. Dr. Kgosientsho Ramokgopa was been appointed the new Minister in the Presidency for Electricity. The designated Minister would have political responsibility, authority, and control over every important aspect of the Energy Action Plan in order to effectively manage the response to the electricity crisis. The Minister is mandated to expedite the procurement of new generation capacity, work



with Eskom leadership to improve the performance of existing power stations, and facilitate coordination of the numerous departments and entities involved in the crisis response plan.

It is against this background that NERSA developed this Annual Performance Plan for 2024/25 to 2026/27. We have acknowledged the fact that as a regulator we have to, more than ever before, remain focused on the role NERSA plays in the aversion and mitigation of a looming energy crises which threatens the growth of South Africa's economy. In any economy, economic growth is closely associated with increasing energy consumption. The availability of secure, reliable and affordable energy supply is essential for industrial processes and the provision of public services such as lighting, heating, cooking, information and communication technology, and mobility. The key role of NERSA is underpinned by its mandate that is enshrined in its founding legislation and is aligned to objectives of our government.

NERSA took cognisance of the fact that globally the energy system is undergoing substantial changes, which are driven by, among others, political and economic instability, changes in consumption patterns, supply dynamics and policy shifts. This should be considered with the parameters of the change in the geopolitical landscape of energy and the impact the environmental issues will have on a country's energy system. In the process to determine our strategic direction for the 2024/25 to 2026/27 planning period, the following factors were duly considered and informed the

development of this Annual Performance Plan for the planning period:

1. Customer and Stakeholder

- Providing a stable and accessible Electricity industry;
- Providing an efficient, safe, effective, sustainable, accessible, competitive and transformed piped-gas and petroleum pipelines industry; and
- Integrated and value added service to customers and stakeholders.

2. Financial

- Improve cost efficiency; and
- Improve revenue sustainability model.

3. Internal Business

- Improve project management execution, regulatory efficiency, turnaround times and stakeholder engagement.

4. Organisational Capacity

- Improve knowledge and skills, work culture, information management and systems and tools; and
- Increase advocacy and monitoring of industry transformation

Due to the nature of NERSA's work and the resultant impact on the economy of the country and energy consumers, we have found ourselves in increased pressure to improve the systems, processes and

procedures we apply in carrying out our mandate by striving towards defensible decisions. NERSA also needs to ensure that it remains relevant in the evolving energy sector. I believe that will require a higher level of being proactive in responding to changes in the regulatory environment. The Energy Regulator considered the aforementioned developments, trends and challenges to effectively plan for improved regulation of South Africa's energy industry.

The Plan's performance targets have been set against each objective outlined in the Strategic Plan. Specific, measurable, achievable, realistic and time-bound – or SMART – key performance indicators for 2024/25, with quarterly targets, will ensure that the strategic objectives are met.

Adequate resourcing of the organisation, as well as the quarterly performance reviews, will ensure the assessment of the overall performance of each programme against this Annual Performance Plan.

The National Energy Regulator of South Africa takes pride in submitting its Annual Performance Plan, which sets out the strategic focus of the Energy Regulator for the 2024/25 to 2026/27 planning period. The Energy Regulator fully endorses this Annual Performance Plan and commits to supporting its implementation.

THEMBANI BUKULA

Chairperson

National Energy Regulator of South Africa



STATEMENT BY THE CHIEF EXECUTIVE OFFICER

It is my honour to present the Annual Performance Plan of the National Energy Regulator of South Africa (NERSA) for the financial years 2024/25 to 2026/27.

NERSA is an independent regulator and was established on 1 October 2005 in terms of the National Energy Regulator Act, 2004 (Act No. 40 of 2004). Its mandate is to regulate the electricity industry in terms of the Electricity Regulation Act, 2006 (Act No. 4 of 2006), the piped-gas industry in terms of the Gas Act, 2001 (Act No. 48 of 2001), and the petroleum pipelines industry in terms of the Petroleum Pipelines Act, 2003 (Act No. 60 of 2003).

In carrying out its mandate, NERSA endeavours to achieve its vision to be a recognised world-class leader in energy regulation. NERSA is expected to implement its mandate and to take the necessary regulatory decisions proactively in anticipation of and in response to the changing delivery environment in the energy industry. Furthermore, NERSA aims to balance the economic interests of all stakeholders by ensuring the growth and sustainability of the petroleum pipelines, electricity, and piped gas industries.

It also endeavours to make these industries more affordable and accessible. This will assist in improving everyone's quality of life and preserving South Africa's

socio-economic growth. The key focus areas of NERSA's mandate, as contained in the relevant legislation, summarised as follows:

- issuing licences and setting pertinent conditions;
- setting and/or approving tariffs and prices;
- monitoring and enforcing compliance with licence conditions;
- dispute resolution, including mediation, arbitration and the handling of complaints;
- gathering, storing and disseminating industry information;
- setting rules, guidelines and codes for the regulation of the three industries;
- the determination of conditions of supply and applicable standards; and
- the registration of import and production activities.

This Annual Performance Plan complies with the minimum requirements stated in the Revised Framework for Strategic Plans and Annual Performance Plans, issued in 2019 by the Department of Planning, Monitoring and Evaluation (DPME). The development of this Annual Performance Plan is informed by the

Amended Strategic Plan for the 2020/21 to 2024/25 period and is aligned with the new outcomes that were formulated in 2022. The outputs and targets stated in the Annual Performance Plan provide the basis for NERSA's support of the following key priorities of Government:

- Priority 1: Capable, ethical and developmental state;
- Priority 2: Economic transformation and job creation; and
- Priority 3: Education, skills and health.

NERSA's performance in the Annual Performance Plan is measured against five programmes, namely Regulatory Service Delivery; Advocacy and Engagement; Innovation; Operational Efficiency and Quality Management; and People and Organisational Culture. Emphasis has been placed on the need to focus on the value added by NERSA's regulation on the Energy Sector. NERSA believes that its services add value to the electricity, piped-gas and petroleum pipeline industries in support of Government's economic and social objectives, which are encompassed by key policies. Value has been added in each of the following impact statements, within



the limits of NERSA's mandate, which will guide the implementation of its five programmes:

- a stable and accessible electricity industry that supports an improved quality of life and economic activity;
- an efficient, safe, effective, sustainable, accessible, competitive and transformed piped-gas industry;
- an efficient, safe, effective, sustainable, competitive and transformed petroleum pipelines industry; and
- NERSA being established and perceived as an efficient, effective and credible regulator.

The deliberate focus by NERSA on the value it adds and is adding to all stakeholders affected by the regulation of the electricity, piped-gas and petroleum pipeline industries resulted in the following being identified as strategic areas of improvement for the organisation:

- Facilitate orderly, sustainable and affordable development within the transient environment.
- Develop and/or provide tools, policies and procedures in support of regulatory and governance activities in an innovative way.
- Manage NERSA's reputation and implement effective stakeholder management and regulatory advocacy.
- Create a high-performance culture to improve turnaround times and efficiencies.

NERSA endeavours to support Government in its efforts to deal with the energy crisis, while regulating the

electricity industry – striving for a balance between the needs of both the consumers and licensees. Through the development of best practice pricing and tariff techniques, NERSA will persist in its efforts to provide affordable energy for South African consumers. NERSA will also focus on the issues the energy sector is facing, particularly in relation to our mandatory role of conducting audits to assess licensees' compliance with terms and enforcing the implementation of measures to address licensees' instances of non-compliance.

Improved turnaround times for applications for tariffs, prices, licensing, and registration; ongoing licensee performance monitoring; handling of non-compliant licences; and periodic evaluations of the sufficiency of competition are just a few examples of the ways in which the regulatory tools and procedures that were initiated in the previous fiscal year will be improved. For the purpose of the energy industry's restructuring, NERSA will collaborate with the Department of Mineral Resources and Energy. NERSA remains committed to collaborating with key role players to work towards affordable energy for all.

The South African electricity industry is facing unprecedented challenges, such as the after-effects of the COVID-19 pandemic and the subsequent uptick in demand; the electricity supply crisis, which has worsened in the past few years; and the transformation to a more liberalised market, which continued at a rapid pace with additional complexities related to energy security. The following are some of NERSA's notable achievements during the planning period:

- NERSA approved the reviewed Cost of Supply (COS) Framework, which seeks to address the numerous challenges faced by the South African electricity supply industry, which include ageing infrastructure, capacity constraints, financial viability, tariff structure and affordability.
- NERSA approved the reward amount of R69.05m to Eskom Distribution for its performance levels achieved for the third year (2021/22) of the fourth Multi-Year Price Determination (MYPD4) control period. The reward amount will be applied through the Regulatory Clearing Account (RCA).
- In support of Government's Policy direction to unbundle Eskom into three separate divisions, Generation, Transmission and Distribution, NERSA approved the issuing of a 25-year non-exclusive licence to the National Transmission Company South Africa (NTCSA) in 2023, to operate transmission electricity infrastructure in South Africa. In addition, NERSA approved a five-year trading and 25-year import/export licence for the NTCSA. The issuing of these licences by the Energy Regulator was a critical step towards Eskom's unbundling and a remarkable milestone in support of the transition to a diverse and competitive electricity sector.
- Completion of the inquiry into the identification of measures to improve competition, investment and security of supply in the gas industry. Through this inquiry, eleven measures were identified that can be implemented by NERSA, as well as other regulators and policymakers, over the short, medium and long term, which will have the effect



of improving competition, investment and security of supply in the gas industry.

- NERSA also hosted a successful stakeholder workshop with licensees in Durban, KwaZulu Natal Province, as part of the drive to educate stakeholders about the Petroleum Pipelines Regulation's mandate. The presentations focused on methodologies and compliance matters and included key stakeholders like Department of Mineral Resources and Energy (DMRE) and the Port Regulator of South Africa.

NERSA aims to improve the organisation's resilience and its ability to be agile and innovative. The automation and digitisation of NERSA's systems will be a focus area going forward, with the implementation of the Budgeting Module and Business Process Automation plan being a priority.

NERSA's strategic focus in this Annual Performance Plan is based on our Mission Statement, namely:

To regulate the energy industry in accordance with government laws and policies, standards and international best practices in support of sustainable and orderly development.

The specific, measurable, achievable, realistic and time-bound – or 'SMART' – key performance indicators for 2024/25, with quarterly targets, will ensure that the strategic objectives are achieved. Our overall aim is that the implementation of this Annual Performance Plan will facilitate a secure, reliable, affordable, sustainable,

competitive and transformed energy industry, which contributes to the economic growth of South Africa. In order to achieve this, NERSA places a high premium on the capacity building of its staff. In addition to the training and development of its staff members, NERSA has been and will continue to run successful internship and learnership programmes.

The Energy Regulator provides strategic direction and support to the management and personnel who are in charge of carrying out this Annual Performance Plan. I would like to use this opportunity to reaffirm our commitment to implementing this Plan, as proven by our track record of clean audits as per the audit reports of the AGSA. Consequently, we are confident that we will be successful in enabling an energy sector that is safe, dependable, reasonably priced, sustainable, competitive, and transformative, and that supports South Africa's economic expansion.

I would like to acknowledge the important work that NERSA's staff are doing and their valuable contribution to NERSA's vision and mission. I would also like to encourage an innovative and collaborative spirit in implementing the legislative mandate of NERSA, as well as any future strategic programmes.

ADV. NOMALANGA SITHOLE

Chief Executive Officer

National Energy Regulator of South Africa



OFFICIAL SIGN-OFF

It is hereby certified that this Annual Performance Plan.

- was developed by the Energy Regulator with inputs from the Executive Management of NERSA;
- takes into account all the relevant policies, legislation and other mandates for which the Energy Regulator is responsible; and
- accurately reflects the outcomes and outputs that the Energy Regulator will endeavour to achieve over the period 2023/24 – 2025/26.

Ms. Amukelani Chauke

Senior Manager: Strategic Planning and Monitoring

Ms. Bulelwa Pono

Chief Financial Officer

Adv. Nomalanga Sithole

Chief Executive Officer (Accounting Officer)

Approved by:

Thembani Bukula

Chairperson (on behalf of the Accounting Authority)

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ACRONYMS AND ABBREVIATIONS

AFDB	African Development Bank	GSA	Gas Supply Agreement	PE(R) STEL	Political, Economic, Regulatory, Social, Technological, Environmental and Legal
AFUR	African Forum for Utility Regulators	GUMP	Gas Utilisation Master Plan	PFMA	Public Finance Management Act, 1999 (Act No. 1 of 1999)
APP	Annual Performance Plan	HDI/ HDSA	Historically Disadvantaged Individuals/South Africans	PPA	Power Purchase Agreement
B-BBEE	Broad-Based Black Economic Empowerment	IBT	Inclining Block Tariff	PPR	Petroleum Pipelines Regulation
CAGR	Compound Annual Growth Rate	IDM	Integrated Demand Management	PV	Photovoltaic
CBM	Coal Bed Methane	IEA	International Energy Agency	REIPP	Renewable Energy Independent Power Producer
CNG	Compressed Natural Gas	IEP	Integrated Energy Plan	REIPPPP	Renewable Energy Independent Power Producer Procurement Programme
CCGT	Combined Cycle Gas Turbines	IPP	Independent Power Producer	RERA	Regional Electricity Regulatory Association
CPI	Consumer Price Index	IRP	Integrated Resource Plan	RIA	Regulatory Impact Assessment
DJP	Durban-to-Johannesburg Pipeline	ISO	International Organisation for Standardisation	ROMP-CO	Republic of Mozambique Pipeline Investment Company
DoE	Department of Energy	LNG	Liquefied Natural Gas	SA-CREEE	SADC Centre for Renewable Energy, Energy and Efficiency
DMRE	Department of Mineral Resources and Energy	MPP	Multi-Product Pipeline	SADC	Southern African Development Community
ELR	Electricity Regulation	MTEF	Medium-term Expenditure Framework	SAPIA	South Africa Petroleum Industry Association
ESI	Electricity Supply Industry	MTSF	Medium-Term Strategic Framework	SAPP	Southern African Power Pool
FBE	Free Basic Electricity	MW	Megawatt	SCOA	Standard Chart of Accounts
FID	Final Investment Decision	NDP	National Development Plan	SIP	Strategic Integrated Project
FLNG	Floating Liquefied Natural Gas	NERSA	National Energy Regulator of South Africa		
GAR	Piped-Gas Regulation	NFI	Non-Financial Information		
GDP	Gross Domestic Product	OCGT	Open Cycle Gas Turbine		
GJ	Gigajoule	OECD	Organisation for Economic Co-operation and Development		

PART A: OUR MANDATE

1. Updates to the relevant Legislative and Policy Mandates

- 1.1. There have been no changes to the legislative or other mandates of the National Energy Regulator of South Africa (NERSA).
- 1.2. NERSA is the regulatory authority established in terms of the National Energy Regulator Act, 2004 (Act No. 40 of 2004) with a mandate:
 - to undertake the functions of the National Electricity Regulator as set out in the Electricity Regulation Act, 2006 (Act No. 4 of 2006),
 - to undertake the functions of the Gas Regulator as set out in the Gas Act, 2001 (Act No. 48 of 2001),
 - to undertake the functions of the Petroleum Pipelines Regulatory Authority as set out in the Petroleum Pipelines Act, 2003 (Act No. 60 of 2003); and
 - to perform such other functions as may be assigned to it by or under these Acts.
- 1.3. The regulatory functions of NERSA, as contained in the legislation relevant for the regulation of the energy industry, are summarised as follows:
 - issuing of licences with conditions;
 - setting and/or approving tariffs and prices;
 - monitoring and enforcing compliance with licence conditions;
 - dispute resolution including mediation, arbitration and the handling of complaints;
 - gathering, storing and disseminating industry information;
 - setting of rules, guidelines and codes for the regulation of the three industries;
 - determining of conditions of supply and applicable standards;
 - consulting with government departments and other bodies with regard to industry development and regarding any matter contemplated in the three industry Acts;
 - expropriating land as necessary to meet the objectives of the relevant legislation;
 - registration of import and production facilities; and
 - performing any activity incidental to the execution of its duties
- 1.4. NERSA derives its revenue by, among others, imposing prescribed levies on the regulated industries, following a prescribed transparent procedure. In this regard, the following Acts govern the imposition of such levies:
 - the Gas Regulator Levies Act, 2002 (Act No. 75 of 2002);
 - the Petroleum Pipelines Levies Act, 2004 (Act No. 28 of 2004); and
 - section 5B of the Electricity Act, 1987 (Act No. 41 of 1987).
- 1.5. Apart from the aforementioned industry specific legislation that anchors NERSA's mandate and the imposition of levies, the following facilitating and foundational legislation are also applicable to NERSA's conduct of its business:
 - the Public Finance Management Act, 1999 (Act No. 1 of 1999) (PFMA), which specifies the accounting of NERSA as a Section 3A Public Entity;
 - the Promotion of Access to Information Act, 2000 (Act No. 2 of 2000) (PAIA), which determines the way that NERSA has to treat access to information;

- the Promotion of Administrative Justice Act, 2000 (Act No. 3 of 2000) (PAJA), which determines the just administrative action of NERSA;
 - the Protection of Personal Information Act, 2013 (Act No 4 of 2013) (POPIA), which determines the way that NERSA has to treat personal information; and
 - all other applicable laws of the Republic of South Africa.
- 1.6. NERSA's mandate is further derived from published government policies and regulations developed by the Minister in terms of the Electricity Regulation Act, Gas Act and Petroleum Pipelines Act. As outlined in these legislative prescripts, NERSA must make decisions that are not at variance with published government policy. The relevant applicable policies are::
- White Paper on Energy Policy for South Africa of 1998;
 - Electricity Pricing Policy (EPP) of the South African Electricity Supply Industry;
 - Free Basic Electricity Policy;
 - White Paper on Renewable Energy Policy for South Africa of 2003;
 - Energy Security Master Plan: Liquid Fuels, published by the Department of Energy in 1998 and 2007;
 - National Development Plan;
 - Industrial Policy Action Plan (IPAP); and
 - Integrated Resource Plan (IRP) 2019.
- 1.7. NERSA advocates for the implementation of the White Paper on Energy Policy of 1998 before the principles enshrined in the policy and the suite of subsequent legislation are overhauled. As the Energy Regulator, we are aware that the policies of 1998 and the consequent suite of legislation (Gas Act, Petroleum Pipelines Act, National Energy Regulator Act and Electricity Regulation Act) that were developed between 2001 and 2006, have been actively implemented since the establishment of NERSA in October 2005. It is
- only now that we are able to give private investors some certainty regarding energy infrastructure investments and the level playing field we are expected to provide. Recent private sector licence applications in the piped-gas and petroleum pipelines industries are a testimony to the success of government's liberalisation policies.
- 1.8. The Electricity Regulation Act gives the mandate for competitive bidding of electricity generation capacity to the Department of Mineral Resources and Energy (DMRE), following a Cabinet decision that private sector participation in the electricity industry be split 70:30 between Eskom and the private sector, with the DMRE procuring the plant and Eskom being the 'off-taker'. Thus, it is competition for the market, but not within the market at this stage.
- 1.9. With the rapid price reduction of solar panels, a situation has arisen where rooftop solar has started to become attractive for residential consumers. This trend is more pronounced in relation to commercial premises. These installations are not effectively dealt with in the current regulatory framework, because the 'Electricity Regulations on New Generation Capacity' are only applicable to state-owned entities.
- 1.10. To license all of these small installations is also onerous for the installer and NERSA. It is a much too expensive and complex process to be a realistic option for dealing with this class of generation. However, in spite of their small size, the large amount of these installations means that collectively, they will make up a significant portion of generation capacity. This will impact allocations made in the Integrated Resource Plan (IRP).
- 1.11. In the previous five-year planning period, NERSA found that there were developments in the three industries that are not covered by the current industry- specific Acts. This requires a review of the regulatory legislation.

2. Updates to Institutional Policies and Strategies

- 2.1. Although policy formulation is outside of NERSA's realm of authority, specific policy gaps are continuously identified. These gaps require ongoing dialogue and strategic engagement with the Department of Mineral Resources and Energy in order to ensure that there is alignment between NERSA's strategic direction and the Department's policy thrusts.
- 2.2. In addition to its mandate, as per the legislation mentioned in the previous section, the Energy Regulator's decisions are informed by published policies of government. Within the parameters of NERSA's mandate and the resultant functions, NERSA contributes towards critical government priorities and programmes. Below is a summary of NERSA's contributions towards the:
- enabling milestones in the National Development Plan (NDP);
 - strategic integrated projects in the National Infrastructure Plan (NIP); and
 - seven priorities announced by the Honourable President, Mr Cyril Ramaphosa during the State of the Nation Address (SONA) in Parliament on 20 June 2019.

2.2.1. NERSA's contribution to the National Development Plan

- 2.2.1.1. The National Development Plan (NDP), sets out the country's ambitious goals for poverty reduction, economic growth, economic transformation and job creation. Its aim is to eliminate poverty and reduce inequality by 2030 through uniting South Africans, unleashing the energies of its citizens, growing an inclusive economy, building capabilities, enhancing the capability of the state and leaders by working together to solve complex problems. The high-level objectives of the NDP are to:

- reduce the number of people who live in households with a monthly income below R419 per person (in 2009 prices) from 39% to zero; and
- reduce inequality, as measured by the Gini Coefficient, from 0.69 to 0.6.

- 2.2.1.2. Chapter 4 of the NDP deals with *Economic infrastructure – the foundation of social and economic development*. This chapter places the emphasis on the need for South Africa to maintain and expand, among others, its electricity infrastructure in order to support economic growth and social development goals. In respect of the regulation of the energy sector, NERSA noted that the NDP calls for greater emphasis on stimulating market competition and promoting affordable access to quality services when issuing licences and setting tariffs.

- 2.2.1.3. In order to achieve the NDP goals by 2030, 19 enabling milestones were identified. Even though NERSA contributes indirectly to most of the enabling milestones, it contributes specifically to four pertinent enabling milestones. Table 1 below summarises NERSA's contribution to the relevant enabling milestones.

Table 1: NERSA's contribution to the NDP

Relevant enabling milestones	NERSA's contribution
1. Increase employment from 13 million in 2010 to 24 million in 2030	• Implementation of the Youth Employment Accord • Implementation of a Learnership Programme, as well as an Internship Programme • Training and development of staff and stakeholders • Techno Girls programme where ten girls from grade 9 to grade 12 are exposed to NERSA's activities through visits to the organisation during school holidays.
4. Establish a competitive base of infrastructure, human resources and regulatory frameworks	• Publication of rules, codes and guides for the regulation of the electricity, piped-gas and petroleum pipelines industries • Setting rules and frameworks that facilitate the building of new infrastructure • Setting and/or approving cost-reflective tariffs and market-related prices that encourage investment • Facilitating and enforcing third-party access to facilities • Monitoring compliance through undertaking technical audits leading to regular maintenance and refurbishment of infrastructure and thus contributing to an increase in quality of supply
5. Ensure that skilled, technical, professional and managerial posts better reflect the country's racial, gender and disability makeup	• NERSA ensures continued compliance with the Skills Development Act • Implementation of an Employment Equity Plan • When recruiting new staff members, NERSA ensures as far as possible that the representation within the relevant department and division reflects the country's racial, gender and disability makeup.
6. Broaden ownership of assets to historically disadvantaged groups	• Licensing and the setting and/or approving of tariffs and prices, as in this manner NERSA creates pre-conditions towards the achievement of this milestone • Issuing licences to eligible applicants to facilitate the meeting of stated socio- economic development targets • Facilitating and enforcing third-party access to facilities • Promoting companies that are owned and controlled by Historically Disadvantaged Individuals (HDIs), enabling them to become competitive • Regulatory advocacy for strengthening the powers of the Regulator.

RELEVANT ENABLING MILESTONES	NERSA'S CONTRIBUTION
10. Produce sufficient energy to support industry at competitive prices, ensuring access for poor households, while reducing carbon emissions per unit of power by about one-third	• Regulating in a manner that facilitates security of supply • Taking affordability into consideration when setting and/or approving tariffs and prices • Determining inclining block tariffs and free basic electricity tariffs to protect the low income electricity consumers • Facilitating the conclusion of Power Purchase Agreements between the buyer and the renewable energy Independent Power Producers • Facilitation of the implementation of the Integrated Resource Plan (IRP) through concurring with determinations made by the Minister in line with section 34 of the Electricity Regulation Act, 2006 (Act No. 4 of 2006) • Development and implementation of the Grid Code for renewable energy to facilitate the introduction of renewable energy power producers • Registration of gas importation and production facilities • Monitor the implementation of the Gas Utilisation Master Plan (once promulgated) • Facilitating access to electricity in setting aside some funds for the Electrification Cross-subsidy as part of determining electricity prices • Incorporating compliance with the National Environmental Management Act, 1998 (Act No. 107 of 1998) into licence conditions • Promoting energy efficiency in general in South Africa and in particular in the NERSA building • Facilitating the transition to a low carbon economy • Regulatory advocacy with regard to cleaner fuels policy.

2.2.2. NERSA's contribution to the Medium Term Strategic Framework 2019-2024

2.2.2.1. The Medium-Term Strategic Framework (MTSF) is a five-year plan by the government that is intended to implement the electoral mandate and the National Development Plan Vision (NDP) 2030.

2.2.2.2. It aims to address the challenges of unemployment, inequality and poverty through the three pillars of the NDP:

- Achieving a more capable State
- Driving a strong and inclusive economy
- Building and strengthening the capabilities of South Africans.

2.2.2.3. The seven priorities, which will be achieved through more focused implementation, coordination and integration by the various levels of government, including state-owned enterprises, the private sector and civil society, are as follows:

- **Priority 1:** A capable, ethical and developmental state
- **Priority 2:** Economic transformation and job creation
- **Priority 3:** Education, skills and health
- **Priority 4:** Consolidating the social wage through reliable and quality basic services
- **Priority 5:** Spatial integration, human settlements and local government
- **Priority 6:** Social cohesion and safe communities
- **Priority 7:** A better Africa and world

2.2.2.4. NERSA identified the following government priorities to which it can contribute as part of implementing its mandate:

- **Priority 2:** Economic transformation and job creation
- **Priority 3:** Education, skills and health
- **Priority 7:** A better Africa and world

2.2.2.5. Table 2 below summarises NERSA's contribution to government's priorities.

Table 2: NERSA's contribution to government's priorities

Relevant Priorities	NERSA's contribution
1. A capable, Ethical and Developmental State	• Transparent regulatory processes • All decisions and reasons for decisions are made public by being published on the NERSA website • The public is invited to make comments prior to decisions being made (written or in public hearing) • Customer education programmes and awareness campaigns • Training and development of staff and stakeholders, including training to electricity distributors on the completion of the forms requesting information from them • Techno Girls programme - where ten girls from grade 9 to grade 12 are exposed to NERSA's activities through visits to the organisation during school holidays.
2. Economic Transformation and Job Creation	By facilitating investment in the energy industry and thereby contributing to economic growth, leading to job creation, NERSA contributes through: • Licensing and the setting and/or approving of tariffs and prices, as in this manner NERSA creates pre-conditions towards the achievement of this priority; • Approving renewable energy licenses to ensure that the socio-economic development commitments specified in the bidding process are met; • Promoting companies that are owned and controlled by Historically Disadvantaged Individuals (HDIs) to become competitive; and • Regulating in a manner that facilitates security of supply. Contributing to a competitive and responsive economic infrastructure network through: • Setting rules and frameworks that facilitate the building of new infrastructure; • Setting and/or approving cost reflective tariffs and prices that encourage efficient investment; • Facilitating and enforcing third-party access to facilities; • Monitoring compliance and undertaking technical audits leading to regular maintenance and refurbishment of the infrastructure and therefore to the improvement in quality of supply; and • Promoting competition and competitiveness in the energy industry.
3. Education, skills and health	• Implementation of the Learnership and Internship Programmes • Implementation of the bursary programme for qualifying external applicants • Coordinating the design of a regulatory course at an accredited institution of higher learning • Coordinating the development of a technical regulatory training and development programme.

2.2.3. NERSA's contribution to the National Infrastructure Plan 2050 – Phase 1

2.2.3.1. The South African Government published the National Infrastructure Plan (NIP) 2050 Phase – in March 2022. The NIP 2050 identifies the most critical actions needed for sustained improvement in public infrastructure delivery. The NIP 2050 will have an impact in the short term, but will also keep longer-term imperatives in view. This phase of the NIP 2050 focuses on four mission critical network sectors that provide a platform: energy, freight transport, water and digital infrastructure.

2.2.3.2. The NIP 2050 provides guidance on themes common to the four sectors, with significant emphasis on building capacity in the following:

- **Knowledge and innovation services**, for capability in planning, monitoring, budgeting, finance, procurement, project preparation, project management and sector-specific innovation. This enables evidence-based decision-making, improves cost-effectiveness, mitigates risk and helps optimise and can contribute significantly to improving infrastructure quality, delivery and sustainability. Building these capabilities will be the NIP's top priority.
- **Public-private cooperation and stimulation of competition**, where appropriate, in the delivery of public infrastructure.
- **Spatial transformation** to promote more inclusive development in line with the National Spatial Development Framework (NSDF).
- **Blended project finance** and innovative green finance.
- **Executive management and technical capability** within the state and its entities, so that they are stable and can lead and deliver with confidence.
- **Economic regulation**.
- **Industrial development and localisation** in the design and approach to implementation. Examples are localisation of supplier industries to infrastructure projects, driving the establishment of Special Economic Zones around intermodal transport linkage nodes, and the stimulation of the civil construction and supplier industries.

- **Efficient modes of delivery**.
- **A safe, secure and ethical environment** for public infrastructure delivery
- **Delivery of an Africa regional infrastructure programme**.
- **South African civil construction and supplier industries**, so that local industry gains from state infrastructure investment.

2.2.3.3. In order to address these challenges and goals, a total of 18 strategic integrated projects (SIPs) have been developed. The following three SIPs were identified for energy:

a) SIP 8: Green energy in support of the South African economy

- Support sustainable green energy initiatives on a national scale through a diverse range of clean energy options as envisaged in the Integrated Resource Plan (IRP2010).
- Support bio-fuel production facilities.

b) SIP 9: Electricity generation to support socio-economic development

- Accelerate the construction of new electricity generation capacity in accordance with the IRP2010 to meet the needs of the economy and address historical imbalances.
- Monitor implementation of major projects such as new power stations: Medupi, Kusile and Ingula.

c) SIP 10: Electricity transmission and distribution for all

- Expand the transmission and distribution network to address historical imbalances, provide access to electricity for all and support economic development.
- Align the 10-year transmission plan, the services backlog, the national broadband roll-out and the freight rail line development to leverage off regulatory approvals, supply chain and project development capacity.

2.2.3.4. Table 3 on the next page summarises NERSA's contribution to the relevant strategic integrated projects (SIPs).

Table 3: NERSA's contribution to the NIP

RELEVANT PRIORITIES	NERSA'S CONTRIBUTION
8. Green energy in support of the South African economy	- Facilitating the conclusion of Power Purchase Agreements between the buyer and the renewable energy Independent Power Producers; - Incorporating compliance with the National Environmental Management Act, 1998 (Act No. 107 of 1998) into licence conditions; - Facilitation of the implementation of the Integrated Resource Plan (IRP) through considering concurring with determinations made by the Minister in line with section 34 of the Electricity Regulation Act, 2006 (Act No. 4 of 2006); - Facilitating the transition to a low carbon economy; and - Regulatory advocacy with regard to cleaner fuels policy.
9. Electricity generation to support socio-economic development	- Regulating in a manner which facilitates security of supply and investment; - Facilitating the conclusion of Power Purchase Agreements between the buyer and the renewable energy Independent Power Producers; - Setting rules and frameworks that facilitate the building of new infrastructure; - Setting and/or approving cost reflective tariffs and prices that encourage investment; - Monitoring compliance through undertaking technical audits leading to regular maintenance and refurbishment of infrastructure and thus contributing to an improvement in quality of supply.
10. Electricity transmission and distribution for all	- Facilitating access to electricity in setting aside some funds for the Electrification Cross-subsidy as part of determining electricity prices; - Taking affordability into consideration when setting and/or approving tariffs and prices, while allowing a provision for expansion of current operations; - Determining inclining block tariffs and free basic electricity tariffs to protect the low income electricity consumers; - Facilitating reliability of supply; - Determining benchmarks and monitoring maintenance of infrastructure; - Auditing of the implementation of the Transmission Development Plan; - Monitoring compliance with licence conditions; and - Dispute resolution, including mediation, arbitration and handling of complaints.

3. Updates to relevant Court Rulings

- 3.1. The Eskom and Others v Emfuleni Local Municipality and Others has continuous impact on electricity distribution compliance framework and administration. It requires an update on our existing framework and also how it should be utilised within the mandate of NERSA.
- 3.2. The Werner Van Wyk & 3 others v the Minister of Employment and Labour also has an impact on NERSA continuous good governance/administration. The judgement requires NERSA to revise its Maternity/ Paternity Leave.

PART B: OUR STRATEGIC FOCUS

4. Updated External Situational Analysis

Energy demand and supply trends and developments in the global, continental, regional and national environments have an impact on the performance environment of NERSA.

4.1. Global Trends

4.1.1. Overview

- 4.1.1.1. The disruption to global energy supplies and the associated energy shortages caused by the Russia–Ukraine war increases the importance attached to addressing all three elements of the energy trilemma: security, affordability, and sustainability (BP Outlook, 2023). The war has long-lasting effects on the global energy system. The heightened focus on energy security increases the demand for domestically produced renewables and other non-fossil fuels, helping to accelerate the energy transition.
- 4.1.1.2. Energy market dynamics for Europe in 2023 will be as challenging as in 2022, although energy commodity prices have declined significantly. Natural gas storage in Europe is likely to be completely depleted before spring, while little new import capacity will yet be available. High energy costs and falling demand are forcing industries across Europe to idle while easing price trajectories in oil, natural gas and electricity markets. Input costs will remain elevated for several years, making some European industrial sectors uncompetitive, and resulting in a loss of global market share. Chemicals and base metals will be the worst affected owing to their high reliance on natural gas as an input, but downstream industries such as the automotive sector will also suffer.

- 4.1.1.3. China is set to benefit at a global level. Within the region, a shift is expected towards production in southern Europe from central Europe, with services holding up better than goods production. The stark increase in energy costs for the European industrial sector in 2023 and beyond will have a significant impact on the competitiveness of the European industry. In addition, the coming recession will reduce domestic demand for industrial products; with global freight costs now declining as pandemic-related supply-chain disruption eases, import substitution is becoming more attractive. The European industry already had a higher cost base than other advanced economies. A further increase will make producing in Europe an unprofitable business strategy for many energy-intensive firms.
- 4.1.1.4. Demand trends that were well established are changing as a result of the present energy crisis. Industries that are vulnerable to global price fluctuations are reducing their output due to actual rationing threats (World Economic Outlook, 2022). In response to increasing prices, consumers are changing their patterns of energy use. Economic growth until 2030 is expected to be slower than projected, which will result in lower levels of activity across all sectors and, in turn, reduce growth in energy demand. According to World Economic Outlook (2022), global Gross Domestic Product (GDP) growth is estimated to decline by 3.3% per year through 2030.
- 4.1.1.5. After recovering from the recent market disruption, it is projected that the global electricity supply will move away from fossil fuels without interruption. As the economic rebound after COVID-19 drove increased electricity demand and as concerns about high natural gas prices and energy security rose, a number of countries signalled a return to coal-fired power in 2022. In the Stated Policies Scenario (STEPS) indicated by the World Energy Outlook, unabated coal in global electricity generation falls from 36% in 2021 to 26% in 2030 and to 12% in 2050, while unabated

natural gas decreased from 23% in 2021 to 20% in 2030 and 13% in 2050. In the Announced Pledges Scenario (APS), unabated coal falls from 23% of total generation in 2030 to 3% in 2050. Unabated natural gas falls to 17% in 2030 and just 6% in 2050, the lowest point in 50 years. Under the Net Zero Emissions (NZE) scenario, unabated coal, natural gas, and oil fall even more rapidly, with electricity achieving net zero emissions globally by 2040.

4.1.2. The Global Petroleum Industry

- 4.1.2.1. TotalEnergies will more than double the production of sustainable aviation fuel from its original 2020 plan as part of its conversion of the former Grandpuits conventional refinery in northern France into a zero-crude industrial platform (Brelsford, 2023). The company has reached final investment decision (FID) to increase Sustainable Aviation Fuel (SAF) production at the Grandpuits platform to 285,000 tonnes/year (tpy), up from the originally planned target of 170,000 tonnes per year announced in 2020.
- 4.1.2.2. In developed countries, oil demand declines over the outlook, driven by the falling use of road transport as the efficiency of the vehicle fleet improves and the electrification of road vehicles accelerates. Even so, oil continues to play a major role in the global energy system for the next 15 to 20 years (BP Outlook, 2023). In terms of the NERSA mandate, this means transitioning will be looked at in terms of cleaner fuels. It further means that there is a need to secure crude oil refining in the country and related infrastructure such as storage tanks and pipelines.
- 4.1.2.3. Global crude oil prices have decreased significantly after March 2023, with projections that the barrel price may continue to decline till the third quarter of 2023 despite global geopolitical hostilities emanating from the Russo–Ukraine War. Insights obtained from various sources through Afriforesight (2023) indicate that oil prices are likely going to decline throughout the first quarter of 2023 and even into the second quarter. Table 4 below represents five key Crude Oil Prices in the World Market from May 2022 to May 2023.

Table 4: Five Key Crude Oil Prices in the World Market (\$/barrel)

MONTH	BRENT	WTI	DUBAI-OMAN	NIGERIA	ANGOLA	LIBYA
May-22	112.0	109.5	108.0	117.0	114.3	107.9
Jun-22	117.5	114.1	112.7	130.5	126.8	118.7
Jul-22	105.1	99.8	102.6	121.7	119.0	107.7
Aug-22	97.7	91.5	97.9	106.4	106.6	97.1
Sep-22	90.6	83.9	91.6	93.2	92.8	87.1
Oct-22	93.6	87.0	92.0	96.6	95.8	90.2
Nov-22	90.9	84.1	86.6	93.5	93.2	88.4
Dec-22	81.5	76.7	78.6	82.7	80.5	77.0
Jan-23	84.0	78.3	81.4	85.0	81.8	76.5
Feb-23	83.5	76.8	82.5	86.1	83.7	76.4
Mar-23	79.2	73.4	79.5	81.2	80.0	73.6
Apr-23	83.5	79.5	84.5	86.8	86.3	80.2
May-23	75.7	71.7	76.0	77.4	77.5	71.6
Jun-23	75.0	70.3	75.7	77.1	76.8	70.5
Jul-23	80.2	75.7	81.8	82.5	82.1	76.3
Aug-23	85.1	81.3	87.4	89.4	88.1	82.9
Sep-23	92.6	89.2	94.2	98.2	96.0	89.0
Oct-23	88.7	85.5	90.5	96.1	93.1	88.2
Nov-23	82.0	77.4	84.1	86.3	85.6	80.1

Sources: Brent (Intercontinental exchange) (UK & Norway); US (New York Mercantile Exchange); Refinitiv (Nigeria, Angola, Libya, UAE & Oman), Afriforesight (July 2022)

From January to October 2023, the average Brent spot price was \$83.83/bbl, representing a 18% decrease y-o-y. Similarly, the average Brent month-ahead price was \$82.73/bbl, representing a 17% decrease y-o-y.

The average Brent Crude oil price decreased from US\$82.14 to US\$75.7 by 31 May 2023. Oil markets were subject to a series of large shocks in 2022, such as Russia's invasion of Ukraine in late February and the ensuing sanctions, embargoes, and the price cap on Russian oil imports; a coordinated response by oil-consuming nations (led by the US) to control prices by a massive release of strategic stocks; recessionary and inflationary pressures weighing on the global economy; China's demand shocks from its strict zero-COVID-19 policy, and the massive transformations in crude and product trade flows, among other factors (Fattouh et al, 2023). Geopolitical tension and war in the Middle East between Israel and Hamas has invoked less volatility in the global oil market with supply side pressures over demand leading subdued oil prices in November 2023.

- 4.1.2.4. The international Liquefied Petroleum Gas (LPG) market has depicted that price discovery in this key commodity is linked to crude oil prices and prices of other fossil fuels or forms of energy (artificial price integration). Table 5 on the next page shows how prices from the five key regions provided by various sources through Afriforesight (2023) have trended.
- 4.1.2.5. Between January and May 2023, global LPG price benchmarks have reduced significantly in response to decreasing crude oil prices. The South African maximum refinery gate price per tonne has declined by over 30% between March and May 2023 in response to the global decline in LPG prices. As at 31 May 2023, the Saudi Arabian price, as a key global benchmark, stood at US\$555 per tonne against the South African price, which was recorded at US\$683 per tonne.
- 4.1.2.6. LPG prices surged early in the first quarter of 2022 due to a tightening market supply that has globally been attributed to refinery closures around

the world. However, with declining oil prices in the first quarter of 2023, prices of LPG have declined significantly, ranging between approximately R431 to R582 per tonne in the global market. The South African market for LPG has responded, showing downward pressures between March and April 2023, with the price decreasing from R933 to R898 per tonne.

- 4.1.2.7. The decrease in crude oil prices is the key driver for the global LPG price formation. However, the South African market is leading in that price trajectory compared to other regional markets in question, as the LPG price is higher in South Africa against global price benchmarks.
- 4.1.2.8. The second edition of the MSGBC (Mauritania, Senegal, Gambia, Guinea Bissau and Guinea Conakry) oil, gas and power 2022 conference has been announced to take place on 16 and 17 December 2023 in the Islamic Republic of Mauritania. Key topics to be covered include world energy financing, regional cooperation in energy development, promotion of cross-border synergies, women in energy and the latest developments in exploration activities.
- 4.1.2.9. The South Sudan Oil & Power (SSOP) – Juba 2023 energy conference, with the theme 'Engine for East African growth', took place from 14 to 16 June 2023 as the gateway to East African Energy development. The fourth Angola Oil & Gas Luanda 2023 conference is set to take place in September 2023.
- 4.1.2.10. The Rand has also weakened significantly against major world currencies. Global investors' confidence has been adversely affected by geopolitical tensions mainly between Ukraine and Russia. This unfortunate global development has been aggravated by the response from the European Union and countries that are part of the North Atlantic Trade Organization (NATO), with various economic sanctions declared on Russia. The Russian-Ukraine aggression has shaken all commodity markets, as well as financial markets, in most jurisdictions around the world.

Table 5: LPG Price trends in key regional markets (\$/tonne)

REGION	SAUDI ARABIA	US	ALGERIA	NORTH SEA	MEDITERRANEAN	JAPAN	SOUTH AFRICA
MIX (PROPANE: BUTANE)	60:40	60:40	60:40	60:40	60:40	60:40	60:40
DATE	\$/T	\$/T	\$/T	\$/T	\$/T	\$/T	\$/T
Jan-22	728	626	706	751	751	784	964
Feb-22	775	672	743	786	786	833	920
Mar-22	905	746	842	908	907	952	971
Apr-22	948	687	894	823	826	858	1 091
May-22	854	650	772	801	796	822	1 067
Jun-22	750	640	768	730	717	766	1 044
Jul-22	725	585	680	658	642	728	885
Aug-22	666	558	654	621	605	673	918
Sep-22	642	497	629	598	600	640	801
Oct-22	578	444	548	580	589	624	786
Nov-22	610	445	579	597	605	702	773
Dec-22	650	385	588	549	559	624	783
Jan-23	596	464	568	575	579	706	828
Feb-23	790	484	688	654	643	712	730
Mar-23	728	421	660	552	538	565	933
Apr-23	551	431	501	521	510	582	898
May-23	555	346	487	433	421	516	683

Sources: Aramco (Saudi Arabia), Refinitive (US, North Sea & Japan), Central Energy Fund (South Africa).

- 4.1.2.11. The Minister of Mineral Resources and Energy has approved the implementation of revised zone differentials into the price structures of petrol, diesel and Illuminating Paraffin (IP) with effect from 5 April 2023. The annual adjustments to road transport tariffs applicable to petrol, diesel and IP price structures will range from an increase of 40.8 c/l in Gordonia Central Magisterial District Pricing Zone to an increase of 7.8 c/l for petrol and diesel, as well as 14.1 c/l for IP (Zone 9C-Gauteng).
- 4.1.2.12. Interest rates have gone up and will affect economic growth. South Africa's value-added exports have not increased significantly. Political instability and hostility in Central Eastern Europe (CEE) is affecting emerging economies. Global energy value chains have been disrupted as Russia's aggression towards Ukraine becomes fortified. Russia has remained adamant that it will not tolerate the expansion of NATO's interests in CEE. Russia is the world's second-largest oil producer and a key supplier of crude oil to European refineries. It has also historically been the largest supplier of natural gas to Europe, providing about 35 per cent of its supply.
- 4.1.3. The Global Gas Industry**
- 4.1.3.1. The year 2022 has been recognised as a year when gas markets have experienced unprecedented challenges and turbulence. Natural gas consumers struggled with abnormally high prices. Through its global annual survey, the International Gas Union (IGU) indicated that there is now a functioning global gas market and that despite the challenges faced by everyone in managing the high-price environment, a functioning market is key to ensuring energy security. An efficient global market has allowed the necessary flexibility to reorganise gas supply flows to where they were required most in a matter of just a few months. Unfortunately, the market remains significantly supply-constrained, resulting in high prices. Ongoing efforts to add new sources of gas supply, together with prudent measures to enhance efficiency and conservation on the energy demand side, are needed to help rebalance energy markets and make energy access secure and affordable to all (IGU, 2023).
- 4.1.3.2. The year 2023 was characterised by a return to the historical price paths. Prices of all regional price benchmarks declined by more than 50% on average. On the supply side, global LNG loadings rose above 1.7 billion in February 2023, aided by a significant increase in production from Australia, Malaysia, and the United States. With the spring maintenance season likely still a month or more away, global supply could continue to rise given the pending restart of Freeport LNG, which now has one of three trains online (S&P Global Platts, 2023).
- 4.1.3.3. On the demand side, global LNG deliveries rose to levels above 1.8 billion cubic meters per day in February 2023, led by an increase in imports to South Korea, Japan, and South/South East Asia, which offset falling demand in China. Despite modest strengthening on physical LNG, the Japan Korea Marker (JKM) derivative continued to settle, with summer contract prices dropping by an average of 66 cents/MMBtu. This is aligned with the downward trend in the Title Transfer Facility (TTF) curve, as the JKM is now trading at approximately US\$1.40/MMBtu below the Standard & Poor (S & P) Global Platts Commodity Insights' Outlook.
- 4.1.3.4. Inflows of LNG into Britain's regasification terminals have increased. Such inflows have exerted downward pressure on natural gas prices on the spot market in Britain compared with northern Europe, which lacks such facilities. This is a positive development for consumers struggling with high bills. However, energy suppliers buy gas on longer-term contracts, where the price has not fallen. Britain's storage capacity is poor, even more so since Rough, its largest facility, closed in 2017. With nowhere to put the gas, it is being exported to Europe or used to generate electricity.
- 4.1.3.5. The European Union could reduce its imports of Russian natural gas by more than one-third within a year through a combination of measures that would be consistent with the European Green Deal and support energy security and affordability (IEA, 2022). While the TTF continued to decline from January 2023 to May 2023, S & P Global Platts indicated that the price curve has been supported by 'bullish coal and oil prices'. Commodity Insights' TTF forecast is now bullish for the market over a

two-year horizon, ranging from US\$1/MMBtu in the summer of 2023 to approximately US\$5.70/MMBtu in the winter of 2023. The price increase in the premium market price between the summer and winter of 2023 is likely going to tighten the LNG market.

- 4.1.3.6. The electricity price in Germany and other EU countries increased significantly in the first half of 2022, with the electricity and gas price developments running visibly parallel to each other (Roeger & Welfens, 2022). A positive correlation between electricity and natural gas prices has been noted. Governments could counter gas price shocks and associated electricity price shocks, in particular, via transfers to private households, whose consumption expenditures can thus be supported; or use subsidies in the area of gas-fired power generation in the electricity market.
- 4.1.3.7. The merit-order approach to pricing in the electricity market, which is in widespread use across the EU27 and the United Kingdom (UK), has proven to be somewhat economically problematic in the context of the Russo–Ukrainian War. The massive increases in gas prices since the summer of 2022 – in the context of Russian supply cuts to the European Union (EU) – has led to an abnormally high electricity price.
- 4.1.3.8. The Henry Hub price ranged between US\$2 and US\$3 since January 2023 until the present, as a result of increases in feed gas volumes ramping up to Freeport LNG, which requested the Federal Energy Regulatory Commission to approve its final liquefaction train in March 2023. US natural gas inventories have risen, together with dry gas production. Less torturous winter weather has suppressed natural gas demand, thus enabling lower price trends.
- 4.1.3.9. In the Americas, imports of LNG have risen to 39 million cubic meters per day in February (S & P Global Platts, March 2023). Intense rainfalls in Brazil have kept the country absent from the LNG market for two months in a row for the first time since the summer of 2020. With the reservoir

levels at 80% full across the country, there were no further projections of LNG imports beyond March 2023. Brazil's Compass Gas has delayed the start-up date for the Santos Floating Storage Regasification Unit project to the end of June 2023. Up in the Andes, Colombia pushed back the bidding deadline for the Pacific LNG project to 8 June 2023. The Colombian government has shown consideration to import natural gas from Venezuela in the medium term.

- 4.1.3.10. In Asia, while the JKM had briefly reverted to an outright premium over the TTF, a net swing of around US\$1.50/MMBtu emerged, hence showing signs of gaining strength over price benchmarks in North West Europe. This has driven the European market back to a modest premium. The reversion of the JKM-TTF price spread back to the historical JKM premium relationship had a direct bearing on the restoration of pre-COVID-19 natural gas global trading patterns.
- 4.1.3.11. The Russians were accused of using their natural gas resources as an economic and political weapon. The International Energy Agency's (IEA's) (2022) 10-Point Plan provided practical steps to cut Europe's reliance on Russian gas imports by over a third within a year, while supporting the shift to clean energy in a secure and affordable manner. Europe must rapidly reduce the dominant role of Russia in its energy markets, and ramp up the alternatives as quickly as possible.
- 4.1.3.12. Kadri Simson (2022), European Commissioner for Energy, recently called for a reduction in the European Union's dependence on Russian gas as a strategic imperative. EU has significantly diversified natural gas supplies, building LNG terminals and new interconnectors. However, Russia's attack on Ukraine has emerged as a watershed moment. The EU has initiated a pathway for Europe to become independent from Russian gas as soon as possible. Various bilateral negotiations between exporters of natural gas in Africa and EU countries have been noted throughout 2022 and early 2023.

1 Werner, R & Welfens, P.J (2022). Gas price caps and electricity production effects in the context of the Russo-Ukrainian War: modelling and new policy reforms (December 2022).

4.1.3.13. Reducing reliance on Russian gas will not be simple for the EU. It will require a concerted and sustained policy effort across multiple sectors, alongside strong international dialogue on energy markets and security. There are multiple links between Europe's policy choices and broader global market balances. Strengthened international collaboration with alternative pipeline and LNG exporters – and with other major gas importers and consumers – has been identified as critical factors. Clear communication among governments, industry and consumers has become an essential element for successful implementation. As the world's leading energy authority, the IEA seeks to continue to serve as a focal point for global dialogue on how to ensure a secure and sustainable energy future.

4.1.3.14. As of 31 May 2023, natural gas prices in the Latin American market were noted as presented in the table below.

Table 6: Natural gas prices in Latin America (2023)

	Jan	Feb	March	April	May
Country	US\$/MNBtu	US\$/MNBtu	US\$/MNBtu US\$/GJ	US\$/MNBtu	US\$/MNBtu
Argentina	17.73	14.33	13.21	12.24	9.75
Brazil	17.33	14.16	12.99	11.99	9.61
Mexico	17.59	13.91	12.73	11.71	9.31

Sources: Refinitiv

4.1.3.15. Price reductions in Latin America from January to May 2023 are attributed to a decline in seasonally lower heating demand in Mexico and other forces suppressing domestic demand for natural gas. In Argentina and Brazil, price decreases are driven by a decline in demand for LNG in Europe and the Asian markets.

4.1.3.16. In North America, between January and May 2023, natural gas prices at various pipeline city gate prices were noted as indicated in Table 7 below.

Table 7: Prices for piped-gas in North American hubs and city gates (2023)

	Jan	Feb	March	April	May
Country	US\$/MNBTU	US\$/MNBTU	US\$/MNBTU US\$/GJ	US\$/MNBTU	US\$/MNBTU
Henry Hub	3.27	2.38	2.30	2.16	2.15
Algonquin City gate	4.82	6.12	2.90	1.92	1.81
Chicago City	3.21	2.35	2.32	2.00	1.93
Dominion South	2.57	2.00	2.07	1.64	1.40
PG& E City Gate	16.76	7.71	7.99	5.65	3.82
SolCal City Gate	18.24	7.42	8.93	7.39	2.26
Z6 New York	3.23	3.39	2.22	1.74	1.46
Waha Hub	2.08	1.79	1.49	1.48	1.52
Canada Alberta	2.83	2.10	2.23	1.87	1.72

Sources: World Bank, S&P Global

4.1.3.17. Natural gas prices in 2023 have so far decreased compared to the benchmarks that were astronomically high in 2022. A rise in LNG export demand has created external demand for US shale gas, especially as the European premium attracts more exports than before, with the war and tension between Russia and Ukraine forcing NATO to look for substitutes for Russian piped-gas as well as LNG.

4.1.3.18. In North-West Europe, Central Europe and the Scandinavian Peninsula, as well as in Mediterranean Europe, key price references, including the Netherlands Titles Transfer (TTF) facility price and the National Balancing Point (NBP) price benchmark, were noted between January and May 2023, as follows.

Table 8: Natural gas prices in North West, Mediterranean and Eastern Europe (2023)

	Jan	Feb	March	April	May
Country	US\$/MNBtu	US\$/MNBtu	US\$/MNBtu US\$/GJ	US\$/MNBtu	US\$/MNBtu
France	18.70	15.91	13.94	13.33	9.64
Germany	20.08	16.92	13.99	14.07	10.35
Netherlands	20.18	16.54	13.81	13.52	10.11
UK	18.65	16.25	13.11	12.55	9.05
Italy	21.48	17.72	14.56	14.77	11.22
Spain	19.07	16.26	13.64	12.16	9.26
Russian Spimex	1.90	1.80	1.73	1.63	1.67
Russia (Piped exports by Gazprom)	7.84	7.78	7.79	7.92	7.76
Canada Alberta	2.83	2.10	2.23	1.87	1.72

Sources: Refinitiv (Germany THE, Italy), World Bank (TTF), PEGAS Spot (France), MIBGAS (Spain), OfGEM (UK-NBP), SPIMEX (Russia Nadym), Gazprom (Russia pipeline exports)

4.1.3.19. Across jurisdictions in the European natural gas market, prices have decreased significantly. With key interest in the NBP and TTF price references that are feeding into the South African price trajectory, the need to have a price cap, as has been argued and adopted from the Iberian Peninsula, has weakened. On 31 May 2023, the TTF recorded US\$10.11/MMBtu, while the NBP futures averaged US\$9.05/MMBtu. This is a clear indication that natural gas prices have stabilised towards the pre-COVID-19 levels.

4.1.3.20. Prices surged throughout the last quarter of 2021 and the beginning of 2022 due to a combination of factors, including the effect of prolonged cold weather in the northern hemisphere and the on-going Russia–Ukraine war with deteriorating geopolitical conditions. The year 2023 brought stabilisation in the European natural gas markets, as prices recorded below US\$12/MMBtu from levels of about US\$25/MMBtu in June 2022.

Table 9: Natural gas prices in the Middle East and Asia (2023)

	Jan	Feb	March	April	May
Country	US\$/MNBTU	US\$/MNBTU	US\$/MNBTU US\$/GJ	US\$/MNBTU	US\$/MNBTU
UAE	18.00	13.93	12.55	11.43	9.61
Pakistan	18.39	14.15	12.78	11.61	9.77
Japan (JCIF)	20.19	18.42	16.03	14.37	14.04
Japan, South Korea, China & Taiwan (JKM)	24.50	16.75	13.59	12.35	10.40
Taiwan	22.02	16.23	13.30	11.98	10.04
Singapore LNG Netback	20.97	15.12	12.67	11.62	9.92
Australia	20.64	14.56	12.44	11.42	9.75

Sources: S&P Global, Refinitiv & Afriforesight (UAE Netback prices, Pakistan, Taiwan, Singapore); World Bank (Japan JCIF); NYMEX (JKM – Japan, China, South Korea & Taiwan); ASX (Australia Wallumbilla)

4.1.3.21. Overall, prices in Asia and the Middle East have shown that heating demand has gone down against rising supply leading to a decline in the Asian premium and low gas prices in general. A decline in Asian prices has also pushed prices in the Middle East downwards, since countries like United American Energy (UAE), and Pakistan are exporters to Asia and the Asian Pacific.

4.1.3.22. Globally, natural gas prices have moved on a trajectory due to persistent geopolitical factors. A solid demand for energy in the United States and worldwide continues to push energy prices higher, therefore traders may soon push natural gas prices to an even higher level. EIA (2022) data reveals that working gas in storage is 15.1% lower compared to the five-year average. Inventories of crude oil in the United States are also 15% below the five-year average, which is a clear indication that energy markets remain tight. According to EIA (2022), LNG prices averaged US\$23.77/MMBtu in East Asia and US\$26.51/MMBtu in Europe. The huge spread between US natural gas prices and international spot prices will likely remain intact, thereby showing that the world gas market remains tight due to developments in the United States.

4.1.3.23. Global gas price formation remains very contentious due to market distortions induced by the unsolved geopolitical conflict emanating primarily from Russia and Ukraine. Weather conditions in Europe and Asia have become significantly warmer. Other potential catalysts, such as oil prices or international spot prices, have surged with no sign of retreat. Oil markets have responded to the easing of global lockdowns and other movement restrictions aligned with COVID-19 protocols. International spot prices have been driven by the global energy crunch and the challenging geopolitical situation. In the outlook, natural gas prices may be driven by tight energy markets and increased electricity consumption as Europe and Asia enjoy warmer summer weather.

4.1.3.24. Highlights from the aggregated total of countries that reported oil data for March 2022 include the following key points:

- Oil demand in March was at 101% of 2019 levels, while crude production was only at 97%.
- Demand exceeded 2019 levels despite sharp declines in China.
- Crude production is materially lower now vs. in 2019 in Nigeria, the US, Angola, Iraq, and the UK.
- Demand for gasoline, diesel, and LPG exceeded pre-COVID-19 levels in March 2022, but jet fuel demand was 25% lower than 2019 levels.
- Product inventories in March 2022 fell by 32.4 mb (slightly more than the seasonal average) and are now 99.8 mb below the five-year average.
- Crude inventories in March increased by 12.7mb (slightly less than the seasonal average) and are now 284 mb below the five-year average.

4.1.3.25. In Saudi Arabia and the Middle East, production of crude oil in March 2022 increased by 75 kb/d to 10.30 mb/d. It is now 2.16 mb/d above the levels from a year ago. Crude oil exports in March declined by 72 kb/d to 7.24 mb/d and product exports declined by 207 kb/d to 1.49 mb/d. Crude oil inventories declined by 1.3 mb in March to 135.8mb.

4.1.3.26. China's total product demand for petroleum products fell by 1.47 mb/d in March to 14.49 mb/d and was 688 kb/d below year-ago levels, with crude oil imports declining by 1.15 mb/d in March 2022 to 10.09 mb/d. China's total product exports increased marginally by 22 kb/d in March to 1.03 mb/d but were 660 kb/d below year-ago levels.

4.1.4. The Electricity Industry

4.1.4.1. The growth of coal consumption in 2021 is a continuation of the rebound in global coal demand that began in the final quarter of 2020. While an exceptionally cold snap in December in northeast Asia was partly to blame for increasing coal demand, the rapid growth of coal-fired electricity generation is a reminder of coal's central role in fuelling some of the world's largest economies.

4.1.4.2. Some countries, including Japan and EU countries, are dealing with energy security to ensure uninterrupted energy access and energy supplies at affordable prices. Japan is restarting nuclear reactors and diversifying sources of supply. Europeans are importing LNG and accelerating the deployment of clean energy technologies. They are also enacting laws to protect consumers from high prices by setting price caps and cutting fuel taxes.

4.1.4.3. Global prices of oil and natural gas began to increase rapidly in 2020, however, the Russian invasion of Ukraine worsened the situation. Furthermore, coal-fired electricity has temporarily returned due to concerns about the price of natural gas and energy security. Some countries, including the USA, China and Japan, through increased support for combinations of renewables, nuclear power, carbon capture, hydrogen, and ammonia, will make progress toward achieving net zero emissions.

4.1.4.4. According to World Economic Outlook (2022), it is anticipated that in the STEPS, CO₂ emissions from electricity generation will decline by more than 10% in 2030 and around 40% by 2050. In the APS, emissions will drop

by 80% in 2050. The use of coal and natural gas is being reduced more rapidly in the APS, which causes a faster decline in emissions.

4.1.4.5. European countries have reduced their reliance on imports of Russian fuel. However, they still depend on fossil fuels as primary source of electricity generation. Over the past years, Europe has made significant progress in its transition to the use of renewable energy sources to generate electricity (Eurostat, 2023).

4.1.4.6. According to Eurostat (2023), in 2021, the European Union (EU) imported 56% of its energy while producing about 44% of its own. Solid fuels, natural gas, crude oil, nuclear energy, and renewables (including hydro, wind, and solar energy) are among the many energy sources used to produce energy.

Table 10: Electricity production in Europe

Source	Electricity Generation (2021)
Renewable energy	40.9%
Nuclear energy	31.3%
Solid fuels	18.1%
Natural gas	6.4%
Crude oil	3.1%
Other	0.2%

Sources: Eurostat

4.1.4.7. The major contributor to energy production was renewables energy, accounting for 41% of all EU energy production. The next-largest source was nuclear energy (31%); followed by solid fuels (18%), natural gas (6%) and crude oil (3%).

4.1.4.8. Petroleum products (like heating oil, gasoline, and diesel fuel), which accounted for 35% of the total energy consumption in 2021, were the most consumed. Electricity and gas (natural and manufactured gas) ranked second with 23% each, followed by direct-use renewables (not transformed into electricity, for example wood, solar or hot water production) (12%), derived heat (such as district heating) (5%) and solid fossil fuels (mostly coal) (31%). Because electricity also contains renewable energy from sources like hydropower, wind power, and solar photovoltaic, the actual consumption of renewable energy is higher than 12%.

Table 11: North West Europe electricity consumption

Country	Electricity Consumption (2021)
Luxembourg	16.0%
Belgium	21.1%
Denmark	20.1%
Germany	21.3%
Sweden	34.3%
Netherlands	21.3%
Ireland	23.1%
Iceland	52.9%

Sources: Eurostat

Table 12: Central Europe electricity consumption

Country	Electricity consumption (2021)
Poland	16.7%
Hungary	19.3%
Austria	21.0%
Slovakia	20.9%
Czechia	20.3%

Sources: Eurostat

Table 13: Eastern Europe electricity consumption

Country	Electricity consumption (2021)
Bulgaria	26.3%
Croatia	20.6%
Poland	16.7%
Romania	15.7%

Sources: Eurostat

4.1.4.9. Tables 10 to 13 provide the electricity consumption of the EU in 2021. Electricity consumption was highest in Iceland at 52%, while Romania consumed the lowest electricity (15.7%). About two-thirds of the energy available in the EU is utilised by end users. Households (energy used in residential buildings), transportation (rail road, domestic aircraft, or Ireland shipping), industry, services (private and public services), agriculture and forestry are just a few of the economic sectors that consume energy.

4.1.4.10. Transport accounted for 29% of total energy consumption in 2021, followed by households (28%), industry (26%), services (14%), and agriculture and forestry (3%) (Eurostat, 2023). In the first half of 2022, household electricity prices, including taxes and levies, in North West Europe were very high per 100 Kilowatt hours (KWh), while the lowest cost was found in Eastern Europe and some parts of Central Europe.

Table 14: Electricity prices in North West Europe

Country	Electricity price per €/100kWh
Luxembourg	€20
Belgium	€34
Denmark	€46
Sweden	€23
Netherlands	€5
Ireland	€30
Iceland	€15
Germany	€33

Sources: Eurostat

Table 15: Electricity prices in Central Europe

Country	Electricity price per €/100kWh
Poland	€15
Hungary	€9
Austria	€23
Slovakia	€18
Czechia	€31

Sources: Eurostat

Table 16: Electricity prices in Eastern Europe

Country	Electricity price per €/100kWh
Bulgaria	€11
Croatia	€14
Romania	€24

Sources: Eurostat

4.1.4.11. Table 14 to 16 shows the electricity prices of North West, Central and Eastern Europe. Household electricity prices were the highest in Denmark in the first half of 2022 (€46 per 100 kWh), followed by Belgium (€34 per 100 kWh) and Germany (€33 per 100kWh), while the lowest costs were found in the Netherlands (€5 per 100kWh) and Hungary (€9 per 100 kWh).

4.1.4.12. According to US Short-Term Outlook (2023), US electricity demand reaches its lowest point during the spring months (March–May) as a result of a decrease in the demand for both space heating and air cooling. In particular, for thermal power plants, power plant operators use this gap in demand to make the necessary repairs to their generating units, nuclear power facilities, as well as coal and gas-fired units. On the other hand, due to greater winds in the spring, the production from renewable sources, especially wind, increases (see Table 17 below).

Table 17: US electricity generation in spring

SOURCES	Share of US electricity generation in spring		
Year	2022	2023	2024
Natural gas	39%	39%	37%
Coal	20%	17%	17%
Renewables	22%	24%	26%
Nuclear	19%	20%	20%

Sources: US Energy Information Administration, short-term energy outlook, April 2023

4.1.4.13. In 2023, coal is anticipated to contribute 17% of the total US generation, down from 20% in 2022. This year (2023), natural gas continues to supply 39% of all electricity generated. This year's nuclear power accounts for 20% of total generation, which is an increase of 1%. The prediction shows that generation from renewable resources will expand the most, from a share of 22% in 2022 to 24% in 2023.

- 4.1.4.14. In the US, the total residential demand increased by 3.5% over 2021, reaching an all-time high of 3.9 Terawatt hours (TWh) in 2022. The average monthly electricity consumption per residential customer increased by 2.4% as a result of both a colder winter and a hotter summer. The average monthly US power usage per residential customer grew from 886 KWh in 2021 to 907 kWh in 2022 (EIA, 2023).
- 4.1.4.15. The US' electric power industry generated 4090 million megawatt hours (MWh) of electricity in 2022. For the first time, coal-fired generation in the electric power industry was surpassed in 2022 by generation from renewable sources (wind, solar, hydro, biomass, and geothermal). For the first time in 2021, renewable energy generation exceeded nuclear energy generation, and this trend continued in 2022 (EIA, 2023).
- 4.1.4.16. Natural gas remained the country's leading source of power generation, increasing from 37% of US generation in 2021 to 39% in 2022. The coal-fired generation's contribution fell from 23% in 2021 to 20% in 2022 as a result of retirements and reduced usage of the remaining units. Due to the retirement of the Palisades nuclear power facilities in May 2022, the share of nuclear energy fell from 20% in 2021 to 19% in 2022. Wind and solar energy together indicated an increase from 12% to 14% of total generation in 2022. In 2022, the production of hydropower remained the same at 6%. At less than 1% each, respectively, the shares of geothermal and biomass remained the same.
- 4.1.4.17. An increase in wind and solar generation was driven by growth in wind and solar generating capacity. From 61 Gigawatt (GW) in 2021 to 71 GW in 2022, utility-scale solar capacity in the US electric power industry rose. Between 2021 and 2022, wind capacity increased from 133 GW to 141 GW.
- 4.1.4.18. According to the EIA (2023), more than 56.1 GW of new utility-scale electric generating capacity was anticipated to be added to the power grid during 2023, which would result in the largest net change in the US electric generating capacity since 2003. 7.8 GW of new natural-gas-fired capacity is expected to be added, 80% of which is from combined-cycled plants. The 1.836 MW Guernsey Power Station in Ohio and the 1.214 MW Concentrator Photovoltaics (CPV) Three Rivers Energy Center in Illinois. 7.5 GW of utility-scale wind capacity is anticipated to be added to the grid and Texas will add the highest wind capacity, at 2.2 GW.
- 4.1.4.19. Fossil fuels accounted for more than 80% of Mexico's entire energy supply. In 2019, oil made up 45.20% of the total energy produced, followed by natural gas (37.84%), coal (6.44%), biofuels (5.02%), wind and solar (2.75%), nuclear (1.62%), and hydropower (1.13%). Fossil fuels made up 62.50% of Mexico's installed electrical capacity and 72.1% of its electrical generation in 2020. Hydro (8.59%), wind (6.31%), solar (4.33%, up from 1.83% in 2019), biomass (3.49%), nuclear (3.48%), and geothermal energy (1.46%) all contributed to the production of additional electricity.
- 4.1.4.20. Mexico had 83.1 GW of installed capacity as of 2020. 53.6 GW of capacity was made up of 64.50% fossil fuels, followed by hydro (12.6 GW, 15.17%), wind (6.5 GW, 7.82%), and solar (5.1 GW, 6.19%). Compared to its 2019 production of 323.8 TWh, Mexico generated 312 TWh of energy in 2020.
- 4.1.4.21. Energy demand in Mexico quickly recovered after the pandemic, but in 2022, it increased by about 4% year over year. The pace is forecast to pick up steadily, reaching 3% by 2025, which amounts to an increase from over 300 TWh in 2022 to approximately 330 TWh by 2025, following an anticipated drop to less than 2% growth in 2023. In 2022, a variety of sources, primarily wind and solar PV, but also a slight absolute increase in coal-fired production, were used to meet the increase in demand.
- 4.1.4.22. The government of Mexico, in partnership with state governments, has launched several clean energy projects despite the slowdown in private sector activity in Mexico's electrical market. These include the ongoing development of the solar PV plant in the northern state of Sonora, which should add 420 MW by the end of February 2023 and reach a total of 1 000 MW by the end of the fourth phase in 2027, as well as a new 18 MW solar PV rooftop installation on top of Mexico City's largest market.

- 4.1.4.23. The demand for electricity in Brazil increased slightly in 2022, by 0.3% year over year, but is expected to climb by over 2% annually from 2023 to 2025. The country experienced its worst drought in 90 years in 2022, but hydropower generation recovered with a year-on-year (y-o-y) rise of roughly 17%. This resulted in a reduction in power system emission intensity from 135 gCO₂/kWh in 2021 to 80 gCO₂/kWh in 2022. The intensity of CO₂ is anticipated to keep dropping until it reaches 30 gCO₂/kWh in 2025.
- 4.1.4.24. Between 2021 and 2025, it is predicted that the proportion of generation from hydropower will rise from 55% to 63%, which will drive the decrease in emissions together with an increase in the share of generation from wind and solar PV, which will go from 11% to 17% and 3% to 11%, respectively.
- 4.1.4.25. The Asia Pacific region predicted a 3.3% increase in electricity demand in 2022, which was driven primarily by India's 8.4% increase, which was largely offset by China's 2.6% decline in demand because of its economy's slowdown as a result of its zero-COVID-19 policy. The demand in the two countries accounted for almost 70% of the region's total power consumption of 13,500 Twh, which is equivalent to around 50% of global consumption. Renewable energy sources provided about 60% of the growth in 2022, accounting for more than half of the increase.
- 4.1.4.26. In 2022, coal accounted for 57% of the region's electricity production, with nuclear power and renewable energy sources making up 32% of the mix. Asia Pacific has the greatest electricity-generating CO₂ intensity of our examined areas due to its heavy reliance on coal, with 590g CO₂/kWh in 2022, as opposed to a global average of 460g CO₂/kWh.
- 4.1.4.27. Electricity demand growth in China fell to an expected 2.6% y-o-y in 2022 (much below pre-pandemic rates) under pressure from the widespread economic lockdowns. The increase was mostly driven by demand from the residential sector, because industrial electricity consumption growth was restrained.
- 4.1.4.28. Demand is expected to increase by 5.2% year on average between 2023 and 2025, driven by further electrification in the energy industry, particularly in buildings and transportation. Despite an increase in the amount of renewable energy (30%), coal still makes up the majority of the country's electricity system in China, accounting for over 62% of power generation in 2022. The majority of renewable electricity is produced by hydropower, with the next largest sources being wind (29%) and solar PV (15%). Even though coal-fired supply grew over the summer to make up for low hydro output, its average growth in 2022 compared to 2021 remained moderate (+1.5%), with renewables accounting for nearly 60% of the increase in electricity demand.
- 4.1.4.29. Due to the rapid growth of wind and solar PV, it is anticipated that low-carbon energy would generate 41% of all electricity by 2025, exceeding the 39% goal set in China's 14th Five-Year Plan. Both technologies' capacity additions in 2022 broke previous records, and by 2025, it is anticipated that, their combined capacity will approach 1250 GW, up from more than 630 GW in 2021.

4.2. Continental and Regional Developments

4.2.1. Continental developments

- 4.2.1.1. There is a need to improve access to modern energy to boost Africa's economic development. Currently, in sub-Saharan Africa, fossil fuels make up about 40% of the total energy mix. We can expect a rise in these numbers as oil and natural gas exploration projects accelerate throughout the continent. Despite the massive impact of the COVID-19 pandemic on fossil fuel development and investment, the discovery of new light oil and shale gas reservoirs has reaffirmed Africa's position and potential to be a key oil-producing region. Major discoveries have been pronounced around jurisdictions such as Mauritania, Senegal, DRC, Mozambique, Zimbabwe and Angola.

- 4.2.1.2. The most famous discovery of 2021 was the discovery of a massive reserve of light oil off the coast of Angola. Discovered by Eni Offshore Angola, the reserve holds the potential for 200 to 250 million barrels of oil. This will make West Africa the second-largest oil-producing country in the sub-Saharan region.
- 4.2.1.3. Natural gas pipelines are required to transport gas from gas wells to processing plants, import–export facilities, and homes. Therefore, an increase in Greenfield and Brownfield exploration activities will prompt the oil companies to establish a system of interconnected natural gas pipelines for the region. As these exploration activities are carried out in harsh environmental conditions, the demand for high-performance gas valves will increase. According to the Future Market Insights report, the sales of high-performance oil and gas valves will surpass \$12 billion by 2022 year-end.
- 4.2.1.4. These new oil and gas valves are made of corrosion-resistant alloy (CRA) and can withstand high temperatures, corrosion and sour gas. The introduction of these pressure regulators in every African country will streamline the gas trade within the continent. We are also seeing a sharp increase in the construction of natural gas pipelines across the continent. As governments are seeking to increase gas exports to international markets, the demand for piping services and oil and gas valves will increase significantly.
- 4.2.1.5. Some of the most popular projects include the following:
- The proposed \$6 billion African Renaissance pipeline project that connects Mozambique’s gas-rich Robma River basin to Gauteng, South Africa.
 - The proposal for a 1,800km Tanzania–Uganda Natural Gas Pipeline Project to transport LNG from Dar-es-Salaam, Tanzania to Kampala, Uganda.
 - The proposed Ajaokuta Kaduna Kano natural gas pipeline, which will serve as part of the broader Trans-Nigeria pipeline project. These pipeline projects will not only promote gas exports from Africa, but also increase gas trade within Africa.
- 4.2.1.6. Proven natural gas reserves in Africa are estimated to be at least 18 Tcm, about 8.8% of the world’s total. They are considerable in Northern (45%) and Western (32%) Africa especially, and commercial quantities of natural gas have been found in many African countries¹. The huge reserves of gas can play a significant role in energy transition since natural gas has a much smaller carbon footprint than oil and coal. Natural gas can be used as an energy source to assist certain regions and countries with power generation, transport, and cooking. Sub-Saharan Africa should be prioritised, accounting for three-quarters of the global energy access deficit.
- Several countries, including Mozambique, Tanzania, Mauritania, and Senegal, are actively marketing the use of natural gas to capitalise on recent discoveries (IGU & Halwiti, 2023²).

4.2.1.7. It is demonstrated that natural gas prices, as calculated by Afriforesight (2023), are co-trending and that they are LNG netback prices with full recognition that the basis of comparison is compromised by the fact that the SA price is strictly a piped-gas price. The South African price is trending with continental benchmark prices over the period from May 2022 to April 2023.

4.2.1.8. As at 31 May 2023, natural gas prices in the five African countries considered and reported by Afriforesight (June 2023) were as follows:

Table 18: Natural Gas Prices in 5 African countries considered (2023)

	Jan	Feb	March	April	May
Country	PRICE (\$/MNBTU)				
Algeria	17.48	14.16	12.78	11.78	9.35
Equatorial Guinea	16.71	13.40	12.39	11.33	8.99
Egypt	17.93	14.15			
Nigeria	16.94	13.25	11.98	11.02	8.94
South Africa	24.89	24.29	22.78	21.76	20.73

Sources: Algeria (S&P Global; Afriforesight), Egypt (S&P Global and Afroforesight), Equatorial Guinea (S&P Global; Refinitiv and Afriforesight)

4.2.1.9. Month-on-month price reductions in the four net LNG exporting countries is primarily driven by downward pressure on natural gas prices between April and May in most European countries and the Asian region due to a decline in demand.

4.2.1.10. Africa's energy crisis continues to serve as a direct constraint to economic growth. Currently, more than 600 million people do not have access to electricity and more than 900 million do not have access to clean fuel for

cooking (International Energy Agency, 2022). Electrification of rural areas has become a priority for many African governments. Infrastructure deficits and untapped resources, combined with a net- zero emissions target have created notable opportunities for foreign investors in the green energy sector.

4.2.1.11. Renewable energy is becoming an important part of Africa's energy sector. It has gained significant traction and investment in recent years. Although the continent's energy composition is almost entirely composed of fossil fuels and biomass, renewable energy is projected to account for a major chunk of total power generation capacity in the next ten years.

4.2.1.12. Early movers in renewable energy development are Egypt, South Africa, and Ethiopia. 2023 was expected to be a very productive and profitable year for Egypt, as the country aims to produce 20% of its electricity from renewable energy sources by the end of the year. With the 1,650MW Benban Solarpark, which went live in 2019, and the ongoing \$37 million solar plus storage project to power Egypt's Sukari gold mine, Egypt has taken steps towards integrating solar in its energy structure.

4.2.1.13. The Grand Ethiopian Renaissance Dam plans to add 6.45GW of installed capacity to the Ethiopian country's power grid. When completed, the dam will be the largest hydroelectric power plant in Africa and the seventh-largest in the world (Kaitwade, 2022).

4.2.1.14. Egypt and the United Arab Emirates (UAE) signed a memorandum of understanding to develop a measurement project on establishing the 10GW onshore wind power mega-project in Egypt. The project will be part of Egypt's Green Corridor initiative (a grid dedicated to renewable energy projects) and will contribute to the North African country's goal of ensuring that renewable energy makes up 42% of its energy mix by 2035. When completed, the 10GW wind farm will produce 47,790GWh of clean energy annually and offset 23.8 million tonnes of carbon dioxide emissions (ESI Africa, 2023).

- 4.2.1.15. Southern Africa has been characterised by supply insecurity in recent years. The electricity generation industry is seen as a critical vehicle for regional integration and industrial development. It is historically defined by national, centralised, state-utility monopoly supply. Decades of low revenue, high debt, under-investment and poor maintenance have left the industry falling behind population growth and electricity demand, while a significant percentage of installed capacity is unavailable due to breakdowns and transmission failures. As a result, the region regularly experiences power cuts.
- 4.2.1.16. The regional industry is well-integrated and well-organised in the Southern African Power Pool and there are significant plans to expand capacity. Regulatory reforms are intended to attract private sector participation in electricity generation. Principal among them is the acceleration of private sector-led renewable energy investment. This has included everything from hydroelectric mega projects to small solar- powered generation.
- 4.2.1.17. Electricity generation in Africa is dominated by coal, oil, and natural gas. The shares of coal and natural gas are the most important, but coal-fired electricity is located in South Africa while gas-fired electricity is produced largely in Egypt and Algeria (IGU & Hawilti, 2023). Sub-Saharan Africa relies heavily on oil to support its decentralised power generation industry, which consists of sparse captive power plants and diesel generators, with distributed diesel capacity alone estimated to be anywhere between 45GW and 100 GW. Several African countries, including Ghana, South Africa, and Senegal, are planning to switch to gas to lower their emissions and improve air quality.
- 4.2.1.18. The adoption of natural gas in Africa's energy mix comes with enormous affordability benefits, as gas is often much cheaper in terms of power, industry, and transport than imported diesel. This has been demonstrated in Nigeria where, between 2020 and 2022, CNG was between two to four times cheaper than diesel. The overall growth trend of global demand over the past decade has enabled African countries to monetise gas in lucrative export markets and have resulted in several LNG export terminals on the continent.
- 4.2.1.19. Africa's export capacity has remained relatively small compared to its potential. With the current market fundamentals signalling a strong need for investments in new LNG supply, the continent has a new window of opportunity to position itself as a strategic and reliable global gas supply hub. However, this window is not exclusive to Africa, nor will it stay open forever. By 2025, Africa's LNG export capacity will already have increased from 77.6 million tonnes per annum (mtpa) today to 83mtpa, with new countries joining the club of African gas exporters, such as Senegal, Mauritania, or Congo. Much more potential exists based on the projects currently being proposed, with a total proposed project pipeline totalling over 55mtpa.
- 4.2.1.20. Natural gas is almost absent from East African economies, whose baseload power has traditionally been met by hydropower. Geothermal energy has also become popular in the region, especially in Kenya, but increasingly in Ethiopia and Djibouti as well. For the same reason, gas will play a different role from elsewhere on the continent. Tanzania and Rwanda are two notable exceptions. Tanzania adopted gas as early as 2004 to generate electricity and support Dar es Salaam's growing industrial base, while Rwanda started extracting methane from Lake Kivu in 2016 to produce power. The latter has since then embarked on additional gas extraction to produce CNG and cooking gas for its domestic market.
- 4.2.1.21. After growing a functioning domestic gas value chain, Tanzania plans to start exporting LNG by developing its vast offshore reserves in the south. The country could also become a strategic hub for the region and is notably considering a gas pipeline to Kenya, where 10% of electricity generation relies on HFO and diesel and could be replaced by gas imports. Additional domestic projects in Tanzania, such as Ntorya, could unlock more supply to increase gas penetration across the economy, while supporting regional supply ambitions. Ethiopia could finally emerge as a gas market this decade, provided it can secure a new investor to develop its 113 Billion Cubic Meters (BCM) of recoverable gas in the Ogaden Basin, with reserves concentrated within the Calub and Hilala fields. Development plans initially involved a pipeline to Djibouti before exporting gas as LNG, however the contract with a Chinese operator was terminated in 2022.

- 4.2.1.22. IGU & Hawilti (2023) have presented that approximately 8.8% of the world's proven reserves of natural gas are in Africa. Statistics show that at the end of 2021, some 18 trillion cubic metres (tcm) of proven gas reserves had been discovered across Africa. The highest proportion of this gas is in Nigeria (over 32%), followed by Algeria (25%), Egypt (12%), and Libya (8.3%). While Nigeria and North Africa hold most of these reserves, gas has been found in commercial quantities across the continent. Hawilti's research in 2022 has shown that more than 20 African markets have discovered gas, including Morocco, Algeria, Tunisia, Libya, Egypt, Mauritania, Senegal, Côte d'Ivoire, Ghana, Nigeria, Equatorial Guinea, Gabon, the Republic of Congo, the Democratic Republic of Congo (DRC), Cameroon, Chad, Angola, Namibia, South Africa, Mozambique, Tanzania, Rwanda, and Ethiopia. Except for Mauritania, Namibia, and Ethiopia, all these countries already produce gas in various quantities for exports or for small consumption in power, transport, industry, and cooking.
- 4.2.1.23. North Africa is expected to remain Africa's biggest producing region, led by Egypt and Algeria. Both countries have increased cooperation with international energy companies in 2022 and are committed to exploring additional gas and hydrogen projects. Both Morocco and Algeria are also involved in multi-billion-dollar transnational gas pipeline projects proposed to distribute gas across West Africa before reaching Europe, namely the offshore Nigeria-Morocco Gas Pipeline (NMGP) and the onshore Trans-Saharan Gas Pipeline (TSGP).
- 4.2.1.24. West Africa's gas reserves are concentrated in Nigeria and offshore Mauritania and Senegal. Gas production is mostly reserved for exports via Nigeria LNG's terminal in the Niger Delta. While Nigeria, Côte d'Ivoire, Ghana, Benin, and Togo have all monetised gas domestically, consumption is limited. The power sector remains the main gas off-taker, but is plagued by liquidity constraints in Ghana and Nigeria. West Africa is expected to emerge as a much bigger gas-producing region, as both Mauritania and Senegal join Nigeria and Ghana in the club of gas exporters. Both countries hope that domestic allocation from export projects could support gas-fired generation and replace existing coal and diesel consumption.

- 4.2.1.25. Despite much smaller gas reserves, Central African markets have successfully developed export facilities in Equatorial Guinea and Cameroon, with additional ones currently in development in the Republic of Congo and Gabon. Domestic consumption has mostly remained limited to power, except for Equatorial Guinea, which has developed a robust and diversified downstream gas infrastructure hub at Punta Europa. However, Central Africa remains the most intensive flaring region on the continent and could benefit from new strategies to monetise flared gas. This has caused several technological and technical challenges given the relatively small size of the domestic markets. In the Gulf of Guinea, where Equatorial Guinea is trying to position itself as a regional gas hub. The country is also trying to revive the Fortuna Floating Liquefied Natural Gas (FLNG) project, while positioning its domestic gas processing facilities at Punta Europa on Bioko Island as a regional centre for the processing of stranded gas in the Gulf of Guinea. Congo plans to start exporting LNG by 2023, once Eni commissions its first of the two planned floating LNG units on its Marine XII block. The country also relies on gas for most of its power production and a few additional units are in the pipeline there. Gabon also hopes to start exporting LNG by 2026 via a project proposed at Cap Lopez by an independent operator.

4.2.2. Developments in Southern Africa

- 4.2.2.1. Southern Africa is loosely divided into a hydro-rich north and a thermal-rich south, which is reflected in the mix of energy sources used in generating electricity. The region has significant and largely untapped natural energy resources that make the transition away from carbon-heavy fossil fuels to renewable energy a real possibility. However, there might be a need to pair transition objectives with developmental goals for this possibility to be beneficial. Just over a quarter of hydroelectric power and biomass electricity potential is realised, as is 1% of solar and wind power, and less than that for geothermal power. There is a small but growing presence of independent power producers supplying the Southern African Grid.

- 4.2.2.2. It is anticipated that by 2030, demand in Southern Africa will rise by about 55%. In Southern African power systems, the main sources of new generation are anticipated to be hydro, solar, wind, and natural gas. By 2030, solar PV will account for 20–25% of installed capacity in Southern Africa, according to the International Renewable Energy Agency (IRENA).
- 4.2.2.3. In 2020, Mozambique will have a total installed capacity of 2,780 MW. The Power Sector Master Plan's forecasts indicate that the total installed capacity will rise to 6,001 MW by 2030. At 2,189 MW, hydropower accounts for 79% of the overall energy mix, followed by gas with 442 MW, heavy fuel oil (HFO) with 108 MW, and solar with 41 MW (2020).
- 4.2.2.4. Hidroelectrica de Cahora Bassa (HCB), a government-owned company, runs the Cahora Bassa hydro dam, which is the largest power-producing facility in the nation. With a capacity of 2,075 MW, this dam ranks second in size in Africa. The remaining 35% is exported to Zimbabwe and the northern Mozambican provinces, while 65% of HCB's current generation is sold to South Africa.
- 4.2.2.5. Mozambique's demand is expected to grow at an average annual rate of around 7 to 8% in the next decade. This means it is necessary to add about 1,500 MW to the generation fleet by 2030. Mozambique is also the country in Southern Africa with the highest potential for producing electricity. The country could produce up to 187 GW of electricity using its resources for wind, coal, gas, hydro, and solar energy. Currently, the majority of the electricity is produced by hydroelectric facilities. The Government intends to boost the proportion of coal and gas generation in the near future. By the end of this decade, natural gas-fired power plants are anticipated to generate 44% of all electricity.

Table 19: Electricity consumption in Mozambique

Electricity consumption	Total (billion/kWh)
Own consumption (Mozambique)	11.57 bn/kWh
Production	18.38 bn/kWh
Import	9.93 bn/kWh
Export	12.88 bn/kWh

Sources: World Bank Data

- 4.2.2.6. In the year 2021, Table 20 shows that in Mozambique 18.38 billion per kWh worth of electricity was produced, to which 12.88 billion per kWh was exported, and 9.93 billion per kWh was imported. According to African Energy Commission (2022), the energy production of Mozambique is dominated by biofuels with a rate of 42.03%, followed by coal with 27.53%. Table 20 below shows that Zimbabwe's Energy Production consists largely of 92.91% biofuels and waste. It also includes hydro with 1.04%, and coal and coal products with 6.05%. It is also importing a share of 16.69% of electricity (African Energy Commission, 2022).

Table 20: Zimbabwe's electricity consumption

Consumption by sector	Consumption: Thousand Tonnes of oil Equivalent (Ktoe)
Industry	385.5
Households	146.8
Commercial and public services	32.6
Agriculture, forestry and fishing	118.3

Sources: African Energy Commission, 2022

- 4.2.2.7. The industry sector in Zimbabwe consumed the highest amount of electricity in 2021 with 146.8 Ktoe, followed by households (146.8 Ktoe) and agriculture, forestry and fishing (118.3 Ktoe), while commercial and public services consumed the lowest electricity (32.6 Ktoe).
- 4.2.2.8. Southern Africa has remained a very limited gas frontier, until recently. Angola used to be the primary gas market, with the development of an LNG export industry purely reliant on associated gas. However, exploration activity has unlocked new reserves in Mozambique and South Africa, redrawing the cards of the region's energy security scenario. Regional cooperation is already thriving and could grow further with gas, especially if new discoveries are confirmed in Namibia and Zimbabwe by Invictus Energy (See Invictus Energy, 2023).
- 4.2.2.9. Southern Africa is on the verge of profound transformations that will shape its energy future. Long dominated by Angola as a key gas-producing and exporting country, the region is seeing the entry of several new local and regional gas players. Mozambique, which has been a key supplier of gas to South Africa for the past two decades, became an LNG exporter in 2022 and hopes to significantly ramp up its LNG export capacity by 2030. While export facilities are being developed in the north, the country's capital in the south could also house an LNG import hub serving the sub-region. But Mozambique is also focusing on its domestic economy, with several gas-to-power projects currently in development along with the growth of its domestic LPG supply.
- 4.2.2.10. In South Africa, natural gas has become the natural alternative to move away from coal, and the country is currently planning LNG import terminals on its coast to switch several gigawatts of coal-based power generation capacity. Recent discoveries onshore and offshore are also providing the foundation to grow the gas value chain and serve already well-established manufacturing and transport industries. The country became a small LNG producer in 2022 and will seek to further develop its domestic reserves to supply industrial facilities and decarbonise its transport sector.

4.3. National Developments

4.3.1. The Regulated Electricity Sector

4.3.1.1. In September 2022, Eskom launched power purchase programmes to mitigate the load-shedding crisis. The programmes aim to ensure that the electricity supply increases by 1000 Megawatt (MW) to the national grid. The programme includes the following:

- Standard offer programme: Enables Eskom to acquire electricity at a predetermined price calculated at the avoided cost of its own generation (including long-term energy purchases from independent power producers).
- Emergency Generator programme: Enables independent generators to consistently supply energy so they can compete with Eskom generators in the domestic market.
- Bilateral power import programme: To ensure that neighbouring countries' imports are secure.

4.3.1.2. Table 21 below shows the NERSA-approved Eskom tariffs for the 2023/24 financial year (FY).

Table 21: Eskom Tariff adjustments for the 2022/23 FY

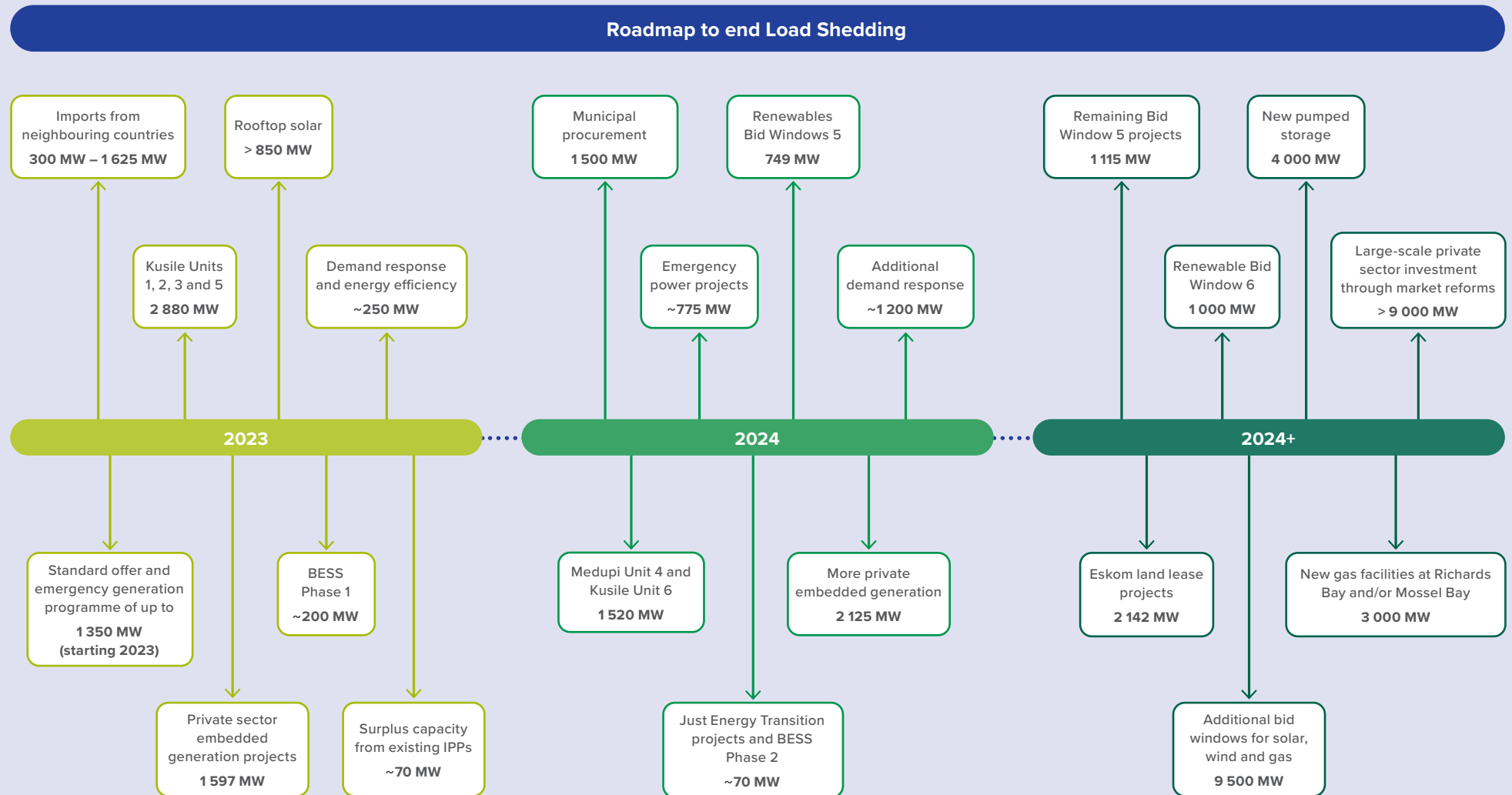
Customer category	Percentage increase
Total standard tariffs	18.65%
Municipal	18.49%
Affordability subsidy(paid by Industrial and urban ESKOM customers)	29.53%
Rural	18.65%
Homelight 20A	10%
Homelight 60A	18.65%

Sources: ESKOM

- 4.3.1.3. During the March 2023 State of the Nation Address (SONA) address, President Cyril Ramaphosa announced the creation of a Ministry of Electricity, in The Presidency. He announced that, the new Minister of Electricity would have political responsibility, authority, and control over every important aspect of the Energy Action Plan to effectively manage the response to the electrical crisis. The Minister would be expected to expedite the procurement of new generation capacity, work with Eskom leadership to improve the performance of existing power stations, and facilitate coordination of the numerous departments and entities involved in the crisis response.
- 4.3.1.4. In May 2023, President Cyril Ramaphosa transferred the administration, powers and functions under section 34(1) of the Electricity Regulation Act to the new Minister of Electricity in the Presidency. The procurement of generation capacity will remain with the Minister of Energy and Mineral Resources. In consultation with NERSA, the Minister may:
- determine that new generation capacity is needed to ensure the continued uninterrupted supply of electricity;
 - determine the types of energy sources from which electricity must be generated, and the percentages of electricity that must be generated from such sources;
 - determine that electricity thus produced may only be sold to the persons or in the manner set out in such notice;
 - determine that electricity thus produced must be purchased by the persons set out in such notice;
 - require that new generation capacity must:
 - be established through a tendering procedure that is fair, equitable, transparent, competitive and cost-effective, and
 - provide for private sector participation.
- 4.3.1.5. President Ramaphosa established a National Energy Crisis Committee (NECOM) which is led by the Director-General in the Presidency, Ms Phindile Baleni. According to the Presidency (2022), NECOM is chaired by The President and includes Eskom, the Minister in the Presidency, the Minister of Mineral Resources and Energy, the Minister of Public Enterprises, the Minister of Finance, the Minister of Forestry, Fisheries and the Environment, and the Minister of Trade, Industry and Competition. NECOM's mandate is to do whatever it takes to end load-shedding. This will be done by overseeing the implementation of an Energy Action Plan to achieve energy security. The Action Plan outlines the following intervention strategies:
- Assist Eskom to improve the availability of existing supply
 - Enable and accelerate private investment in generation capacity
 - Accelerate procurement of new capacity from renewables, gas and battery storage
 - Accelerate the rooftop solar investment by businesses and households
 - Fundamentally transform the electricity sector to achieve long-term energy security.

Figure 1: Roadmap for NECTOM Activities to End Load-shedding

Source: Business Tech (2023)



4.3.1.6. NECOM is therefore executing several initiatives to respond to the energy crisis (see Figure 1 above). Another intervention was the development of a net billing framework. This framework has been submitted to NERSA for the development of Net-billing Rules, to enable distributed generation facilities that have excess capacity to export onto the grid.

4.3.1.7. Due to slow progress in new generation projects and the declining availability of its ageing coal fleet due to energy insufficiency, South Africa's electricity usage decreased by less than 4% year over year in 2022. Over 8 Twh (or 5% of the anticipated annual demand) will be lost through load-shedding in 2022, which is an unprecedented level of load-shedding. Compared to 2021, this amounts to a fourfold increase in unfulfilled demand.

4.3.1.8. According to the Transmission Development Plan (TDP) (2023–2032), by 2032, it is predicted that there will be 6.6 GW of battery storage. While conventional generation is predicted to lose 5GW and see a decline in its share of total energy capacity from 87% in 2021 to 47% in 2032, renewable capacity is predicted to increase from 13% in 2022 to 46% in 2032.

Table 22: Capacities concurred with by the Energy Regulator

No.	Technology	Volume (MW)	Date of Regulator Decision	Developer of Capacity
1	Risk Mitigation Capacity	2 000	14 May 2020	IPPs
2	Renewable (PV & Wind)	6 800	29 July 2020	IPPs
	Storage	513		
	Gas & Diesel	3 000		
	Coal	1 500		
3	Nuclear	2 500	27 October 2021	Eskom
4	Battery Storage and PV	404	25 February 2022	Eskom
5	*STPPP determination	202	25 November 2021	Eskom
6	Renewable (PV, Wind & Storage)	14 771	15 December 2022	IPPs
Total		31 690		

4.3.1.9. It is also expected that by 2032, the system would have around 45 GW of renewable energy, 27 GW of coal, and 8.5 GW of gas installed. Solar PV is predicted to reach 18 326 MW by 2032, with wind generation reaching 26 437 MW. Due to the Risk Mitigation Independent Power Producer Procurement Programme (RMIPPPP), 75 MW of wind power will be added in 2022, and 22 799 MW will be added during the TDP period.

Table 23: Total hourly energy demand forecast between May and July 2023

Months	Contracted Forecast	Residual Forecast
May	27.367.19 MW	25.739.00 MW
June	32.385.11 MW	31.097.49 MW
July	32.825.47 MW	31.362.20 MW

Sources: ESKOM

4.3.1.10. Table 24 reveals that electricity demand reaches its highest peak during winter months as a result of an increase in the demand for space heating. In April 2023, stage 6 load-shedding was implemented due to a shortage of generation capacity and it will continue until further notice. During April 2023, breakdowns were at 17 481MW of generating capacity, while the amount of generating capacity that was out of service for planned maintenance was 5 657MW.

Table 24: Renewable Energy

Current Installed Capacity (MW)	
CSP	500.0
PV	2,287.1
Wind (Eskom+ IPP)	3,442.6
Total (Including other renewable energies)	6,280.2

Sources: ESKOM

4.3.1.11. According to Eskom (2023), in 2022, Concentrated Solar Power (CSP) contributed 1,448,276 MWh in total energy, PV contributed 4,844,736 MWh, Wind (Eskom and IPP) contributed 9,692,373 MWh. In 2023, CSP was 915,623 MWh and lastly, wind (Eskom and IPP) to contribute 2,985,031 MWh.

4.3.1.12. There has been an uptake of distributed generation after the removal of the licensing threshold, as per the amended Schedule 2 of the Electricity Regulation Act (ERA). Table 25 below shows the capacities registered between 2020 and 2023, to date.

Table 25: Capacities registered between 2020 and 2023 as per amended Schedule 2 of the ERA

Capacity	2020	2021	2022	2023
< 1MW	54,02	103,89	107,76	35,06
> 1MW	-	30,51	1547,04	2755,67
Total	54,02	134,41	1654,80	2790,73

4.3.1.13. The court has found the current methodology, Benchmarks and Municipal Guidelines, used for the setting of municipal tariffs to be irrational and unlawful. NERSA has been given a reprieve of 12 months to use them for the 2023/24 FY. However, from the 2024/25 FY, NERSA must develop a new methodology that makes use of the cost of supply to set tariffs for municipalities.

4.3.1.14. NERSA is busy with the development of a new pricing methodology, namely the Electricity Price Determination Methodology, to be used for the electricity industry to ensure cost reflectivity and the transparency of tariffs.

4.3.1.15. NERSA is currently processing the National Transmission Company of South Africa (NTCSA) for the unbundled Transmission company. The NTCSA is; it is a separate legal entity division wholly owned by Eskom Holdings, established through the Department of Public Enterprise (DPE) Roadmap in terms of the Reformed Electricity Supply Industry issued on 29 October 2019, including government objectives. On 28 September 2022, NERSA received the following three licence applications from NTCSA in line with Section 7(1)(a) of the ERA:

- trading licence;
- transmission licence and
- import and export licence.

4.3.2. Oil and Natural Gas in South Africa

4.3.2.1. In December of 2022, a fuel tanker carrying around 60 000 litres of liquefied petroleum gas (LPG) got stuck under a low-lying bridge in Boksburg, east of Johannesburg, and exploded. The explosion claimed 41 lives and injured around 321 people, substantially damaging the emergency and X-ray units of the Tambo Memorial Hospital, located about 100 meters away. Several houses and vehicles were also damaged. Among the dead were 10 hospital staff members, 4 children and residents from the surrounding area who had gathered to see the burning truck. It is believed that the truck was transporting LPG used in homes and industries for heating and cooking to Botswana from South Africa's Indian Ocean port of Richards Bay. The driver was found not to have been negligent, as he had been following directions to the highway, however, the bridge had not been visibly marked. This incident calls for increased compliance measures on safety standards in terms of handling and/or transporting oil and gas products.

4.3.2.2. As at the end of February 2023, the cumulative slate amounted to a negative balance of R2.43 billion for petrol and diesel. In line with the provisions of the Self- Adjusting Slate Levy Mechanism, there has been a decrease of 4.38 c/l in the slate levy, which will be implemented into the price structures of petrol and diesel, with effect from 5 April 2023. The slate levy applicable will be 17.54 c/l.

4.3.2.3. Based on current local and international factors, the fuel prices in Gauteng (Zone 9C) for April 2023 will be adjusted as follows:

- Petrol (93 ULP & LRP): One cent per litre (1.00 c/l) decrease.
- Petrol (95 ULP & LRP): Two cents per litre (2.00 c/l) increase.
- Diesel (0.05% sulphur): Seventy-three point five eight cents per litre (73.58 c/l) decrease.
- Diesel (0.005% sulphur): Seventy-four point five eight cents per litre (74.58 c/l) decrease.

- Illuminating Paraffin (wholesale): One hundred and twenty-four point nine cents per litre (124.90 c/l) decrease.
- SMNRP for IP: Two hundred and one cents per litre (201.00 c/l) decrease.
- Maximum LP Gas Retail Price: Ninety-two cents per kilogram (92.00 c/kg) decrease.

4.3.2.4. The Astron refinery was shut down following an explosion on 2 July 2020 that resulted in two fatalities of employees leading to an unplanned shutdown (Reuters, 2023). However, in the first quarter of 2023, the refinery was set to be undergoing final touches before reopening, which means the initial plan of reopening by the end of Quarter 4 of 2022 was not successful. The reopening of this refinery would provide a boost to the South African petroleum refining space against the backdrop of the closure of several other refineries.

4.3.2.5. In terms of Oil and Gas, the Upstream Petroleum Resources Development Bill (UPRD) was tabled to Parliament; and iGas, a subsidiary of the Central Energy Fund, has acquired an additional 40% ownership of the ROMPCO pipeline. South Africa and Mozambique jointly own 80% of the pipeline, with Sasol owning the remaining 20%. This is of benefit to South Africa in its gas industrialisation strategy to support the economy. CGS has confirmed the verification of the shale gas samples that were tested internationally. We are now awaiting environmental assessment approval before the next phase commences.

4.3.2.6. A gas master plan, yet to be presented to Cabinet, is at an advanced stage of development following the publication of the Base Case Report late last year. The plan will consolidate all this work. Notably, oil and gas projects continue to be under threat from well-funded lobby groups, which also misinform unsuspecting communities. Since February, there has therefore been consultations with some of the Traditional Councils in the Eastern Cape to help them understand the possible benefits of the upstream petroleum industry to their communities and the South African economy in general. More traditional leaders and communities will be engaged in future.

- 4.3.2.7. The Gas-To-Liquids (GTL) Refinery is located in Mossel Bay, Cape Town and has a capacity of about 45 000 b/d. The Refinery processes both the gas and condensates to produce liquid fuels and chemicals. It was commissioned in 1992 as the world's first GTL refinery and remains the third-largest GTL refinery among the five now operating worldwide. This refinery has been offline since 2020 due to shortages of feedstock. However, a long-term feedstock solution is under development, which is expected to provide feedstock to enable full production capacity. It is anticipated that the refinery will be brought back online by 2027/28 at the latest (Creamer Media, 2023).
- 4.3.2.8. LNG is a cleaner and more affordable source of energy. It is available now to power various industries, enabling a transition towards a net-zero emitting future. The gas- to-power business is important for South Africa and the whole region, as the Southern African Development Community (SADC) is battling energy shortages, which have crippled the operations of economies. While bold steps have been taken to include renewable energy into the regional energy mix, natural gas provides baseload capacity, as well as flexibility, thereby giving investors certainty in terms of energy security and reliable supply.
- 4.3.2.9. In South Africa, current efforts to stop capacity shortages are not enough to yield an adequate reserve margin, as the decommissioning of some of the old coal-fired power stations has begun. Technologies that could be deployed quickly and safely are required. Gas-powered plants can be built far more quickly than other baseload plants, such as coal or nuclear.
- 4.3.2.10. Preparations for bunkering operations in Coega have begun, and the first LNG molecule was earmarked to land in September 2021, at a 38-million gigajoule (GJ) annual capacity. DNG is planning to utilise bunkering vessels to receive, store, transfer and regasify LNG. They are in the process of acquiring these vessels while considering building their own in future. However, the process requires various regulatory permits from regulatory authorities, such as the Transnet National Ports Authority (TNPA), NERSA, the South African Maritime Safety Authority (SAMSA), the Department of Environmental Affairs, the Department of Fisheries and Forestry (DEFF), the Department of Human Settlements and Water Affairs (DHSWA) and the Department of Transport. In addition, collaborative efforts among regulators are required to achieve energy development or advancement in terms of resources or sources and security.
- 4.3.2.11. Bunkering operations will include a floating storage unit (FSU), which will be moored at the Port of Coega. This vessel will receive LNG from carriers coming from their Nigerian suppliers. Small-scale LNG carriers will shuttle between the FSU and ships being bunkered, performing ship-to-ship transfers. An important point is that the bunkering operation is driven by the delivery of LNG to on-shore customers in the region, where the small-scale LNG carriers will be utilised. As these vessels are in a position to be fully utilised on other projects, they are readily available to deliver bunkers as and when required.
- 4.3.2.12. The DNG LNG investment project is estimated at USD150-million. DNG will be able to deliver affordable LNG since the required infrastructure will be in place for other projects and therefore the costs for delivering a first-class bunkering service will mostly be attributed to those projects' costs. While South Africa's capacity shortages tend to be mostly unmanageable at peak, it is common knowledge that when the electricity system is under pressure due to plant breakdowns, peak power plants (pumped storage and diesel plants) are utilised outside of peak at an excruciatingly high cost to the country.
- 4.3.2.13. The inclusion of LNG in South Africa's energy mix should bring some relief to many customers, notably to those who can be supplied from a source, not on the grid. LNG will become the main contributor to an efficient and cost-effective deliverable, while other deliverables can be described as too costly, too dirty, or too small to be of a major significance. LNG will be available for all industries, gas-to-power (large and small customers), industrial processes, logistics and mining activities in South Africa. Gas-to-power will become a major contributor to the national grid.

- 4.3.2.14. LNG has become the accepted fuel for vessels since the 2020 requirement of 0.05% emissions of sulphur. Many new ships are equipped with dual-fuel main engines. There are currently 385 vessels that use LNG, and a further 185 new builds are on order. This number will grow. Currently, 3 MTPA of LNG is being consumed. By 2025, this volume is expected to grow to between 8 and 10 MTPA and by 2040, the volume is predicted to reach between 35 and 40 MTPA. As can be seen, the marine sector is becoming an important sector in LNG.
- 4.3.2.15. DNG is positioning itself to be an important part of the bunker sector by offering bunkers to all LNG burners, being placed in the middle of the main world trade routes, east and west. By offering bunkers in this important position, ship owners will be able to carry more cargo and reduce the number of bunkers to be carried per long-haul passages, creating more profitable voyages.
- 4.3.2.16. As indicated by the Minister of Higher Education, Science and Innovation during the launching ceremony of the hydrogen society roadmap on 17 February 2022, South Africa remains a promisingly significant contributor to global hydrogen output. South Africa currently produces 2% of the global demand for hydrogen, mostly made from natural gas by Sasol. Given the projected global demand for green hydrogen, South Africa has the opportunity to convert its current global supply to green hydrogen and the potential to increase the country's share of the green hydrogen market. Projects such as the Platinum Valley, the Limpopo Science and Technology Park, and the CoalCO₂-X Research Development and Innovation (RDI) Programme are all significant drives in the economy to increase South Africa's capacity to produce hydrogen.

4.4. Macroeconomic Outlook in South Africa

- 4.4.1. **During the 2023 budget speech, the Finance Minister allocated R254bn over the medium term to assist Eskom to settle or pay its debts. However, this will also improve its Balance Sheet in order to be in a position to go to the market to borrow funds for maintenance and other operational issues that will result into the improvement of the grid.** Key macroeconomic developments in the fourth quarter include the major budget announcements of 22 February 2023. The government debt trajectory is expected to peak at 74% in 2025/26, given Eskom's need to borrow to cater for its financial needs as the country faces acute electricity shortages. Load-shedding has persisted as the country's energy crisis deepens.
- 4.4.2. Statistics South Africa has revealed that CPI stood at 5.5% in November 2023, down from 5.9% in October 2023. After declining significantly in the third quarter of 2023, the average headline CPI inflation expectations for 2024 and 2025 of analysts, business people and trade unions increased again in the fourth quarter. Unchanged at 6.1% for 2023, respondents expect inflation to be 5.7% in 2024 and 5.6% in 2025. Similarly, the forecast of five-year inflation expectations ticked up from 5.1% to 5.2%, largely driven by trade union officials raising their forecast. Both the one- and five-year-ahead inflation expectations of households edged up marginally in the fourth quarter, to 7.2% and 10.2% respectively. The survey respondents expect GDP growth to be 0.9% this year, then accelerate to 1.3% next year (compared to 0.8% and 1.4% before). The three social groups are in general agreement that salaries and wages will increase by around 5% in 2023 and in 2024.

- 4.4.3. Manufacturing production has declined by 4.7% year on year, as last noted in December, after a decrease of 1.8% in November 2022. A performance analysis of the manufacturing sector (one of the key sectors to economic growth) includes the biggest notable drag on growth emanating from petrochemicals (-12%), iron, metal products and machinery (-7%) and food and beverages (-4.4%). The productivity of the mining sector declined by 3.5% year on year in December 2022. Output has now declined consecutively for 11 months and as such, production decreased by 7.2% in 2022 following a post-lockdown rebound of 11.6% in 2021. Still, total mineral sales were 2.8% higher in 2022 compared to 2021, on the back of higher commodity prices. Unemployment was estimated at 32.7% as of March 2023.
- 4.4.4. Given the weak outlook for job growth in 2023/24, the rising cost of living, the elevated interest rates, and the decline in buying power in 2022, it is not unexpected that South African consumers are despondent about their personal financial outlook. While employment increased significantly in 2022, the outlook for job creation so far in 2023 is a lot more conservative due to multiple economic headwinds. At the same time, the buying power of salaried workers is rapidly declining due to lower take-home pay as a result of elevated inflation. This decline in buying power is also stoking fears about social risk.
- 4.4.5. There is no end in sight to Russia's war in Ukraine and recession risks remain elevated in the Euro Area, even though energy constraints have eased. Growth prospects for the United States in 2023 are lower. The growth outlook for China has improved, but is likely to remain modest by historical standards. In the developing world, several economies face debt distress, which has been exacerbated by tighter global financial conditions. Taking these and other factors into account, the South African Reserve Bank's (SARB's) forecast for global growth in 2023 is revised slightly lower to 1.6% (from 1.9%), as expectations point to better growth in 2024 at 2.6% (up from 2.4%).

5. Internal Situational Analysis

5.1. Introduction

- 5.1.1. NERSA's operating environment was analysed using the SWOT (Strengths, Weaknesses, Opportunities and Threats) and PERSTEL (Political, Economic, Regulatory, Social, Technology, Environment and Legal) factor analysis. The outcome of the analysis is presented as follows:
- 5.1.1.1. The identified external factors (threats and opportunities) are linked to the relevant PERSTEL factors and enhancing and mitigating strategies have been proposed.
- 5.1.1.2. The identified internal factors (strengths and weaknesses) are listed, with proposed enhancing and mitigating strategies.
- 5.1.2. As part of the analysis of the operating environment, key challenges were identified that could have an impact on NERSA's ability to achieve some of the planned outputs in the Annual Performance Plan for 2024/25 – 2026/27. Initiatives are proposed to mitigate the challenges and key issues that must be addressed when implementing the proposed initiatives are stated, where applicable.
- 5.1.3. The last element of the internal situational analysis was the identification of key priorities to focus on in the new planning period for the new planning period (2024/25 – 2026/27).

5.2. Threats and opportunities in Operating Environment

Opportunities	Enhancing Strategies	Perstel Factor	Impact if factor is not address
1. Restructuring of the ESI	a) Regulate according to existing legislation	Political	1.2.1. How will we deal with licensees who do not cooperate or who do not have the technology for such a portal. (This could be identified as a risk for such a project and be dealt with accordingly)
2. Amendment of all enabling legislation and policies	a) Provide inputs to Draft Amendment Bills and Policies b) Regulatory and policy advocacy in line with the approved Regulatory Stakeholder Engagement and Policy Advocacy Strategy	Legislation	None
3. Introduction of new technologies in the regulated industries	a) Regulatory advocacy to influence political will towards new technologies b) Sensitise stakeholders and policy makers on the benefits of new technologies in the regulated industries c) Fast tracking of amendment of legislation d) NERSA to be adequately capacitated (enhance current skills) and resources (human and financial) to deal with the complexities resulting from the new technologies	Technology	3.1.1. How will external people access this system?
4. Defragmentation of regulation policies	a) Fast track regulatory and policy advocacy in line with the approved Stakeholder Engagement and Advocacy Strategy	Regulatory	• Impact on effectiveness of regulation. • Regulatory burden due to the concurrent jurisdiction, as well as overlapping • Functions/roles between NERSA, relevant government departments and other • Regulatory authorities. • Regulatory uncertainty. • Risk of legal challenges
5. Collaboration and partnerships with national, regional and international agencies	a) Ensure approvals of partnership collaborations at national, regional and international levels. b) Implementation and monitoring of engagement strategy (capacity building, development/enhancement of regulatory tools, participation in specific projects). c) Ensure and monitor compliance with DMRE's guidelines and Government's positions.	Regulatory Legal Political Social	• Loss of opportunities for skills development, which will have an impact on effective regulation. • Loss of exposure to practical national, regional and international best practices to enhance effective regulation • Loss of opportunity to contribute towards regional projects • NERSA's relevancy could be compromised

Threats	Mitigating Strategies	Perstel Factor	Impact If Factor Is Not Addressed
1. Continued perception of the deregulation of regulated industries	a) Improved regulatory advocacy to differentiate between deregulation and unbundling of the ESI. b) Intensify research on regulatory issues that fall within NERSA's mandate and publish papers on NERSA's website c) Intensify stakeholder engagement with a focus on value creation to emphasise the value of regulation of the industries and the role of NERSA d) Address the above in the strategy to profile NERSA appropriately – also highlight the impact of deregulation on the socio-economy of SA e) Assess the reasons for the call for deregulation – did NERSA contribute to this call through the way NERSA regulates – reduce the perception that NERSA is a bottleneck/red tape. f) Improve process and frequency of conducting regulatory impact assessments. g) Clarify NERSA's role in regulating the energy industry	Regulatory Legal	• Job losses • Exploitation of consumers, interested and affected parties • Impact on NERSA's role
2. Successful legal challenges of NERSA's	a) Independent peer review of all regulatory tools to ensure that they are aligned with our mandate b) Undertake an analysis of the reasons why legal challenges were successful c) Improvement of quality of reasons for decisions (time management and quality assurance) d) Improve document management linked to regulatory decisions	Legal	• Regulatory uncertainty
3. Vandalism and theft of infrastructure in regulated industries	a) Collaboration with law enforcement agencies and community structures b) Establishment of stakeholder forums (Regulator, law enforcement, Licensees and community) to discuss mitigation measures and strategies to the challenges	Legal Social Economic	• Negative impact on security of supply - Raises the cost of energy due to the repairs of the damages and theft – this is passed through to the consumer - Poses safety risk to the environment and the surroundings

Threats	Mitigating Strategies	Perstel Factor	Impact If Factor Is Not Addressed
4. Security of supply – poor maintenance of infrastructure/ system, gas supply from other countries, refineries closures	a) Increased engagement with demand side of the ESI b) Develop methodologies to encourage alternative supply of gas c) Regulatory advocacy to encourage government to secure gas supplies from other countries and to address the closures of the refineries d) Develop performance standards for licensees in the three regulated energy industries	Political Regulator Economic	• Inadequate import infrastructure • Increase in prices • Continued lack of competition in markets where competition is possible • Collapse of the grid • Inadequate investment in Energy infrastructure
5. Developments in the international markets, which have a direct impact on SA prices and may have unintended consequences – i.e. impact of war between Russia and Ukraine, international production constraints, etc.	a) Delink the NERSA's methodologies to international prices b) Where licensees purchase LNG from an international market at an 'arm's length' transaction, NERSA must implement methodologies that ensure that the licensees fully recover all their costs c) Emulating prices that would persist in competitive markets d) Facilitating the establishment of conditions for effective competition in the relevant trading markets in the gas industry. e) Conduct regulatory advocacy and utilise methodologies that will encourage the development of national	Economic Political	• Increase in prices. • Shortage of energy products. • Increase in OCGT prices (due to increase of diesel prices) • Inadequate investment in energy infrastructure
6. Natural disasters and pandemics – impact on infrastructure, operations and affordability of energy	a) Annual review and testing of business continuity plans b) Consider the impact of Natural disasters and pandemics on regulatory activities c) Adapt regulatory processes to ensure that the regulated industries can develop as normally as possible	Environmental Economic Political	• Loss of security of supply energy • Investments in infrastructure impacted because projects are placed on hold • Operations of regulator and licensees negatively affected • Increase in cost of energy

Threats	Mitigating Strategies	Perstel Factor	Impact If Factor Is Not Addressed
7. Abuse of dominance by the monopolies that NERSA regulates	a) Regulate in a manner that incentivises investment b) Approve competitive pricing and tariffs c) Enforcing third-party access	Regulatory	- Lack of competition in markets where competition is possible - Excessive prices - Exclusionary conduct
8. Continuous load-shedding	a) Development of service quality incentives (SQI) related to the energy availability factor (EAF) b) Timeously develop regulatory tools to facilitate the implementation of Government's interventions to alleviate load-shedding i.e. net billing, demand response programmes and participation by private generators c) Review demand response tariff to incentivise participation d) Improve licensing and registration processes e) Develop a regulatory tool that will allow the entry of new players, especially for generation	Regulatory Economic	Increasing energy prices
9. Decline in volumes and more registrations than licences	a) Monitor implementation of the approved financial sustainability strategy and plan	Economic	Decline in NERSA revenue

5.3. Analysis of Strengths and Weaknesses

Strengths	Enhancing Strategies
1. Skilled workforce	1.1. Continuous enhancement of leadership and management skills 1.2. Review Dual Career Path Policy 1.3. Review Secondment Policy (externally and internally)
2. Diversity of staff (skills, gender)	2.1. Continuous improvement of recruitment processes to address this matter
3. Professionalism	3.1. Continuous improvement of NERSA's code of conduct
4. Research based products (policies, methodologies, guidelines, etc.) and services	4.1. Implementation and enforcement 4.2. Improve synergy across all divisions
5. Strong stakeholder engagement	5.1. Continuous improvement of methods for stakeholder engagements 5.2. Improve areas of focus for stakeholder engagements
6. High percentage of staff retention	6.1. Continuous improvement of HR policies, e.g. remuneration policy (market related salary survey) and succession policy 6.2. Implement the formalised organisational culture to ensure a conducive working environment
7. Ability to execute our mandate through effective decision-making	7.1. Ensure that all decisions taken are compliant with approved procedures
8. Transparent processes	8.1. Portal on website to indicate where applications for tariffs/price and licensing/registration
9. Ability to share knowledge and information through NERSA's capacity building programme	9.1. Maintenance of capacity building programme
10. Financial health	10.1. Monitor operating environment 10.2. Monitor and analyse the volumes of activities of the licensees relevant to NERSA's funding
11. Good governance	11.1. Monitor compliance with all relevant legislation and policies
12. Relationships with regulators at national, regional and international level	12.1. Maintain and/or improve cooperation with local and international regulators
13. Ability to adapt to changes that have an impact on our internal operating environment (e.g. reaction to COVID-19 restrictions)	13.1. Continuously adjust policies to deal with changes

Weaknesses	Mitigating Strategies
1. Decision-making processes too slow (turnaround time to deal with applications is too long and causes a bottleneck)	1.1. Vigorous regulatory and stakeholder advocacy on regulatory processes 1.2. Review regulatory rules and processes to improve turn-around-times 1.3. Implementation of the approved to-be business processes 1.4. Automation of approved business processes 1.5. Review delegation matrix to improve efficiency in decision-making.
2. Non-recognition of internal competencies outside of regulatory divisions which are needed to elevate regulatory outputs and deal with challenges	2.1. Develop a framework to guide the formal interaction between regulatory divisions and relevant support services 2.2. Review TBNS Policy and Implementation Plan.
3. Lack of understanding of NERSA's mandate, role and regulatory decision-making processes	3.1. Intensification of stakeholder engagements (external and internal) 3.2. Review of the Communication Strategy 3.3. Customer education programmes 3.4. Communicate regulatory decisions.
4. Operating mostly in manual manner	4.1. Implement the proposed plan for the automation of the approved to-be business processes 4.2. Develop a collaboration portal between NERSA and licensees 4.3. Implement technology to improve data-driven decisions

5.4. Organisational capacity

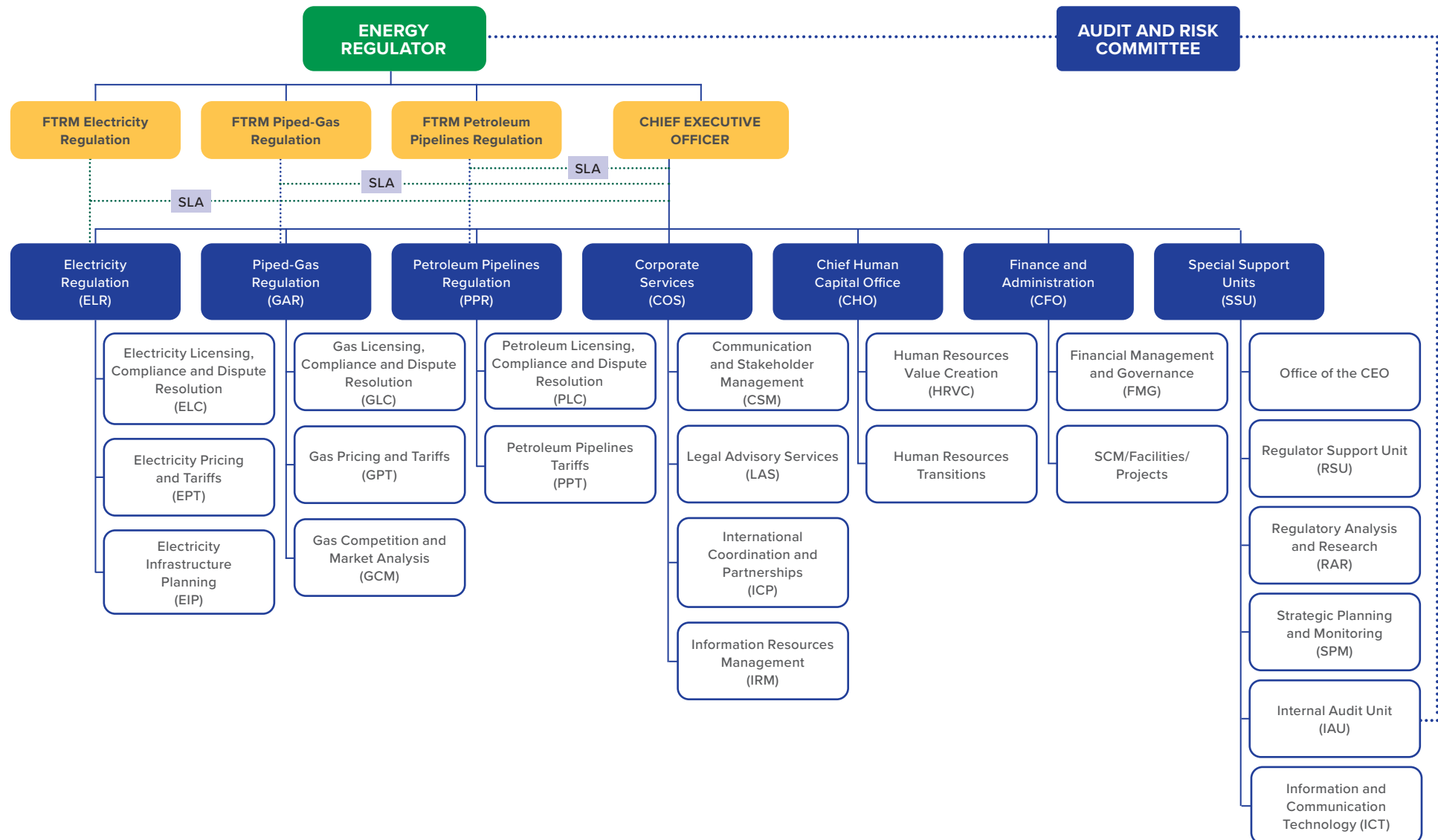
5.4.1. NERSA has an approved structure of 252 staff members. The staff strength is sitting at 237. This consists of 233 permanent employees and four Full-Time Regulator Members (FTRMs). Table 26 below summarises the staff complement of NERSA.

Table 26: NERSA Staff complement

Division	Department	Complement
Electricity Regulation (ELR)	FTRM	3
	Executive	3
	Electricity Pricing and Tariffs (EPT)	35
	Electricity Licensing, Compliance and Dispute Resolution (ELC)	34
	Electricity Infrastructure Planning (EIP)	13
Total		88
Piped-Gas Regulation (GAR)	FTRM	3
	Executive	5
	Gas Pricing and Tariffs (GPT)	8
	Gas Licensing, Compliance and Dispute Resolution (GLC)	11
	Gas, Competition and Market Analysis (GCM)	4
Total		31
Petroleum Pipelines Regulation (PPR)	FTRM	3
	Executive	6
	Petroleum Pipelines Tariffs (PPT)	9
	Petroleum Licensing, Compliance and Dispute Resolution (PLC)	9
Total		27

Division	Department	Complement
Finance and Administration (CFO)	Executive	3
	Financial Management and Governance (FMG)	7
	Supply Chain Management	12
Total		22
Human Resources (CHO)	Executive	2
	Human Resources – Value Creation	8
	Human Resources -Transactions	3
Total		13
Corporate Services (COS)	Executive	3
	Legal Advisory Services (LAS)	6
	Communication and Stakeholder Management (CSM)	9
	International Co-ordination and Partnerships (ICP)	3
	Information Resources Management (IRM)	7
Total		28
Specialised Support Units (SSU)	Internal Audit (IAU)	7
	Strategic Planning and Monitoring (SPM)	4
	Regulator Support (RSU)	12
	CEO's Office Operations (COO)	4
	Regulatory Analysis and Research (RAR)	6
	Information and Communication Technology (ICT)	10
Total		43
Grand Total NERSA Staff Complement		252

5.4.2. Below is the approved NERSA Organisational Structure:



5.5. Status regarding compliance with the B-BBEE Act

For 2021/22, NERSA was classified as a compliant entity under the Broad-Based Black Economic Empowerment, 2003 (Act No. 53 of 2003) (B-BBEE Act), having been certified a Level eight (8) B-BBEE contributor, valid until 30 March 2023. This was due to the fact that the priority element of skills development did not meet the minimum threshold.

NERSA has successful learning programmes that include learnerships and internships, as well as training programmes. In the next financial year, NERSA will seek to address this outcome by implementing appropriate interventions, and to improve the participation of black unemployed people in training. A significant improvement has been noted in the Enterprise and Supplier Development element. The Enterprise Development Strategy, approved by the Energy Regulator, is being implemented to improve procurement from designated groups.

PART C: MEASURING OUR PERFORMANCE

6. Institutional Programme Performance Information

6.1. Introduction

6.1.1. As stated in the Strategic Plan, NERSA's overall impact statement is as follows:

Secure, reliable, affordable, sustainable, competitive and transformed energy industry, which contributes to the economic growth of South Africa.

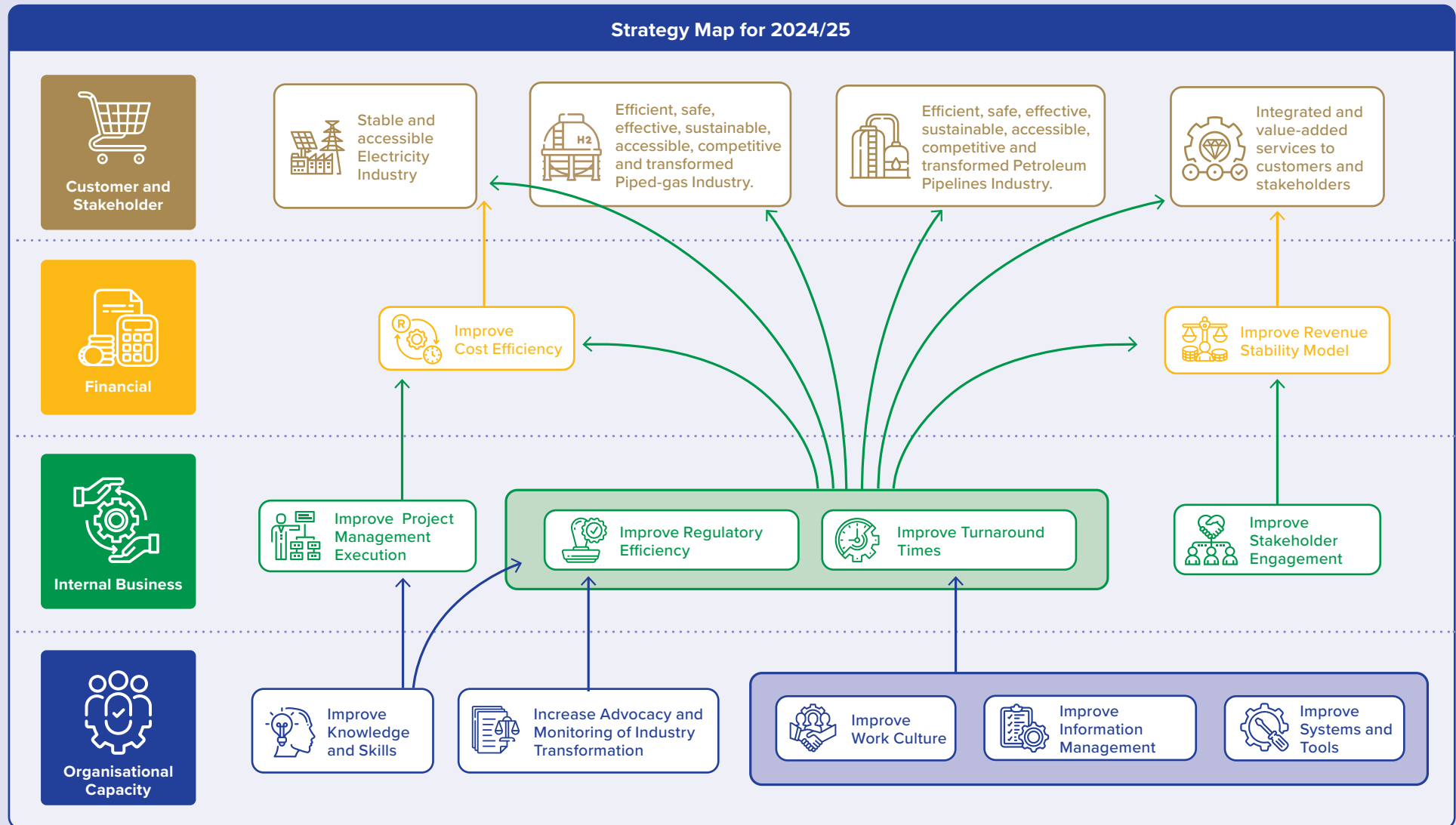
6.1.2. NERSA adopted the following strategic objectives:

- a) Facilitate entry, setting prices and resolving disputes through efficient regulatory tools on an annual basis
- b) Enable a stable and diverse energy sector system and pricing regime which supports access through regulatory services that are delivered on time (annually) and to quality standards
- c) Review regulatory tools periodically to ensure that they are innovative in response to the transformation of the energy industry
- d) Review the Regulatory Framework periodically to ensure that it is relevant for effective regulation, for the benefit of customers and stakeholders, through regulatory advocacy and stakeholder engagement
- e) Enable the provision of integrated and value-added services in an equitable/a fair manner to customers.

6.1.3. The table below indicates the link between NERSA's programmes, the impact statements and outcomes stated in the Strategic Plan, as well as Government's priorities.

Impact Statements	Outcomes	Programme	Link to MTSF Priority
- A stable and accessible Electricity Industry that supports an improved quality of life and economic activity - Efficient, safe, effective, sustainable, accessible, competitive and transformed piped-gas industry - Efficient, safe, effective, sustainable, competitive and transformed petroleum pipelines industry	- Efficiency in facilitating entry, setting prices and resolving disputes - A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards - Innovation drives our response to the transition of the Industry	Programme 1: Regulatory Service Delivery	MTSF Priority 2: Economic Transformation and Job Creation.
NERSA established and perceived as an efficient, effective and credible regulator	- Energy industry regulatory framework is relevant for effective regulation, for the benefit of the customers and stakeholders. - Integrated and value-added services to customers	Programme 2: Advocacy and Engagement	MTSF Priority 2: Economic Transformation and Job Creation MTSF Priority 2: A better Africa and world.
NERSA established and perceived as an efficient, effective and credible regulator	Innovation drives our response to the transition of the Industry	Programme 3: Innovation	MTSF Priority 1: Capable, Ethical and Developmental State
NERSA established and perceived as an efficient, effective and credible regulator	Integrated and value-added services to customers	Programme 4: Operational Efficiency and Quality Management	MTSF Priority 1: Capable, Ethical and Developmental State MTSF Priority 3: Education, skills and health
- Efficiency in facilitating entry, setting prices and resolving disputes - A stable and diverse energy sector system and pricing regime which supports access through regulatory services that are delivered on time and to quality standards - Innovation drives our response to the transition of the Industry	Integrated and value-added services to customers	Programme 5: People and Organisational Culture	MTSF Priority 3: Education, Skills And Health.

6.1.4. Below is NERSA's Strategy Map, which forms the basis for this Annual Performance Plan



6.2. Planned Performance

6.2.1. Programme 1: Regulatory Service Delivery

a) The programme's purpose is to:

- set and/or approve tariffs and prices in order to ensure a fair balance between the needs of the customer and the regulated entity;
- ensure the orderly development of the energy industry and to ensure that all activities related to all operations are licensed and registered as required by the Electricity Regulation Act, 2006 (Act No. 4 of 2006), Gas Act, 2001 (Act No. 48 of 2001) and the Petroleum Pipelines Act, 2003 (Act No. 60 of 2003);
- ensure that all licensees in the three regulated industries fully comply with their licence conditions, including those relating to health, safety, security and environmental standards and requirements, as well as any other standards and requirements prescribed by the relevant industry-specific legislation;
- ensure compliance with directives to govern relations between a licensee and its end users;
- ensure that disputes and complaints between licensees or between licensees and customers or end-users are managed effectively and settled in a manner that is appropriate; and that when needed, any mediation or arbitration required will be done within prescribed procedures; and
- ensure the setting of appropriate rules, guidelines and codes of best practices in the quest to promote uniformity and standardise practices in the regulation of the three energy industries.

b) This programme will therefore contribute to the following:

- A fair balance between the needs of the customer and the regulated entity. While the customer must be protected against misuse of monopolistic powers and unnecessary price hikes, the regulated entities must have sufficient income to ensure that they can continue operating as a going concern and have enough revenue for the maintenance and refurbishment of infrastructure.
- The creation of investor confidence and lessening the regulatory burden on licensees. In order to achieve orderly investor confidence in the energy industries, there must be standardised practices that are the same for all participants, and NERSA must maintain and safeguard these standards. This will facilitate investment in the energy industries, as investors and developers need a sound regulatory framework to ensure that they receive the expected returns for their investment.

6.2.1.1. Sub-Programme: Electricity Industry Regulation

The purpose of this subprogramme is to carry out the following functions for the Electricity industry, Setting and/or approval of tariffs and prices

1. Licensing and registration
2. Compliance monitoring and enforcement
3. Dispute resolution, including mediation, arbitration and handling of complaints
4. Setting of rules, guidelines and codes for regulation.

NERSA embarked on a process to develop a strategy for the regulation of the Electricity Industry. This process will be continued for the regulation of the Piped-Gas Industry and Petroleum Pipeline Industry. Therefore, the planned performance for this sub- programme looks different from the sub-programmes for the other two regulated industries.

i. Setting and/or approval of tariffs and prices

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS							
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD			
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	
MTSF Priority 2: Economic Transformation and Job Creation										
Efficiency in facilitating entry, setting prices and resolving disputes	1. Electricity pricing for Eskom	Energy Regulator decision on Eskom and municipal electricity prices within the stated timeframe	New target	New target	Regulator decision by 28 February 2023	Regulator decision by 28 February 2024	Regulator decision by 28 February 2025	Regulator decision by 28 February 2026	Regulator decision by 28 February 2027	
	2. Electricity pricing for municipalities	Energy Regulator decision on municipal electricity prices within the stated time frame	New target	New target	Regulator decision by 28 February 2023	Regulator decision by 28 February 2024	Regulator decision by 15 March 2025	Regulator decision by 15 March 2026	Regulator decision by 15 March 2027	
	3. Improved pricing system for the electricity industry	Energy Regulator decision on the improved pricing system for the electricity industry within the stated time frame	New target	New target	New target	New target	Regulator decision by 31 March	-	-	

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS							
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD			
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	
MTSF Priority 2: Economic Transformation and Job Creation										
Efficiency in facilitating entry, setting prices and resolving disputes	4. One progress report on the implementation of the Regulatory Reporting Manuals regarding the Standard Chart of Accounts (SCOA) for municipalities	Number of reports on the implementation of the Regulatory Reporting Manuals regarding the Standard Chart of Accounts (SCOA) for municipalities considered by the relevant committee or the Energy Regulator within the stated time frame	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March

b) *Indicators, Annual and Quarterly Targets*

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. Electricity pricing for Eskom	Energy Regulator decision on Eskom's electricity prices within the stated time frame	Regulator decision by 28 February 2025	-	-	-	Regulator decision by 28 February 2025
2. Electricity pricing for municipalities	Energy Regulator decision on municipal electricity prices within the stated time frame	Regulator decision by 15 March 2025	-	-	-	Regulator decision by 15 March 2025
3. Improved pricing system for the electricity industry	Energy Regulator decision on the improved pricing system for the electricity industry within the stated time frame	Regulator decision by 31 March 2025	-	-	-	Regulator decision by 15 March 2025
4. One progress report on the implementation of the Regulatory Reporting Manuals regarding the Standard Chart of Accounts (SCOA) for municipalities	Number of reports on the implementation of the Regulatory Reporting Manuals regarding the Standard Chart of Accounts (SCOA) for municipalities considered by the relevant committee or the Energy Regulator within the stated time frame	1 considered annually by the REC by 31 March	-	-	-	1 considered annually by the REC by 31 March

ii. Licensing and Registration

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
Efficiency in facilitating entry, setting prices and resolving disputes	1. Improved licensing processes to facilitate entry	Energy Regulator decision on improved licencing processes within stated time frame	New target	New target	New target	New target	Regulator decision by 30 December 2024	-	-
		Percentage of complete licence applications considered by the relevant subcommittee or the Energy Regulator within the stated time frame	New target	New target	New target	120 working days	90% within 110 working days	90% within 110 working days	90% within 100 working days
	2. Improved registration processes to facilitate entry	Percentage of complete applications for the registration of electricity generation facilities considered by the relevant subcommittee or the Energy Regulator within the stated time frame	New target	New target	94% within 45 working days	45 working days	90% within 45 working days	90% within 40 working days	90% within 40 working days

b) *Indicators, Annual and Quarterly Targets*

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. Improved licencing processes to facilitate entry	Energy Regulator decision on improved licencing processes within stated time frame	Regulator decision by 30 December 2024	-	-	Regulator decision by 30 December 2024	-
	Percentage of complete licence applications considered by the relevant subcommittee or the Energy Regulator within the stated time frame	90% within 110 working days	90% within 110 working days	90% within 110 working days	90% within 110 working days	90% within 110 working days
2. Improved registration processes to facilitate entry	Percentage of complete applications for the registration of electricity generation facilities considered by the relevant subcommittee or the Energy Regulator within the stated time frame	90% within 45 working days	90% within 45 working days	90% within 45 working days	90% within 45 working days	90% within 45 working days

iii. Compliance Monitoring and Enforcement

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS							
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD			
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	
MTSF Priority 2: Economic Transformation and Job Creation										
Efficiency in facilitating entry, setting prices and resolving disputes	1.	Improved compliance audit processes and standards	Energy Regulator decision on improved compliance audit processes within the stated time frame	New target	New target	New target	New target	Regulator decision by 31 March 2025	-	-
	2.	Compliance audit plans executed with quarterly reports in place	Percentage variance of planned versus actual compliance audit plans executed	New target	New target	80%	80%	80%	80%	80%
			Number of analysis reports on audits conducted within the stated time frame	New target	New target	New target	4	4	4	4
	3.	Enforcement plan in place and executed, with non-compliance findings compiled quarterly	Percentage variance of planned versus actual enforcement plan executed	New target	New target	80%	80%	80%	80%	80%
			Number of reports on non-compliance findings compiled within the stated time frame	New target	New target	New target	4	4	4	4

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
Efficiency in facilitating entry, setting prices and resolving disputes	4. One audit report on the review of the annual performance of IDM to be considered by the ELS/REC/ER	Number of reports on audit findings on the review of the annual performance of IDM to be considered by the relevant subcommittee or the Energy Regulator within the stated time frame	New target	New target	New target	New target	1 within 6 months after receipt of Eskom’s IDM Annual Performance Report	1 within 6 months after receipt of Eskom’s IDM Annual Performance Report	1 within 6 months after receipt of Eskom’s IDM Annual Performance Report
	5. One audit report on the Transmission Network Development projects for compliance with the South African Grid Code considered by the ELS/REC/ER	One report on audit findings on the Transmission Network Development projects for compliance with the South African Grid Code considered annually by the relevant subcommittee or the Energy Regulator within the stated time frame	New target	New target	New target	New target	1 by 31 March	1 by 31 March	1 by 31 March

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
Efficiency in facilitating entry, setting prices and resolving disputes	6. One audit report on the Distribution Network Development projects for compliance with the South African Grid Code considered by the ELS/REC	Number of reports on the Distribution Network Development projects for compliance with the South African Grid Code considered annually by the relevant subcommittee or the Energy Regulator within the stated time frame, subject to all information being available	New target	New target	New target	New target	1 by 31 March	1 by 31 March	1 by 31 March
	7. Two monitoring reports on the performance and progress of Renewable Energy projects considered by the ELS/REC	Two monitoring reports on the performance and progress of Renewable Energy projects for consideration bi-annually by the relevant subcommittee or the Energy Regulator within the stated time frame, subject to all information being available	New target	New target	New target	New target	2 by 30 September and 31 March respectively	2 by 30 September and 31 March respectively	2 by 30 September and 31 March respectively

b) Indicators, Annual and Quarterly Targets

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. Improved compliance audit processes and standards	Energy Regulator decision on improved compliance audit processes within the stated time frame	Regulator decision by 31 March	-	-	-	Regulator decision by 31 March 2025
2. Compliance audit plans executed with quarterly reports in place	Percentage variance of planned versus actual compliance audit plans executed	80%	10%	35%	35%	Audit plan for 2025/26
	Number of analysis reports on audits conducted within the stated time frame	4	1	1	1	1
3. Enforcement plan in place and executed, with non- compliance findings compiled quarterly	Percentage variance of planned versus actual enforcement plan executed	80%	Enforcement plan for 2024/25	10%	30%	40%
	Number of reports on non-compliance findings compiled within the stated time frame	4	1	1	1	1
4. One audit report on the review of the annual performance of IDM to be considered by the ELS/ REC/ER	Number of reports on audit findings on the review of the annual performance of IDM to be considered by the relevant subcommittee or the Energy Regulator within the stated time frame	1 within 6 months after receipt of Eskom's IDM Annual Performance Report	-	-	-	1 within 6 months after receipt of Eskom's IDM Annual Performance Report
5. One audit report on the Transmission Network Development projects for compliance with the South African Grid Code considered by the ELS/ REC/ER	One report on audit findings on the Transmission Network Development projects for compliance with the South African Grid Code considered annually by the relevant subcommittee or the Energy Regulator, within the stated time frame	1 by 31 March	-	-	-	1 by 31 March

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
6. One audit report on the Distribution Network Development projects for compliance with the South African Grid considered by the ELS/REC	Number of reports on the Distribution Network Development projects for compliance with the South African Grid considered annually by the relevant subcommittee or the Energy Regulator, within the stated time frame, subject to all information being available	1 by 31 March	-	-	-	1 by 31 March
7. Two monitoring reports on the performance and progress of Renewable Energy projects considered by the ELS/REC	Two monitoring reports on the performance and progress of Renewable Energy projects for considered bi-annually by the relevant subcommittee or the Energy Regulator within the stated time frame, subject to all information being available	2 by 30 September and 31 March respectively	-	1 by 30 September	-	1 by 31 March

iv. Dispute resolution, including mediation, arbitration and handling of complaints

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
Efficiency in facilitating entry, setting prices and resolving disputes	1. Investigations Framework and Process for complaints and dispute resolution in place and executed	% of categorised disputes/ complaints closed in line with the approved Investigations Framework and Process for complaints and dispute resolution	New target	New target	90%	90%	90%	90%	90%

b) Indicators, Annual and Quarterly Targets

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. Investigations Framework and Process for complaints and dispute resolution in place and executed	% of categorised disputes/ complaints closed in line with the approved Investigations Framework and Process for complaints and dispute resolution	90%	90%	90%	90%	90%

v. Setting of rules, guidelines and codes for regulation

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
Innovation drives our response to the transition of the Industry	1. Review and develop targeted tools and systems for a changing electricity industry	Percentage variance between planned versus actual targeted tools and systems reviewed and developed	New target	New target	80%	80%	80%	80%	80%
	2. Process Draft Section 34 Determinations from the Minister of Electricity (MOE)	100% of draft Section 34 Determinations received, with complete information, considered by the relevant subcommittee or the Energy Regulator within the stated time frame	New target	New target	New target	New target	100% within 4 months	100% within 4 months	100% within 4 months
	3. Complete applications from the ESI <u>requiring exemption</u> from the South African Grid Code, considered by the ELS/REC	Percentage of complete applications requiring exemption from the South African Grid Code considered by the <u>relevant subcommittee</u> or the Energy Regulator within the stated time frame	New target	New target	New target	New target	100% within 3 months	100% within 3 months	100% within 3 months
	4. Complete applications from the ESI <u>requiring amendment</u> to the South African Grid Code, considered by the ELS/REC	Percentage of complete applications <u>requiring amendment</u> to the South African Grid Code considered by the relevant subcommittee or the Energy Regulator within the stated time frame	New target	New target	New target	New target	100% within 3 months	100% within 3 months	100% within 3 months

b) Indicators, Annual and Quarterly Targets

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. Review and develop targeted tools and systems for a changing electricity industry	Percentage variance between planned versus actual targeted tools and systems reviewed and developed	80%	-	20%	30%	30%
2. Process Draft Section 34 Determinations from the Minister of Electricity (MOE)	100% of draft Section 34 Determinations received, with complete information, considered by the relevant subcommittee or the Energy Regulator within the stated time frame	100% within 4 months	100% within 4 months	100% within 4 months	100% within 4 months	100% within 4 months
3. Complete applications from the ESI <u>requiring exemption</u> from the South African Grid Code, considered by the ELS/REC	Percentage of complete applications <u>requiring exemption</u> from the South African Grid Code considered by the relevant subcommittee or the Energy Regulator within the stated time frame	100% within 3 months	100% within 3 months	100% within 3 months	100% within 3 months	100% within 3 months
4. Complete applications from the ESI <u>requiring amendment</u> to the South African Grid Code, considered by the ELS/REC	Percentage of complete applications <u>requiring amendment</u> to the South African Grid Code considered by the relevant subcommittee or the Energy Regulator within the stated time frame	100% within 3 months	100% within 3 months	100% within 3 months	100% within 3 months	100% within 3 months

6.2.1.2. Sub-Programme 2: Piped-Gas Industry Regulation

The purpose of this subprogramme is to carry out the following functions for the Piped-Gas industry;

1. Setting and/or approval of tariffs and prices
2. Licensing and registration
3. Compliance monitoring and enforcement
4. Dispute resolution, including mediation, arbitration and handling of complaints
5. Setting of rules, guidelines and codes for regulation.

i. Setting and/or approval of tariffs and prices

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	1. 100% of complete maximum price applications considered by the ER within 120 working days after date of publication of the preliminary assessment of the maximum price applications	% of complete maximum price applications considered by the Energy Regulator within the stated time frame	100%	100%	100%	100%	100%	100%	100%

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	2. 100% of complete applications on distinguishing features considered by the ER within 120 working days after the date of the publication of preliminary assessment of the applications	% of complete applications on distinguishing features considered by the Energy Regulator within the stated time frame	New target	100%	100%	100%	100%	100%	100%
	3. 100% of complete transmission tariff applications considered by the ER within 120 working days after the date of publication of preliminary assessment of tariff applications	% of complete transmission tariff applications considered by the Energy Regulator within the stated time frame	No applications received	100%	100%	100%	100%	100%	100%

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	4. Four quarterly calculations of the ROMPCO tariff for gas volumes below 120 million Gigajoules	Number of calculations of the ROMPCO tariff for gas volumes below 120 million Gigajoule considered by the Energy Regulator within the stated time frame	4 considered quarterly by the PGS	4 considered quarterly by the PGS	4 considered quarterly by the PGS	4 considered quarterly by the PGS	4 considered quarterly by the PGS	4 considered quarterly by the PGS	4 considered quarterly by the PGS
	5. One report on the assessment of the adequacy of competition	Number of reports on the assessment of the adequacy of competition considered by the Energy Regulator within the stated time frame	-	1	-	-	1 considered by the PGS by 31 March 2025	-	-
	6. Reviewed Gas Tariff Guidelines	Energy Regulator decision on reviewed gas tariff guidelines within the stated time frame	New target	New target	New target	New target	-	Regulator decision by 31 March 2026	-

b) *Indicators, Annual and Quarterly Targets*

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. 100% of complete maximum price applications considered by the ER within 120 working days after date of publication of the preliminary assessment of the maximum price applications	% of complete maximum price applications considered by the Energy Regulator within the stated time frame	100%	100%	100%	100%	100%
2. 100% of complete applications on distinguishing features considered by the ER within 120 working days after the date of the publication of preliminary assessment of the applications	% of complete applications on distinguishing features considered by the Energy Regulator within the stated time frame	100%	100%	100%	100%	100%
3. 100% of complete transmission tariff applications considered by the ER within 120 working days after the date of the publication of preliminary assessment of the tariff applications	% of complete transmission tariff applications considered by the Energy Regulator within the stated time frame	100%	100%	100%	100%	100%
4. Four quarterly calculations of the ROMPCO tariff for gas volumes below 120 million Gigajoules	Number of calculations of the ROMPCO tariff for gas volumes below 120 million Gigajoule considered by the Energy Regulator within the stated time frame	4 considered quarterly by the PGS	1 considered by the PGS	1 considered by the PGS	1 considered by the PGS	1 considered by the PGS
5. One report on the assessment of the adequacy of competition	Number reports on the assessment of the adequacy of competition considered by the Energy Regulator within the stated time frame	1 considered by the PGS by 31 March 2025	-	-	-	1 considered by the PGS by 31 March 2025

ii. Licensing And Registration

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	1. 100% of complete licence applications considered by the PGS/REC/ER within 60 working days from the closing date of the public comment period or period of applicant's response to objections received	% of complete licence applications considered by the Energy Regulator within the stated time frame	100%	100%	100%	100%	100%	100%	100%
	2. 100% of complete applications for licence amendments/ revocations/ conversions considered by the PGS/REC within 60 working days from the closing date of the public comment period or period of applicant's response to objections received	% of complete applications for licence amendments/ revocations/ conversions considered by the Energy Regulator within the stated time frame	100%	100%	100%	100%	100%	100%	100%

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
	3. 100% of complete applications for the registration of gas activities considered by the PGS within 60 working days from the closing date of the public comment period	% of complete applications for the registration of gas activities considered by the Energy Regulator within the stated time frame	100%	100%	100%	100%	100%	100%	100%

b) *Indicators, Annual and Quarterly Targets*

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. 100% of complete licence applications considered by the PGS/REC/ER within 60 working days from the closing date of the public comment period or period of applicant's response to objections received	% of complete licence applications considered by the Energy Regulator within the stated time frame	100%	100%	100%	100%	100%
2. 100% of complete applications for licence amendments/revocations/conversions considered by the PGS/REC within 60 working days from the closing date of the public comment period or period of applicant's response to objections received	% of complete applications for licence amendments/revocations/conversions considered by the Energy Regulator within the stated time frame	100%	100%	100%	100%	100%
3. 100% of complete applications for the registration of gas activities considered by the PGS within 60 working days from the closing date of the public comment period	% of complete applications for the registration of gas activities considered by the Energy Regulator within the stated time frame	100%	100%	100%	100%	100%

iii. Compliance Monitoring and Enforcement

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	1. Twelve monthly volume balance reports assessed	Number of monthly volume balance reports considered by the relevant subcommittee within the stated time frame	12 considered annually by the PGS	12 considered annually by the PGS	12 considered annually by the PGS	12 considered annually by the PGS	12 considered annually by the PGS	12 considered annually by the PGS	12 considered annually by the PGS
	2. One audit report on the compliance of ROMPCO pipeline	Number of audit reports on compliance of the ROMPCO pipeline considered by the relevant subcommittee within the stated time frame	1 considered annually by the PGS by 31 March	1 considered annually by the PGS by 31 March	1 considered annually by the PGS by 31 March	1 considered annually by the PGS by 31 March	1 considered annually by the PGS by 31 March	1 considered annually by the PGS by 31 March	1 considered annually by the PGS by 31 March
	3. One report on compliance with licence conditions	Number of reports on licensees' compliance with licence conditions considered by the relevant subcommittee within the stated time frame	45 inspected facilities	1 considered annually by the PGS by 31 March	1 considered annually by the PGS by 31 March	1 considered annually by the PGS by 31 March	1 considered annually by the PGS by 31 March	1 considered annually by the PGS by 31 March	1 considered annually by the PGS by 31 March

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	4. Monitoring reports on the implementation of transmission tariffs, after one year, following the approval of the transmission tariff	% of monitoring reports on the implementation of transmission tariffs considered by the relevant subcommittee within the stated time frame	3	3	3	100% considered annually by the PGS by 31 March	100% considered annually by the PGS by 31 March	100% considered annually by the PGS by 31 March	100% considered annually by the PGS by 31 March
	5. Four reports (one for each licensee – SASOL, ROMPCO, Transnet and SLG) on the implementation of the RRM for the preceding financial year	Number of reports on the implementation of the RRM for the preceding financial year considered by the relevant subcommittee or the Energy Regulator within the stated time frame	4 considered annually by the PGS/ REC by 31 March	4 considered annually by the PGS/ REC by 31 March	4 considered annually by the PGS/ REC by 31 March	4 considered annually by the PGS/ REC by 31 March	4 considered annually by the PGS/ REC by 31 March	4 considered annually by the PGS/ REC by 31 March	4 considered annually by the PGS/ REC by 31 March

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	6. Monitoring reports per licensee on the implementation of Maximum Prices, after one year, following the approval of the maximum price	% of monitoring reports per licensee on the implementation of Maximum Prices considered by the relevant subcommittee within the stated time frame	1 report for each of the 8 licensees	1 Report for each of the 7 licensees	100% considered annually by the PGS by 31 March	100% considered annually by the PGS by 31 March	100% considered annually by the PGS by 31 March	100% considered annually by the PGS by 31 March	100% considered annually by the PGS by 31 March
	7. One report on the assessment of the maximum gas price's competitiveness and impact	Number of reports on the assessment of the maximum gas price's competitiveness and impact considered by the relevant subcommittee within the stated time frame	New target	New target	New target	New target	-	1 considered annually by the PGS by 31 March	1 considered annually by the PGS by 31 March

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	8. One report on the applicability of third-party access to new mobile gas infrastructure	Number of reports on applicability of third-party access to new mobile gas infrastructure considered by the relevant subcommittee within the stated time frame	New target	New target	New target	New target	-	-	1 considered by the PGS by 31 March
	9. One report on the assessment of the ability to supply gas at competitive prices and the impact of distribution exclusivity on the market	Number of reports on the assessment of the ability to supply gas at competitive prices and the impact of distribution exclusivity on the market considered by the relevant subcommittee within the stated time frame	New target	New target	New target	New target	-	1 considered by the PGS by 31 March	-

b) *Indicators, Annual and Quarterly Targets*

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. Twelve monthly volume balance reports assessed	Number of monthly volume balance reports considered by the relevant subcommittee within the stated time frame	12 considered annually by the PGS	3 considered by the PGS	3 considered by the PGS	3 considered by the PGS	3 considered by the PGS
2. One audit report on the compliance of ROMPCO pipeline	Number of audit reports on compliance of the ROMPCO pipeline considered by the relevant subcommittee within the stated time frame	1 considered annually by the PGS by 31 March	-	-	-	1 considered by the PGS by 31 March
3. One report on compliance with licence conditions	Number of reports on licensees' compliance with licence conditions considered by the relevant Subcommittee within the stated time frame	1 considered annually by the PGS by 31 March	-	-	-	1 considered by the PGS by 31 March
4. Monitoring reports on the implementation of transmission tariffs, after one year, following the approval of the transmission tariff	% of monitoring reports on the implementation of transmission tariffs considered by the relevant subcommittee within the stated time frame	100% considered annually by the PGS by 31 March	-	-	-	100% considered by the PGS by 31 March
5. Four reports (one for each licensee – SASOL, ROMPCO, Transnet and SLG) on the implementation of the Regulatory Reporting Manuals (RRM) for the preceding financial year	Number of reports on the implementation of the RRM for the preceding financial year considered by the relevant subcommittee or the Energy Regulator within the stated time frame	4 considered annually by the PGS/REC by 31 March	-	-	-	4 considered by the PGS/REC by 31 March
6. Monitoring reports per licensee on the implementation of maximum prices, after one year following the approval of the maximum price	% of monitoring reports per licensee on the implementation of maximum prices considered by the relevant subcommittee within the stated time frame	100% considered annually by the PGS by 31 March	-	-	-	100% considered by the PGS by 31 March

iv. Dispute resolution, including mediation, arbitration and handling of complaints

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS							
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD			
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	
MTSF Priority 2: Economic Transformation and Job Creation										
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	1.	60% of complaint investigations completed within 12 months and a report on findings considered by the PGS	% of complaint investigations completed and a report on findings considered by the relevant subcommittee or the Energy Regulator within the stated time frame	50%	50%	60%	60%	60%	70%	70%
	2.	60% of initiated investigations and inquiries completed within 12 months and a report on findings considered by the PGS	% of initiated investigations completed and a report on findings considered by the relevant subcommittee or the Energy Regulator within the stated time frame	50%	50%	60%	60%	60%	70%	70%

b) *Indicators, Annual and Quarterly Targets*

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. 60% of complaint investigations completed within 12 months and a report on findings considered by the PGS	% of complaint investigations completed and a report on findings considered by the relevant subcommittee or the Energy Regulator within the stated time frame	60%	60%	60%	60%	60%
2. 60% of initiated investigations and inquiries completed within 12 months and a report on findings considered by the PGS	% of initiated investigations completed and a report on findings considered by the relevant subcommittee or the Energy Regulator within the stated time frame	60%	60%	60%	60%	60%

v. Setting of rules, guidelines and codes for regulation

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS							
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD			
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	
MTSF Priority 2: Economic Transformation and Job Creation										
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	1. Two reports on new developments in the gas industry	Number of reports on new developments in the gas industry considered by the relevant committee or the Energy Regulator within the stated time frame	2 considered bi-annually by the PGS by 30 September and 31 March	2 considered bi-annually by the PGS by 30 September and 31 March	2 considered bi-annually by the PGS by 30 September and 31 March	2 considered bi- annually by the PGS by 31 March	2 considered bi-annually by the PGS by 31 March	2 considered bi-annually by the PGS by 31 March	2 considered bi-annually by the PGS by 31 March	
	2. One report on the impact of developments on competition in the gas industry	Number of reports on the impact of developments on competition in the gas industry considered by the relevant committee or the Energy Regulator within the stated time frame	New target	1 considered annually by the PGS by 31 March	1 considered annually by the PGS by 31 March	1 considered annually by the PGS by 31 March	-	1 considered annually by the PGS by 31 March	1 considered annually by the PGS by 31 March	

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	3. One report on progress made with the implementation of the measures to improve competition and investment in the gas industry	Number of reports on the progress made with the implementation of the measures to improve competition and investment in the gas industry considered by the relevant committee or the Energy Regulator within the stated time frame	New target	New target	New target	New target	1 considered annually by the ER by 31 March	1 considered annually by the ER by 31 March	1 considered annually by the ER by 31 March
	4. One report on the monitoring of the implications of the lack of supply of methane rich gas by Sasol Gas	Number of reports on the monitoring of the implications of the lack of supply of methane rich gas by Sasol Gas considered by the relevant committee or the Energy Regulator within the stated time frame	New target	New target	New target	New target	-	1 considered annually by the PGS by 31 March	-

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	5. One report on the identification of potential points of network interconnection locally and cross border with new gas sources supplies	Number of reports on the identification of potential points of network interconnection locally and cross border with new gas sources supplies considered by the relevant committee or the Energy Regulator within the stated time frame	New target	New target	New target	New target	-	-	1 considered annually by the PGS by 31 March
	6. One report on the regulatory impact assessment of gas regulatory activities since the implementation of the new maximum price methodology	Number of reports on the regulatory impact assessment of gas regulatory activities since the implementation of the new maximum price methodology considered by the relevant committee or the Energy Regulator within the stated time frame	New target	New target	New target	New target	-	1 considered annually by the PGS by 31 March	-

b) *Indicators, Annual and Quarterly Targets*

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. Two reports on new developments in the gas industry	Number of reports on new developments in the gas industry considered by the relevant subcommittee or the Energy Regulator within the stated time frame	2 considered bi-annually by the PGS by 31 March	-	-	-	2 considered by the PGS 31 March
2. One report on progress made with the implementation of the measures to improve competition and investment in the gas industry	Number of reports on the progress made with the implementation of the measures to improve competition and investment in the gas industry considered by the relevant committee or the Energy Regulator within the stated time frame	1 considered annually by the PGS by 31 March	-	-	-	1 considered by the PGS by 31 March

6.2.1.3. Sub-programme 3: Petroleum Pipeline Industry Regulation

The purpose of this subprogramme is to carry out the following functions for the Piped-Gas industry;

1. Setting and/or approval of tariffs and prices
2. Licensing and registration
3. Compliance monitoring and enforcement
4. Dispute resolution, including mediation, arbitration and handling of complaints
5. Setting of rules, guidelines and codes for regulation.

i. Setting and/or approval of tariffs and prices

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF PRIORITY 2: ECONOMIC TRANSFORMATION AND JOB CREATION									
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	1. 80% of complete pipeline, storage and loading facility tariff applications considered by the REC/PPS/ER within 6 months from receipt of complete/adequate application	% of complete pipeline, storage and loading facility tariff applications considered by the relevant subcommittee or the Energy Regulator within the stated time frame	90%	97%	93%	80%	85%	85%	85%

b) Indicators, Annual and Quarterly Targets

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. 80% of complete pipeline, storage and loading facility tariff applications considered by the REC/PPS/ER within 6 months from receipt of complete/adequate application	% of complete pipeline, storage and loading facility tariff applications considered by the relevant subcommittee or the Energy Regulator within the stated time frame	85%	85%	85%	85%	85%

ii. Licensing and Registration

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS							
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD			
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	
MTSF Priority 2: Economic Transformation and Job Creation										
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	1.	100% of complete licence applications considered by the PPS/REC/ER within 60 working days under the conditions as prescribed in section 19(1) of the Petroleum Pipelines Act	% of complete licence applications considered by the relevant subcommittee or the Energy Regulator within the stated time frame	75%	100%	100%	100%	100%	100%	100%
	2.	100% of complete applications for licence amendments/ revocations considered by the PPS/REC/ER within 60 working days from the closing date of the public comment period or period of applicant's response to objections received	% of complete applications for licence amendments/ revocations considered by the relevant subcommittee or the Energy Regulator within the stated time frame	100%	100%	100%	100%	100%	100%	100%

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	3. One report on investigations into suspected unlicensed activities	Number of reports on investigations done into suspected unlicensed activities considered by the relevant subcommittee or the Energy Regulator within the stated time frame	1 ³ considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March
	4. One report on the geographic spread of licences issued for petroleum pipelines infrastructure and new entrants	Number of reports on the geographic spread of licences issued for petroleum pipelines infrastructure and new entrants considered by the relevant subcommittee or the Energy Regulator within the stated time frame	New target	New target	1 considered annually by the PPS by 31 March	1 considered annually by the PPS by 31 March	1 considered annually by the PPS by 31 March	1 considered annually by the PPS by 31 March	1 considered annually by the PPS by 31 March

³ It was determined that 4 quarterly reports are too many. It was therefore decided that one consolidated report on all the investigations would be done.

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	5. Two reports on the inland security of supply	Number of reports on the inland security of supply considered by relevant subcommittee or the Energy Regulator within the stated time frame	2 considered bi-annually by the PPS by 30 September and 31 March	2 considered bi-annually by the PPS by 30 September and 31 March	2 considered bi-annually by the PPS by 30 September and 31 March	2 considered bi-annually by the PPS by 30 September and 31 March	2 considered bi-annually by the PPS by 31 March	2 considered bi-annually by the PPS by 31 March	2 considered bi-annually by the PPS by 31 March
	6. Report on prudency review of identified licensees	Number of reports on prudency reviews of identified licensees considered by relevant subcommittee or the Energy Regulator within the stated time frame	New target	New target	New target	1 considered by ER by 31 March 2024	1 considered by ER by 31 March 2025	1 considered by ER by 31 March 2026	1 considered by ER by 31 March 2027

b) Indicators, Annual and Quarterly Targets

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. 100% of complete licence applications considered by the PPS/REC/ER within 60 working days under the conditions as prescribed in section 19(1) of the Petroleum Pipelines Act	% of complete licence applications considered by the relevant subcommittee or the Energy Regulator within the stated time frame	100%	100%	100%	100%	100%
2. 100% of complete applications for licence amendments/ revocations considered by the PPS/REC/ER within 60 working days from the closing date of the public comment period or period of applicant's response to objections received	% of complete applications for licence amendments/ revocations considered by the relevant subcommittee or the Energy Regulator within the stated time frame	100%	100%	100%	100%	100%
3. One report on investigations into suspected unlicensed activities	Number of reports on investigations done into suspected unlicensed activities considered by the relevant subcommittee or the Energy Regulator within the stated time frame	1 considered annually by the REC by 31 March	-	-	-	1 considered by the REC by 31 March
4. One report on the geographic spread of the licences issued for petroleum pipelines infrastructure and new entrants considered annually by the PPS by 31 March	Number of reports on the geographic spread of licences issued for petroleum pipelines infrastructure and new entrants considered by the relevant subcommittee or the Energy Regulator	1 considered annually by the PPS by 31 March	-	-	-	1 considered by the PPS by 31 March

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
5. Two reports on the inland security of supply considered bi-annually by the PPS by 30 September and 31 March	Number of reports on the inland security of supply considered by relevant subcommittee or the Energy Regulator within the stated time frame	2 considered bi-annually by the PPS by 31 March	-	-	-	2 considered by the PPS by 31 March
6. Report on prudency review of identified licensees	Number of reports on prudency reviews of identified licensees considered by relevant subcommittee or the Energy Regulator within the stated time frame	1 considered by ER by 31 March 2024	-	-	-	1 considered by ER by 31 March 2024

iii. Compliance Monitoring and Enforcement

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	1. Two reports on trends regarding utilisation of storage facilities and third-party access	Number of reports on trends regarding utilisation of storage facilities and third-party access, considered by the relevant subcommittee or the Energy Regulator within the stated time frame	2 considered bi-annually by the PPS by 30 September and 31 March	2 considered bi-annually by the PPS by 30 September and 31 March	2 considered bi-annually by the PPS by 30 September and 31 March	2 considered bi-annually by the PPS by 31 March	2 considered bi-annually by the PPS by 31 March	2 considered bi-annually by the PPS by 31 March	2 considered bi-annually by the PPS by 31 March

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	2. One report on the implementation of the methodology to determine uncommitted capacity	Number of reports on the implementation of the methodology to determine uncommitted capacity considered by the relevant subcommittee or the Energy Regulator within the stated time frame	1 considered annually by the PPS by 31 March	1 considered annually by the PPS by 31 March	1 considered annually by the PPS by 31 March	1 considered annually by the PPS by 31 March	1 considered annually by the PPS by 31 March	1 considered annually by the PPS by 31 March	1 considered annually by the PPS by 31 March
	3. Two reports on the construction of new facilities	Number of reports on the construction of new facilities considered by the relevant subcommittee or the Energy Regulator within the stated time frame	4 considered quarterly by the PPS	2 considered bi-annually by the PPS by 30 September and 31 March	2 considered bi-annually by the PPS by 30 September and 31 March	2 considered bi-annually by the PPS by 31 March	2 considered bi-annually by the PPS by 31 March	2 considered bi-annually by the PPS by 31 March	2 considered bi-annually by the PPS by 31 March
	4. Two reports on licensees' compliance with statutory reporting requirements	Number of reports on licensees' compliance with statutory reporting requirements considered by the relevant subcommittee or the Energy Regulator within the stated time frame	4 considered quarterly by the PPS	2 considered bi-annually by the PPS by 30 September and 31 March	2 considered bi-annually by the PPS by 30 September and 31 March	2 considered bi-annually by the PPS by 31 March	2 considered bi-annually by the PPS by 31 March	2 considered bi-annually by the PPS by 31 March	2 considered bi-annually by the PPS by 31 March

b) Indicators, Annual and Quarterly Targets

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. Two reports on trends regarding utilisation of storage facilities and third-party access	Number of reports on trends regarding utilisation of storage facilities and third-party access, considered by the relevant subcommittee or the Energy Regulator within the stated time frame	2 considered bi-annually by the PPS by 31 March	-	-	-	2 considered bi-annually by the PPS by 31 March
2. One report on the implementation of the methodology to determine uncommitted capacity	Number of reports on the implementation of the methodology to determine uncommitted capacity considered by the relevant subcommittee or the Energy Regulator within the stated time frame	1 considered annually by the PPS by 31 March	-	-	-	1 considered by the PPS by 31 March
3. Two reports on the construction of new facilities	Number of reports on the construction of new facilities considered by the relevant subcommittee or the Energy Regulator within the stated timeframe	2 considered bi-annually by the PPS by 31 March	-	-	-	2 considered bi-annually by the PPS by 31 March
4. Two reports on licensees' compliance with statutory reporting requirements	Number of reports on licensees' compliance with statutory reporting requirements considered by the relevant subcommittee or the Energy Regulator within the stated timeframe	2 considered bi-annually by the PPS by 31 March	-	-	-	2 considered bi-annually by the PPS by 31 March

iv. Dispute resolution, including mediation, arbitration and handling of complaints

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 2: Economic Transformation and Job Creation									
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	1. 100% of complaints investigated and report considered by the PPS within 12 months of receipt of adequate information from relevant parties	% of complaints investigated and report considered by the relevant subcommittee or the Energy Regulator within the stated time frame of receipt of adequate information from relevant parties	100%	100%	100%	100%	100%	100%	100%

b) Indicators, Annual and Quarterly Targets

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. 100% of complaints investigated and report considered by the PPS within 12 months of receipt of adequate information from relevant parties	% of complaints investigated and report considered by the relevant subcommittee or the Energy Regulator within the stated time frame of receipt of complete information from relevant parties	100%	100%	100%	100%	100%

v. Setting of rules, guidelines and codes for regulation

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS							
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD			
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	
MTSF Priority 2: Economic Transformation and Job Creation										
A stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time and to quality standards	1. One report on the monitoring of the implementation of the revised tariff methodology	Number of reports on the monitoring of the implementation of the tariff methodology considered by the relevant subcommittee or the Energy Regulator within the stated time frame	1 considered annually by the PPS by 31 March	1	1 considered annually by the PPS by 31 March	1 considered annually by the PPS by 31 March	1 considered annually by the PPS by 31 March	1 considered annually by the PPS by 31 March	1 considered annually by the PPS by 31 March	1 considered annually by the PPS by 31 March
	2. Reviewed pipelines tariff methodology	Energy Regulator decision on the Reviewed pipelines tariff methodology within the stated time frame	1	-	-	Regulator decision by 31 March 2024	-	-	Regulator decision by 31 March 2027	

b) Indicators, Annual and Quarterly Targets

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. One report on the monitoring of the implementation of the revised methodology	Number of reports on the monitoring of the implementation of the tariff methodology considered by the relevant subcommittee or the Energy Regulator within the stated time frame	1 considered annually by the PPS by 31 March	-	-	-	1 considered annually by the PPS by 31 March

6.2.1.4. Explanation of Planned Performance

The planned outputs are in line with the regulatory functions of NERSA, as contained in relevant legislation.

6.2.1.5. Programme Resource Considerations

The budget for activities relating to the regulation of the energy industry is based on a ring-fencing methodology that was approved to comply with section 13 of the National Energy Regulator Act, 2004 (Act No. 40 of 2004). The methodology is based on direct employment cost as a basis of common cost apportionment. Direct costs are allocated directly to the respective industry.

The table below indicates the approved staff complement and the approved budget for 2024/25 for Programme 1: Regulatory Service Delivery.

DIVISIONS	RELEVANT STRUCTURES	STAFF COMPLEMENT	BUDGET (R)	% ALLOCATION
Electricity Regulation (ELR)	Executive Manager	3	8 657 182	80%
	Electricity Pricing and Tariffs (EPT)	35	36 389 123	80%
	Electricity Licensing, Compliance and Dispute Resolution (ELC)	34	36 937 710	80%
	Electricity Infrastructure Planning (EIP)	13	18 599 176	80%
Piped-Gas Regulation (GAR)	Executive Manager	5	11 497 694	90%
	Gas Pricing and Tariffs (GPT)	8	12 540 911	95%
	Gas Licensing, Compliance and Dispute Resolution (GLC)	11	15 165 681	100%
	Gas Competition and Market Analysis (GCM)	4	6 198 696	95%
Petroleum Pipelines Regulation (PPR)	Executive Manager	6	6 448 837	50%
	Petroleum Pipelines Tariffs (PPT)	9	9 870 079	80%
	Petroleum Licensing, Compliance and Dispute Resolution (PLC)	9	10 572 498	80%

Note: The % allocation is based on the staff complement of the Organisation in line with the ring-fencing methodology.

Please refer to Part D: Funding for NERSA for the detailed budget.

6.2.1.6. Key Risks

Please refer to Section 6.2.6 below for NERSA's detailed Strategic Risk Register.

6.2.2. Programme 2: Advocacy and Engagement

The programme's purpose is to contribute towards relevant legislation and policies; government's transformation as well as to inform customers and stakeholders.

6.2.2.1 Sub-programme: Regulatory and Policy Advocacy

The purpose of this sub programme is to coordinate advocacy with the shareholders and stakeholders regarding regulatory and policy matters.

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS							
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD			
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	
MTSF Priority 1: Capable, Ethical and Developmental State										
Energy industry regulatory framework is relevant for the effective regulation for the benefit of the customers and stakeholders	1. Two reports on regulatory advocacy, one each for the piped- gas and petroleum pipelines regulated industries, aimed at the improvement of the regulatory framework provided through legislation, regulation and government policies	Number of reports on regulatory advocacy considered by the relevant subcommittee or the Energy Regulator within stated time frame	New target	New target	4 considered annually by the PGS (2) and PPS (2) by 31 March	2 considered annually by the PGS (1) and PPS (1) by 31 March	2 considered annually by the PGS (1) and PPS (1) by 31 March	2 considered annually by the PGS (1) and PPS (1) by 31 March	2 considered annually by the PGS (1) and PPS (1) by 31 March	2 considered annually by the PGS (1) and PPS (1) by 31 March
Innovation drives our response to the transition of the Industry	2. Regulatory Advocacy in line with the approved annual ESI Advocacy and Stakeholder Engagement Plan aimed at influencing the effective functioning of the electricity industry	Percentage variance of planned versus actual annual ESI Advocacy and Stakeholder Engagement Plan executed	New target	65%	65%	65%	65%	65%	65%	65%

b) *Indicators, Annual and Quarterly Targets*

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. Two reports on regulatory advocacy, one each for the piped-gas and petroleum pipelines regulated industries, aimed at the improvement of the regulatory framework provided through legislation, regulation and government policies	Number of reports on regulatory advocacy considered by the relevant subcommittee or the Energy Regulator within stated time frame	2 considered annually by the PGS (1) and PPS (1) by 31 March	-	-	-	2 considered annually by the PGS (1) and PPS (1) by 31 March
2. ESI Regulatory Advocacy in line with the approved annual ESI Advocacy and Stakeholder Engagement Plan aimed at influencing legislative and policy changes	Percentage variance of planned versus actual annual ESI Advocacy and Stakeholder Engagement Plan executed	65%	Annual ESI Advocacy and Stakeholder Engagement Plan	10%	25%	30%

6.2.2.2 Subprogramme: Customer and Stakeholder Engagement

The purpose of this sub programme is to coordinate customer and stakeholder engagements regarding regulatory matters.

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 7: A better Africa and world									
Integrated and value-added services to customers	1. Two reports on stakeholder engagements for the piped-gas and petroleum pipelines regulated industries	Number of reports on stakeholder workshops / meetings for the piped-gas and petroleum pipelines regulated industries considered by the relevant subcommittee or the Energy Regulator within the stated timeframe	New target	New target	New target	2 considered annually by the PGS (1) and PPR (1) by 31 March	2 considered annually by the PGS (1) and PPR (1) by 31 March	2 considered annually by the PGS (1) and PPR (1) by 31 March	2 considered annually by the PGS (1) and PPR (1) by 31 March
	2. ESI Stakeholder engagement in line with the approved annual ESI Advocacy and Stakeholder Engagement Plan	Percentage variance of planned versus actual annual ESI Stakeholder Engagement Plan executed	New target	New target	New target	65%	65%	65%	65%

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 7: A better Africa and world									
Integrated and value-added services to customers	3. Seventy five ESI customer education programmes undertaken annually by 31 March	Number of ESI customer education programmes undertaken within the stated time frame	30 ⁴	50	75	75	75	75	75
	4. Two reports on partnership creation to position NERSA as a recognised regulator nationally, regionally and internationally	Number of reports on partnership creation considered by the relevant subcommittee or the Energy Regulator within the stated time frame	2 considered bi-annually by the REC by 30 September and 31 March	2 considered bi-annually by the REC by 30 September and 31 March	2 considered bi-annually by the REC by 30 September and 31 March	2 considered bi-annually by the REC by 31 March	2 considered bi-annually by the REC by 31 March	2 considered bi-annually by the REC by 31 March	2 considered bi-annually by the REC by 31 March
	5. One report on the implementation of the stakeholder perception plan	Number of reports on the implementation of the stakeholder perception plan considered by the relevant subcommittee or the Energy Regulator within the stated time frame	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March

4 The **target** was changed from 60 to 30 customer education programmes due to the restrictions on traveling and gathering of people that makemade conducting customer education workshops difficult. NERSA will utilise radio interviews and the distribution of brochures to conduct some level of customer education programmes.

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 7: A better Africa and world									
Integrated and value-added services to customers	6. One report on the analysis of the Stakeholder Management/ Reputational Risk survey considered by the Energy Regulator by 31 March 2025	Number of reports on the analysis of the Stakeholder Management/ Reputational Risk survey considered by the relevant subcommittee within the stated time frame	New target	New target	New target	New target	1 considered by the ER by 31 March 2025	-	-
	7. One report on communication and reputation management	Number of reports on communication and reputation management considered by the relevant subcommittee or the Energy Regulator within the stated time frame	New target	New target	New target	New target	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March	1 considered annually by the REC by 31 March

b) *Indicators, Annual and Quarterly Targets*

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. Two reports on stakeholder engagements for the piped-gas and petroleum pipelines regulated industries	Number of reports on stakeholder workshops/meetings for the piped-gas and petroleum pipelines regulated industries considered by the relevant subcommittee or the Energy Regulator within the stated time frame	2 considered annually by the PGS (1) and PPR (1) by 31 March	-	-	-	2 considered by the PGS (1) and PPR (1) by 31 March
2. ESI Stakeholder engagement in line with the approved annual ESI Advocacy and Stakeholder Engagement Plan	Percentage variance of planned versus actual annual ESI Stakeholder Engagement Plan executed	65%	Annual ESI Advocacy and Stakeholder Engagement Plan	10%	25%	30%
3. Seventy five ESI customer education programmes undertaken annually by 31 March	Number of ESI customer education programmes undertaken within the stated time frame	75	Annual ESI customer education programme plan 20	25	15	15
4. Two reports on partnership creation to position NERSA as a recognised regulator nationally, regionally and internationally	Number of reports on partnership creation considered by the relevant subcommittee or the Energy Regulator within the stated time frame	2 considered biannually by the REC by 31 March	-	-	-	2 considered by the REC by 31 March
5. One report on the implementation of the stakeholder perception plan	Number of reports on the implementation of the stakeholder perception plan considered by the relevant subcommittee or the Energy Regulator within the stated time frame	1 considered annually by the REC by 31 March	-	-	-	1 considered annually by the REC by 31 March

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
6. One report on the analysis of the Stakeholder Management/ Reputational Risk survey considered by the Energy Regulator by 31 March 2025	Number of reports on the analysis of the Stakeholder Management/ Reputational Risk survey considered by the relevant subcommittee within the stated time frame	1 considered by the ER by 31 March 2025	-	-	-	1 considered by the REC by 31 March 2025
7. One report on communication and reputation management	Number of reports on communication and reputation management considered by the relevant subcommittee or the Energy Regulator within the stated time frame	1 considered annually by the REC by 31 March	-	-	-	1 considered annually by the REC by 31 March

6.2.2.3 Explanation of Planned Performance

The planned output is in support of NERSA's regulatory functions, as contained in the relevant legislation.

6.2.2.4 Programme Resource Considerations

The budget for the activities relating to the regulation of the energy industry is based on a ring-fencing methodology that was approved to comply with section 13 of the National Energy Regulator Act, 2004 (Act No. 40 of 2004). The methodology is based on direct employment cost as a basis for common costs apportionment. Direct costs are allocated directly to the respective industry.

The table below indicates the approved staff complement and the approved budget for 2024/25 for Programme 2: Advocacy and Engagement.

DIVISIONS	RELEVANT STRUCTURES	STAFF COMPLEMENT	BUDGET (R)	% ALLOCATION
Electricity Regulation (ELR)	Executive Manager	3	2 164 295	20%
	Electricity Pricing and Tariffs (EPT)	35	9 097 281	20%
	Electricity Licensing, Compliance and Dispute Resolution (ELC)	34	9 234 428	20%
	Electricity Infrastructure Planning (EIP)	13	4 649 794	20%
Piped-Gas Regulation (GAR)	Executive Manager	5	1 277 522	10%
	Gas Pricing and Tariffs (GPT)	8	660 048	5%
	Gas Licensing, Compliance and Dispute Resolution (GLC)	0	-	0%
	Gas Competition and Markets (GCM)	4	326 247	5%
Petroleum Pipelines Regulation (PPR)	Executive Manager	6	6 448 837	50%
	Petroleum Pipelines Tariffs (PPT)	9	2 467 520	20%
	Petroleum Licensing, Compliance and Dispute Resolution (PLC)	9	2 643 124	20%
Corporate Services	Executive Manager	3	248 834	5%
	Communication and Stakeholder Management (CSM)	9	17 457 309	100%
	International Co-ordination and Partnerships (ICP)	3	6 847 780	100%

Note: The % allocation is based on the staff complement of the Organisation in line with the ring-fencing methodology.

Please refer to Part D: Funding for NERSA for the detailed budget.

6.2.2.5 Key Risks

Please refer to Section 6.2.6 below for NERSA's detailed Strategic Risk Register.

6.2.3. Programme 3: Innovation

The programme aims to ensure a technology solution that supports the business in delivering integrated and value-added services to customers internally and externally.

6.2.3.1 Subprogramme: Integrated and Value-Added Services

The purpose of this sub programme is to facilitate integrated and value added services aimed at improving NERSA's service delivery.

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF PRIORITY 1: CAPABLE, ETHICAL AND DEVELOPMENTAL STATE									
Innovation drives our response to the transition of the industry	1. Two reports on the implementation of the approved ICT Strategy	Number of reports on the implementation of the approved ICT Strategy considered by the relevant committee or the Energy Regulator within the stated time frame	New target	2 considered bi-annually by the ITGC by 30 September and 31 March	2 considered bi-annually by the ITGC by 30 September and 31 March	2 considered bi-annually by the 31 March	2 considered bi-annually by the ITGC by 31 March	2 considered bi-annually by the ITGC by 31 March	-

b) Indicators, Annual and Quarterly Targets

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. Two reports on the implementation of the approved ICT Strategy	Number of reports on the implementation of the approved ICT Strategy considered by the relevant committee or the Energy Regulator within the stated time frame	2 considered bi-annually by the ITGC by 31 March	-	-	-	2 considered bi-annually by the ITGC by 31 March

6.2.3.2 Explanation of Planned Performance

The planned output is in support of NERSA's regulatory functions of NERSA, as contained in relevant legislation.

6.2.3.3 Programme Resource Considerations

The budget for activities relating to the regulation of the energy industry is based on a ring-fencing methodology that was approved to comply with section 13 of the National Energy Regulator Act, 2004 (Act No. 40 of 2004). The methodology is based on direct employment costs as a basis of common costs apportionment. Direct costs are allocated directly to the respective industry.

The table below indicates the approved staff complement and the approved budget for 2024/25 for Programme 3: Innovation.

DIVISIONS	RELEVANT STRUCTURES	STAFF COMPLEMENT	BUDGET (R)	% ALLOCATION
Specialised Support Units (SSU)	Information and Communication Technology (ICT)	10	9 996 123	20%
Corporate Services	Information Resources Management (IRM)	7	767 867	5%

Note: The percentage allocation is based on the staff complement of the organisation in line with the ring-fencing methodology.

Please refer to Part D: Funding for NERSA for the detailed budget.

6.2.3.4 Key Risks

Please refer to Section 6.2.6 below for NERSA's detailed Strategic Risk Register.

6.2.4. Programme 4: Operational Efficiency and Quality Management

The programme aims to ensure that NERSA's integrated operational processes, improved planning and project management remain relevant in supporting core business.

6.2.4.1 Subprogramme: Integrated Operations and Research and Analysis

The purpose of this sub programme is to ensure that:

- a) relevant research and information analysis is conducted; and
- b) organisational and operational processes contribute to improved service delivery

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS							
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD			
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	
MTSF Priority 1: Capable, ethical and developmental state										
Integrated and value-added services to customers	1. One report on the environmental scanning	Number of reports on the environmental scanning considered by the relevant subcommittee or the Energy Regulator within the stated time frame	1 considered annually by the REC by 30 June	1 considered annually by the REC by 30 June	1 considered annually by the REC by 30 June	1 considered annually by the REC by 30 June	1 considered annually by the REC by 30 April	1 considered annually by the REC by 30 April	1 considered annually by the REC by 30 April	

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 1: Capable, ethical and developmental state									
Integrated and value-added services to customers	2. Reviewed Regulatory Reporting Manuals for Non-Financial and Financial Information	Energy Regulator’s decision on the Reviewed Regulatory Reporting Manuals for Non-Financial and Financial Information	New target	New target	New target	New target	Energy Regulator’s decision on Reviewed Regulatory Reporting Manuals for Non-Financial and Financial Information (Vol 1) by 31 March 2025	Reviewed Regulatory Reporting Manuals for Non-Financial and Financial Information (Vol 2-3)	Reviewed Regulatory Reporting Manuals for Non-Financial and Financial information (Vol 3-4)
	3. Audit opinion that is not qualified implementation of the Regulatory Reporting Manuals for Non-financial and financial information	Unqualified audit opinion	Unqualified audit opinion	Clean audit opinion	Unqualified audit opinion	Unqualified audit opinion	Unqualified audit opinion	Unqualified audit opinion	Unqualified audit opinion

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 1: Capable, ethical and developmental state									
Integrated and value-added services to customers	4. One report on the implementation of the measures to improve NERSA's B-BBEE rating per financial year	Number of reports on the implementation of the measures to improve NERSA's B-BBEE rating per financial year considered by the relevant subcommittee or the Energy Regulator within the stated time frame	New target	New target	New target	New target	1 considered annually by the ER by 31 March	1 considered annually by the ER by 31 March	1 considered annually by the ER by 31 March
	5. One report on the implementation of gender mainstreaming initiatives	Number of reports on the implementation of the gender mainstreaming plan considered by the relevant subcommittee within the stated time frame	New target	New target	1 considered annually by the HRRC by 31 March	1 considered annually by the HRRC by 31 March	1 considered annually by the HRRC by 31 March	1 considered annually by the HRRC by 31 March	1 considered annually by the HRRC by 31 March

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 1: Capable, ethical and developmental state									
Integrated and value-added services to customers	6. One report on benchmarking of NERSA's operational processes controls against international standards	Number of reports on benchmarking of NERSA's operational processes controls against international standards considered by the relevant subcommittee within the stated time frame	New target	New target	New target	New target	1 considered by the REC by 31 March 2025	-	-
	7. One report on the implementation of the Hybrid Working Strategy	Number of reports on the implementation of the Hybrid Working Strategy considered by the relevant subcommittee within the stated time frame	New target	New target	New target	New target	1 considered by the REC by 31 March 2025	1 considered by the REC by 31 March 2026	-

b) *Indicators, Annual and Quarterly Targets*

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. One report on the environmental scanning	Number of reports on environmental scanning considered by the relevant subcommittee or the Energy Regulator within the stated time frame	1 considered annually by the REC by 30 April	1 considered annually by the REC by 30 April	-	-	
2. Reviewed Regulatory Reporting Manuals for Non-Financial and Financial Information	Energy Regulator decision on the Reviewed Regulatory Reporting Manuals for Non-Financial and Financial Information	Energy Regulator's decision on Reviewed Regulatory Reporting Manuals for Non-Financial and Financial Information (Vol 1) by 31 March 2025	-	-	-	Energy Regulator's decision on Reviewed Regulatory Reporting Manuals for Non-Financial and Financial Information (Vol 1) by 31 March 2025
3. Audit opinion that is not qualified	Unqualified audit opinion	Unqualified audit opinion	-	Unqualified audit opinion	-	-
4. One report on the implementation of the measures to improve NERSA's B-BBEE rating per financial year	Number of reports on the implementation of the measures to improve NERSA's B-BBEE rating per financial year considered by the relevant subcommittee or the Energy Regulator within the stated time frame	1 considered annually by the ER by 31 March	-	-	-	1 considered annually by the ER by 31 March

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
5. One report on the implementation of gender mainstreaming initiatives	Number of reports on the implementation of the gender mainstreaming plan considered by the relevant subcommittee within the stated time frame	1 considered annually by the HRRC by 31 March	-	-	-	1 considered annually by the HRRC by 31 March
6. One report on benchmarking of NERSA's operational processes controls against international standards	Number of reports on benchmarking of NERSA's operational processes controls against international standards considered by the relevant subcommittee within the stated time frame	1 considered by the REC by 31 March 2025	-	-	-	1 considered by the REC by 31 March 2025
7. One report on the implementation of the Hybrid Working Strategy	Number of reports on the implementation of the Hybrid Working Strategy considered by the relevant subcommittee within the stated time frame	1 considered by the REC by 31 March 2025	-	-	-	1 considered by the REC by 31 March 2025

6.2.4.2 Explanation of Planned Performance

The planned outputs are in support of NERSA's regulatory functions, as contained in relevant legislation.

6.2.4.3 Programme Resource Considerations

The budget for activities relating to the regulation of the energy industry is based on a ring-fencing methodology that was approved to comply with section 13 of the National Energy Regulator Act, 2004 (Act No. 40 of 2004). The methodology is based on direct employment cost as a basis of common costs apportionment. Direct costs are allocated directly to the respective industry.

The table below indicates the approved staff complement and the approved budget for 2024/25 for Programme 4: Operational Efficiency and Quality Management.

DIVISIONS	RELEVANT STRUCTURES	STAFF COMPLEMENT	BUDGET (R)	% ALLOCATION
Finance and Administration (CFO)	Chief Financial Officer	3	8 092 966	100%
	Financial Management and Governance (FMG)	7	12 700 843	100%
	Supply Chain Management, Facilities and Projects	12	24 371 264	100%
Corporate Services (COS)	Executive Manager	3	4 727 850	95%
	Legal Advisory Services (LAS)	6	30 957 007	100%
	Information Resources Management (IRM)	7	14 589 480	95%
Specialised Support Units (SSU)	Internal Audit (IAU)	7	13 384 966	100%
	Strategic Planning and Monitoring (SPM)	4	7 413 115	100%
	Regulator Support (RSU)	11	33 223 305	100%
	CEO's Office Operations (COO)	6	8 102 325	100%
	Regulatory Analysis and Research (RAR)	6	11 518 863	100%
	Information and Communication Technology (ICT)	10	39 984 493	80%

Note: The percentage allocation is based on the staff complement of the Organisation in line with the ring-fencing methodology.

Please refer to Part D: Funding for NERSA for the detailed budget.

6.2.4.4 Key Risks

Please refer to Section 6.2.6 below for NERSA's detailed Strategic Risk Register.

6.2.5. Programme 5: People and Organisational Culture

The programme aims to ensure a conducive work culture and human capacity balanced between specialised skills and generic skill requirements, as well as system development to deliver value to customer and stakeholder expectations.

6.2.5.1 Sub-programme: Human Resources and Capacity

The purpose of this sub programme is to facilitate the availability of appropriate skills, organisational culture and human capacity.

a) Outcomes, Outputs, Performance Indicators and Targets

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS							
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD			
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	
MTSF Priority 1: Capable, ethical and developmental state										
MTSF Priority 3: Education, skills and health										
Integrated and value-added services to customers	1. One report on Organisational Culture Assessment	Number of reports on Organisational Culture Assessment considered by the relevant committee or the Energy Regulator within the stated time frame	New Target	New Target	1 considered annually by the HRRC by 31 March	1 considered annually by the HRRC by 31 March	1 considered annually by the HRRC by 31 March	1 considered annually by the HRRC by 31 March	1 considered annually by the HRRC by 31 March	1 considered annually by the HRRC by 31 March

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS							
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD			
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	
MTSF Priority 1: Capable, ethical and developmental state										
MTSF Priority 3: Education, skills and health										
Integrated and value-added services to customers	2. Two reports on the implementation of the Employment Equity Plan	Number of reports on the implementation of the Employment Equity Plan considered by the relevant committee or the Energy Regulator within the stated time frame	2 considered bi-annually by the HRRC by 30 September and 31 March	2 considered bi-annually by the HRRC by 30 September and 31 March	2 considered bi-annually by the HRRC by 30 September and 31 March	2 considered bi-annually by the HRRC by 31 March	2 considered bi-annually by the HRRC by 31 July and 30 January	2 considered bi-annually by the HRRC by 31 July and 30 January	2 considered bi-annually by the HRRC by 31 July and 30 January	2 considered bi-annually by the HRRC by 31 July and 30 January
	3. 50% of women in management positions	Percentage of women in management positions	50%	50%	50%	50%	50%	50%	50%	50%
	4. 2% of people with disabilities employed	Percentage of people with disabilities employed	2%	2%	2%	2%	2%	2%	2%	2%

OUTCOMES	OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS						
			AUDITED PERFORMANCE			ESTIMATED PERFORMANCE	MTEF PERIOD		
			2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
MTSF Priority 1: Capable, ethical and developmental state									
MTSF Priority 3: Education, skills and health									
Integrated and value-added services to customers	5. Four reports on the implementation of the Youth Employment Accord	Number of reports on the implementation of the Youth Employment Accord considered by the relevant committee or the Energy Regulator within the stated time frame	4 considered quarterly by the HRRC	4 considered quarterly by the HRRC	4 considered quarterly by the HRRC	4 considered quarterly by the HRRC	4 considered quarterly by the HRRC	4 considered quarterly by the HRRC	4 considered quarterly by the HRRC
	6. One report on the implementation of the bursary programme for qualifying external applicants	Number of reports on the implementation of the bursary programme for qualifying external applicants considered by the relevant committee or the Energy Regulator within the stated time frame	1 considered annually by the HRRC by 31 March	1 considered annually by the HRRC by 31 March	1 considered annually by the HRRC by 31 March	1 considered annually by the HRRC by 31 March	1 considered annually by the HRRC by 31 March	1 considered annually by the HRRC by 31 March	1 considered annually by the HRRC by 31 March

b) Indicators, Annual and Quarterly Targets

OUTPUTS	OUTPUT INDICATORS	ANNUAL TARGETS	QUARTERLY TARGETS			
			Q1	Q2	Q3	Q4
1. One report on Organisational Culture Assessment considered annually by the HRRC by 31 March	Number of reports on Organisational Culture Assessment considered by the relevant committee or the Energy Regulator within the stated time frame	1 considered annually by the HRRC by 31 March	TORs and procurement for OCA	Roll out of OCA and produce report. Report approved by HRRC	Implement recommendations as per OCA	1 considered annually by the HRRC by 31 March
2. Two reports on the implementation of the Employment Equity Plan considered bi-annually by the HRRC by 30 September and 31 March	Number of reports on the implementation of the Employment Equity Plan considered by the relevant committee or the Energy Regulator within the stated time frame	2 considered bi-annually by the HRRC by 31 March	-	-	-	2 considered by the HRRC by 31 March
3. 50% of women in management positions	Percentage of women in management positions	50%	-	-	-	50%
4. 2% of people with disabilities employed	Percentage of people with disabilities employed	2%	-	-	-	2%
5. Four reports on the implementation of the Youth Employment Accord considered quarterly by the HRRC	Number of reports on the implementation of the Youth Employment Accord considered by the relevant committee or the Energy Regulator within the stated time frame	4 considered quarterly by the HRRC	1 considered by the HRRC	1 considered by the HRRC	1 considered by the HRRC	1 considered by the HRRC
6. One report on the implementation of the bursary programme for qualifying external applicants considered annually by the HRRC by 31 March	Number of reports on the implementation of the bursary programme for qualifying external applicants considered by the relevant committee or the Energy Regulator within the stated time frame	1 considered annually by the HRRC by 31 March	-	-	-	1 considered by the HRRC by 31 March

6.2.5.2 Explanation of Planned Performance

The planned outputs are in support of NERSA's regulatory functions, as contained in relevant legislation.

6.2.5.3 Programme Resource Considerations

The budget for activities relating to the regulation of the energy industry is based on a ring-fencing methodology that was approved to comply with section 13 of the National Energy Regulator Act, 2004 (Act No. 40 of 2004). The methodology is based on direct employment costs as a basis of common cost apportionment. Direct costs are allocated directly to the respective industry.

The table below indicates the approved staff complement and the approved 2024/25 budget for Programme 5: People and Organisational Culture.

DIVISIONS	RELEVANT STRUCTURES	STAFF COMPLEMENT	BUDGET (R)	% ALLOCATION
Human Resources (CHO)	Chief Human Capital Officer	2	4 407 114	100%
	Human Resources – Value Creation	8	26 811 994	100%
	Human Resources – Transactions	3	4 964 451	100%

Note: The % allocation is based on the staff complement of the organisation in line with the ring-fencing methodology.

Please refer to Part D: Funding for NERSA for the detailed budget.

6.2.5.4 Key Risks

Please refer to Section 6.2.6 below for NERSA's detailed Strategic Risk Register.

6.2.6. Strategic Risks

- 6.2.6.1 NERSA's approved Risk Management Policy requires that strategic risks are updated, reviewed and evaluated annually.
- 6.2.6.2 The approach on assessing risks is vital, as it assists in the customisation of the risk identification process. The ER approved the following risk identification approach techniques:
- a) Objective-by-objective approach – risks linked to each strategic objective are identified
 - b) Risk-based approach – a list of risks is generated from all relevant stakeholders, and then ranked in accordance to their priority
 - c) Process-based approach – each process is analysed to identify inherent risks
 - d) Control-based approach – internal controls are evaluated to identify inherent risks.
- 6.2.6.3 At its Strategic Risk Assessment Workshop on 31 October 2022, the ER resolved that the objective-by-objective approach be adopted in the identification of the proposed strategic risks for NERSA. The following strategic objectives were adopted:
- a) Facilitate entry, set prices and resolve disputes through efficient regulatory tools on an annual basis
 - b) Enable a stable and diverse energy sector system and pricing regime that supports access through regulatory services that are delivered on time (annually) and to quality standards
 - c) Review regulatory tools periodically to ensure that it is innovative in response to transformation of the energy industry
 - d) Review the Regulatory Framework periodically to ensure that it is relevant for effective regulation for the benefit of customers and stakeholders, through regulatory advocacy and stakeholder engagement
 - e) Enable provision of integrated and value-added services in an equitable/a fair manner to customers.
- 6.2.6.4 The following strategic risks were identified:
- a) Perceived regulatory irrelevance;
 - b) Rising energy costs - high energy prices and tariffs;
 - c) Damage to NERSA's reputation as perceived by its stakeholders;
 - d) Lack of growth, sustainability and transformation of energy industry;
 - e) Disruption in the functioning of the business;
 - f) Supply instability;
 - g) The probability of exposure or loss resulting from a cyber-attack or data breach in the organisation;
 - h) Limitations to implement NERSA's governing legislation; and
 - i) Inadequate maturity level of the organisational system, process and financial resources and skills and capacity of human resources.

6.2.6.5 The detailed strategic risk register is below.

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING	LIKELIHOOD RATING	INHERIT RISK
R1	Establish and position NERSA as a credible and reliable regulator	Regulatory	Perceived regulatory irrelevance	The shift in the markets such as evolving market conditions, the energy industry transformation and a change in the environmental requirements due to energy transformation poses a risk to the country. We also need to check if the current legal framework is fit for purpose. There is no clarity given to the market regarding the future of the industry. There is a proposal for the new Electricity Regulation Act	1. NERSA losing court cases due to: 1.1. NERSA regulatory tools such as methodologies and processes not being adhered to 1.2. Lack of proper document management and review 1.3. Sharing of confidential documents 2. Outdated regulatory tools and legislation and rules due to: 2.1. Not keeping up with changing environment 2.2. Gaps in regulatory legislation e.g. enforcement	1. Loss of credibility; established uncertainty in industry players and public 2. Decisions questionable and possible review of such (is like a circular reference/spiral) 2.1 Reduced sustainability of the organisation 2.2 Downsizing the organisation threatening the existence of the organisation 2.3 Job losses 3. Reputational damage 3.1 Non-optimal decisions/outcomes 3.2. Operational inefficiencies 4. Out of touch with customers' needs/expectations	5	4	20
							CRITICAL	LIKELY	EXTREME

CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
1. to 7. Regulate according to existing legislation 1.1 Nersa follows due process, including public consultation when making decisions. 1.1 & 2 Finalise review of Petroleum Pipelines tariff methodology 1.1. Gas Methodology approval - the maximum price of gas 1.2. Regulatory training and awareness 1.2. SharePoint document management system 1.3. Employment Contract addresses confidentiality 1.4. Disciplinary Code for breach of confidentiality 1.5 Environmental scan reports 2. Petroleum Storage and Loading methodology reviewed in 2021/22	Good	0,65	HIGH	16	EM:P PR EM: ELR EM: GAR SM: ICT EM: COS 1.1. Finalisation of Electricity Pricing Methodology 1.2, 1.3 & 4: Maintain current controls 2. Continue with effective and efficient implementation of existing controls 2.1 Regulatory advocacy to influence political will towards new technologies 2.1 Provide inputs to Draft Amendment bills and Policies 2.2, 2.3 Improved regulatory advocacy to differentiate between deregulation and unbundling of the electricity supply industry (ESI) 3.1 Review the SOP to strengthen the pre application process 3.2. Automation of the application process 4. Continue with effective and efficient implementation of existing controls	HOD: EPT HODs: EPT, GPT & PPT All HODs of ELR & GAR HODs: EPT, ELC &EIP SM: ICT All HODs of ELR & GAR and HOD: CSM	1.1 31 March 2024 1.2, 1.3 & 4: Quarterly monitoring of controls 2. Quarterly monitoring of controls 3.1 28 February 2024 3.2 31 March 2024 4. Quarterly monitoring of controls 7. 30 June 2025

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R1					2.3. Regulatory overlaps (Fragmentation of Regulations (Non-alignment of regulatory activities)) 2.4. Evolving market conditions 3. NERSA processes perceived to be a bottleneck in the development of the industry e.g. Inability to convene tribunal expeditiously 4. Inadequate engagement with stakeholders 5. Possible deregulation of the energy industry 6. Perceived irrelevance of the regulator 7. Outdated ICT infrastructure (hardware and software)	5. Consumers and interested and affected parties may be exploited and/or exposed to possible abuse by industry players 7. Lack of integrated ICT systems leading to incorrect and untimely decisions 8. Impact on NERSA's role 8.1 Decline in NERSA's revenue 9. Regulatory uncertainty						

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
2.1	Regular review of developments in the gas, petroleum pipeline and electricity industries, and the impact of those developments on the structure of, and competition within the industry. This serves to inform the Regulator in advance when changes to methodologies are required as a result, or where advocacy should be conducted to change legislation where required.					7. Implement approved ICT strategic projects 8. Monitor the implementation of the approved financial sustainability strategy and plan 1. to 9. Independent peer review of all regulatory tools to ensure that it is aligned with our mandate 1. to 9. Undertake an analysis of the reasons why legal challenges were successful 1. to 9. Improvement of quality of reasons for decisions (time management and quality assurance) 1. to 9. Improve document management linked to regulatory decisions		
2.1	Regular determination of the adequacy of competition in the gas industry to ensure that NERSA has a mandate to approve maximum prices of gas.							

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING	LIKELIHOOD RATING	INHERIT RISK
R1					8. Decline in volumes and more registrations than licensees				

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	2.2. & 2.3.& 5. Regulatory advocacy (Advocacy with the DMRE on measures to improve competition, including filling regulatory gaps and other stakeholders) 2.3. MoUs, Regulatory advocacy and stakeholder engagements. 3. Regulatory advocacy to influence policy/legislation 3.1 Reasons are published in respect of all decisions made. 3.2 Policies and Standard Operating Procedures 4. Regular engagement with stakeholders through customer/stakeholder education and workshops. 4.1 Frequently Asked Questions on Tariff Methodologies							

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R1												

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	4.2 Standard Operating Procedures 4.3. Public participation process 4.4. Organisational stakeholder engagement and advocacy strategy 4.5. Stakeholder survey 6. Regular review of the developments in the regulated industries and the impact of those developments on the structure of, and competition within the industry. This serves to inform the Regulator in advance when changed 7. Manual way of working 7.1 Silo systems							

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING	LIKELIHOOD RATING	INHERIT RISK
R2	Promote accessible and affordable energy for all citizens	Socio economic	Rising energy costs - high energy prices and tariffs	The risk is related to the socio-economic issues facing the country including the rise in electricity theft, vandalism of energy infrastructure amongst others. The economic risk of slow investment in infrastructure in the country as a result of slow economic growth and political risk as the country heads to the National Elections in 2024	1.1 Imprudent investments, Cost Overrun on Capital Projects and Inefficient operations by licensees 2. Change in technology (Migration by users) and substitution due to high prices 3. Inadequate maintenance, theft and vandalism of infrastructure resulting in dilapidated and unavailability of infrastructure 4. Inadequate regulatory tools not responsive to international developments	1., - 7 Collapse of the energy industry leading to economic decline. 1.1., 4., 5. Decline in energy service and increase in customer dissatisfaction 2., 3, Demand for product reducing. 2.1., 6. Inadequate price competition 2.2. Incorrect or suboptimal decisions 2.3. Inability to regulate effectively 3. Rise in the cost of energy due to the repairs of the damages and theft - this is passed through to the consumer 3.1 Poses safety risk to the environment and the surroundings	CRITICAL 5	LIKELY 4	EXTREME 20

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	1. Prudency Assessment and Prudence Guidelines approved in August 2018. 1. EPDM methodologies reviewed 1.1. & 4.2. Regular review of Methodologies. 2. Regulatory advocacy 3. Regulatory Tool for technical and non-technical losses and compliance enforcement for maintenance 3. Implementation plan for a formal approach to collaboration with licensees and law enforcement agencies 4. Advocacy with the DMRE on measures to improve competition, including filling regulatory gaps 4.1. Independent peer review of NERSA regulatory Tools	Good	0,65	HIGH	13 EM: ELR EM: GAR EM: PPR SM: RAR EM: COS	1. Regular review of EPDM methodologies 2., 4., 6, 7. Continue with effective and efficient implementation of existing controls 3. Collaboration with law enforcement agencies and community structures 3.1 Establishment of stakeholder forums (Regulator, law enforcement, licensees and community) to discuss mitigation measures and strategies to the challenges 4.1 Energy Data procurement 5.1 Establishment of the Tribunal for Enforcement of compliance 5.1 Review Terms of Reference for setting tariffs 5.2 Establishment of the Tribunal for penalties for non- compliance in Piped-Gas and Petroleum Pipelines Industries	HOD: EPT All HODs of ELR, GAR & PPR HOD: IRM All HODs of ELR All HODs of GAR & PPR	1. Ongoing 2, 6, 7. Quarterly monitoring of controls 4. Ongoing 5.1. As and when case identified 5.1 Align process with the organisational structure review 5.2 As and when case identified

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R2					5. Electricity funds cross- subsidize other municipal services 6. Lack of energy supply diversification and competition 7. Dependency on imports							

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	5. Ring fencing of bulk amount and enforcement of distribution licence conditions 5.1. Benchmark and guidelines for Municipal Tariff 5.2. Individual Municipal Tariff Reviews 5.3. Compliance Audits 5.4. Compliance Reporting various stakeholders including DMRE, NT, SALGA and COGTA 6. Conducting analysis of the impact of NERSA's decisions in the gas, petroleum pipeline and electricity industries on competition to ensure that NERSA's mandate to improve competition and transformation in the energy industries are considered in its decision-making processes							

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R2												

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	6.1. Continued advocacy and engagements with the Competition Commission to align and strengthen the two regulators' approaches to improve competition and transformation in the gas industry 6.2. Advocacy with the DMRE on measures to improve competition, including filling regulatory gaps 6.3. Enforcing third-party access to infrastructure to improve growth, competition and transformation in the gas industry and implementing measures identified to address such bottlenecks 6.4. Conducting inquiries to identify bottlenecks to competition, growth and transformation in the gas industry.							

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R2												

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	6.5. Licencing new entrants 6.6. Registrants and IPP 6.7. NERSA gives input in the formulation of the Integrated Resource Plan (IRP) by DMRE. This Plan ensures that the country's energy demand is met through diversified Energy Sources. Further to this, NERSA is also mandated to process section 34 Determinations. Concurrence to the Section 34 Determination by the Regulator is the mechanism that operationalises the IRP 7. Regulatory Advocacy							

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING	LIKELIHOOD RATING	INHERIT RISK
R3	Establish and position NERSA as a credible and reliable Regulator	Strategic	Damage to NERSA's reputation as perceived by its stakeholders	NERSA's reputation as perceived by its stakeholders is key to delivering on its mandate	1. Ineffective dissemination of information 2. Inadequate regulatory advocacy 3. Inability to reach communities in rural areas due to a lack of online platforms in rural areas 4. Poor stakeholder participation in consultation processes 5. Stakeholder s' perception that their views are not considered during decision- making 6. Outdated regulated tools (methodologies) 7. Prolonged legislation review process 8. Inadequate stakeholder management	1. Loss of trust and credibility 2. Reputational damage 3. Poor stakeholder participation in the consultation process	MAJOR 4	LIKELY 4	HIGH 20

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
1.	Regulatory Advocacy	Fair	0,80	13	EM: ELR EM: GAR EM: PPR EM: COS CHCO SM: CEO's office	1.- 3. Continue with effective and efficient implementation of existing controls 1. -3. Appointment of a service provider to assist with reputation management 1. Develop a framework for consequence management 1.3 Fast-track the amendment of legislation 2. Roll out of interventions as recommended by the ER member skills gap analysis. 2.1 Review Dual Career Path Policy 2.2 Review Secondment Policy (externally and internally) 2.3 Continuous improvement of NERSA's Code of Conduct Continuous improvement of HR policies, e.g. remuneration policy and succession policy 2.4 Implement the formalised organisational culture to ensure a conducive working environment	HOD: CSM Regulatory divisions	1. - 3. Ongoing 1. - 3. From 1 April 2024 for 12 months 1. 31 March 2024 2. 31 March 2024 3. 31 March 2024
1.1	Regular review of developments in the electricity and gas industries and the petroleum pipeline, and the impact of those developments on the structure of, and competition within the industry. This serves to inform the Regulator in advance when changes to methodologies are required as a result, or where advocacy should be conducted to change legislation where required			HIGH				
1.2	Roll out of the Skills Audit recommendations							
1.3	New competency framework							

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R3											EXTREME	20

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	2. Completion of ER members' skills gap analysis. 2.1 Recruiting individuals with adequate skills and experience to be able to conduct sound and improved analysis based on sound economic principles and international best practice 2.2 Regulatory training and awareness 2.3 Disciplinary policy and codes 2.4 Methodologies are reviewed periodically 3. Stakeholder Engagement and Advocacy Strategy 3.1. Stakeholder perception survey 3.2. Customer education 3.3 Communication Strategy 3.4 Regulatory advocacy			16 HIGH	EM: ELR EM: GAR EM: PPR EM: COS CHCO SM: CEO's office	2.5 Maintenance of the capacity building programme 2.6 Develop a framework to guide the information interaction between regulatory divisions and relevant support services 2.7 Review TBNS Policy and Implementation Plan 3. Review the Communication Strategy 3.1 Ensure that all decisions taken are compliant with approved procedures 3.2 Sensitise stakeholders and policymakers on the benefits of new technologies in the regulated industries 3.3 Portal on the website to indicate applications for tariffs/price and licensing/ registration		

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING	LIKELIHOOD RATING	INHERIT RISK
R4	Promote competition and competitiveness within the energy industry	Strategic	Lack of growth, sustainability and transformation of energy industry	The energy industry is stagnant to some extent with the lack of growth, sustainability and transformation that needs to be managed as a risk	- Insufficient infrastructure - Limited supply of energy resources - Gaps in legislation with regard to third party access - Potential impact of the Environmental, Social and Governance standards (ESG) and Carbon tax on the operations of the regulated energy industry - Change in environmental requirements due to energy transformation - Onerous requirements for new entrants - Lack of coordination amongst other authorities - Impact of vandalism on energy supply	1&7&8. Lack of competitive outcomes, including (i) lack of new entry and expansion; (ii) lack of bargaining power of customers due to limited number of suppliers; (iii) uncompetitive prices and tariffs; and (iv) lack of quality of service and products - Shortage of supply can drive up prices for resources - Low levels of 3rd party access persists 3.1 Perceived regulatory irrelevance - Negative macro-economic outcomes (rising inflation, lack of growth in the economy, increased poverty) 4.1. Inability to access energy due to high costs 5. Increase in energy costs	MAJOR 4	LIKELY 4	HIGH 16

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	1. Setting and approval of maximum prices and tariffs that incentivise investments in infrastructure that would facilitate increases in supplies 1.1 Enforcing third-party access to infrastructure to improve growth, competition and transformation in the energy industries 1.2. Regulatory advocacy 1.3. Licencing and registration of energy facilities 2. Research on industry developments. 2.1. Setting maximum prices and tariffs that incentivise investments in infrastructure that would facilitate increases in supplies	Fair	0,80	HIGH	13	EM: ELR E M: GAR EM: PPR SM: RAR 1. - 8. Continue with effective and efficient implementation of existing Controls 1. - 8. Identification of improvement strategies for existing controls 1. - 8. Intensify research on regulatory issues that fall within NERSA's mandate and publish papers on NERSA's website 1. - 8. Intensify stakeholder engagement with a focus on value creation to emphasise the value of regulation of the industries and the role of NERSA 1. - 8. Address the above in the strategy to profile NERSA appropriately – also highlight the impact of deregulation on the socio-economy of SA	All HODs in ELR, GAR & PPR SM: RAR All HODs in ELR, GAR & PPR SM: RAR	1-8. Quarterly monitoring of controls 1-8. Ongoing

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R4						6&7. Limited new entry and continued dominance of monopoly suppliers 7.1. Discourage and inhibit new investment. 8. Huge budgets spent on the repair and replacement of infrastructure 8.1 Delays in the implementation of projects reducing the performance of infrastructure						

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	2.2. Enforcing third-party access to infrastructure to improve growth, competition and transformation in the gas industry 2.3. Regulatory advocacy 3. Advocacy with the DMRE on measures to improve competition, including filling regulatory gaps 3.1. Established cooperative relationships with specific regulatory authorities through MoUs. 4. Regulatory advocacy 5. Environmental scan reports 6. Conducting inquiries to identify bottlenecks to entry, competition, growth and transformation in the gas industry					1. - 8. Assess the reasons for the call for deregulation – did we contribute to this call in the way we regulate – reduce the perception that NERSA is a bottleneck/red tape 1.- 8. Improve the process and frequency of conducting regulatory impact assessments 1.- 8. Clarify NERSA's role in regulating the energy industry 1.- 8. Monitor the operating environment 1.- 8. Monitor and analyse the volumes of activities of the licensees relevant for NERSA's funding 1.- 8. Monitor compliance with all relevant legislation and policies		

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R4												

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
7.	Continued advocacy and engagements with the Competition Commission to align and strengthen the two regulators' approaches to improve competition and transformation in the gas industry							
8.	Advocacy Plan was approved by ELS on 9 May 2023							

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK		
R5	Promote energy supply that is certain and secure for current and future user needs	ICT	Disruption in the functioning of the business	External factors such as natural disasters, pandemics and labour and civil unrest may disrupt the functioning of the business.	1. Inadequate Business Continuity Measures (Disasters - i.e. Epidemic, Labour and Civil Unrest) 2. System downtime due to hardware failures 3. Loss and/or corruption of data and records 4. Outdated technologies and tools (Out of warrant servers and storage) 5. Inadequate safe working procedures 6. "Working from home" (stolen computers, unauthorised access to information) 7. Inadequate back-up processes	1.- 3. Inability to carry out NERSA's mandate (delay in processing applications/taking decisions) 1.- 3. Loss of security of supply energy 1.- 3. Investments in infrastructure impacted because projects are put on hold 1.- 3. Operations of the Regulator and licensees are negatively affected 1.- 3. increase in the cost of energy 3. Need to recreate institutional knowledge. 4. Organisational resilience compromised to function optimally during an event that disrupts operations 4.1. Poor computer performance	CRITICAL	5	POSSIBLE	3	HIGH	15	

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	1.1 Business Continuity Management Policy; BCM Strategy; BCM Plans 1.2. Disaster Recovery Plan 1.3. Generators and UPS 1.4. Conduct Disaster Recovery Test- The Disaster Recovery Test was conducted from 14 February 2023 to 28 February 2023 1.5. Additional site identified for business continuity 2.&4. Procure solutions with maintenance agreements in place 3.1 SharePoint document management system 3.2 Backups implemented 5.& 6. End user policy 5.1. Health and Safety Reps, Policy and Committee 5.2. Health and Safety Surveys 5.3.&6.1 NERSA security policy (Data in use, Assets etc)	Fair	0,80	HIGH	12	SM: SPM SM: ICT EM: COS CHCO EM: COS 1.1. Review the Governance documents for BCM 1.2. Consider the impact of natural disasters and pandemics on regulatory activities 1.3. Adapt regulatory processes to ensure that the regulated industries can develop as normal as possible 2. Replacement of outdated hardware and ensure warrant and service level agreement with service providers 3. & 4. Continue with effective and efficient implementation of existing Controls 5.- 5.3. Review of policy documentation and align with new normal strategy 6. Finalise New Normal strategy 6.1. Procure storage facilities for working at home 7. Enhance security features and ensure email is accessed through VPN	SM: SPM SM: ICT HOD: IRM & SM: ICT SM: ICT, HR HODs & HOD: IRM SM: SPM HOD: IRM SM: ICT	1.1. 30 September 2024 2. Ongoing 3&4. Quarterly monitoring of controls 5.- 5.3. Six months post- approval of the New Normal Strategy (Approximately June 2024) 6. 31 March 2024 6.1. 31 March 2024 7. 30 September 2024

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING	LIKELIHOOD RATING	INHERIT RISK	
R5						5. Safety of staff compromised which could lead to financial claims against NERSA. 6. Compromised data confidentiality, integrity and availability. 6.1. Exposure to security vulnerabilities. 6.2. Financial losses for NERSA/injury/loss of assets 6.3 ICT security policies 7. Loss of data and critical information. 7.3 Offsite records storage facility				

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	6.2. Operational procedures for Working from Home 7.1 Laptops are backed up with Cibecs. The server environment is backed up with Veem software. 7.2 Completed and continued awareness creation regarding the link between the Cibecs backup solution that is dependent on a consistent VPN connection.							

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING	LIKELIHOOD RATING	INHERIT RISK
R6	Create a regulatory environment that facilitates investment in energy infrastructure	Security of Supply	Supply instability	Geopolitical factors such as the Russia/ Ukraine war and Israeli- Palestinian conflict are posing a security supply risk to South Africa. We need to change or shift the way we regulate the sector to ameliorate this risk	1. Lack of sufficient local energy resources resulting in heavy import reliance 2. Lack of transformation and growth of industry 3. Poor performance of Eskom generation fleets (failure of IPP to come on line) 4. Closure of petroleum refineries 5. Lack of adequate investment into energy infrastructure 6. Lack of diversification of energy supply impacting the energy mix 7. Vandalism to infrastructure 8. Lack of decisive policy direction	1. Shortage of energy products 1 & 2. Lack of competitive outcomes, including (i) lack of new entry and expansion due to limited supply; (ii) lack of bargaining power of customers due to limited number of suppliers; (iii) uncompetitive prices and tariffs; and (iv) lack of quality of service and products; (v) lack of growth and transformation in the energy sector 2. Lack of new entrants, growth and diversification of the energy mix 3 & 4. Increase exposure to supply disruptions of specific feedstocks 5. Job losses 5.1 Ageing assets 6. Increase in regulatory uncertainty	MAJOR 4	LIKELY 4	HIGH 16

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	1. Interaction with neighbouring countries to explore opportunities for supply. 2. Analyzing the impact of NERSA's decisions in the energy industries on competition to ensure that NERSA's mandate to improve competition and transformation in the energy industries are considered in its decision-making processes 2.1&4. Continued advocacy and engagements with the Competition Commission to align and strengthen the two regulators' approaches to improve competition and transformation in the gas industry	Good	0,65	MODERATE	10	EM: ELR EM: GAR EM: PPR 1. - 6. and 8.- 10. Continue with effective and efficient implementation of existing controls 7. Implementation plan for formal approach for Collaboration with Licencees and Law Enforcement Agencies 1. to 11. increased engagement with the demand side of the ESI 2. Develop methodologies to encourage alternative supply of gas 2.1 Regulatory advocacy to encourage the government to secure gas supplies from other countries and to address the closure of the refineries 6. Develop performance standards for licensees in the three regulated energy industries	SM: SPM SM: ICT SM: ICT HOD: IRM & SM: ICT SM: ICT, HR HODs & HOD: IRM SM: SPM HOD: IRM SM: ICT	1. - 6., 8-10. Quarterly monitoring of controls

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R6					9. Geopolitical factors 10. Lack of regional integration 11. Inadequate methodologies to respond to the changing environment Developments in the international markets which have a direct impact on SA prices and may have unintended consequences - i.e. impact of war between Russia and Ukraine, international production constraints amongst others	7. Disruptions/downtimes, repair costs and environmental damage (spills) 8. Rolling blackouts (a more advanced form of load shedding) 8.1 Collapse of the grid 9. The collapse of the utility, civil strife and protests. 9. Loss of stature as an investment destination. 10. Further fragmentation and loss of comparative advantage that an interconnected regional energy infrastructure can provide. 11. Inadequate import infrastructure 11.1 Increase in OCGT prices (due to the increase of diesel prices)						

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	2.2.&4. Advocacy with the DMRE on measures to improve competition, including filling regulatory gaps 2.3. Enforcing third-party access to infrastructure to improve growth, competition and transformation in the gas industry 2.4. Conducting inquiries to identify bottlenecks to competition, growth and transformation in the gas industry and implementing measures identified to address such bottlenecks. 3. Enforcement of licence conditions 5. Licencing and registration of new entrants 5.1 Approving prices and tariffs that incentivise investment					2.2 Delink the NERSA's methodologies to international prices 2.3 Where licensees purchase LNG from an international market at an 'arm's length' transaction, NERSA must implement methodologies which ensures the licensee fully recovers all its costs 1. - 11. Emulating prices that would persist in competitive markets 1. - 11. Facilitating the establishment of conditions for effective competition in the relevant trading markets in the gas industry. 1. - 11. Conduct regulatory advocacy and utilise methodologies that will encourage the development of national supplies		

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R6												

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
6.	Analyzing the impact of NERSA's decisions in the gas industry on competition to ensure that NERSA's mandate to improve competition and transformation in the gas industry is taken into consideration in its decision- making processes					1. - 11. Annual review and testing of business continuity plans		
6.1.	Continued advocacy and engagements with the Competition Commission to align and strengthen the two regulators' approaches to improve competition and transformation in the gas industry					1. - 11. Consider the impact of natural disasters and pandemics on regulatory activities		
						1. - 11. Adapt regulatory processes to ensure that the regulated industries can develop as normal as possible		
6.2.	Advocacy with the DMRE on measures to improve competition, including filling regulatory gaps					1. - 11. Development of service quality incentives (SQI) related to the energy availability factor (EAF)		

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R6												

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	6.3. Enforcing third-party access to infrastructure to improve growth, competition and transformation in the gas industry 6.4. Conducting inquiries to identify bottlenecks to competition, growth and transformation in the gas industry. 6.5. Licencing new entrants 6.6. Registrants and IPPs 6.7. NERSA gives input in the formulation of the Integrated Resource Plan (IRP) by DMRE. This Plan ensures that the country's energy demand is met through diversified Energy Sources. Further to this, NERSA is also mandated to process section 34 Determinations.					1. - 11. Timeously develop regulatory tools to facilitate the implementation of Government's interventions to alleviate load shedding i.e. Net billing, Demand Response Programmes and Participation by private generators 1. - 11. Review Demand response tariff to incentivise participation 1. - 11. Improve licensing and registration processes 1. - 11. Develop regulatory tools which will allow the entry of new players especially for the generation		

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R6												

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	Concurrence to the Section 34 Determination by the Regulator is the mechanism that operationalises the IRP 7. Regulatory Tool for technical and non-technical losses 8. Regulatory advocacy 8,9. & 10. Current platforms such as RERA and Gas Commission are used to interact with neighbouring countries within SADEC to discuss matters of common interest and advance regional integration. 11. Develop and review methodologies to respond to the changing environment							

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING	LIKELIHOOD RATING	INHERIT RISK
R7	Establish and position NERSA as a credible and reliable regulator	ICT	The probability of exposure or loss resulting from a cyber attack or data breach in the organisation	This risk covers the issue of cybersecurity, the changes in technology and the emerging global security vulnerabilities	1. Emerging global security vulnerabilities (Cyber threats, ransomwares, hacking) 2. Technology not keeping abreast with the fast-emerging cyber threat 3. Lack of policy enforcement 4. Users not security conscious 5. Inadequate cyber security measures (No multi-factor authentication)	1. Unauthorised access to information leading to loss of information or compromised integrity and confidentiality 2. Having to pay after suffering ransomware attack. 3. Downtime, disrupting business processes. 4. Destruction or corruption of data 5. Lack of data integrity and operational inefficiencies.	CRITICAL 5	POSSIBLE 3	HIGH 15

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	1. Deployed Perimeter Security (Firewall, Virtual Private Network, Email Gateway, Anti-virus etc.). Checks are conducted quarterly. Internal audit reports indicate the effectiveness of the controls. 2. Continuously upgrading operating systems. Checks are conducted quarterly. Internal audit reports indicate the effectiveness of the controls. 3. Enforcement automatic by way of security tools through firewall and antivirus. Checks are conducted quarterly. Internal audit reports indicate the effectiveness of the controls.	Good	0,65	MODERATE	SM: ICT	1. Enhance security features and ensure email is accessed through VPN 2-4 Continue with effective and efficient implementation of existing Controls 5.1 Appoint panel of cyber security experts 5.2. Development of Implementation plan for Data Loss Prevention 5.3. Deploy multi-factor authentication 5.4. Improve user awareness campaign	SM: Contracts	1. Ongoing 2.-4. Quarterly monitoring of controls 5.1. Ongoing 5.2. 31 December 2023 5.3. 31 December 2023 5.4. Ongoing

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R7												

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
3.1.	Network Penetration Testing. Checks are conducted quarterly Internal audit reports indicate the effectiveness of the controls.							
4.	Continuous awareness campaigns. Checks are conducted quarterly. Internal audit reports indicate the effectiveness of the controls.							
5.	No current control							

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK		
R8	Promote regulatory certainty within the energy industry	Service Delivery	Limitations to implement NERSA's	The shift in the markets such as evolving market conditions, the energy industry transformation and a change in the environmental requirements due to energy transformation poses a risk to the country	1. Litigious environment 2. Energy Industry Transformation 3. New technology developments not addressed by legislation 4. Misalignment in policy direction 5. Inconsistent regulatory decisions and implementation of regulatory tools 6. Inefficient regulatory tools 7. Inefficient collaboration with other regulators 8. Regulatory Framework unresponsive to customer and stakeholder needs	1. Consumes resources to challenge decisions 2. Falling behind on emission targets 3. Unable to meet demands on changing environment. 4. Taking decisions not aligned with Government policy 5. Creates uncertainty in industry 6. Improper decisions 6.1. Lose opportunity to benchmark/do peer reviews 6.2. Not able to operate optimally 7. Not meeting expectations of customers/ stakeholders. 8. Create uncertainty in industry 8.1 Failure to ensure orderly development of the industry	CRITICAL	5	POSSIBLE	3	HIGH	15	

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
1.1	Analyzing the impact of NERSA's decisions in the energy industries on competition to ensure that NERSA's mandate to improve competition and transformation in the industries are considered in its decision- making processes	Good	0,65	7.8	EM: GAR EM: ELR EM: PPR SM: RAR	1. - 4. & 7. - 8. Continue with effective and efficient implementation of existing controls 5.1. Review of Electricity Pricing Methodology 6. Ensure and monitor compliance with the government's positions 7. Maintain and/or improve cooperation with local and international regulators	HOD: GCM HOD: HRT HOD: PPT HOD: GPT HOD: GLC HOD: ELR	1. - 4. & 7. -8.: Quarterly monitoring of controls 5.1. 31 March 2025
1.1.	Recruiting individuals with adequate skills and experience to be able to conduct sound and improved analysis based on sound economic principles and international best practices			MODERATE				

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R8												

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	1.3. Regular determination of the adequacy of competition in the gas industry to ensure that NERSA has a mandate to approve maximum prices of gas (if this is not done, NERSA will face litigation). 2. Licencing Conditions 2.1. Compliance monitoring 3., 5.&6., 8 Advocacy with the DMRE on measures to improve competition, including filling regulatory gaps 4. Regulatory Advocacy 5.1. Independent peer review of NERSA regulatory Tools 5.2. Gas Methodology implementation and monitoring - the maximum price of gas 5. & 6. Finalise review of Petroleum Pipelines tariff methodology							

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R8												

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	7. Continued advocacy and engagements with the Competition Commission 8.1. Conducting inquiries to identify bottlenecks (which may include regulatory gaps) to competition, growth and transformation in the gas industry. 8.2. Ex-post Regulatory Impact Assessment of NERSA decisions							

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK		
R9	Promote energy supply that is certain and secure for current and future user needs	Strategic	Inadequate maturity level of the organisational systems, processes and financial resources and skills and capacity of human resources	This risk follows from scarce skills that were identified as a weakness. Employees need cutting-edge training opportunities	1. Inability for NERSA to fully fund its operations due to declining industry volumes in supply of energy (declining economic status, Inflation) 2. Inadequate cash flow 3. Outdated Regulator methodologies and legislation 4. Inadequate skills-base within NERSA - not aligned to changing energy industry 5. Rise in demand for scarce skills within NERSA 6. Technological advancements - need for reskilling. 7. Weakening value proposition, uncompetitive salaries compared to industry	1&2. Budget constraints/ sustainability 3. Proper/optimal decisions not possible 4. Decisions questionable and possible review of such 5 & 9. Unable to meet demands on changing environment. 6. Inflexible skills based that is unresponsive to future needs 7. Loss of talent to competitor organisations 8. Loss of opportunities for skills development which will impact negatively on regulation 9. Operational inefficiencies 10. Loss of exposure to practical national, regional and international best practices to enhance effective regulation	MODERATE	3	POSSIBLE	3	MODERATE	9	

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
1.	Three-year volume projections are provided by licensees during the annual budget process which is useful for monitoring risks associated with volumes.	Good	0,65	MODERATE	6	1.2. Implementation of the Financial Sustainability Strategy	CFO HOD: FMG HOD: EPT CHCO/ SM: RSU HOD: HRT/H RV ICT All HODs and SMs	1.1. Annually 31 March
1.1.	Implementation of the approved NERSA financial sustainability strategy and implementation plan with the aim to preserve current revenue streams and identify additional sources of funds - implemented in 1st quarter - Budgets submitted to DMRE on 31 August - implemented in 2nd quarter - Implemented a process to confirm the budget for each expenditure request. - Quarterly reports considered by the FIC and ER					2. Monitor actual spending to remain within budget and revenue collection		1.2. Annually 30 June
						3.2 Implementation of Electricity Pricing Methodology 4, 5 & 6: Roll out of interventions as recommended by the ER member skills gap analysis.		2. Quarterly
						5 & 6 Enhancement of the learnership and internship programme to ensure an adequate supply of skills and employability by extending the contract term.		3.2 Ongoing
						5 & 6 Finalisation of the development of regulatory programmes at academic institutions		4,5&6: Ongoing
								5&6: Ongoing
								7. Quarterly monitoring of controls
								8. 31 March 2024
								9. Quarterly monitoring of controls

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R9					8. Lack of process automation 9. Inadequate and lack of policies and procedures to meet changing legislation	11. NERSA's relevancy can be compromised						

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
2.	Expenditure management to be within budget. Debtors' days are managed within 30 days. NERSA has a cash mitigation reserve fund and investments of surplus funds with the SARB Corporation for Public Deposits.					7. Continue with effective and efficient implementation of existing controls		
3.	Regular review of developments in the gas industry and Petroleum pipeline, and the impact of those developments on the structure of, and competition within the industry. This serves to inform the Regulator in advance when changes to methodologies are required as a result, or where advocacy should be conducted to change legislation where required					7.1 Implementation and monitoring engagement strategy (capacity building, development or enhancement of regulatory tools, participation in specific projects)		
3.1	Implementation on the revised Petroleum Pipelines tariff methodology					8. Automation of the business process		
3.3.	Implementation of the Gas Methodology - the maximum price of gas					9. Continue with effective and efficient implementation of existing controls		
						10. Ensure approvals of partnership collaborations at national, regional and international levels		

RANKING	STRATEGIC OBJECTIVES	RISK CATEGORY	RISK DESCRIPTION	EXPLANATORY NOTES	CAUSES OF THE RISK (BACKGROUND)	CONSEQUENCES	IMPACT RATING		LIKELIHOOD RATING		INHERIT RISK	
R9												

	CURRENT CONTROLS	CONTROL EFFECTIVENESS	CONTROL EFFECTIVENESS RATING	RESIDUAL RISK	RISK OWNER	RISK RESPONSE STRATEGIES (ACTION PLANS)	ACTION OWNER	TIME SCALE
	4, 5&6 Roll out of the Skills Audit recommendations. 4, 5&6 New competency framework. 4, 5&6 Completion of ER members skills gap analysis. 4, 5, 6 and 9 Implementation of the Workplace Skills Plan and the Organisational Training Plan 5 & 6. Learnership and internship programme. 5 & 6. External Bursary Programme 7. Engage employees on value proposition and salary benchmark 8. Manual Business Processes 9. Policy reviews and Workshops of users on policies, procedures and processes							

Part D: Funding for NERSA

NERSA's approved budget is attached as Annexure A.

Part E: Technical Indicator Descriptions

1. Performance Indicators

These indicators are divided in the following programmes:

- a) **Programme 1: Regulatory Service Delivery**
- b) **Programme 2: Advocacy And Engagement**
- c) **Programme 3: Innovation**
- d) **Programme 4: Operational Efficiency and Quality Management**
- e) **Programme 5: People and Organisational Culture**



1.1. PROGRAMME 1: REGULATORY SERVICE DELIVERY

1.1.1. Subprogramme: Electricity Industry Regulation

1.1.1.1. Setting and/or approval of tariffs and prices

Indicator title	1. Energy Regulator decision on Eskom and municipal electricity prices within the stated timeframe	2. Energy Regulator decision on municipal electricity prices within the stated time frame
Definition	This is the decision of the Energy Regulator on Eskom and municipal electricity prices within the stated timeframe..	This is the decision of the Energy Regulator on municipal electricity prices within the stated time frame
Source of data	Tariff Applications and D Forms; Tariff analysis schedules	Tariff applications and D Forms; tariff analysis schedules
Method of calculation/ assessment	Energy Regulator decision per annum	Energy Regulator decision per annum
Means of verification	Applications; Reasons for Decisions; Minutes of REC and ELS meetings	Applications, Reasons for Decisions, minutes of REC and ELS meetings
Assumptions	Complete applications received from licensees	Complete applications received from licensees
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation type	Non-cumulative	Non-cumulative
Reporting cycle	Annual	Annual
Desired performance	Regulator decision by 28 February 2025	Energy Regulator decision by 15 March 2025
Indicator responsibility	EM (ELR) and HOD (EPT)	EM (ELR) and HOD (EPT)

1.1.1.1. Setting and/or approval of tariffs and prices

Indicator title	3. Energy Regulator decision on the improved pricing system for the electricity industry within the stated time frame	4. Number of reports on the implementation of the Regulatory Reporting Manuals regarding the Standard Chart of Accounts (SCOA) for municipalities considered by the relevant committee or the Energy Regulator within the stated time frame
Definition	This is the decision of the Energy Regulator improved pricing system for the electricity industry within the stated time frame	This is the number of progress reports on the implementation of the Regulatory Reporting Manuals regarding the Standard Chart of Accounts (SCOA) for the municipalities considered by the relevant subcommittee
Source of data	Pricing system	Analysis of the implementation of the Regulatory Reporting Manuals regarding the Standard Chart of Accounts (SCOA) for the municipalities
Method of calculation/ assessment	Energy Regulator decision per annum	Number of reports
Means of verification	Reasons for Decisions, minutes of REC and ELS meetings	Submissions to REC; Minutes of REC
Assumptions	Complete applications received from licensees	Analysis completed
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation type	Non-cumulative	Cumulative
Reporting cycle	Annual	Annual
Desired performance	Energy Regulator decision by 15 March 2025	One progress report by 31 March
Indicator responsibility	EM (ELR) and HOD (EPT)	SM (RAR)

1.1.1.2. Licensing and Registration

Indicator title	1. Energy Regulator decision on improved licencing processes within stated time frame	2. Percentage of complete licence applications considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Definition	This is the decision of the Energy Regulator improved licencing processes within the stated time frame	This is the percentage of complete licence applications that are considered by the Regulator Executive Committee or the Electricity Subcommittee (depending
Source of data	Licencing processes	Licensing processes, applications for licences
Method of calculation/ assessment	Energy Regulator decision per annum	Number of working days from receipt of a complete application to decision by the Energy Regulator
Means of verification	Reasons for Decisions, minutes of REC and/or ELS meetings	Reviewed processes, applications, Reasons for Decisions, minutes of ELS and/or ER meetings
Assumptions	Complete information received from licensees	Complete application received
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation type	Non-cumulative	Non-cumulative
Reporting cycle	Annual	Quarterly
Desired performance	Regulator decision by 30 December 2024	90% within 110 working days
Indicator responsibility	EM (ELR) and HOD (ELC)	EM (ELR) and HOD (ELC)

1.1.1.2. Licensing and Registration

Indicator title	3. Percentage of complete applications for the registration of electricity generation facilities considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Definition	This is the percentage of complete applications for registration of electricity generation activities that are considered by the Regulator Executive Committee or the Electricity Subcommittee (depending on delegation) and submitted to the Energy Regulator for a final decision, in compliance with the legislated timeframes
Source of data	Registration processes, applications for registration
Method of calculation/ assessment	Number of working days from receipt of a complete application to decision by the Energy Regulator
Means of verification	Applications, Reasons for Decisions, minutes of ELS and ER meetings
Assumptions	Complete application received
Disaggregation of beneficiaries (where applicable)	Not applicable
Spatial transformation (where applicable)	Not applicable
Calculation type	Non-cumulative
Reporting cycle	Quarterly
Desired performance	90% within 45 working days
Indicator responsibility	EM (ELR) and HOD (ELC)

1.1.1.3. Compliance monitoring and enforcement

Indicator title	1. Energy Regulator decision on improved compliance audit processes within the stated time frame
Definition	This is the decision of the Energy Regulator improved compliance audit processes within the stated time frame
Source of data	Compliance audit reports
Method of calculation / assessment	Energy Regulator decision per annum
Means of verification	Reasons for Decisions, minutes of REC and/or ELS meetings
Assumptions	Audits completed and reports compiled as planned
Disaggregation of beneficiaries (where applicable)	Not applicable
Spatial transformation (where applicable)	Not applicable
Calculation Type	Non-cumulative
Reporting cycle	Annual
Desired performance	Regulator decision by 30 December 2024
Indicator Responsibility	EM (ELR) and HOD (ELC)

1.1.1.3. Compliance monitoring and enforcement

Indicator title	2. Percentage variance of planned versus actual compliance audit plans executed	3. Number of analysis reports on audits conducted within the stated time frame
Definition	This is the percentage variance between the planned compliance audits completed in line with the annual Compliance Audit Plan and the number of actual compliance audits conducted	This is the number of quarterly reports on actual compliance audits conducted in line with the annual Compliance Audit Plan
Source of data	Compliance Audit Plan, list of actual audits conducted	Compliance Audit Plan, audit reports
Method of calculation/ assessment	$(\text{Number of actual audits conducted}) / (\text{number of planned audits}) * 100$	Number of quarterly reports on actual compliance audits conducted per annum
Means of verification	Minutes of ELS meeting	Minutes of ELS meeting
Assumptions	Audits conducted as planned	Reports on completed compliance audits conducted
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation type	Cumulative	Cumulative
Reporting cycle	Quarterly	Quarterly
Desired performance	80% of planned audits conducted	4 quarterly reports
Indicator responsibility	EM (ELR) and HOD (ELC)	EM (ELR) and HOD (ELC)

1.1.1.3. Compliance monitoring and enforcement

Indicator title	4. Percentage variance of planned versus actual enforcement plan executed	5. Number of reports on non-compliance findings compiled within the stated time frame
Definition	This is the percentage variance between the planned enforcement actions on non-compliance in line with the annual Enforcement Plan and the number of actual enforcement actions taken	This is the number of quarterly reports on actual enforcement actions taken in line with the annual Enforcement Plan.
Source of data	Enforcement Plan, compliance findings	Enforcement Plan, compliance findings
Method of calculation/ assessment	$(\text{Number of actual enforcement actions on non-compliance}) / (\text{number of planned enforcement actions}) * 100$	Number of reports on actual enforcement actions taken per quarter
Means of verification	Minutes of ELS and ER meetings	Minutes of ELS and ER meetings
Assumptions	Enforcement actions conducted as planned	Audits conducted as planned, reports completed
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation type	Cumulative	Cumulative
Reporting cycle	Quarterly	Quarterly
Desired performance	80% planned enforcement actions on non-compliance conducted	4 quarterly reports
Indicator responsibility	EM (ELR) and HOD (ELC)	EM (ELR) and HOD (ELC)

1.1.1.3. Compliance monitoring and enforcement

Indicator title	6. Number of reports on audit findings on the review of the annual performance of IDM to be considered by the relevant subcommittee or the Energy Regulator within the stated time frame	7. One report on audit findings on the Transmission Network Development projects for compliance with the South African Grid Code considered annually by the relevant subcommittee or the Energy Regulator within the stated time frame
Definition	This is a number of reports of the IDM performance based on Eskom's breakdown of all IDM programmes/ technologies with their estimated costs, demand and energy savings that was submitted to the Energy Regulator with the MYPD application.	These are reports regarding the audits conducted on projects included in Eskom's approved Transmission Development Plan in order to evaluate the compliance of these projects with the approved Grid Code
Source of data	Approved audit report that details the assessment and evaluation of IDM for compliance with the South African Grid Code	Approved audit report that details the assessment and evaluation of projects in Eskom's approved Transmission Development Plan for compliance with the South African Grid Code
Method of calculation/ assessment	Number of reports within 6 months after receipt of Eskom's IDM Annual Performance Report.	Number of reports
Means of verification	Minutes of ELS and ER meetings	Submissions to ELS/REC; minutes of ELS/REC
Assumptions	Audits conducted as planned	Audits completed as planned
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation type	Non-cumulative	Cumulative
Reporting cycle	Annual	Annual
Desired performance	1 report within 6 months after receipt of Eskom's IDM Annual Performance Report.	1 report by 31 March
Indicator responsibility	EM (ELR) and HOD (EIP)	EM (ELR) and HOD (EIP)

1.1.1.3. Compliance monitoring and enforcement

Indicator title	8. Number of reports on the Distribution Network Development projects for compliance with the South African Grid considered annually by the relevant subcommittee or the Energy Regulator, within the stated time frame, subject to all information being available	9. Two monitoring reports on the performance and progress of Renewable Energy projects for considered bi-annually by the relevant subcommittee or the Energy Regulator within the stated time frame, subject to all information being available
Definition	These are reports regarding the audits conducted on projects included in Eskom's approved Distribution Development Plan in order to evaluate the compliance of these projects with the approved Grid Code	These are monitoring reports on the performance of and progress made with renewable energy aimed at informing all stakeholders and decision makers on the status.
Source of data	Approved audit report that details the assessment and evaluation of projects in Eskom's approved Distribution Development Plan for compliance with the South African Grid Code	Reports on the performance and progress of Renewable Energy
Method of calculation/ assessment	Number of audit reports per year	Number of reports per year
Means of verification	Submissions to ELS/REC; minutes of ELS/REC	Submissions to ELS/REC; minutes of ELS/REC
Assumptions	Audits completed as planned	These are monitoring reports on the performance of and progress made with renewable energy aimed at informing all stakeholders and decision makers on the status.
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation type	Cumulative	Cumulative
Reporting cycle	Annual	Bi-annually
Desired performance	One audit report on the Distribution Network Development projects for compliance with the South African Grid considered by the ELS/REC	Two monitoring reports considered annually by the ELS/REC by 30 September and 31 March respectively
Indicator responsibility	EM (ELR) and HOD (EIP)	EM (ELR) and HOD (EIP)

1.1.1.4. Dispute resolution, including mediation, arbitration and handling of complaints

Indicator title	1. % of categorised disputes/ complaints closed in line with the approved Investigations Framework and Process for complaints and dispute resolution
Definition	This is the percentage of categorised complaints/disputes, which includes initiated investigations, that are closed within in line with the approved Complaint/ Dispute Resolution/Investigation Framework and Process
Source of data	Approved Complaint/Dispute Resolution/ Investigations Framework and Process, records of disputes/complaints received
Method of calculation/ assessment	$(\text{Number of closed categorised disputes and/or complaints within 180 working days of receipt}) / (\text{number of received categorised complaints and/or disputes}) * 100$
Means of verification	Database of all complaints/disputes received and closed
Assumptions	Complete information is received from complainants
Disaggregation of beneficiaries (where applicable)	Not applicable
Spatial transformation (where applicable)	Not applicable
Calculation type	Non-cumulative
Reporting cycle	Quarterly
Desired performance	90% of categorised disputes/complaints, including initiated investigations closed within the agreed upon turnaround time
Indicator responsibility	EM (ELR) and HOD (ELC)

1.1.1.5. Setting of rules, guidelines and codes for the regulation of the electricity industry

Indicator title	1. Percentage variance between planned versus actual targeted tools and systems reviewed and developed	2. 100% of draft Section 34 Determinations received, with complete information, considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Definition	This is the percentage of planned review and/or development of targeted regulatory tools completed in line with an annual plan for the reviews and/or developments to be concluded in the planning period	This is the percentage of draft Section 34 Determinations received, with complete information, considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Source of data	Analysis reports of reviewed tools and systems	Analysis reports of Section 34 Determinations received
Method of calculation/ assessment	(Number of actual reviews and/or development of targeted regulatory tools)/(planned number of reviews and/or development of targeted regulatory tools)*100	(Number of draft Section 34 Determinations received /(planned number of draft Section 34 Determinations received)*100
Means of verification	Minutes of ELS and ER meetings	Minutes of ELS and ER meetings
Assumptions	Reviews and/or developments concluded as planned	Complete information received
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation type	Cumulative	Non-Cumulative
Reporting cycle	Quarterly	Quarterly
Desired performance	80% of planned reviews and/or development of targeted regulatory tools conducted	100% within 4 months
Indicator responsibility	EM (ELR) and HOD (EIP)	EM (ELR) and HOD (EIP)

1.1.1.5. Setting of rules, guidelines and codes for the regulation of the electricity industry

Indicator title	3. Percentage of complete applications <u>requiring exemption</u> from the South African Grid Code considered by the relevant subcommittee or the Energy Regulator within the stated time frame	4. Percentage of complete applications <u>requiring amendment</u> to the South African Grid Code considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Definition	This is the percentage of decisions taken regarding applications for exemption from the Grid Code made by the relevant Subcommittee within 60 days from receipt of application	This is the percentage of decisions taken regarding applications for amendment of the Grid Code made by the relevant Subcommittee within 60 days from receipt of application
Source of data	Applications for <u>exemptions</u> of the grid code	Applications for <u>amendments</u> to the grid code
Method of calculation/ assessment	(number of applications requiring <u>exemptions</u> completed within 60 days / number of applications for exemptions received)*100	(Number of applications requiring <u>amendments</u> completed within 60 days / number of applications for <u>amendments received</u>)*100
Means of verification	Applications; recommendations from the Grid Code Advisory Committee	Applications; recommendations from the Grid Code Advisory Committee
Assumptions	Recommendations from Grid Code Advisory Committee submitted with all required supporting documents	Recommendations from Grid Code Advisory Committee submitted with all required supporting documents
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation type	Non-cumulative	Non-cumulative
Reporting cycle	Quarterly	Quarterly
Desired performance	100% of complete applications from the ESI requiring exemption to the South African grid code, considered by the ELS/REC within 3 months from receipt of complete information	100% of complete applications from the ESI requiring amendment to the South African grid code, considered by the ELS/REC within 3 months from receipt of complete information
Indicator responsibility	EM (ELR) and HOD (EIP)	EM (ELR) and HOD (EIP)

1.1.2. Piped-Gas Industry Regulation

1.1.2.1. Setting and/or approval of tariffs and prices

Indicator title	1. Percentage of complete maximum price applications considered by the relevant subcommittee or the Energy Regulator within the stated time frame	2. Percentage of complete applications on distinguishing features considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Definition	This is the percentage of complete applications for maximum prices of piped-gas considered by the relevant subcommittee within a set time frame, subject to a finding that there is inadequate competition	This is the percentage of complete applications on distinguishing features considered by the relevant subcommittee within a set time frame
Source of data	Applications for maximum prices of gas	Applications for distinguishing features
Method of calculation / assessment	(Number of complete applications for maximum prices considered within 120 working days after date of publication of the preliminary assessment of the maximum price applications)/(number of complete applications for maximum prices received)*100	(Number of complete applications on distinguishing features considered within 120 working days)/(number of complete applications on distinguishing features received)*100
Means of verification	Reason for decisions, minutes of ER meeting	Reason for decisions, minutes of ER meeting
Assumptions	Complete applications received from licensees	Complete applications received from licensees
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Non-cumulative	Non-cumulative
Reporting cycle	Quarterly	Quarterly
Desired performance	100% of complete maximum price applications considered by the ER within 120 working days after date of publication of the preliminary assessment of the maximum price applications	100% of complete applications on distinguishing features considered by the ER within 120 working days after the date of the publication of preliminary assessment of the applications
Indicator Responsibility	EM (GAR) and HOD (GPT)	EM (GAR) and HOD (GPT)

1.1.2.1. Setting and/or approval of tariffs and prices

Indicator title	3. Percentage of complete transmission tariff applications considered by the relevant subcommittee or the Energy Regulator within the stated time frame	4. Number of calculations of the ROMPCO tariff for gas volumes below 120 million gigajoule considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Definition	This is the percentage of transmission tariff applications considered by the relevant subcommittee within a set time frame, subject to a finding that there is inadequate competition	This is the number of calculations of the ROMPCO tariff for gas volumes below 120 million gigajoules considered by the relevant subcommittee within a set time frame
Source of data	Applications for transmission tariff	Schedule One to the Agreement and PPI from StatsSA, report containing the ROMPCO tariffs for volumes below 120 GJ
Method of calculation / assessment	(Number of complete transmission tariff applications considered within 120 working days)/number of complete applications for transmission tariffs received)*100	Actual number of calculations and publication of the ROMPCO tariff for volumes below 120 gigajoule considered per annum
Means of verification	Reason for decisions, minutes of ER meeting	Submissions to PGS, minute of the PGS meeting
Assumptions	Complete applications received from licensees	Information received timeously from ROMPCO
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Non-cumulative	Cumulative
Reporting cycle	Quarterly	Quarterly
Desired performance	100% of complete transmission tariff applications considered by ER within 120 working days after date of publication of preliminary assessment of tariff applications	Four quarterly calculations of the ROMPCO tariff for gas volumes below 120 million gigajoules considered quarterly by the PGS
Indicator Responsibility	EM (GAR) and HOD (GPT)	EM (GAR) and HOD (GPT)

1.1.2.1. Setting and/or approval of tariffs and prices

Indicator title	5. Number of reports on the assessment of the adequacy of competition considered by the Energy Regulator within the stated time frame
Definition	This is the number of reports complied annually on the assessment of the adequacy of competition in the piped-gas industry considered by the relevant Subcommittee, within a set time frame.
Source of data	Competition assessment and analysis reports
Method of calculation / assessment	Number of reports per year
Means of verification	Submissions to PGS; minutes of the PGS
Assumptions	Assessment and analysis completed on time.
Disaggregation of beneficiaries (where applicable)	Not applicable
Spatial transformation (where applicable)	Not applicable
Calculation Type	Cumulative
Reporting cycle	Annually
Desired performance	One report on the assessment of the adequacy of competition considered by the PGS by 31 March 2025
Indicator Responsibility	EM (GAR) and HOD (GPT)

1.1.2.2. Licensing and Registration

Indicator title	1. Percentage of complete licence applications considered by the relevant subcommittee or the Energy Regulator within the stated time frame	2. Percentage of complete applications for licence amendments/revocations/conversions considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Definition	This is the percentage of the licence applications considered by the REC or PGS (depending on the delegation) within a set time frame	This is the percentage of the applications for licence amendment, considered by the relevant subcommittee within a set time frame
Source of data	Licence applications	Applications for licence amendments
Method of calculation / assessment	(Number of complete licence applications considered within 60 working days after the end of the objection period or period of applicant's response to objections received)/(total number of complete applications received) * 100	(Number of complete applications for amendments/revocations/conversions considered within 60 working days from receipt of complete application)/(total number of complete applications received)*100
Means of verification	Reasons for decision, minutes of REC/PGS (depending on delegation) meeting	Reasons for decision, minutes of REC/PGS (depending on delegation) meeting
Assumptions	Complete applications submitted	Complete applications submitted
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Non-cumulative	Non-cumulative
Reporting cycle	Quarterly	Quarterly
Desired performance	100% of complete licence applications considered by the PGS/REC/ER within 60 working days from date of close of public comment period or period of applicant's response to objections received	100% of complete applications for licence amendments/revocations/conversions considered by the PGS/REC within 60 working days from date of close of public comment period or period of applicant's response to objections received
Indicator Responsibility	EM (GAR) and HOD (GLC)	EM (GAR) and HOD (GLC)

1.1.2.2. Licensing and Registration

Indicator title	3. Percentage of complete applications for the registration of gas activities considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Definition	This is the percentage of the registration applications for operations or activities related to the production and importation of gas considered by the relevant subcommittee within a set time frame
Source of data	Registration applications
Method of calculation / assessment	$(\text{Number of complete registration applications considered within 60 working days from receipt of complete application}) / (\text{total number of complete applications received}) * 100$
Means of verification	Reasons for decision, minutes of PGS meeting
Assumptions	Complete applications submitted
Disaggregation of beneficiaries (where napplicable)	Not applicable
Spatial transformation (where applicable)	Not applicable
Calculation Type	Non-cumulative
Reporting cycle	Annual
Desired performance	100% of complete applications for the registration of gas activities are processed and considered by the PGS within 60 working days from date of close of public comment period
Indicator Responsibility	EM (GAR) and HOD (GLC)

1.1.2.3. Compliance monitoring and enforcement

Indicator title	1. Number of monthly volume balance reports assessed and analysis reports considered by the relevant subcommittee or the Energy Regulator within the stated time frame	2. Number of audit reports on compliance of the ROMPCO pipeline considered by the relevant subcommittee within the stated time frame
Definition	This is the number of reports on the assessment and analysis of Sasol's monthly volume balance reports considered by the relevant subcommittee, within 60 days from date of receipt of information from Sasol, in order for NERSA to have regular, systematic, consistent, and sufficient non-financial information relevant to economic regulation, to enhance the efficiency and transparency of the regulatory process	This is the number of reports on audits conducted on the ROMPCO pipeline according to the compliance framework and non-compliance notices issued (where necessary), considered by the relevant committee by the end of the financial year
Source of data	Volume balance assessment reports	Audit reports
Method of calculation / assessment	Actual number of reports considered per annum	Actual number of reports considered per annum
Means of verification	Submissions to PGS, minutes of PGS meeting	Submissions to PGS, minutes of PGS meeting
Assumptions	Information received timeously from Sasol	Approval received to travel to Mozambique to conduct audit
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Cumulative
Reporting cycle	Quarterly	Annually
Desired performance	Twelve monthly volume balance reports assessed and analysis reports considered quarterly by the PGS	One audit report on the compliance of ROMPCO pipeline considered annually by the PGS by 31 March
Indicator Responsibility	EM (GAR) and HOD (GLC)	EM (GAR) and HOD (GLC)

1.1.2.3. Compliance monitoring and enforcement

Indicator title	3. Number of reports on licensees' compliance with licence conditions considered by the relevant subcommittee within the stated time frame	4. Percentage of monitoring reports on the implementation of transmission tariffs considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Definition	This is the number of reports on inspections conducted aimed at enforcing monitoring and compliance of licensed entities with licence conditions considered by the relevant committee within stated time frame	This is the percentage of reports on the monitoring of the implementation of transmission tariffs by ROMPCO, Transnet and Sasol Gas, respectively, considered by the relevant committee within stated time frame
Source of data	Approved plan for annual inspections, Inspection reports	Monitoring reports of ROMPCO, Transnet and Sasol Gas
Method of calculation / assessment	Actual number of reports considered per annum	(Actual number of reports)/(number of licensees with approved transmission tariffs)*100
Means of verification	Submissions to PGS, minutes of PGS meeting	Submissions to PGS, minutes of PGS meeting, date of approval of transmission tariffs
Assumptions	Inspection reports completed after each inspection	Analysis of implementation of transmission tariffs completed
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Non-cumulative
Reporting cycle	Quarterly	Quarterly
Desired performance	One report on compliance with licence conditions considered annually by the PGS by 31 March	100% of monitoring reports on the implementation of transmission tariffs considered annually by the PGS by 31 March, after one year, following the approval of the transmission tariff
Indicator Responsibility	EM (GAR) and HOD (GLC)	EM (GAR) and HOD (GLC)

1.1.2.3. Compliance monitoring and enforcement

Indicator title	5. Number of reports on the implementation of the RRM for the preceding financial year considered by the relevant subcommittee or the Energy Regulator within the stated time frame	6. Percentage of monitoring reports per licensee on the implementation of maximum prices considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Definition	This is the number of reports on the implementation of the RRM aimed at achieving uniformity and consistent reporting of information required for tariff setting/approval and performance monitoring, considered by the relevant subcommittee	This is the percentage of reports on the implementation of maximum prices aimed at evaluating compliance, considered by the relevant subcommittee
Source of data	Analysis on the implementation of the RRM	Analysis on the implementation of maximum prices
Method of calculation / assessment	Actual number of reports considered per annum	(Actual number of reports)/(number of licensees with approved maximum prices)*100
Means of verification	Submission to PGS/REC, minutes of PGS/REC meeting	Submission to PGS/REC, minutes of PGS/REC meeting, date of approval of transmission tariffs
Assumptions	Analysis of the implementation of the RRM completed	Analysis of the implementation of the Maximum Prices completed
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Cumulative
Reporting cycle	Quarterly	Quarterly
Desired performance	Four reports (one for each licensee – SASOL, ROMPCO, Transnet and SLG) on the implementation of the RRM for the preceding financial year considered annually by the PGS/REC by 31 March	100% of monitoring reports per licensee on the implementation of maximum prices, after one year following the approval of the maximum price considered annually by the PGS by 31 March
Indicator Responsibility	EM (GAR) and HOD (GLC)	EM (GAR) and HOD (GPT)

1.1.2.4. Dispute resolution, including mediation, arbitration and handling of complaints

Indicator title	1. Percentage of complaint investigations completed and a report on findings considered by the relevant subcommittee or the Energy Regulator within the stated time frame	2. Percentage of initiated investigations completed and a report on findings considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Definition	This is the percentage of investigations into complaints and disputes received, completed within the stated time frames and a report on the findings considered by the relevant subcommittee	This is the percentage of initiated investigations within the stated time frames and a report on the findings considered by the relevant subcommittee
Source of data	Records of complaints received	Records of complaints initiated, RFD, minutes of the relevant subcommittee meeting
Method of calculation / assessment	(Number of complaints received completed within 12 months after receipt)/ (total number of applications received)*100	(Number of initiated investigations completed within 12 months after receipt)/ (total number of initiated investigations)* 100
Means of verification	RFD, minutes of PGS meeting	RFD, minutes of PGS meeting
Assumptions	Complete information received from complainant, report on findings	Initiated investigations completed, report on findings
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Non-cumulative	Non-cumulative
Reporting cycle	Annual	Annual
Desired performance	60% of complaint investigations completed within 12 months and a report on findings considered by the PGS	60% of initiated investigations and inquiries completed within 12 months and a report on findings considered by the PGS
Indicator Responsibility	EM (GAR) and (HOD (GPT) or HOD (GLC))	EM (GAR) and (HOD (GPT) or HOD (GLC))

1.1.2.5. Setting of rules, guidelines and codes for the regulation of the piped-gas industry

Indicator title	1. Number of reports on new developments in the gas industry considered by the relevant subcommittee or the Energy Regulator within the stated time frame	2. Number of reports on the progress made with the implementation of the measures to improve competition and investment in the gas industry considered by the relevant committee or the Energy Regulator within the stated time frame
Definition	This is the number of reports relating to the new developments in piped-gas industry	This is the number of reports on progress made with the implementation of the measures to improve competition and investment in the gas industry
Source of data	Reports considered, minutes of the relevant subcommittee meeting	Reports considered, minutes of the relevant subcommittee meeting
Method of calculation / assessment	Actual number of reports considered per annum	Actual number of reports considered per annum
Means of verification	Submissions to PGS, minutes of PGS meeting	Submissions to PGS, minutes of PGS meeting
Assumptions	Reports compiled	Reports compiled
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Cumulative
Reporting cycle	Annual	Annual
Desired performance	Two reports on new developments in the gas industry considered bi-annually by the PGS by 31 March	One report on progress made with the implementation of the measures to improve competition and investment in the gas industry
Indicator Responsibility	EM (GAR), HOD (GLC) and HOD (GPT)	EM (GAR), HOD (GLC) and HOD (GPT)

1.1.3. Petroleum Pipelines Industry Regulation

1.1.3.1. Setting and/or approval of tariffs and prices

Indicator title	1. % of complete pipeline, storage and loading facility tariff applications considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Definition	This is the percentage of all the pipeline, storage and loading facility tariff applications considered by the relevant subcommittee within eight months of receipt of complete application
Source of data	Applications for tariffs
Method of calculation / assessment	(Number of complete tariff applications considered by the relevant subcommittee within eight months of receipt of complete application)/(total number of complete tariff applications received)*100
Means of verification	Reasons for Decision and minutes of PPS meeting
Assumptions	Complete applications received
Disaggregation of beneficiaries (where applicable)	Not applicable
Spatial transformation (where applicable)	Not applicable
Calculation Type	Non-cumulative
Reporting cycle	Annually
Desired performance	85% of complete pipeline, storage and loading facility tariff applications considered by the REC/PPS/ER within six months from receipt of complete/adequate application
Indicator Responsibility	EM (PPR) and HOD (PPT)

1.1.3.2. Licensing and Registration

Indicator title	1. % of complete licence applications considered by the relevant subcommittee or the Energy Regulator within the stated time frame	2. % of complete applications for licence amendments/ revocations considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Definition	This is the percentage of licence applications that will be decided upon within the timelines prescribed in section 19(1) of the Petroleum Pipelines Act	This is the percentage of applications for licence amendments/ revocations that will be decided upon within the timelines prescribed in section 19(1) of the Petroleum Pipelines Act
Source of data	Licence applications	Licence amendment applications
Method of calculation / assessment	(Number of complete applications decided upon within statutory turn-around/ number of received complete licence applications) * 100	(Number of complete applications decided upon within statutory turn-around/ number of received complete licence applications)*100
Means of verification	Reasons for decision (RFD) and minutes of PPS/REC/ER meeting	Reasons for decision (RFD) and minutes of PPS/REC/ER meeting
Assumptions	Complete applications	Complete applications
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Non-cumulative	Non-cumulative
Reporting cycle	Quarterly	Quarterly
Desired performance	100% of complete licence applications considered by the PPS/ REC/ER within 60 working days under the conditions prescribed in section 19(1) of the Petroleum Pipelines Act	100% of complete applications for licence amendments/ revocations considered by the PPS/REC/ER within 60 working days from date of close of public comment period or period of applicant's response to objections received
Indicator Responsibility	EM (PPR) and HOD (PLC)	EM (PPR) and HOD (PLC)

1.1.3.2. Licensing and Registration

Indicator title	3. Number of reports on investigations done into suspected unlicensed activities considered by the relevant subcommittee or the Energy Regulator within the stated timeframe	4. Number of reports on the geographic spread of licences issued for petroleum pipelines infrastructure and new entrants considered annually by the relevant subcommittee or the Energy Regulator
Definition	This is the number of reports on investigations done into suspected unlicensed activities considered by the relevant subcommittee	This is the number of reports on the geographic spread of licences issued for petroleum pipelines infrastructure and new entrants considered by the relevant subcommittee
Source of data	Data based on suspected unlicensed activities	Reports considered, minutes of the relevant subcommittee meeting Number of reports considered per annum
Method of calculation / assessment	Actual number of reports considered per annum	Actual number of reports considered per annum
Means of verification	Submissions to REC, minutes of REC meeting	Submissions to PPS, minutes of PPS meeting
Assumptions	Investigations completed	Reports compiled
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Cumulative
Reporting cycle	Annual	Annual
Desired performance	One report on investigations done into suspected unlicensed activities considered annually by the REC by 31 March	One report on the geographic spread of licences issued for petroleum pipelines infrastructure and new entrants considered annually by the PPS by 31 March
Indicator Responsibility	EM (PPR) and HOD (PLC)	EM (PPR) and HOD (PLC)

1.1.3.2. Licensing and Registration

Indicator title	5. Number of reports on the inland security of supply considered by relevant subcommittee or the Energy Regulator within the stated time frame	6. Number of reports on prudency reviews of identified licensees considered by relevant subcommittee or the Energy Regulator within the stated time frame
Definition	This is the number of reports on the inland security of supply considered by the relevant subcommittee	This is the number of reports on prudency reviews of identified licensees considered by the relevant subcommittee
Source of data	Reports considered, minutes of the relevant subcommittee meeting	Reports considered, minutes of the relevant subcommittee meeting
Method of calculation / assessment	Actual number of reports considered per annum	Actual number of reports considered per annum
Means of verification	Submissions to PPS, minutes of PPS meeting	Submissions to PPS, minutes of PPS meeting
Assumptions	Reports compiled	Reports compiled
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Cumulative
Reporting cycle	Bi-annual	Bi-annual
Desired performance	Two reports on the inland security of supply considered bi-annually by the PPS by 30 September and 31 March	One report on prudency reviews of identified licensees considered annually by the PPS by 31 March
Indicator Responsibility	EM (PPR) and HOD (PLC)	EM (PPR) and HOD (PLC)

1.1.3.3. Compliance monitoring and enforcement

Indicator title	1. Number of reports on trends regarding utilisation of storage facilities and third-party access, considered by the relevant committee or the Energy Regulator within the stated time frame	2. Number of reports on the implementation of the methodology to determine uncommitted capacity considered by the relevant committee or the Energy Regulator within the stated time frame
Definition	This is the number of reports on trends regarding the utilisation of storage facilities and 3rd party access, considered by the relevant subcommittee, aimed at promoting competition in the industry	This is the number of reports on the analysis of the implementation of the methodology to determine uncommitted capacity, considered by the relevant Subcommittee, aimed at promoting 3rd party access
Source of data	Analysis reports	Analysis of the implementation of the methodology to determine uncommitted
Method of calculation / assessment	Actual number of reports considered per annum	Actual number of reports considered per annum
Means of verification	Submissions to PPS, minutes of PPS meeting	Submissions to PPS, minutes of PPS meeting
Assumptions	Analysis of trends completed	Analysis of trends completed
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Cumulative
Reporting cycle	Bi-annual	Annual
Desired performance	Two reports on trends regarding utilisation of storage facilities and third-party access considered bi-annually by the PPS by 31 March	One report on the implementation of the methodology to determine uncommitted capacity considered annually by the PPS by 31 March
Indicator Responsibility	EM (PPR) and HOD (PLC)	EM (PPR) and HOD (PLC)

1.1.3.3. Compliance monitoring and enforcement

Indicator title	3. Number of reports on the construction of new facilities considered by the relevant committee or the Energy Regulator within the stated time frame	4. Number of reports on licensees' compliance with statutory reporting requirements considered by the relevant committee or the Energy Regulator within the stated time frame
Definition	This is the number of reports detailing the compliance of construction licensees with licence conditions. The reports are compiled and considered by the relevant subcommittee on bi-annually	This is a report on the compliance of the licensees with all the statutory reporting requirements considered by the relevant subcommittee on bi-annually
Source of data	Database of identified construction of new facilities	Database on licensees' compliance with statutory reporting requirements
Method of calculation / assessment	Actual number of reports considered per annum	Actual number of reports considered per annum
Means of verification	Submissions to PPS, minutes of PPS meeting	Submissions to PPS, minutes of PPS meeting
Assumptions	Analysis of construction of new facilities completed	Analysis of licensees' compliance with statutory reporting requirements completed
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Cumulative
Reporting cycle	Bi-annual	Bi-annual
Desired performance	Two reports on the construction of new facilities considered bi-annually by the PPS by 31 March	Two reports on licensees' compliance with statutory reporting requirements considered bi-annually by the PPS by 31 March
Indicator Responsibility	EM (PPR) and HOD (PLC)	EM (PPR) and HOD (PLC)

1.1.3.4. Dispute resolution, including mediation, arbitration and handling of complaints

Indicator title	1. % of complaints investigated and report considered by the relevant subcommittee or the Energy Regulator within the stated time frame of receipt of adequate information from relevant parties
Definition	This is the percentage of the complaints investigated and considered by the relevant subcommittee within 60 days of receipt of complete information from relevant parties
Source of data	Records of complaints received
Method of calculation / assessment	$(\text{Number of finalised complaints within 60 days of receipt}) / (\text{number of received complaints}) * 100$
Means of verification	Submissions for PPS, minutes of PPS meeting
Assumptions	Complete information received from relevant parties, investigations completed as planned
Disaggregation of beneficiaries (where applicable)	Not applicable
Spatial transformation (where applicable)	Not applicable
Calculation Type	Non-cumulative
Reporting cycle	Annually
Desired performance	100% of complaints investigated and report considered by the PPS within six months of receipt of complete information from relevant parties
Indicator Responsibility	EM (PPR) and HOD (PLC)

1.1.3.5. Setting of rules, guidelines and codes for the regulation of the petroleum pipelines industry

Indicator title	1. Number of reports on the monitoring of the implementation of the tariff methodology considered by the relevant committee or the Energy Regulator within the stated time frame	2. Energy Regulator decision on the Reviewed pipelines tariff methodology within the stated time frame
Definition	This is the number of reports the monitoring of the implementation of the tariff methodology considered by the relevant committee annually	This is the Energy Regulator decisions on the reviewed pipelines tariff methodology considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Source of data	Analysis of the implementation of the tariff methodology by licensees	Methodology, review reports
Method of calculation / assessment	Actual number of reports considered per annum	Reviewed pipelines tariff methodology considered by the relevant subcommittee
Means of verification	Submissions to PPS, minutes of PPS meeting	Submissions to PPS/ER, minutes of PPS/ER meeting
Assumptions	Analysis completed	Review of pipelines tariff methodology completed as planned
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Non-cumulative
Reporting cycle	Annual	Annual
Desired performance	One report on the monitoring of the implementation of the revised methodology considered annually by the ER by 31 March	Regulator decision by 31 March 2027
Indicator Responsibility	EM (PPR) and HOD (PPT)	EM (PPR) and HOD (PLC)

1.2. Programme 2: Regulatory Service Delivery

1.2.1. Subprogramme: Regulatory and Policy Advocacy

Indicator title	1. Number of reports on regulatory advocacy considered by the relevant subcommittee or the Energy Regulator within stated time frame	2. Percentage variance of planned versus actual annual ESI Advocacy and Stakeholder Engagement Plan executed
Definition	This is the number of reports on regulatory advocacy engagements with decision-makers, aimed at improving the regulatory framework provided through legislation, regulation and government policies – one each for the piped-gas and petroleum pipelines regulated industries	This is the percentage variance between the actual regulatory advocacy and stakeholder engagements and the planned regulatory advocacy and stakeholder engagements in line with the annual ESI Advocacy and Stakeholder Engagement Plan
Source of data	Reports on each engagement, indicating the reason for and outcome of the engagement	ESI Advocacy and Stakeholder Engagement Plan, reports on advocacy and stakeholder engagements conducted
Method of calculation / assessment	Actual number of reports considered per annum	(Number of actual regulatory advocacy and stakeholder engagements)/ planned number of regulatory advocacy and stakeholder engagements in line with the annual ESI Advocacy and Stakeholder Engagement Plan)*100
Means of verification	Submissions to PPS and PGS, minutes of PPS and PGS meetings	Minutes of ELS meetings
Assumptions	Reports on each engagement compiled	ESI Advocacy and Stakeholder Engagement Plan implemented as planned
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Cumulative
Reporting cycle	Annual	Quarterly
Desired performance	Two reports considered annually by the PGS (1) and PPS (1) by 31 March	65% of planned regulatory advocacy and stakeholders engagements conducted
Indicator Responsibility	EM (GAR): EM (PPR)	EM (ELR) and HOD (ELC)

1.2.2. Subprogramme: Customer and Stakeholder Engagement

Indicator title	1. Number of reports on stakeholder workshops/meetings for the piped- gas and petroleum pipelines regulated industries considered by the relevant subcommittee or the Energy Regulator within the stated time frame	2. Percentage variance of planned versus actual annual ESI Advocacy and Stakeholder Engagement Plan executed
Definition	This is the number of reports on stakeholder workshops/ meetings and engagements within the three regulated industries – one each for the piped-gas and petroleum pipelines regulated industries – considered by the relevant subcommittee or the Energy Regulator within the stated time frame	This is the percentage variance between the actual regulatory advocacy and stakeholder engagements and the planned regulatory advocacy and stakeholder engagements in line with the annual ESI Advocacy and Stakeholder Engagement Plan
Source of data	Reports on each engagement indicating the reason for and outcome of the engagement	ESI Advocacy and Stakeholder Engagement Plan, reports on advocacy and stakeholder engagements conducted
Method of calculation / assessment	Actual number of reports considered per annum	(Number of actual regulatory advocacy and stakeholder engagements)/planned number of regulatory advocacy and stakeholder engagements in line with the annual ESI Advocacy and Stakeholder Engagement Plan)*100
Means of verification	Submissions to PPS and PGS, minutes of PPS and PGS meetings	Minutes of ELS meetings
Assumptions	Report on each engagement compiled	ESI Advocacy and Stakeholder Engagement Plan implemented as planned
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Cumulative
Reporting cycle	Annual	Quarterly
Desired performance	Two reports considered by the PGS (1) and PPR (1) by 31 March	65% of planned regulatory advocacy and stakeholders engagements conducted
Indicator Responsibility	EM (GAR): EM (PPR)	EM (ELR) and HOD (ELC)

1.2.2. Subprogramme: Customer and Stakeholder Engagement

Indicator title	3. Number of ESI customer education programmes undertaken within the stated time frame	4. Number of reports on partnership creation considered by the relevant committee or the Energy Regulator within the stated time frame
Definition	This is the number of ESI customer education programmes conducted, where NERSA engages its stakeholders in a number of ways, including education programmes	This is the number of reports on partnership creation, which includes engagements with other regulators; participation in regulatory associations, events and conferences; and partnerships with other institutions for capacity- building purposes, aimed at positioning NERSA as a recognised regulator
Source of data	Annual plan for ESI customer education programmes	Reports on an overview of international engagements and partnerships activities
Method of calculation / assessment	Actual number of ESI customer education programmes held per annum	Actual number of reports considered per annum
Means of verification	Submissions to ELS, minutes of ELS meeting	Submissions to REC, Minutes of REC meeting
Assumptions	Programmes conducted as planned in the annual plan for ESI customer education programmes	Analysis completed
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Cumulative
Reporting cycle	Quarterly	Bi-annual
Desired performance	Seventy-five ESI customer education programmes undertaken annually by 31 March	Two reports on partnership creation to position NERSA as a recognised regulator nationally, regionally and internationally considered bi-annually by the REC by 30 September and 31 March
Indicator Responsibility	EM (ELR) and HOD ELC	EM (COS) and HOD (ICP)

1.2.2. Subprogramme: Customer and Stakeholder Engagement

Indicator title	5. Number of reports on the implementation of the stakeholder management plan considered by the relevant subcommittee or the Energy Regulator within the stated time frame	6. Number of reports on the analysis of the Stakeholder Management/Reputational Risk survey considered by the relevant subcommittee within the stated time frame
Definition	This is the number of reports on the implementation of the stakeholder management plan	This is the number of reports on the analysis of the Stakeholder Management/Reputational Risk survey
Source of data	Reports on implemented actions	Project plan and progress reports
Method of calculation / assessment	Actual number of reports considered per annum	Number of reports per annum
Means of verification	Submissions to REC, minutes of REC meeting	Submissions to REC; Minutes of REC
Assumptions	Individual reports are completed for each action completed	Analysis completed
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Cumulative
Reporting cycle	Annually	Annually
Desired performance	One report on the implementation of the stakeholder management plan considered annually by the REC by 31 March	One report on the analysis of the Stakeholder Management/Reputational Risk survey considered by the Energy Regulator by 31 March 2025
Indicator Responsibility	EM (COS) and HOD (CSM)	EM (COS) and HOD (CSM)

1.2.2. Subprogramme: Customer and Stakeholder Engagement

Indicator title	7. Number of reports on communication and reputation management considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Definition	Number of reports on communication and reputation management considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Source of data	Reports on implemented actions
Method of calculation / assessment	Actual number of reports considered per annum
Means of verification	Submissions to REC, minutes of REC meeting
Assumptions	Individual reports are completed for each action completed
Disaggregation of beneficiaries (where applicable)	Not applicable
Spatial transformation (where applicable)	Not applicable
Calculation Type	Cumulative
Reporting cycle	Annually
Desired performance	One report considered annually by the REC by 31 March
Indicator Responsibility	EM (COS) and HOD (CSM)

1.3. Programme 3: Innovation

1.3.1. Subprogramme: Integrated And Value-Added Services

Indicator title	1. Number of reports on the implementation of the approved ICT Strategy considered by the relevant committee or the Energy Regulator within the stated time frame
Definition	This is the number of reports to indicate how the implementation of the ICT strategy is progressing
Source of data	ICT Strategy, implementation reports
Method of calculation / assessment	Actual number of reports considered per annum
Means of verification	Submissions to REC; Minutes of REC
Assumptions	Progress reports are compiled
Disaggregation of beneficiaries (where applicable)	Not applicable
Spatial transformation (where applicable)	Not applicable
Calculation Type	Cumulative
Reporting cycle	Bi-annually
Desired performance	Two reports considered bi-annually by the ITGC by 31 March
Indicator Responsibility	CIO

1.4. Programme 4: Operational Efficiency and Quality Management

Indicator title	1. Number of reports on environmental scanning considered by the relevant subcommittee or the Energy Regulator within the stated time frame	2. Energy Regulator decision on the Reviewed Regulatory Reporting Manuals for Non-Financial and Financial Information
Definition	This is the number of reports scanning NERSA's business environment considered by the relevant subcommittee	This is the Energy Regulator Decision Reviewed Regulatory Reporting Manuals for non-financial and financial information considered by the relevant subcommittee
Source of data	International reports, research report	Analysis on the progress made with the implementation of the RRM's for financial and non-financial information
Method of calculation / assessment	Actual number of reports considered per annum	Actual number of reports considered per annum
Means of verification	Submissions to REC, minutes of REC meetings	Energy Regulator's decision
Assumptions	Analysis completed	Analysis completed
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Cumulative
Reporting cycle	Annually	Annually
Desired performance	1 report considered annually by the REC by 30 April	Energy Regulator's decision on Reviewed Regulatory Reporting Manuals for Non-Financial and Financial Information (Vol 1) by 31 March 2025
Indicator Responsibility	SM (RAR) and SM (SPM)	SM (RAR)

1.4. Programme 4: Operational Efficiency and Quality Management

Indicator title	3. Unqualified audit opinion	4. Number of reports on the implementation of the measures to improve NERSA's B-BBEE rating per financial year considered by the relevant subcommittee or the Energy Regulator within the stated time frame
Definition	This is the outcome of audit conducted by the Auditor-General on NERSA annually	This is the number of reports on implementation of the measures to improve NERSA's B-BBEE rating
Source of data	Final management report from the AG	Progress Report and minutes of the relevant Subcommittee
Method of calculation / assessment	Unqualified audit opinion – yes/no	One report considered per annum
Means of verification	Audit report	Submission to REC, minutes of REC meeting
Assumptions	Collaboration of management	B-BBEE measures implemented as planned
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Non-cumulative	Cumulative
Reporting cycle	Annually	Annual
Desired performance	Unqualified audit	One report on the implementation of the measures to improve NERSA's B-BBEE rating per financial year
Indicator Responsibility	CFO and HOD (FAD)	SM: CEO's Office

1.4. Programme 4: Operational Efficiency and Quality Management

Indicator title	5. Number of reports on the implementation of the gender mainstreaming plan considered by the relevant subcommittee within the stated time frame	6. Number of reports on benchmarking of NERSA's operational processes controls against international standards considered by the relevant subcommittee within the stated time frame
Definition	This is the number of reports on the implementation of the gender mainstreaming plan	This is the number of reports on benchmarking of NERSA's operational processes controls against international standards
Source of data	Gender mainstreaming plan	Approved operational processes
Method of calculation / assessment	Actual number of reports considered per annum	Actual number of reports considered per annum
Means of verification	Implementation reports, minutes of REC meeting	Submissions to REC, minutes of REC meeting
Assumptions	Implementation of the gender mainstreaming plan actioned as planned	Benchmarking of NERSA's operational processes controls as planned
Disaggregation of beneficiaries (where applicable)	The plan that will be developed by the end of 2024/25 will provide information	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Cumulative
Reporting cycle	Annually	Annually
Desired performance	One report on the implementation of gender mainstreaming initiatives considered annually by the REC by 31 March	One report considered by the REC by 31 March 2025
Indicator Responsibility	EM: COS	SM: SPM

1.4. Programme 4: Operational Efficiency and Quality Management

Indicator title	7. Number of reports on the implementation of the Hybrid Working Strategy considered by the relevant subcommittee within the stated time frame
Definition	This is the number of reports on the implementation of the Hybrid Working Strategy
Source of data	Hybrid Working Strategy
Method of calculation / assessment	Actual number of reports considered per annum
Means of verification	Implementation reports, minutes of REC meeting
Assumptions	Implementation of the Hybrid Working Strategy actioned as planned
Disaggregation of beneficiaries (where applicable)	Not applicable
Spatial transformation (where applicable)	Not applicable
Calculation Type	Cumulative
Reporting cycle	Annually
Desired performance	One report on the Hybrid Working Strategy considered annually by the REC by 31 March
Indicator Responsibility	EM: COS

1.5. Programme 5: People and Organisational Culture

Indicator title	1. Number of reports on Organisational Culture Assessment considered by the relevant committee or the Energy Regulator within the stated time frame	2. Number of reports on the implementation of the Employment Equity Plan considered by the relevant committee or the Energy Regulator within the stated time frame
Definition	This is the number of reports that state the outcome of the annual Organisational Culture Assessment conducted	This is the number of progress reports on the implementation of the Employment Equity Plan considered by the relevant subcommittee
Source of data	Organisational Culture Assessment Report	Analysis of the implementation of the Employment Equity Plan
Method of calculation / assessment	Actual number of progress reports considered per annum	Actual number of progress reports considered per annum
Means of verification	Submissions to HRRC, minutes of HRRC meeting	Submissions to HRRC, minutes of HRRC meeting
Assumptions	Assessment concluded	Analysis completed
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Cumulative
Reporting cycle	Annually	Bi-annually
Desired performance	One report on Organisational Culture Assessment considered annually by the HRRC by 31 March	Two reports considered bi-annually by the HRRC by 31 March
Indicator Responsibility	CHCO and HOD (HR)	CHCO and HOD (HR)

1.5. Programme 5: People and Organisational Culture

Indicator title	3. Percentage of women in management positions	4. Percentage of people with disabilities employed
Definition	Analysis of staff complement to determine percentage of women in management positions	Analysis of staff complement to determine percentage of people with disabilities employed
Source of data	Staff statistical information	Staff statistical information
Method of calculation / assessment	$(\text{Number of women in management positions}) / (\text{number of management positions}) * 100$	$(\text{Number of people with disabilities employed}) / (\text{number of all positions}) * 100$
Means of verification	Submissions to HRRC, minutes of HRRC meeting	Submissions to HRRC, minutes of HRRC meeting
Assumptions	Analysis completed	Analysis completed
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Non-cumulative	Non-cumulative
Reporting cycle	Annually	Annually
Desired performance	50% of women in management positions	2% of people with disabilities employed
Indicator Responsibility	CHCO and HOD (HR)	CHCO and HOD (HR)

1.5. Programme 5: People and Organisational Culture

Indicator title	5. Number of progress reports on the implementation of the Youth Employment Accord considered by the relevant committee or the Energy Regulator within the stated time frame	6. Number of reports on the implementation of the bursary programme for qualifying external applicants considered by the relevant committee or the Energy Regulator within the stated time frame
Definition	This is a report on the status of the percentage of people with disabilities employed	This is a report on monitoring the implementation of the bursary programme for qualifying external applicants
Source of data	Report on the status of the percentage of people with disabilities employed	Approved bursary programme
Method of calculation / assessment	Actual number of progress reports considered per annum	Actual number of progress reports considered per annum
Means of verification	Submissions to HRRC, minutes of HRRC meeting	Submissions to HRRC, minutes of HRRC meeting
Assumptions	Analysis completed	Analysis completed
Disaggregation of beneficiaries (where applicable)	Not applicable	Not applicable
Spatial transformation (where applicable)	Not applicable	Not applicable
Calculation Type	Cumulative	Cumulative
Reporting cycle	Quarterly	Annually
Desired performance	Four reports on the implementation of the Youth Employment Accord considered quarterly by the HRRC	One report on the implementation of the bursary programme for qualifying external applicants considered annually by the HRRC by 31 March
Indicator Responsibility	CHCO and HOD (HR)	CHCO and HOD (HR)

ANNEXURES
BUDGET 2024/25
AND FORECAST
2025/26 – 2026/27



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Funding for the National Energy Regulator of South Africa

1. Funding for the National Energy Regulator of South Africa

1.1. NERSA Mandate and Applicable Funding Legislation

- 1.1.1. The National Energy Regulator of South Africa (NERSA) is the regulatory authority established in terms of the National Energy Regulator Act, 2004 (Act No. 40 of 2004). According to Section 4 of this Act NERSA's mandate is to regulate the electricity industry in terms of the Electricity Regulation Act, 2006 (Act No. 4 of 2006), regulate the piped-gas industry in terms of the Gas Act, 2001 (Act No. 48 of 2001), and regulate the petroleum pipelines industry in terms of the Petroleum Pipelines Act, 2003 (Act No. 60 of 2003).
- 1.1.2. In terms of the National Energy Regulator Act, 2004 (Act No. 40 of 2004), the National Energy Regulator of South Africa (NERSA) will be funded through money appropriated by Parliament; levies and licence fees imposed by or under separate legislation; charges for dispute resolution and other services rendered in terms of the National Energy Regulator Act. However, it suffices to say that the most prudent form of funding for all three industries is through levies and license fees.
 - 1.1.3. For the electricity industry regulation, it is further provided that its funding will include donations or contributions received from any person or entity, as provided for in section 5B of the Electricity Act, 1987 (Act No. 47 of 1987).
 - 1.1.4. Levies from the petroleum pipelines and piped-gas industries are imposed in terms of the Petroleum Pipelines Levies Act, 2004 (Act No. 28 of 2004) and the Gas Regulator Levies Act, 2002 (Act No. 75 of 2002) respectively.
 - 1.1.5. Under section 5B of the Electricity Act [which has not been repealed with the operationalisation of the Electricity Regulation Act, 2006 (Act No. 4 of 2006)], the Minister of Mineral Resources and Energy prescribes the electricity licence fees by Notice in the Government Gazette.
 - 1.1.6. Under the Petroleum Pipeline Levies Act and the Gas Regulator Levies Act, the Energy Regulator first publishes the intended levies for the public to make representations, considers the representation and submits a report to the Minister of Mineral Resources and Energy on the representations and how the representations affected the levies. After the approval by the Minister of Mineral Resources and Energy in concurrence with the Minister of Finance, the Energy Regulator publishes a Notice in the Government Gazette of the approved levies for at least thirty days before the piped-gas and petroleum pipelines industries start paying the levies.



1.2. Method of calculating levies and license fees

- 1.2.1. In order to calculate the licence fees, the estimated energy production figures for the 2023 calendar year for the Electricity Industry are used. From the energy production figures, the net energy that has been sent out by every electricity generator is calculated (i.e. the difference between the gross energy produced and the amount of energy used during the generation process). The unit for this figure is kilowatt-hour. The licence fees are determined by dividing the leviable amount for the electricity industry by the total amount of energy sent out. This provides a figure measured in cents per kilowatt-hour.
- 1.2.2. With regard to the piped-gas industry, the levies imposed are based on the amount of gas, measured in gigajoules, delivered by importers and producers to inlet flanges of transmission or distribution pipelines. The levy is determined by dividing the leviable amount for piped gas regulation by the total gigajoules amount. This provides a figure measured in cents per gigajoules.
- 1.2.3. With regard to the petroleum pipelines industry, the levies imposed are based on the amount of petroleum, measured in litres, delivered by importers, refiners and producers to inlet flanges of petroleum pipelines. The levy is determined by dividing the leviable amount for petroleum pipelines regulation by the total litres amount. This provides a figure measured in cents per litre.

1.3. Method for Ring-fencing of funds and refunds to industry

- 1.3.1. In terms of section 13 (2) and (3) of the National Energy Regulator Act 2004 (Act No. 40 of 2004), NERSA must keep separate accounts for each of the three industries regulated. Section 13 (3) (a) also states that the costs of the Energy regulator must be shared between the electricity, piped-gas and petroleum pipeline regulatory functions in proportion to the costs incurred by the Energy Regulator in respect of each of those regulatory functions. NERSA

has over the years determined a ring fencing methodology to give effect to the accounting of funds of the regulator.

- 1.3.2. The primary accounting principles upon which the ring-fencing methodology is based are the following:
 - a) all direct costs attributable to an industry are allocated to that industry using the time based method of allocation;
 - b) the ratio of employment costs allocated to the three industries is used to allocate the remaining (common) costs to the respective industries; and
 - c) the common costs allocation for the next three years for the electricity, petroleum pipelines and piped-gas industries is 58%:21%:21%.
- 1.3.3. Section 13(1)(c) of the National Energy Regulator Act (Act No 30 of 2004) state that Money received by the Energy Regulator other than money appropriated by Parliament must be paid into the account that is kept in terms of subsection (2) for the industry from which such money was received, and must be used for the sole benefit of that industry.
- 1.3.4. Through section 13 (1) (c) of the NER Act, previously accumulated surpluses are used to reduce the leviable amount to be charged to licensees as refunds to industries. The refunds to industry are maintained per industry and are normally allocated over a three-year period to minimise drastic changes in the license fee/levy rate thus refunding the accumulated surplus to industry.
- 1.3.5. NERSA will be budgeting for a deficit over the 2023 MTEF period in order to minimise increases in the license fee/levy rate.
- 1.3.6. As prescribed by The Public Finance Management Act, NERSA can only budget for a deficit and retain surpluses only with prior approval from the National Treasury. This approval will be sought once the Energy Regulator has approved the proposed budget. NERSA has budgeted for a deficit in the previous three financial years.



2. Basis of Preparation for the Budget (Budgetary Basis)

- 2.1. In terms of the accounting standard on the Presentation of Budget Information in Financial Statements (GRAP 24). This Standard requires a comparison of budget amounts and the actual amounts arising from execution of the budget to be included in the financial statements of entities. Compliance with the requirements of this Standard will also ensure that entities discharge their accountability obligations and enhance the transparency of their financial statements by demonstrating compliance with the approved budget(s) for which they are held publicly accountable.
- 2.2. The standard also defines a Budgetary basis which means the accrual, cash or other basis of accounting adopted in the budget approved by the legislative body.
- 2.3. NERSA elects to prepare its Budget on the Cash basis. This is a basis of accounting that recognises transactions and other events only when cash is received or paid. This means non-cash items such as depreciation, impairments, accruals etc. are not budgeted for. These items are not budgeted for as they are excluded when determining the leviable amounts. Funding requirements also include capital expenditure at full cost, and therefore exclude, depreciation.

Budget 2024/25 Executive Summary

3. Executive Summary

3.1. Economic factors

In terms of National Treasury's 2024 Medium Term Expenditure Framework (MTEF), the 2023 economic outlook has worsened, fiscal revenues are weaker than expected, and the financing of the government borrowing requirement is under renewed pressure. Several key factors are contributing to ongoing uncertainty and volatility. The country continues to grapple with a high unemployment rate, slow economic growth, and persistent structural challenges, including inefficiency in key sectors such as energy and transportation. Moreover, persistent power cuts, deteriorating rail and port infrastructure, have contributed to a weaker domestic outlook. The state wage bill and the weak financial position of State-Owned Companies (SOCs) continue to pose a risk to the fiscus, including those posed by global events and natural disasters.

NERSA has to budget with these factors in mind and maintain a balanced view of ensuring effectiveness in performing its regulatory functions, managing its funding requirements while levying rates and license fees in a responsible and sustainable manner.



3.2. Organisational factors

Strategic Focus Areas

During the Energy Regulator strategic planning sessions, there were strategic improvement areas identified with an aim of continuing to focus on the value add of NERSA in regulating the Electricity, Piped-Gas and Petroleum pipeline energy sectors. The Energy Regulator identified key projects that are critical in performing regulatory functions, to drive innovation through the ICT projects, promote advocacy in the transitioning energy sector and introduce efficiencies in regulation. These strategic focus areas were incorporated in the Annual Performance Plan and while others will be included in Annual Operational plans for 2024/25.

Key Projects

The NERSA funding requirements for 2024/25 take into account the following key projects that should advance NERSA towards improving efficiencies and adding value in the sectors it regulates:

- a) Implementation of the online collaboration portal between NERSA and licensees – (R 15 million)
- b) Implementation of the data warehouse as a business intelligence tool – (R5 million)
- c) Acquisition of online energy markets data for benchmarking studies and data informed decision making – (R4 million)
- d) Implementation of stakeholder management and reputational risk survey and acquisition of reputation management services (R 1.6 million)

- e) Implementation of project management office (R1,4 million). This budget is for a temporary resource. Approval is being sought from the energy regulator to budget for the temporary position.
- f) Institutionalising gender mainstreaming and programmes (R2 million). This includes a temporary resource and training for staff and regulator members. Approval is being sought from the energy regulator to budget for the temporary position.

The total additional budget in 2024/25 to advance key projects is R29 million. The ICT projects are will be implemented in a phased manner, where additional budget requirements to complete the projects are catered for in the forecast years.

3.3. Funding Requirements

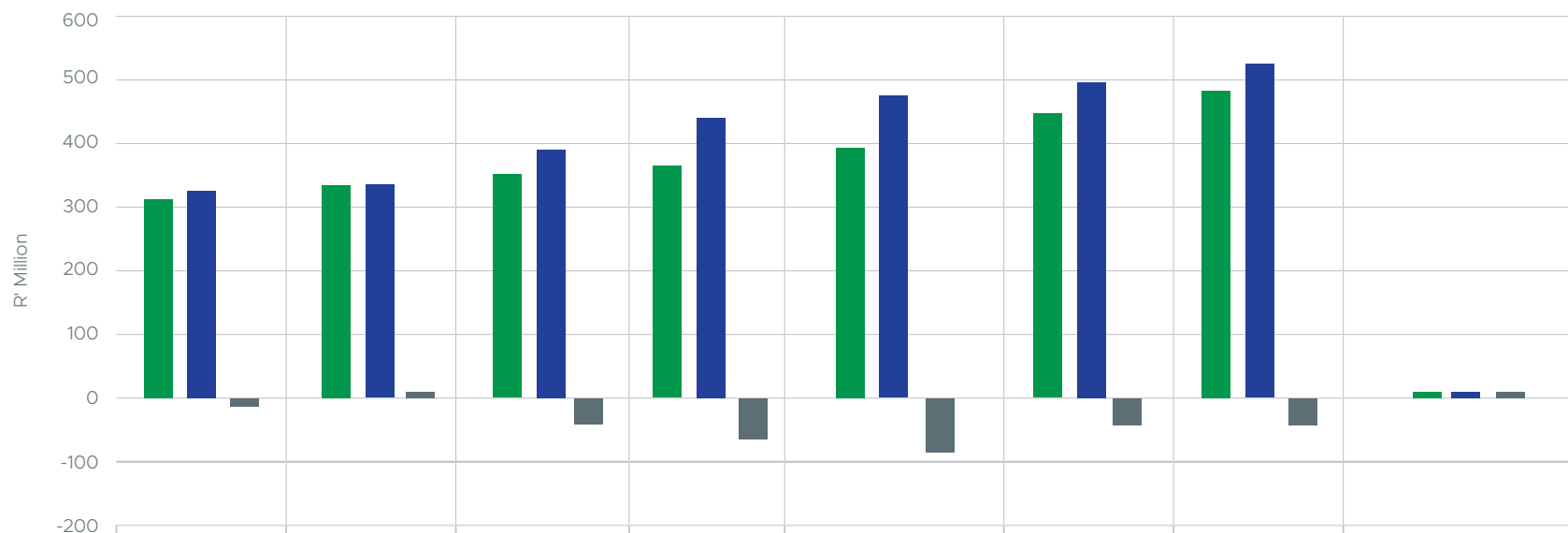
The total funding requirement for the 2024/25 budget is R485 million which is 8.8% higher than the 2023/24 funding requirement. The forecast requirements are projected to increase by 2,9% in the 2025/26 financial year and by 5.7% for the 2026/27 financial year.

The funding requirements will be obtained by charging levies/licence fees to the regulated electricity, piped gas and petroleum pipelines industries. In order to limit increases in the levy/ license fee rate as well as manage surplus funds, accumulated surplus funds from previous financial years have been allocated and deducted from the leviable amount as refunds to the regulated industries.

Total available surpluses for refund are R170 million as at 31 March 2023. Planned refunds to industries are allocated over a three-year period, with R87.2 million to be allocated in 2024/25, R41,4 million in 2025/26 and R41,4 million in 2026/27.



Figure 1: 2024/25 MTEF Revenue and funding requirements



	Target	Actual	Actual	Current	Proposed Budget	Forecast	Forecast	% Change
	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	2023/24 – 2024/25
● Total Income	319	345	353	376	398	458	486	5,9%
● Total Expenditure	328	345	398	446	485	499	528	8,%
● Surplus Utilisation	(8,7)	0,1	(44,8)	(70,2)	(87,2)	(41,4)	(41,4)	24,1%



3.4. Industry Volume Estimates

Volume estimates for the 2024/25 budget period indicate a decline compared to the 2023/24 budget and over the MTEF. Figure 2 outlines a historical, current and projected view of volumes for regulated industries.

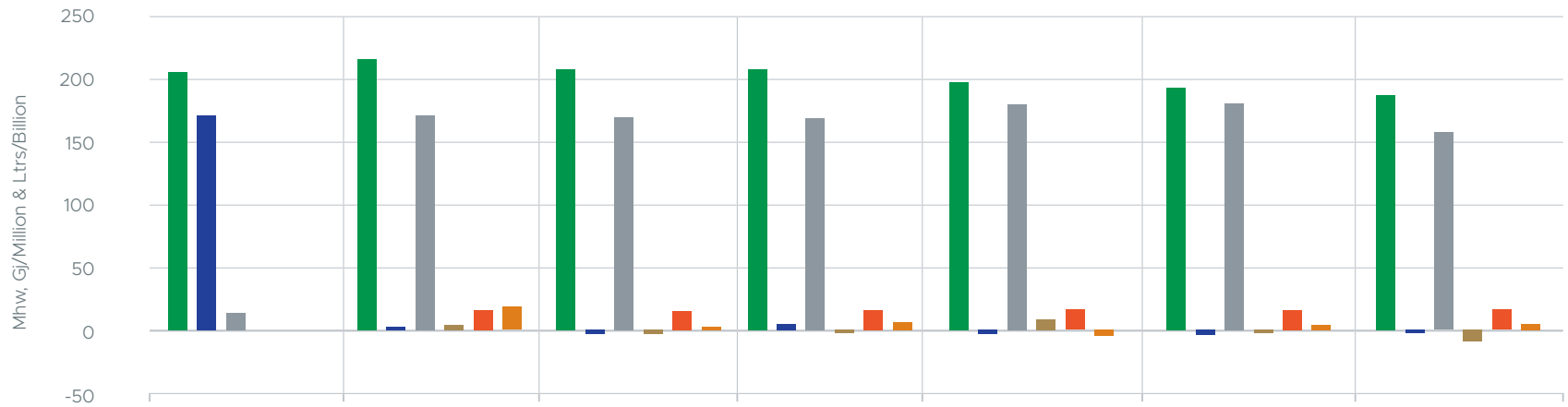
Electricity generation volumes are projected to decrease by 4.61% based on Eskom's energy production plans for the medium term. The production plan factors import and IPP's projections as per their FY 2024 corporate plan and the generation system energy availability factor (EAF) of 60.0% in FY 2024, 65% in FY 2025 and 70% in FY 2026. These estimates are also subject to risks such as Eskom not having adequate capacity to meet demand which may result in load shedding.

Piped gas volumes are increasing by 8.4% as resellers are planning increase usage due to the commissioning for Wax plant in KwaZulu Natal that would be operational from FY24. Commissioning of the plant is assumed to increase gas usage in the budgeting period.

Petroleum pipelines are decreasing by 4% due to the uncertainty of fuel demand, lack of future infrastructure investments from the oil majors' perspective and the anticipated energy transition.



Figure 2: Projected Volumes for 2024/25 MTEF



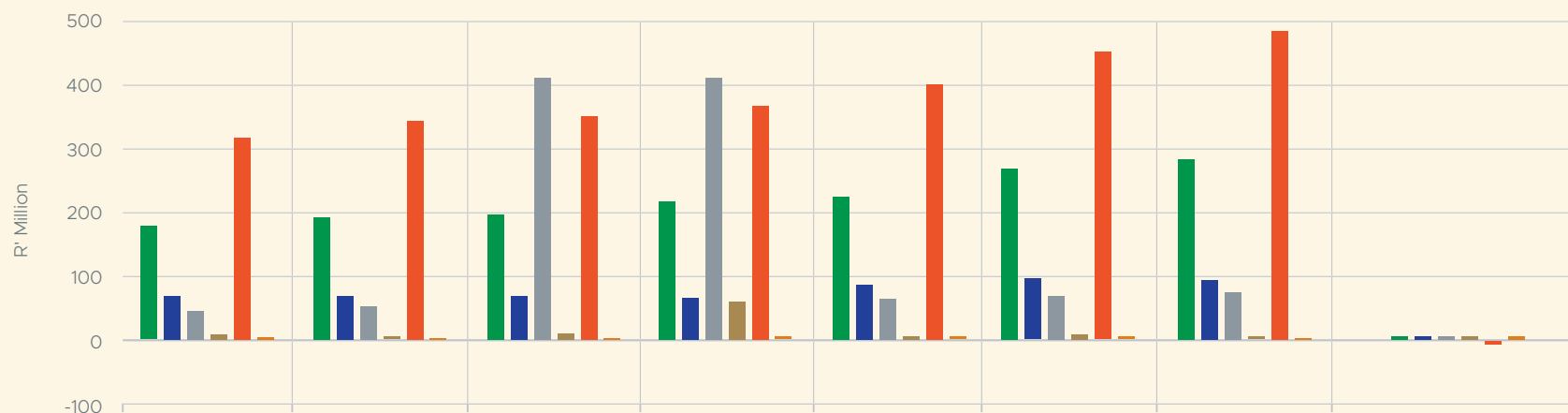
	Actual	Actual	Actual	Current Budget	Proposed Budget	Forecast	Forecast
	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
Electricity (MwHh)	207	215	209	210	200	194	189
% Change		4,33	(2,97)	0,41	(4,61)	(2,88)	(2,61)
Piped Gas (Gj)	173	175	172	168	182	182	158
% Change		1,50	(1,88)	(2,48)	8,4	(0,10)	(12,83)
Petroleum Pipelines (ltrs)	13	15,3	15,5	16,3	15,6	16,0	16,1
% Change		17,73	1,31	5,10	(4,0)	2,36	0,81

3.5. Revenue

Revenue for the 2024/25 financial year amounts to R398 million, which is 5.9% higher than the 2023/24 budgeted amount of R376 million. Interest income is projected to decrease slightly as interest rates normalise as well as NERSA surplus funds are utilised. Figure 3 provides a breakdown of the NERSA sources of revenue historically and the projected values over the 2024 MTEF.



Figure 3: Projected Revenue for 2024 MTEF



	Actual	Actual	Actual	Current Bud- get	Proposed Budget	Forecast	Forecast	% Change
	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27	2023/24 – 2024/25
Electricity License Fees	184	200	206	220	231	276	293	4,93%
Piped Gas Levies	74	75	74	77	88	96	102	14,36%
Petroleum Pipelines Levies	52	61	62	67	68	77	83	1,29%
Interest and Other Income	9	9	12	12	11	9	8	(4,1%)
Total	319	345	353	376	395	458	486	5,93%
% Change		8,1%	2,4%	6,4%	5,9%	15,0%	62,%	

The required revenue is also supplemented by the utilisation of previously accumulated surpluses as a form of refund of levies to the regulated industries. Total available refunds at 31 March 2023 amount to R170 million. Table 1 below provides a breakdown of accumulated surpluses available as refunds per industry and the proposed refund of R87.2 million for the 2024/25 financial year.

**Table 1: Refunds to Industry**

Regulated Industry	Accumulated Surplus as at 31 March 2023	Commitment as at 31 March 2023	Cash Flow Mitigation Reserve for the 2023/24 Financial Year Budget	Refunds Available as at 31 March 2023	Refunds Balance 2024/25	Refunds Balance	% Refunds allocated
Electricity Regulation	R115,260,853	R87,036	R46,418,327	R68,755,491	(R46,500,770)	R22,254,722	67.63
Piped-Gas Regulation	R53,521,170	R31,513	R17,622,428	R35,867,165	(R14,953,482)	R20,913,683	41.69
Petroleum Pipelines Regulation	R81,182,966	R31,513	R15,709,419	R65,442,034	(R25,715,013)	R39,727,021	39.29
Total	R249,964,926	R150,062	R79,750,175	R170,064,690	(R87,169,265)	R82,895,425)	51.69

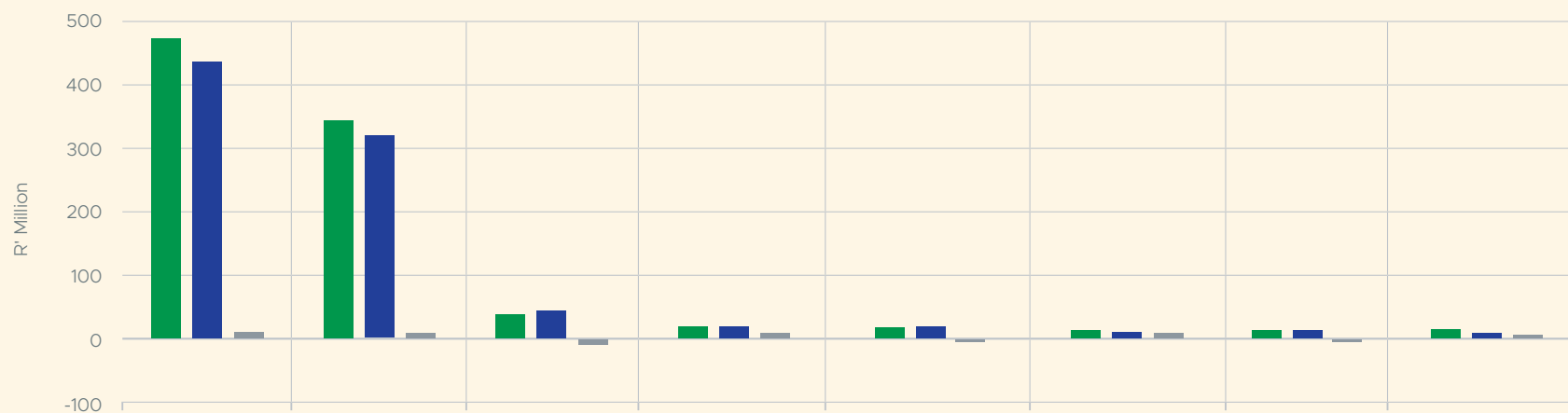
Refunds to regulated industries is allocated after capital commitments (as audited) and the cash-flow risk mitigation reserve held by the Energy Regulator to overcome timing differences between the start of the financial year and the start of levy payment by the industries and possible delay in the approval of the budget by the Minister. The reserve is based on 3 months' employment cost and 4.5% of the annual operating expenditure.

3.6. Operating Expenditure

Operating expenditure for the 2024/25 financial year amounts to R468 million, which is 6.7% higher than the R439 million of 2023/24. Figure 4 below provides a year on year view of operating costs per the aggregated expenditure categories.



Figure 4: 2024/25 Operating Expenditure



	Total Operating Expenditure	Employment cost	Professional fees	ICT and Office Administration	Travel Accommodation and Training	Advertising, Promotion and Communication	Facilities Maintenance	Other Expenses
Proposed Budget 2024/25	468	345	40	27	21	12	11	13
Current Budget 2023/24	439	320	47	21	22	11	11	7
% Change Budget 2023/24 - 2024-25	6,7%	7,8%	(14,1%)	27,7%	(3,0%)	3,6%	(2,1%)	83.1%



The variance per category of operating expenditure can be summarised as follows

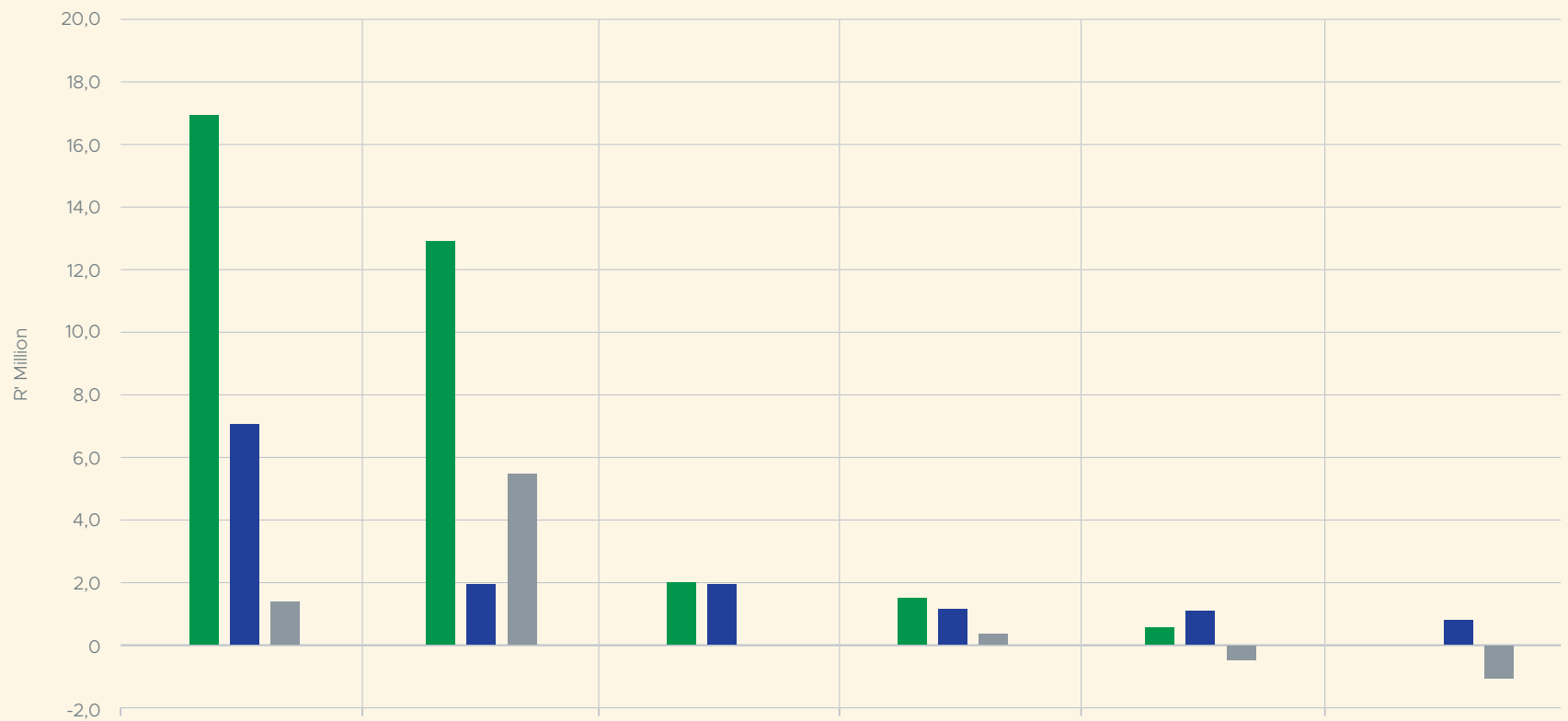
- **Employment cost** will be increasing by **7.8%** due to the projected salary adjustment of 7.2% on gross salaries and new additional fixed term contract appointments for the setup of the gender mainstreaming office and the project management office to operationalise the revised operating model.
- Professional fees will be decreasing by **14.1%** due to fewer planned consulting projects. Professional fees also include provision for on-going legal defense costs for various litigations against NERSA.
- ICT and Office Administration will be increasing by **27.7%** due to the implementation of two new ICT projects, which are the online collaboration portal and data warehouse project to cover hosting fees and requirements. The application development costs are allocated to capital expenditure.
- Travel, Accommodation and Training will be decreasing by **3%** due to reprioritisation of budget from travel costs to key projects and training. The overall reduction in the travel budget is 6.2%. The training budget however increases by 11% to continuous staff development.
- Advertising, Promotion and Communication will be increasing by **3.6%** due to expenditure related to advertising, publication and communication and stakeholder meetings.
- Facilities Maintenance will be decreasing by **2.1%** due to building operating expenses and insurance still required to maintain and secure facilities.
- Other expenses will be increasing by **83.1%** due to the addition of new data requirements on electricity energy markets for regulatory divisions.

3.7. Capital Expenditure

Capital expenditure for the 2024/25 financial year amounts to R18 million, which is **142%** higher than the R7 million for 2023/24 due to a significant increase in budget requirements for Computer software. The software acquisition relates to the development of an online collaboration portal between NERSA and licensees and the development of data warehouse as a business intelligence tool.



Figure 5: Capital Expenditure 2024/25



	Capital Expenditure	Computer Software	Computer Hardware	Building Improvement	Office Furniture and Equipment	Motor Vehicles
Proposed Budget 2024/25	17,1	13,0	2,0	1,5	0,6	–
Current Budget 2023/24	7,0	2,0	2,0	1,1	1,1	0,8
% Change	142,8%	550,0%	0,0%	33,7%	(46,5%)	(100%)



Annual Budget for the year 2024/25: Consolidated Summary

INCOME AND EXPENDITURE BUDGET 2023/24 AND FORECAST FOR THE PERIOD 2024/25 AND 2025/26

DESCRIPTION	NOTES	A	B		C	E	1	F	H
		BUDGET	ACTUAL	% Variance (A / B)	APPROVED BUDGET	CONSOLIDATED BUDGET	% Variance (C/E)	FORECAST	FORECAST
		2022/23	2023/24		2023/24	2024/24		2025/26	2026/27
TOTAL INCOME		337,019,635	353,314,741	4.8%	398,085,195	5.9%	457,804,314	486,204,691	420 660 577
License fees from Electricity Industry	1	200,427,830	205,648,339	2.6%	219,911,150	230,760,934	4.9%	275,647,658	293,035,373
Levies from Piped-Gas Industry	2	71,971,726	73,996,486	2.8%	76,849,794	87,887,092	14.4%	95,937,599	102,380,406
Levies from Petroleum Pipeline Industry	3	58,408,397	61,613,928	5.5%	67,344,007	68,214,207	1.3%	77,217,982	83,120,187
Registration fees	4	100,000	83,200	(16.8%)	70,000	70,000	0.0%	70,000	70,000
Interest received	5	6,050,378	11,718,527	93.7%	11,601,149	11,118,145	(4.2%)	8,894,516	7,560,339
Rental Income	6	61,304	31,581	(48.5%)	33,160	34,818	5.0%	36,559	38,387
Other Income			222,681	0.0%	0.0%	0.0%	0.0%	-	-
TOTAL OPERATING EXPENDITURE		396,817,219	388,015,478	2.2%	438,833,926	468,154,460	6.7%	489,252,026	517,652,403
Advertising, Promotion and Communication	7	140,087	297,621	155.4%	140,087	140,087	0.0%	149,893	159,936
Employment cost	8	1,255,180	1,432,205	101.3%	1,343,883	1,343,883	(3.3%)	1,423,172	1,444,520
Facilities Maintenance	9	36,769,997	39,173,170	12.6%	41,061,838	41,061,838	8.2%	43,936,167	46,879,890
Office Administration	10	-	3,969,417	0.0%	4,320,840	4,320,840	0.0%	4,320,840	4,320,840
Professional fees	11	-	260,400	0.0%	247,200	247,200	0.0%	247,200	247,200
Travel, Accommodation and Training	12	8,965,576	9,432,228	4.1%	9,599,166	9,599,166	6.7%	10,165,517	10,318,000
Other Expenses136,135,838	13	75,000	-	40.0%	75,000	75,000	0.0%	80,250	85,627
NET SURPLUS/(DEFICIT) before Depreciation		(59,797,584)	(34,700,737)	0.0%	(63,024,665)	(70,069,265)	11.2%	(31,447,713)	(31,447,713)



Annual Budget for the year 2024/25: Consolidated Summary

INCOME AND EXPENDITURE BUDGET 2023/24 AND FORECAST FOR THE PERIOD 2024/25 AND 2025/26

DESCRIPTION	NOTES	A	B		C	E	1	F	H
		BUDGET	ACTUAL	% Variance (A /B)	APPROVED BUDGET	CONSOLIDATED BUDGET	% Variance (C/E)	FORECAST	FORECAST
		2022/23	2023/24		2023/24	2024/24		2025/26	2026/27
Depreciation		-	5 354 469	0.0%	-	-	0.0%	-	-
NET SURPLUS/ (DEFICIT) for the period		(59 797 584)	(40 055 206)	0.0%	(63 024 665)	(70 069 265)	(11.2%)	(31 447 713)	(31 447 713)
TOTAL CAPITAL EXPENDITURE	14	13,632,353	10,085,820	26.0%	7 202 000	17 100 000	137.4%	10 000 000	10 000 000
Motor vehicles	14.1	1,850,000	1,882,673	(1.8%)	800,000	-	(100.0%)	1,000,000	1,000,000
Computer software	14.2	1,900,000	1,883,901	0.8%	2,000,000	13,000,000	550.0%	5,000,000	5,000,000
Office furniture and equipment	14.3	2,083,000	116,823	94.4%	1,280,000	600,000	(53.1%)	500,000	500,000
Building improvements	14.4	699,353	-	100.0%	1,122,000	1,500,000	33.7%	1,500,000	1,500,000
Computer hardware		7,100,000	6,202,422	12.6%	2,000,000	2,000,000	0.0%	2,000,000	2,000,000
Funding Requirement (Opex excl Dep+Capex)		410,449,572	398,101,298	(3.0%)	446,035,926	485,254,460	8.8%	499,252,026	527,652,403
Cash Flow Mitigating Reserve		66,806,283	66,430,010	0.6%	74,325,836	79,750,175	7.3%	85,584,877	92,155,796



Annual Budget for the year 2024/25: Consolidated Detail

INCOME AND EXPENDITURE BUDGET 2023/24 AND FORECAST FOR THE PERIOD 2024/25 AND 2025/26

DESCRIPTION	NOTES	A	B		C	E	1	F	H
		BUDGET	ACTUAL	% Variance (A / B)	APPROVED BUDGET	CONSOLIDATED BUDGET	% Variance (C/E)	FORECAST	FORECAST
		2022/23	2023/24		2023/24	2024/24		2025/26	2026/27
TOTAL INCOME		337,019,635	353,314,741	4.8%	375,809,261	398,085,195	5.9%	457,804,314	486,204,691
License fees from Electricity Industry	1	200 427 830	205,648,339	2.6%	219,911,150	230,760,934	4.9%	275,647,658	293,035,373
Levies from Piped-Gas Industry	2	71,971,726	73,996,486	2.8%	76,849,794	87,887,092	14.4%	95,937,599	102,380,406
Levies from Petroleum Pipeline Industry	3	58,408,397	61,613,928	5.5%	67,344,007	68,214,207	1.3%	77,217,982	83,120,187
Registration fees	4	100,000	83,200	(16.8%)	70,000	70,000	0.0%	70,000	70,000
Interest received	5	6,050,378	11,718,527	93.7%	11,601,149	11,118,145	(4.2%)	8,894,516	7,560,339
Rental Income	6	61,304	31,581	(48.5%)	33,160	34,818	5.0%	36,559	38,387
Other Income			222,681	0.0%	0.0%	0.0%	0.0%	-	-
TOTAL OPERATING EXPENDITURE		396 817 219	388 015 478	2.2%	438 833 926	468 154 460	6.7%	489 252 026	517 652 403
National/International/Initiatives	7.1	-	-	0.0%	70,000	-	(100.0%)	-	-
Publications and Communications	7.2	2,560,000	,2,430,450	5.1%	2,635,000	1,800,000	(31.7%)	1,884,600	1,973,176
Sponsorships	7.3	100,000	23,349	76.7%	100,000	200,000	100.0%	209,400	219,242
Advertising	7.4	8,405,000	8,450,301	(0.5%)	6,765,000	7,965,000	17.7%	8,339,355	8,731,305
Stakeholder Meetings		790,000	662,068	16.2%	890,000	890,000	0.0%	931,830	975,626
Tribunals and Hearings	7.2	985,000	214,112	78.3	1,010,000	1,030,000	2.0%	1,078,410	1,129,095
Advertising, Promotion and Communication	7	12 840 000	11 780 280	8.3%	11 470 000	11 885 000	3.6%	12 443 595	13 028 444



Annual Budget for the year 2024/25: Consolidated Detail

INCOME AND EXPENDITURE BUDGET 2023/24 AND FORECAST FOR THE PERIOD 2024/25 AND 2025/26

DESCRIPTION	NOTES	A	B		C	E	1	F	H
		BUDGET	ACTUAL	% Variance (A /B)	APPROVED BUDGET	CONSOLIDATED BUDGET	% Variance (C/E)	FORECAST	FORECAST
		2022/23	2023/24		2023/24	2024/24		2025/26	2026/27
Gross Salaries	8.1	229,812,514	229,442,471	(0.2%)	256,636,488	256,636,488	7.6%	299,206,459	324,339,802
Learnership Allowance		1,639,241	499,841	69.5%	1,638,636	1,638,636	0.0%	1,776,281	1,925,489
Internship Allowance		1,639,241	602,144	63.3%	1,638,636	1,638,636	0.0%	1,776,281	1,925,489
Leave Pay: Staff		811,277	533,100	34.3%		811,277	0.0%	811,277	811,277
Leave pay: Regulator Members		140,087	297,621	(112.5%)	140,087	140,087	0.0%	140,087	140,087
Performance Bonus: FTRM		1,255,180	1,432,205	(14.1%)	1,343,883	1,343,883	(3.3%)	1,380,957	1,467,958
Performance Bonus: Staff	8.2	36,769,997	39,173,170	(6.5%)	41,061,838	41,061,838	8.2%	48,182,331	52,229,647
Cellphone and data allowance Staff	8.3	-	3,969,417	0.0%	4,320,840	4,320,840	0.0%	4,320,840	4,320,840
Cellphone and data allowance FTRM	8.4	-	260,400	0.0%	247,200	247,200	0.0%	247,200	247,200
Remuneration: FTRM		8,965,576	9,432,228	(5.2%)	9,599,166	9,599,166	6.7%	10,883,961	11,569,651
Publication Incentives		75,000	-	40.0%	100.0%	75,000	0.0%	75,000	75,000
Salaries Temporary Staff		2,900,000	879,712	69.7%	2,637,768	4 100 000	55.4%	1,300,000	1,300,000
Employment cost	8	284,008,113	286,522,308	(0.9%)	320,150,820	344,979,062	7.8%	370,100,675	400,352,439
Maintenance	9.1	3,800,000	2,003,743	47.3%	6,294,926	3,800,000	(39.6%)	3,978,600	4,165,594
Office Rental	9.2	-	923,836	0.0%	-	600,000	0.0%	628,200	657,725
Motor Vehicle Expenses	9.3	400,000	459,445	(14.9%)	180,000	450,000	150.0%	471,150	493,294
Facility Management Operating Expenses	9.4	3,150,000	2,049,834	0.0%	1,022,729	2,090,310	104.4%	2,188,555	2,291,417
Municipal Charges	9.5	2,750,00	2,699,196	1.8%	2,500,000	2,820,000	12.8%	2,952,540	3,091,309
Insurance		1,059,104	992,109	6.3%	1,400,000	1,400,000	0.0%	1,465,800	1,534,693
Facilities Maintenance	9	11,159,104	9,128,163	18.2%	11,397,655	11,160,310	(2.1%)	11,684,845	12,234,032



Annual Budget for the year 2024/25: Consolidated Detail

INCOME AND EXPENDITURE BUDGET 2023/24 AND FORECAST FOR THE PERIOD 2024/25 AND 2025/26

DESCRIPTION	NOTES	A	B		C	E	1	F	H
		BUDGET	ACTUAL	% Variance (A / B)	APPROVED BUDGET	CONSOLIDATED BUDGET	% Variance (C/E)	FORECAST	FORECAST
		2022/23	2023/24		2023/24	2024/24		2025/26	2026/27
Office operational expenses-Lease Payments	10.1	1,003,385	608,111	39.4%	1,053,554	1,159,723	10.1%	1,214,230	1,271,299
Postage & Courier Services	10.2	54,401	2,211	4.0%	55,500	67,500	21.6%	70,673	73,994
Personal Protective Equipment		175,000	-	100.0%	277,500	475,000	71.2%	497,325	520,699
Information Technology Operations	10.3	8,250,000	4,476,627	45.7%	8,058,000	13,383,576	66.1%	10,000,000	10,000,000
Software License Fees		7,675,245	7,901,154	(2.9%)	7,989,597	8,389,385	5.0%	7,500,000	7,500,000
Stationery and Printing		762,250	218,379	71.4%	768,250	289,000	(62.4%)	302,583	316,804
Organizational Membership Subscriptions	10.4	1,659,244	1,463,327	11.8%	1,674,244	1,884,274	12.5%	1,972,835	2,065,558
Professional Membership Subscriptions	10.5	146,985	95,736	34.9%	143,090	131,027	(8.4%)	137,185	143,633
Telephone and fax		767,408	596,351	22.3%	769,908	760,000	(1.3%)	795,720	833,119
Office Administration	10	20,493,918	15,411,895	24.8%	20,789,644	26,539,485	27.7%	22,490,551	22,725,107
Consultants' Fees	11.1	15,228,742	13,422,493	11.9%	18,557,414	11,820,358	(36.3%)	12,375,915	12,957,583
External Auditors	11.2	2,520,308	2,301,394	8.7%	3,209,390	3,369,860	5.0%	3,528,243	3,694,071
Remuneration - PTRM and External Members	11.3	2,599,507	2,446,476	5.9%	3,957,749	3,957,749	0.0%	4,143,764	4,338,521
Recruitment costs		800,000	487,344	39.1%	600,000	600,000	0.0%	628,200	657,725
Legal fees		19,500,000	20,190,046	(3.5%)	18,000,000	18,000,000	0.0%	18,846,000	19,731,762
Co-sourced internal audit function		2,200,000	2,126,825	3.3%	2,200,000	2,200,000	0.0%	2,303,400	2,411,660
Professional fees	11	42,848,557	40,974,577	4.4%	46,524,554	39,947,967	(14.1%)	41,825,522	43,791,321



Annual Budget for the year 2024/25: Consolidated Detail

INCOME AND EXPENDITURE BUDGET 2023/24 AND FORECAST FOR THE PERIOD 2024/25 AND 2025/26

DESCRIPTION	NOTES	A	B		C	E	1	F	H
		BUDGET	ACTUAL	% Variance (A /B)	APPROVED BUDGET	CONSOLIDATED BUDGET	% Variance (C/E)	FORECAST	FORECAST
		2022/23	2023/24		2023/24	2024/24		2025/26	2026/27
Learnership programme		950,760	338,500	64.4%	1,250,760	1,250,760	0.0%	1,309,546	1,371,094
Study fees		1,392,000	1,618,436	(16.3%)	1,307,000	1,307,000	0.0%	1,368,429	1,432,745
External Bursaries		525,000	423,555	19.3%	551,000	551,000	0.0%	576,897	604,011
Training WEGE	12.1	-	193,000	0.0%	250,000	400,000	60.0%	418,800	438,484
Train. & Dev. Full Time Regulator Members		194,484	178,864	8.0%	400,000	400,000	0.0%	418,800	438,484
Train. & Dev. Part Time Regulator Members		57,519	24,777	56.9%	500,000	500,000	0.0%	523,500	548,105
Train. & Dev. Staff	12.2	4,837,190	4,768,128	1.4%	3,447,190	4,140,311	20.1%	4,334,905	4,538,646
Travel Costs Regulator Members		1,132,396	1,023,371	9.6%	1,800,000	1,932,182	7.3%	2,022,995	2,118,076
Travel Costs Staff	12.3	10,242,340	9,163,364	10.5%	12,015,304	10,383,882	(13.6%)	10,871,925	11,382,905
Travel, Accommodation and Training	12	19,331,689	17,731,995	8.3%	21,521,254	20,865,135	(3.0%)	21,845,797	22,872,549
Bank charges and Forex	13.1	65,000	72,320	(11.3%)	65,000	80,000	23.1%	83,760	87,697
Catering	13.2	500,000	431,180	13.8%	315,000	497,500	57.9%	520,883	545,364
Employees Wellness	13.3	400,000	243,362	39.2%	800,000	500,000	(37.5%)	523,500	548,105
Health and Safety	13.4	900,000	877,822	2.5%	800,000	700,000	(12.5%)	732,900	767,346
Knowledge Centre	13.5	4,270,838	4,485,772	(5.0%)	5,000,000	11,000,000	120.0%	7,000,000	700,000
Loss on disposal of assets		-	355,804	-	-	-	0.0%	-	-
Other Expenses	13	6,135,838	6,466,260	(5.4%)	6,980,000	12,777,500	83.1%	8,861,043	2,648,511
NET SURPLUS/(DEFICIT) before Depreciation		(59,797,584)	(34,700,737)	0.0%	(63,024,665)	(70,069,265)	11.2%	(31,447,713)	(31,447,713)



Annual Budget for the year 2024/25: Consolidated Detail

INCOME AND EXPENDITURE BUDGET 2023/24 AND FORECAST FOR THE PERIOD 2024/25 AND 2025/26

DESCRIPTION	NOTES	A	B		C	E	1	F	H
		BUDGET	ACTUAL	% Variance (A /B)	APPROVED BUDGET	CONSOLIDATED BUDGET	% Variance (C/E)	FORECAST	FORECAST
		2022/23	2023/24		2023/24	2024/24		2025/26	2026/27
Depreciation Building		-	1,322,978	0.0%	-	-	0.0%	-	-
Depreciation Hardware		-	2,106,743	0.0%	-	-	0.0%	-	-
Depreciation Motor Vehicles		-	104,153	0.0%	-	-	0.0%	-	-
Depreciation Office Equipment		-	1,369,128	0.0%	-	-	0.0%	-	-
Depreciation Software		-	451,467	0.0%	-	-	0.0%	-	-
Depreciation		-	5,354,469	0.0%	-	-	0.0%	-	-
NET SURPLUS/ (DEFICIT) for the period		(59,797,584)	(40,055,206)	0.0%	(63,024,665)	(70,069,265)	(11.2%)	(31,447,713)	(31,447,713)
TOTAL CAPITAL EXPENDITURE	14	13,632,353	10,085,820	26.0%	7,202,000	17,100,000	137.4%	10,000,000	10,000,000
Motor vehicles	14.1	1,850,000	1,882,673	(1.8%)	800,000	-	(100.0%)	1,000,000	1,000,000
Computer software	14.2	1,900,000	1,883,901	0.8%	2,000,000	13,000,000	550.0%	5,000,000	5,000,000
Office furniture and equipment	14.3	2,083,000	116,823	94.4%	1,280,000	600,000	(53.1%)	500,000	500,000
Building improvements	14.4	699,353	-	100.0%	1,122,000	1,500,000	33.7%	1,500,000	1,500,000
Computer hardware		7,100,000	6,202,422	12.6%	2,000,000	2,000,000	0.0%	2,000,000	2,000,000
Funding Requirement (Opex excl Dep+Capex)		410,449,572	398,101,298	(3.0%)	446,035,926	485,254,460	8.8%	499,252,026	527,652,403
Cash Flow Mitigating Reserve		66,806,283	66,430,010	0.6%	74,325,836	79,750,175	7.3%	85,584,877	92,155,796



Notes to the Budget

REVENUE

1. Licence fees from Electricity Industry

The licence fees from the electricity Industry increased by **4,9%** from R220 million in 2023/24 to R231 million in 2024/25

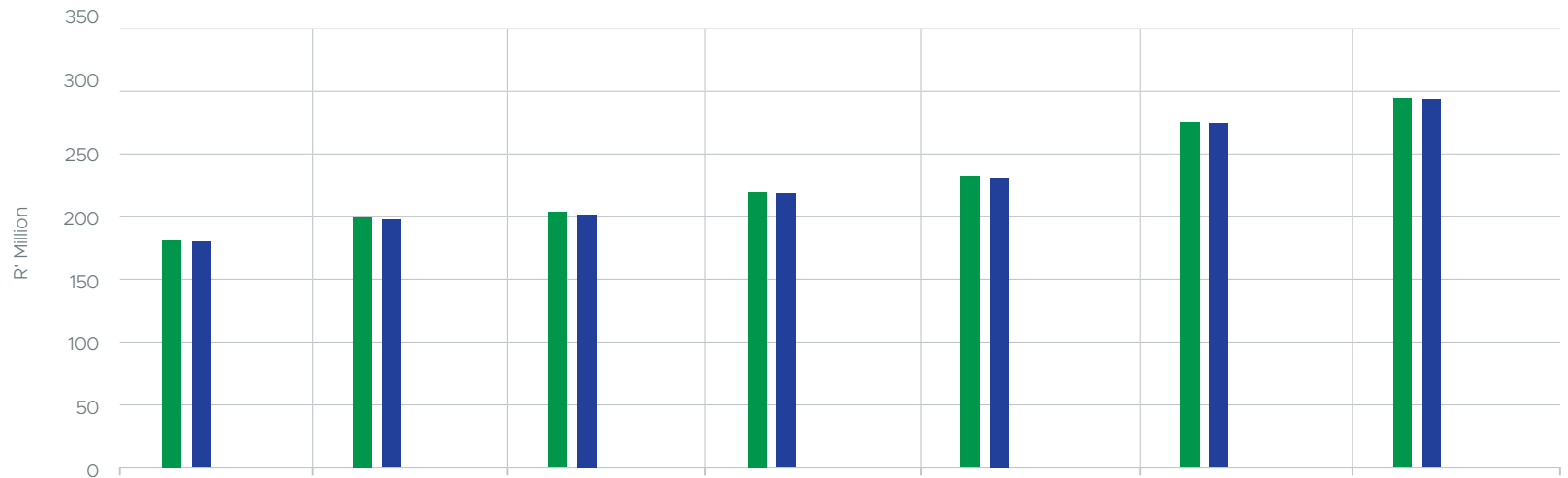
The electricity licence fee rate increases by 10.00% and the changes in the component of the licence fee rate are as follows:

Table 2: Changes in the electricity licence fee rate

Industry	Volumes	Division Operating Expenditure	Support Service Allocation	Capital Expenditure	Interest and Other Income	Refunds to Industry	TOTAL
Electricity Licence fee rate	5.07%	2.73%	5.86%	2.61%	0.31%)	(6.40%)	10.00%



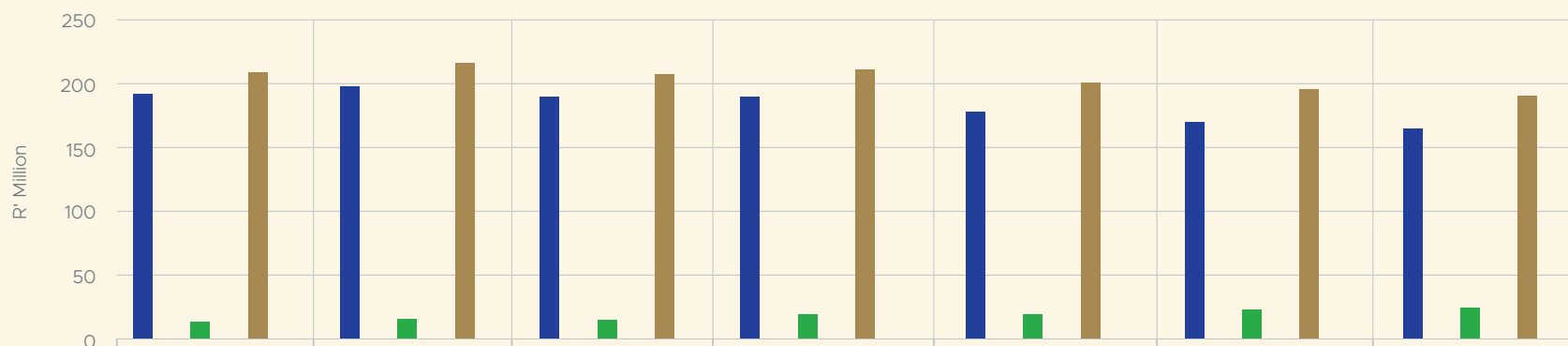
Figure 6: Electricity license fees



	Actual	Actual	Actual	Approved Budget	Budget	Forecast	Forecast
	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
● License Fees	184	200	206	220	231	276	293
● Registration	0.03	0.06	0.08	0.07	0.07	0.07	0.07
● Total	184	200	206	220	231	276	293
● % Change Licenses		8.5%	2.8%	6.9%	4.9%	19.55	6.3%
● % Change Registrations		126.1%	33.3%	(15.9%)	0.0%	0.0%	0.0%
● % Change Total		8.5%	2.8%	6.9%	4.9%	19.4%	6.3%



Figure 7: Electricity volumes



	Actual	Actual	Actual	Approved Budget	Budget	Forecast	Forecast
	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
● Eskom	192	198	191	189	178	171	163
● % Change Eskom		2.8%	(3.3%)	(1.3%)	(5.5%)	(4.3%)	(4.1%)
● SSEG	14	18	18	21	22	24	26
● % Change Other		25%	1%	18%	3%	9%	8%
● Total	207	215	209	210	200	194	189
● % Total		4.3%	(3.0%)	0.4%	(4.6%)	(2.9%)	(2.6%)



2. Levies from Piped-Gas Industry

The piped-gas levies are increased by **14.4%** from R76.8 million in 2023/24 to R88 million in 2024/25

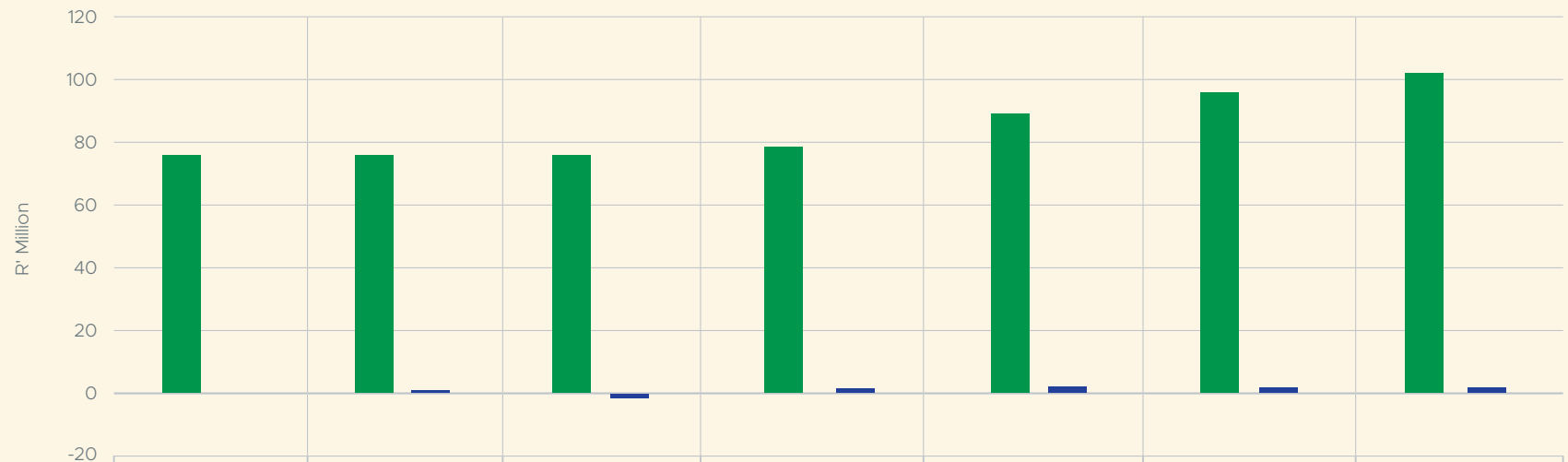
The piped gas levy rate increases by **5.50%** and the changes in the component of the levy are as follows:

Table 3: Changes in the piped gas levy

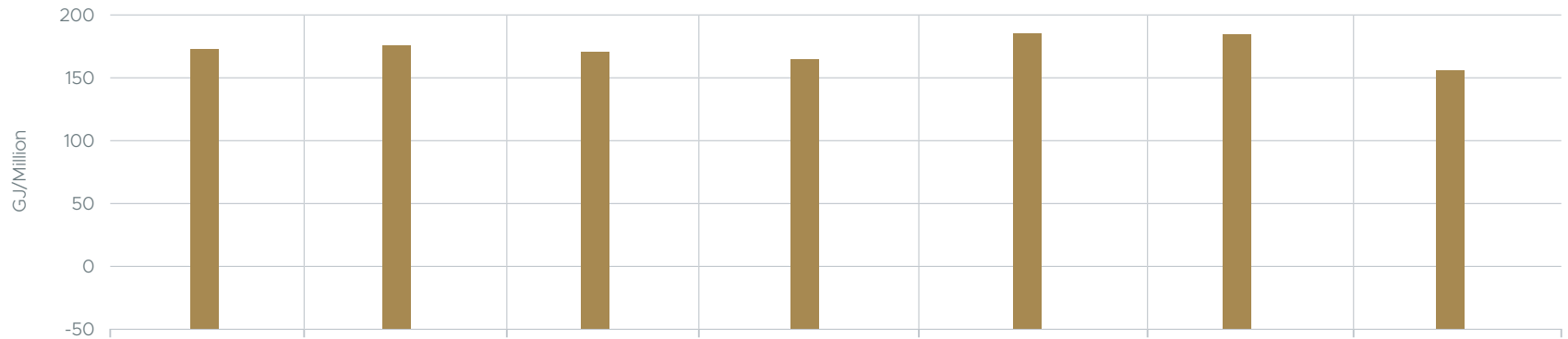
Industry	Volumes	Division Operating Expenditure	Support Service Allocation	Capital Expenditure	Interest and Other Income	Refunds to Industry	TOTAL
Piped Gas levy	(8.86%)	1.76%	6.08%	2.70%	(0.13%)	3.69%	5.50%



Figure 8: Piped-Gas levy income



	Actual	Actual	Actual	Approved Budget	Budget	Forecast	Forecast
	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
● Levy Income (R)	74	75	74	77	88	96	102
● % Change		1.5%	(1.9%)	3.9%	14.4%	9.2%	6.7%

**Figure 9: Piped Gas Volumes**

	Actual	Actual	Actual	Approved Budget	Budget	Forecast	Forecast
	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
● Gigajoules	173	175	172	168	182	182	158
● % Change		1.5%	(1.9%)	(2.5%)	8.4%	(0.1%)	(12.8%)

3. Levies from the Petroleum Pipelines Industry

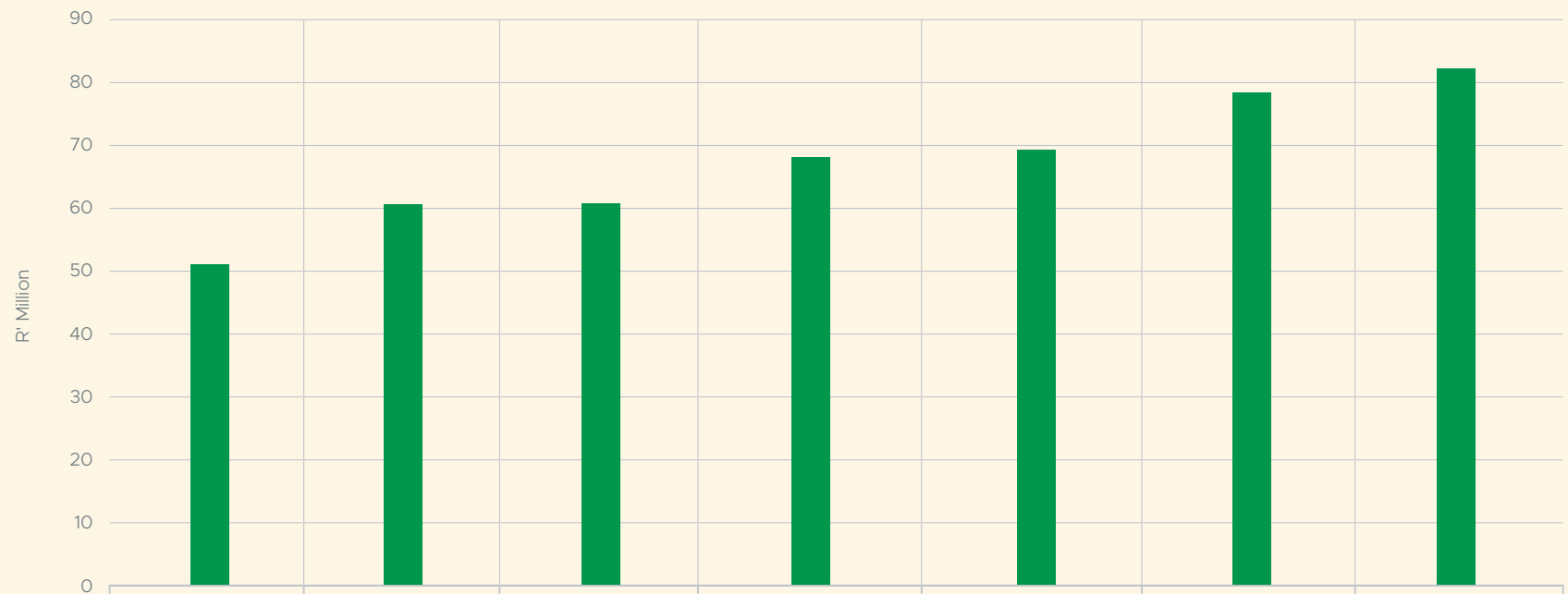
The petroleum pipelines levies are increased by **1.3%** from R67.3 million in 2023/24 to R68.2 million in 2024/25.

Table 4: Changes in the petroleum pipeline levy

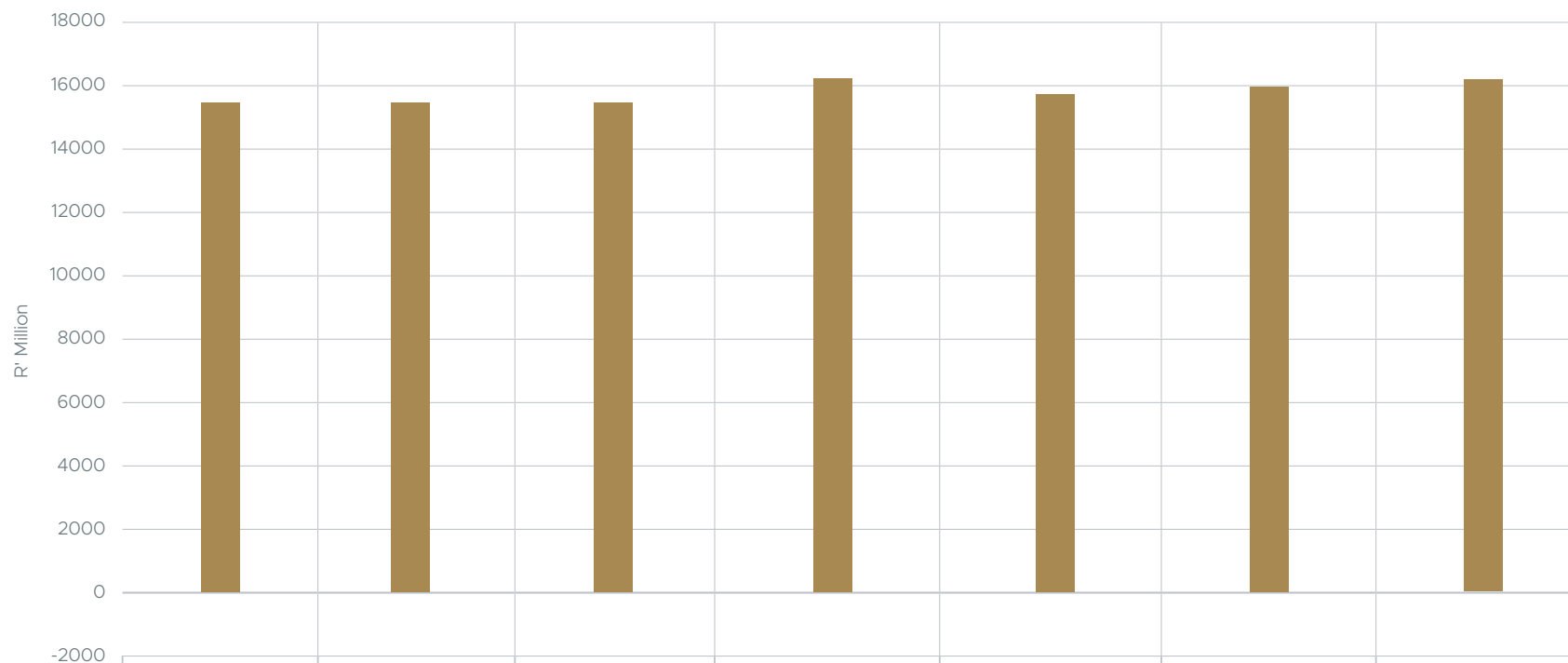
Industry	Volumes	Division Operating Expenditure	Support Service Allocation	Capital Expenditure	Interest and Other Income	Refunds to Industry	TOTAL
Petroleum Pipeline levy	4.21%	(0.40%)	6.93%	3.09%	0.15%	(8.48%)	5.50%



Figure 10: Petroleum Pipelines levy income



	Actual	Actual	Actual	Approved Budget	Budget	Forecast	Forecast
	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
● Levy Income (R)	52	61	62	67	68	77	83
● % Change		17.7%	1.3%	9.3%	1.3%	13.2%	7.6%

**Figure 10: Petroleum Pipelines volumes**

		Actual	Actual	Approved Budget	Budget	Forecast	Forecast
	2020/21	2021/22	2022/23	2023/24	2024/25	2025/26	2026/27
● Million Litres	15,307	15,307	15,507	16,297	15,647	16,017	16,147
● % Change		0.0%	1.3%	5.1%	(4.0%)	2.4%	0.8%



4. Registration fee

Registration fee is for small scale embedded generators of electricity and payable upon the applicants meeting minimum requirements. The registration fees amounts R200 per application and NERSA expects to process approximately 350 applications per year.

5. Interest received

Interest received is budgeted at an average cash level of R98 million at 8.17% (based on the projected deposit rate)

6. Rental income

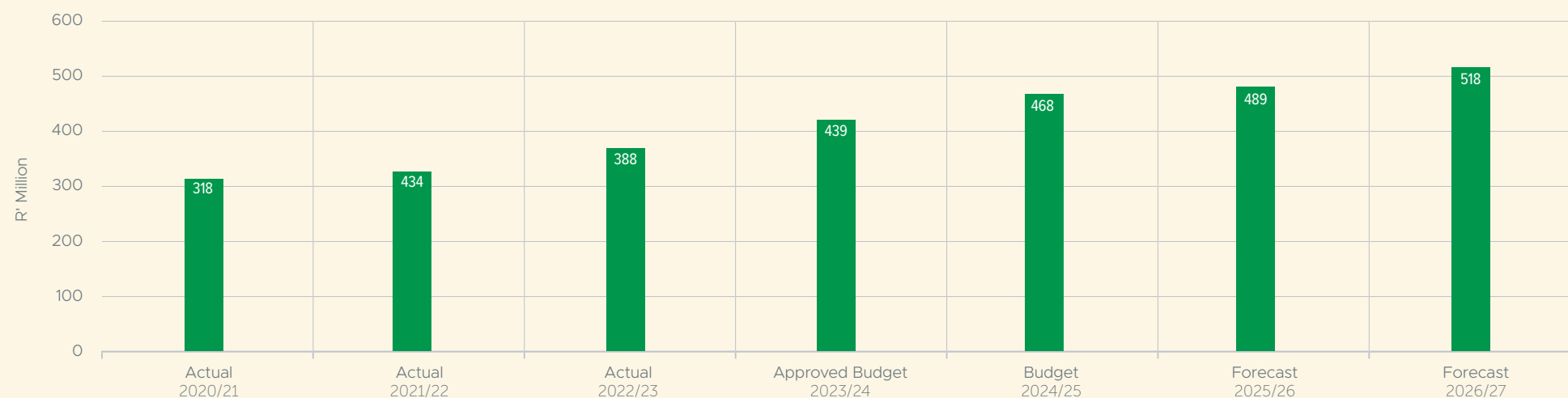
The rental income is for the leasing of office space by African Forum for Utilities. The increase of 5% is due to the 5% annual escalation per rental agreement.

Operating Expenditure

The total operating expenditure budget for 2024/25 is **R468 million**, which is **6.7%** increase compared to the 2023/24 budget. The budget takes into account the new strategic projects relating to Information Communication Technology, Customer Stakeholder management, Communication strategy and gender mainstreaming.

The variances of 10% or more are explained in the notes.

Figure 12: Operational expenditure trends





7. Advertising, Promotion And Communication

There is an overall increase of **3.6%** in the budget and this is attributable to increase in the budget for sponsorships and Advertising

7.1 National/International/Initiatives

This line item is used for International Co-ordination and Partnership activities, which include International Relations and Protocol Training, Protocol Related Activities for Africa Forum for Utility Regulators (AFUR) and Regional Electricity Regulators Association (RERA).

The 100% decrease in the budget is due to a budget no longer required for contribution in the collaboration of a conference.

7.2 Publications and Communications

This line item is used for the publication of NERSA materials (annual report, joint strategic and performance plans and customer education), corporate gifts and promotional materials, and exhibition space for hire and stands.

The budget is decreasing by **31.7%** due to the NERSA initiatives to have online publications as opposed to printing.

7.3 Sponsorships

NERSA receives on a regular basis request for sponsorship from various charitable organizations and institutions. The budget is to respond positively to some of these requests.

The budget is increasing by **100%** due to planned initiatives relating to gender mainstreaming for example donations to GBV Shelters

7.4 Advertising

The increase of 17.7% is due to increased use of advertising services and placing of regulatory notices required, as well as the impact of the use of Media Buying agencies panel for advertising for recruitment and regulatory legal notices. The use of Media Buying agencies includes cost of designing and commissions, while the benefit is the placing of an advert in shorter notice time.

8. Employment Cost

There is an overall increase of **7.8%** in the budget and this is attributable to Gross Salaries and temporary employees.

8.1 Gross Salaries

Gross salaries are increasing by of **7.6%** due to the following:

- NERSA staff and members of management are budgeted to receive 7.4% cost of living adjustment,
- 1% pay progression after the above adjustment

The staff complement remains unchanged at 248 (28 management and 220 staff members).

8.2 Salaries Temporary Staff

The increase of 55.4% is due to budgeting for the three positions, one is relating to gender mainstreaming and two are for the project management office.



9. Facilities Maintenance

There is an overall decrease of **2.1%** in the budget and this is attributable decrease in maintenance which is set off by increase in municipal charges, facilities management, and motor vehicle expenses.

9.1 Maintenance

The decrease in the budget by **39.6%** is due reallocation of costs between maintenance and facility management operational expenses. Monthly routine maintenance for air-con, lift maintenance, fire detection, electrical lighting, pest control and standby generator were previously allocated to maintenance instead of facility management operational expenses.

9.2 Rental Parking Bays

The 100% increase in the budget is due to reallocation of leases for parking bays from office lease payments for accurate classification for financial reporting.

9.3 Motor Vehicle Expenses

The increase by 150% is due to more staff members utilising new vehicles procured when travelling for site visits and audits as opposed to hiring rental cars. The pool vehicles are also used for regular deliveries by NERSA drivers.

9.4 Facilities Management Operating Expenses

The increase in the budget by **104%** is due reallocation of costs between maintenance and facility management operational expenses. Monthly routine maintenance for air-con, lift maintenance, fire detection, electrical lighting, pest control and standby generator were previously allocated to maintenance instead of facility management operational expenses. The budget also includes additional security for public hearing held outside NERSA Offices.

9.5 Municipal charges

The increase of 12, 8% is due increase in the electricity and municipal rates paid to City of Tshwane.

10. Office Administration

There is an overall increase of **27.7%** in the budget and this is attributable to lease payments and information technology operations.

10.1 Office Operational Expenses Lease Payments

The increase in the budget by 10.1% is to cater for increase in the new contract to be entered into. The contract for leasing of multifunctional machines is expiring and the new procurement process has to commence in the 2023/24 financial year.

10.2 Postage and Courier Service

The increase of 21.6% is due to the planned additional exhibitions and customer education workshops. Material needed for these events are sometimes couriered to the venue where these events are to be held.

10.3 Information Technology Operations

The increase of 66.1% is due to online collaboration portal, data warehouse and cyber security projects to be implemented. The procurement process for the online portal has commenced in 2023/24 financial year and the development is planned mainly in 2024/25 financial year.

10.4 Stationary and Printing

The decrease of 62.4% is due to further reductions in the budget as staff are still mainly working from home.



10.5 Organisaional Subscriptions

The increase of 12.5% is due to a NERSA's possible membership of the Energy Regulators Regional Association (ERRA). The proposed membership is aimed at world-class visibility, information sharing, building capacity, and exposure within NERSA. NERSA employees have attended ERRA training courses such as the ERRA Summer School: Introduction to Energy Regulation and Setting the Weighted Average Cost of Capital (WACC), which have allowed the NERSA staff to be exposed to international experts in their respective fields.

11. Professional Fees

There is an overall decrease of **14.1%** in the budget and this is attributable to a decrease in the number of planned projects.

11.1 Consulting Fees

There is a decrease in budget by **36.3%** due to less planned new projects. In the prior year there were many projects on the regulatory divisions that were stopped for example a base-case report to develop a strategy for a Gas, Renewable and Hydrogen Energy Partnership in South Africa. Some continuing projects are budgeted for lesser as they are in the further stages of implementation.

The detail on the consulting fees budget is based on the number of planned projects as indicated in the table 5 below.



Table 5: Planned Project

COST CENTRE	NAME OF PROJECT	PURPOSE OF THE PROJECT	ESTIMATED AMOUNT	EXISTING OR NEW	TERMS OF REFERENCE/ BUSINESS CASE
ELC	Distribution Independent Technical Audits	To assess whether the business operations of the Electricity Departments of licensees are sustainable for compliance monitoring against licence conditions	R1,500,000	New	Yes
EPT	Electricity Price Determination Methodology (EPDM)	Assisted with the new methodology implementation	R1,500,000	Existing	Yes
TOTAL ELECTRICITY			R3,000,000		
CSM	Media Monitoring	Monitoring and analysis of and report on, media articles that affect NERSA.	R300,000	Existing	Yes
CSM	Reputation Management	Provide reputation management services and to complement the work of CSM	R1000,000	New	Contract
CSM	Stakeholder perception and reputation survey	Survey on Stakeholder Perception and Reputation of NERSA	R600,000	New	Yes
CSM	Peer Review Exercise for Regulatory CEO's in the continent	Peer Review Exercise for Regulatory CEO's in the continent	R100,000	New	Yes
CSM	Translation service	As part of the implementation of NERSA's Language Policy. NERSA public materials will be translated into four other languages.	R100,000	Existing	
TOTAL CORPORATE SERVICE			R2,100,000		
CFO	GRAP Technical Review	Review of GRAP reporting standards applicable to NERSA, Training, Technical opinions and AFS review	R120,000	Existing	Contract
FMG	SAGE 300 (ACCPAC) Support	Support and maintenance the Financial Management systems	R603,333	Existing	Contract
FMG	VIP System Support	Support and maintenance the Payroll system	R133,333	Existing	Contract
SCM	Strategic Sourcing	Implementation of Strategic Sourcing	R133,333	New	Contract
SCM	Building improvement	Consulting engineers to assist with Building Improvement Projects	R400,000	Existing	Contract
SCM	Enterprise supplier development project	Development and Implementation of Enterprise and Supplier Development Programme.	R1,000,000	Existing	Contract



COST CENTRE	NAME OF PROJECT	PURPOSE OF THE PROJECT	ESTIMATED AMOUNT	EXISTING OR NEW	TERMS OF REFERENCE/ BUSINESS CASE
TOTAL FINANCE AND ADMINISTRATION			R2,656,666		
HRT	HRIS systems implementation	The project will entail the launch of new HRIS which include data mining modules as well as integrated PMS, Recruitment, Training and other HR modules. The current system is inadequate for NERSA needs	R150,000	Existing	Contract
HRV	Remuneration Consulting	Interventions emanating from the EE Pay Equity Study and will include Pay Equity Validation, advise on remuneration management for outliers as well as General Survey to support findings in the study. General Quarterly Industry-wide Surveys and benchmarking services.	R400,000	Ongoing	Yes
HRV	Pay scale development/ review	The Pay Equity Study identified anomalies in the current pay scale. A service provider is required to assist with the review of the NERSA Pay Scales and conduct an industry benchmark and market positioning	R250,000	New	
HRV	Pay equity Re analysis	Assessment of the compensating employees who have similar job functions with comparably equal pay, regardless of their gender, race, ethnicity or other status.	R350,000	New	
HRV	Competency assessment and LMS(GAR & PPR)	Regulatory Competency assessment and development of Learning Management System	R1000,000	New	
TOTAL HUMAN RESOURCES			R2,150,000		YES



COST CENTRE	NAME OF PROJECT	PURPOSE OF THE PROJECT	ESTIMATED AMOUNT	EXISTING OR NEW	TERMS OF REFERENCE/ BUSINESS CASE
COO	Transformation Initiatives	BBB-EE verification	R200,000	Existing	
IAU	Fraud Hotline	Monthly fees payable to the Service Provider for Management of the NERSA Fraud Hotline.	R33,692	Existing	Contract
SPM	Strategic planning facilitation	Facilitators for the Annual NERSA planning process for a 3-year prod with the aim to develop a new Annual Performance Plan - and if needed the review of the approved Strategic Plan (ER planning sessions and divisional planning sessions)	R200,000	Existing	Contract
SPM	Strategic risk assessment	Suitable and qualified service provider to facilitate the Strategic Risk Assessment for NERSA during the 2023 planning cycle (ER session and management session-EXCO)	R250,000	Existing	Contract
SPM	NERSA ISO Certification readiness	NERSA ISO Certification readiness	R430,000		
RSU	Compliance review within NERSA	Compliance review within NERSA	R400,000		
RSU	Board & Committee Annual Assessment	Independent annual assessment of the Energy Regulator	R200,000	Existing	Contract
RAR	Economic Impact Assessment Models	Assessment of the impact of all major tariff/pricing decisions covering the electricity, piped-gas and petroleum pipelines industries. The SLA between NERSA and the preferred service provider will be finalised before 30 September 2022. This will be a 5-year contract. The total budget for this contract is R3 000 000.	R600,000	Existing	Contract
TOTAL DIRECT SUPPORT			R2,313,692		
TOTAL NERSA			R11,820,358		



12. Travel, Accommodation And Training

There is an overall decrease of **3%** in the budget and this is attributable to travel and accommodation for staff.

12.1 Training Wege

The increase in budget of 60% is due to Gender Mainstreaming workshops for Energy Regulator members and staff planned in 2024/25.

12.2 Training and Development Staff

The increase of 20.1% is due to number of training initiatives NERSA is undertaking. The budget is based on 1.5% of employee costs to align it in prior year spending. In the 2023/24 budget training was reduced across all divisions. Human Resources department is coordinating a training and development plan for staff that will be based on staff performance contracts and also Enterprise wide.

12.3 TRAVEL COST STAFF

Regulatory division's licensing and compliance audit teams and dispute resolutions teams are the main drivers of NERSA travel costs for staff.

The decrease of 13.6% is due to reduction in travel for staff to align with 2022/23 spending and to limit travel costs to only essential travel.

13. Other Expenses

There is an overall increase of **83.1%** in the budget for other expenses mainly due to increases on new Knowledge Centers subscriptions.

13.1 Bank Charges

The increase of 23.1 % is due to increase in the number of planned international travel which requires purchasing of foreign subsistence allowances and increase in the foreign subscriptions.

13.2 Catering

The increase of 57.9% is due to additional catering for customer education meeting in the Electricity, Licensing, Compliance and Dispute Resolution department and stakeholder engagements.

13.3 Employee Wellness

The decrease of 37.5% is due to reduction in the employee wellness activities as staff are still working from home. The budget is aligned to prior year spending.

13.4 Health and Safety

The budget is mainly for the occupational health and safety in the NERSA building. The decrease of 12.5% is due to reduction of activities as staff are still working from home. The budget is aligned to prior year spending.

13.5 Knowledge Centre

The increase of 120% is due to new subscriptions and procurement of data. The major project relates to the Online electricity market insights publications and statistics (data), which includes the following: Coal Reports and data, Water Usage and Plant's Performance Statistics, Electricity Supply Side and Demand Side, Cost of Generating Electricity, Electricity Demand Side Data, Commodity Price and Financial data and information.



14. Capital Expenditure

Capital expenditure for the 2024/25 financial year amounts to R17.1 million, which is **137.4%** lower than the R13.6 million for 2022/23 due to reduction of budget on building improvements and computer software.

14.1 Motor Vehicles

The reduction by 100% is due to no further need to procure vehicles in the 2024/25 financial year. All vehicles required in the pool were procured in the 2022/23 financial year.

14.2 Computer Software

The increase in budget by 550% is due to the two key projects for acquisition and development of the licensee collaboration portal, data warehouse that are budgeted for in the 2024/25 financial year.

14.3 Office Furniture and Equipment

The budget decrease of 53.1% is due to security equipment procured in the prior year not required in the 2024/25 budget. The current year includes provision for the replacement of office furniture and equipment when damaged.

14.4 Building Improvements

The budget increase of 33.7% is due to building improvements needed resulting from close out report on the building refurbishment project. The engineering consultants were appointed in the 2023/24 financial year and the project is commencing.



Annual Budget for the year 2024/25: Electricity Regulation

INCOME AND EXPENDITURE BUDGET FOR 2024/25

	A	B		C	1	D	2
	ELECTRICITY REGULATION BUDGET	ELECTRICITY REGULATION ACTUAL	% Variance (A/B)	ELECTRICITY REGULATION APPROVED BUDGET	% Variance (A/C)	ELECTRICITY REGULATION PROPOSED BUDGET	% Variance (E/C)
DESCRIPTION	2022/23	2022/23		2023/24		2024/25	
TOTAL INCOME	204,072,606	212,679,537	4.2%	226,729,049	11.1%	237,299,652	4.7%
License fees for Electricity Industry	200,427,830	205,648,339	2.6%	219,911,150	9.7%	230,760,934	4.9%
Registration fee	100,000	83,200	(16.8%,)	70,000	(30.0%,)	70,000	0.0%
Interest received	3,509,220	6,804,312	93.9%	6,728,666	91.7%	6,448,524	(4.2%,)
Rental Income	35,556	16,043	(54.9%,)	19,233	(45.9%,)	20,195	5.0%
Other Income	-	127,644	0.0%	-	0.0%	-	0.0%
TOTAL OPERATING EXPENDITURE	103,378,755	104,585,152	(1.2%)	116,002,384	12.2%	122,007,597	5.2%
National/International/Initiatives	-	-	0.0%	-	0.0%	-	0.0%
Publications and Communications	40,000	-	100.0%	-	(100.0%,)	-	0.0%
Sponsorships	-	-	0.0%	-	0.0%	-	0.0%
Advertising	3,500,000	2,882,187	17.7%	2,600,000	(25.7%,)	2,850,000	9.6%
Stakeholder Meetings	490,000	613,370	(25.2%,)	490,000	0.0%	490,000	0.0%
Tribunals and Hearings	780,000	-	100.0%	780,000	0.0%	780,000	0.0%
Advertising, Promotion and Communication	4,810,000	3,495,558	27.3%	3,870,000	(19.5%)	4,120,000	6.5%



Annual Budget for the year 2024/25: Electricity Regulation

INCOME AND EXPENDITURE BUDGET FOR 2024/25

	A	B		C	1	D	2
	ELECTRICITY REGULATION BUDGET	ELECTRICITY REGULATION ACTUAL	% Variance (A/B)	ELECTRICITY REGULATION APPROVED BUDGET	% Variance (A/C)	ELECTRICITY REGULATION PROPOSED BUDGET	% Variance (E/C)
DESCRIPTION	2022/23	2022/23		2023/24		2024/25	
Gross Salaries	74,788,189	75,719,406	(1.2%,)	83,346,247	11.4%	89,738,684	7.7%
Learnership Allowance	-	-	0.0%	-	0.0%	-	0.0%
Internship Allowance	-	-	0.0%	-	0.0%	-	0.0%
Leave Pay: Staff	-	601,973	0.0%	-	0.0%	-	0.0%
Leave pay: Regulator Members	-	-	0.0%	-	0.0%	-	0.0%
Performance Bonus: FTRM	-	-	0.0%	-	0.0%	-	0.0%
Performance Bonus: Staff	11,966,109	13,420,316	(12.2%,)	13,335,400	11.4%	14,442,087	8.3%
Cellphone and data allowance Staff	-	1,395,881	0.0%	1,455,420	0.0%	1,455,420	0.0%
Cellphone and data allowance FTRM	-	-	0.0%	-	0.0%	-	0.0%
Remuneration: FTRM	-	-	0.0%	-	0.0%	-	0.0%
Publication Incentives	-	-	0.0%	-	0.0%	-	0.0%
Salaries Temporary Staff	-	-	0.0%	-	0.0%	-	0.0%
Employment cost	86,754,298	91,137,575	(5.1%)	98,137,066	13.1%	105,636,191	7.6%
Maintenance	-	-	0.0%	-	0.0%	-	0.0%
Motor Vehicle Expenses	-	-	0.0%	-	0.0%	-	0.0%
Facility Management Operating expenses	-	1,900	0.0%	-	0.0%	-	0.0%
Municipal Charges	-	-	0.0%	-	0.0%	-	0.0%
Insurance	-	17,140	0.0%	-	0.0%	-	0.0%
Facilities Maintenance	-	19,040	0.0%	-	0.0%	-	0.0%



Annual Budget for the year 2024/25: Electricity Regulation

INCOME AND EXPENDITURE BUDGET FOR 2024/25

DESCRIPTION	A	B		C	1	D	2
	ELECTRICITY REGULATION BUDGET	ELECTRICITY REGULATION ACTUAL	% Variance (A/B)	ELECTRICITY REGULATION APPROVED BUDGET	% Variance (A/C)	ELECTRICITY REGULATION PROPOSED BUDGET	% Variance (E/C)
	2022/23	2022/23		2023/24		2024/25	
PPE Tools	75,000	-	100.0%	75,000	0.0%	150,000	100.0%
Office operational expenses - Lease Payments	-	-	0.0%	-	0.0%	-	0.0%
Postage & Courier Services	12,500	4,489	64.1%	20,500	64.0%	5,000	(75.6%,)
Information Technology Operations	-	-	0.0%	-	0.0%	-	0.0%
Software License Fees	1,200,000	1,247,922	(4.0%,)	1,200,000	0.0%	1,200,000	0.0%
Stationery and Printing	35,000	1,730	95.1%	32,500	(7.1%,)	16,500	(49.2%,)
Organizational Membership Subscriptions	-	-	0.0%	-	0.0%	-	0.0%
Professional Membership Subscriptions	18,000	19,691	(9.4%,)	23,000	27.8%	24,000	4.3%
Telephone and fax	-	-	0.0%	-	0.0%	-	0.0%
Office Administration	1,340,500	1,273,832	5.0%	1,351,000	0.8%	1,395,500	3.3%
Consultants' Fees	3,320,000	1,576,835	52.5%	5,500,000	65.7%	3,000,000	(45.5%,)
External Auditors	-	-	0.0%	-	0.0%	-	0.0%
Remuneration - PTRM and External Members	-	-	0.0%	-	0.0%	-	0.0%
Recruitment	-	-	0.0%	-	0.0%	-	0.0%
Legal fees	-	-	0.0%	-	0.0%	-	0.0%
Internal Audit	-	-	0.0%	-	0.0%	-	0.0%
Professional fees	3,320,000	1,576,835	52.5%	5,500,000	65.7%	3,000,000	(45.5%)



Annual Budget for the year 2024/25: Electricity Regulation

INCOME AND EXPENDITURE BUDGET FOR 2024/25

DESCRIPTION	A	B		C	1	D	2
	ELECTRICITY REGULATION BUDGET	ELECTRICITY REGULATION ACTUAL	% Variance (A/B)	ELECTRICITY REGULATION APPROVED BUDGET	% Variance (A/C)	ELECTRICITY REGULATION PROPOSED BUDGET	% Variance (E/C)
	2022/23	2022/23		2023/24		2024/25	
Learnership programme	-	-	0.0%	-	0.0%	-	0.0%
Study fees	265,000	428,570	(61.7%,)	185,000	(30.2%,)	185,000	0.0%
External Bursaries	-	-	0.0%	-	0.0%	-	0.0%
Training WEGE	-	-	0.0%	-	0.0%	-	0.0%
Train. & Dev. Full Time Regulator Members	-	-	0.0%	-	0.0%	-	0.0%
Train. & Dev. Part Time Regulator Members	-	-	0.0%	-	0.0%	-	0.0%
Train. & Dev. Staff	1,121,823	825,678	26.4%	1,121,823	0.0%	1,346,080	20.0%
Travel Costs Regulator Members	-	-	0.0%	-	0.0%	-	0.0%
Travel Costs Staff	5,607,134	5,554,386	0.9%	5,662,495	1.0%	6,109,825	7.9%
Travel, Accommodation and Training	6,993,957	6,808,634	2.6%	6,969,318	(0.4%)	7,640,905	9.6%
Bank Charges	-	-	0.0%	-	0.0%	-	0.0%
Catering & Entertain	160,000	273,678	(71.0%,)	175,000	9.4%	215,000	22.9%
Employees Wellness	-	-	0.0%	-	0.0%	-	0.0%
Health and Safety	-	-	0.0%	-	0.0%	-	0.0%
Loss on Disposal of assets	-	-	0.0%	-	0.0%	-	0.0%
Knowledge Centre	-	-	0.0%	-	0.0%	-	0.0%
Other Expenses	160,000	273,678	(71.0%)	175,000	9.4%	215,000	22.9%



Annual Budget for the year 2024/25: Electricity Regulation

INCOME AND EXPENDITURE BUDGET FOR 2024/25

	A	B		C	1	D	2
	ELECTRICITY REGULATION BUDGET	ELECTRICITY REGULATION ACTUAL	% Variance (A/B)	ELECTRICITY REGULATION APPROVED BUDGET	% Variance (A/C)	ELECTRICITY REGULATION PROPOSED BUDGET	% Variance (E/C)
DESCRIPTION	2022/23	2022/23		2023/24		2024/25	
NET SURPLUS/ (DEFICIT) before Depreciation	100,693,851	108,094,385	(7.3%)	110,726,665	10.0%	115,292,056	4.1%
Support Service	128,488,874	122,746,876	4.5%	138,977,934	8.2%	151,874,825	9.3%
Depreciation		3,105,595	0.0%		0.0%		
NET SURPLUS/ (DEFICIT) for the period	(27,795,023)	(17,758,087)	36.1%	(28,251,268)	0.0%	(36,582,770)	29.5%
TOTAL CAPITAL EXPENDITURE	7,906,765	5,849,776	26.0%	4,177,160	(47.2%)	9,918,000	137.4%
Motor vehicles	1,073,000	1,091,951	(1.8%,)	464,000	(56.8%,)	-	(100.0%,)
Computer software	1,102,000	1,092,663	0.8%	1,160,000	5.3%	7,540,000	550.0%
Office furniture and equipment	1,208,140	67,757	94.4%	742,400	(38.6%,)	348,000	(53.1%,)
Building improvements	405,625	-	100.0%	650,760	60.4%	870,000	33.7%
Computer hardware	4,118,000	3,597,405	12.6%	1,160,000	(71.8%,)	1,160,000	0.0%



Annual Budget for the year 2024/25: Piped Gas Regulation

INCOME AND EXPENDITURE BUDGET FOR 2024/25

	A	B		C	1	D	2
	PIPED GAS REGULATION BUDGET	PIPED GAS REGULATION ACTUAL	% Variance (A/B)	PIPED GAS REGULATION APPROVED BUDGET	% Variance (A/C)	PIPED GAS REGULATION PROPOSED BUDGET	% Variance (E/C)
DESCRIPTION	2022/23	2022/23		2023/24		2024/25	
TOTAL INCOME	73,255,179	76,524,675	4.5%	79,292,999	8.2%	90,229,214	13.8%
Levies from Piped-Gas Industry	71,971,726	73,996,486	2.8%	76,849,794	6.8%	87,887,092	14.4%
Interest received	1,270,579	2,457,108	93.4%	2,436,241	91.7%	2,334,810	(4.2%,)
Rental Income	12,874	7,769	(39.7%,)	6,964	(45.9%,)	7,312	5.0%
Other Income	-	63,313	0.0%	-	0.0%	-	0.0%
TOTAL OPERATING EXPENDITURE	38,916,083	38,519,303	1.0%	45,252,389	16.3%	46,602,535	3.0%
National/International/Initiatives	-	-	0.0%	-	0.0%	-	0.0%
Publications and Communications	-	-	0.0%	-	0.0%	-	0.0%
Sponsorships	-	-	0.0%	-	0.0%	-	0.0%
Advertising	1,950,000	2,711,325	(39.0%,)	1,950,000	0.0%	2,300,000	17.9%
Stakeholder Meetings	-	-	0.0%	-	0.0%	-	0.0%
Tribunals and Hearings	100,000	-	100.0%	100,000	0.0%	100,000	0.0%
Advertising, Promotion and Communication	2,050,000	2,711,325	(32.3%)	2,050,000	0.0%	2,400,000	17.1%
Gross Salaries	29,392,991	28,424,595	3.3%	32,685,377	11.2%	35,938,231	10.0%
Learnership Allowance	-	-	0.0%	-	0.0%	-	0.0%
Internship Allowance	-	-	0.0%	-	0.0%	-	0.0%
Leave Pay: Staff	-	99,449	0.0%	-	0.0%	-	0.0%



Annual Budget for the year 2024/25: Piped Gas Regulation

INCOME AND EXPENDITURE BUDGET FOR 2024/25

DESCRIPTION	A	B		C	1	D	2
	PIPED GAS REGULATION BUDGET	PIPED GAS REGULATION ACTUAL	% Variance (A/B)	PIPED GAS REGULATION APPROVED BUDGET	% Variance (A/C)	PIPED GAS REGULATION PROPOSED BUDGET	% Variance (E/C)
	2022/23	2022/23		2023/24		2024/25	
Leave pay: Regulator Members	-	-	0.0%	-	0.0%	-	0.0%
Performance Bonus: FTRM	-	-	0.0%	-	0.0%	-	0.0%
Performance Bonus: Staff	4,702,878	4,174,821	11.2%	5,229,660	11.2%	5,791,423	10.7%
Cellphone and data allowance Staff	-	440,886	0.0%	480,720	0.0%	480,720	0.0%
Cellphone and data allowance FTRM	-	-	0.0%	-	0.0%	-	0.0%
Remuneration: FTRM	-	-	0.0%	-	0.0%	-	0.0%
Publication Incentives	-	-	0.0%	-	0.0%	-	0.0%
Salaries Temporary Staff	-	59,620	0.0%	1,337,768	0.0%	-	100.0%
Employment cost	34,095,869	33,199,371	2.6%	39,733,526	16.5%	42,210,375	6.2%
Maintenance	-	-	0.0%	-	0.0%	-	0.0%
Motor Vehicle Expenses	-	-	0.0%	-	0.0%	-	0.0%
Facility Management Operating expenses	-	-	0.0%	-	0.0%	-	0.0%
Municipal Charges	-	-	0.0%	-	0.0%	-	0.0%
Insurance	-	-	0.0%	-	0.0%	-	0.0%
Facilities Maintenance	-	-	0.0%	-	0.0%	-	0.0%



Annual Budget for the year 2024/25: Piped Gas Regulation

INCOME AND EXPENDITURE BUDGET FOR 2024/25

DESCRIPTION	A	B		C	1	D	2
	PIPED GAS REGULATION BUDGET	PIPED GAS REGULATION ACTUAL	% Variance (A/B)	PIPED GAS REGULATION APPROVED BUDGET	% Variance (A/C)	PIPED GAS REGULATION PROPOSED BUDGET	% Variance (E/C)
	2022/23	2022/23		2023/24		2024/25	
PPE Tools	50,000	-	100.0%	52,500	5.0%	150,000	185.7%
Office Operational Expenses--Lease Payments	-	-	0.0%	-	0.0%	-	0.0%
Postage & Courier Services	750	1,620	(116.0%,)	500	(33.3%,)	2,000	300.0%
Information Technology Operations	-	-	0.0%	-	0.0%	-	0.0%
Software License Fees	-	-	0.0%	-	0.0%	-	0.0%
Stationery and Printing	31,500	5,537	82.4%	31,500	0.0%	18,500	(41.3%,)
Organizational Membership Subscriptions	-	-	0.0%	-	0.0%	-	0.0%
Professional Membership Subscriptions	25,500	17,178	32.6%	18,615	(27.0%,)	21,500	15.5%
Telephone and fax	-	-	0.0%	-	0.0%	-	0.0%
Office Administration	107,750	24,335	77.4%	103,115	(4.3%)	192,000	86.2%
Consultants' Fees	1,000,000	983,350	1.7%	1,200,000	20.0%	-	(100.0%,)
External Auditors	-	-	0.0%	-	0.0%	-	0.0%
Remuneration - PTRM and External Members	-	-	0.0%	-	0.0%	-	0.0%
Recruitment	-	-	0.0%	-	0.0%	-	0.0%
Legal fees	-	-	0.0%	-	0.0%	-	0.0%
Internal Audit	-	-	0.0%	-	0.0%	-	0.0%
Professional fees	1,000,000	983,350	1.7%	1,200,000	20.0%	-	(100.0%)



Annual Budget for the year 2024/25: Piped Gas Regulation

INCOME AND EXPENDITURE BUDGET FOR 2024/25

DESCRIPTION	A	B		C	1	D	2
	PIPED GAS REGULATION BUDGET	PIPED GAS REGULATION ACTUAL	% Variance (A/B)	PIPED GAS REGULATION APPROVED BUDGET	% Variance (A/C)	PIPED GAS REGULATION PROPOSED BUDGET	% Variance (E/C)
	2022/23	2022/23		2023/24		2024/25	
Learnership programme	-	-	0.0%	-	0.0%	-	0.0%
Study fees	225,000	371,200	(65.0%,)	225,000	0.0%	225,000	0.0%
External Bursaries	-	-	0.0%	-	0.0%	-	0.0%
Training WEGE	-	-	0.0%	-	0.0%	-	0.0%
Train. & Dev. Full Time Regulator Members	-	-	0.0%	-	0.0%	-	0.0%
Train. & Dev. Part Time Regulator Members	-	-	0.0%	-	0.0%	-	0.0%
Train. & Dev. Staff	440,896	484,385	(9.9%,)	440,896	0.0%	539,073	22.3%
Travel Costs Regulator Members	-	-	0.0%	-	0.0%	-	0.0%
Travel Costs Staff	996,568	745,337	25.2%	1,494,852	50.0%	1,031,087	(31.0%,)
Travel, Accommodation and Training	1,662,464	1,600,922	3.7%	2,160,748	30.0%	1,795,160	(16.9%)
Bank Charges	-	-	0.0%	-	0.0%	-	0.0%
Catering & Entertain	-	-	0.0%	5,000	0.0%	5,000	0.0%
Employees Wellness	-	-	0.0%	-	0.0%	-	0.0%
Health and Safety	-	-	0.0%	-	0.0%	-	0.0%
Loss on Disposal of assets	-	-	0.0%	-	0.0%	-	0.0%
Knowledge Centre	-	-	0.0%	-	0.0%	-	0.0%
Other Expenses	-	-	0.0%	5,000	0.0%	5,000	0.0%



Annual Budget for the year 2024/25: Piped Gas Regulation

INCOME AND EXPENDITURE BUDGET FOR 2024/25

DESCRIPTION	A	B		C	1	D	2
	PIPED GAS REGULATION BUDGET	PIPED GAS REGULATION ACTUAL	% Variance (A/B)	PIPED GAS REGULATION APPROVED BUDGET	% Variance (A/C)	PIPED GAS REGULATION PROPOSED BUDGET	% Variance (E/C)
	2022/23	2022/23		2023/24		2024/25	
NET SURPLUS/ (DEFICIT) before Depreciation	34,339,096	38,005,372	(10.7%)	34,040,611	(0.9%)	43,626,679	28.2%
Support Service	46,521,832	43,490,552	6.5%	50,319,597	8.2%	54,989,161	9.3%
Depreciation		1,124,449	0.0%	-	0.0%		
NET SURPLUS/ (DEFICIT) for the period	(12,182,736)	(6,609,629)	45.7%	(16,278,986)	0.0%	(11,362,482)	(30.2%)
TOTAL CAPITAL EXPENDITURE	2,862,794	2,118,022	26.0%	1,512,420	(47.2%)	3,591,000	137.4%
Motor vehicles	388,500	395,361	(1.8%,)	168,000	(56.8%,)	-	(100.0%,)
Computer software	399,000	395,619	0.8%	420,000	5.3%	2,730,000	550.0%
Office furniture and equipment	437,430	24,533	94.4%	268,800	(38.6%,)	126,000	(53.1%,)
Building improvements	146,864	-	100.0%	235,620	60.4%	315,000	33.7%
Computer hardware	1,491,000	1,302,509	12.6%	420,000	(71.8%,)	420,000	0.0%



Annual Budget for the year 2024/25: Petroleum Pipelines

INCOME AND EXPENDITURE BUDGET FOR 2024/25

	A	B		C	1	D	2
	PETROLEUM PIPELINES REGULATION BUDGET	PETROLEUM PIPELINES REGULATION ACTUAL	% Variance (A/B)	PETROLEUM PIPELINES REGULATION APPROVED BUDGET	% Variance (A/C)	PETROLEUM PIPELINES REGULATION PROPOSED BUDGET	% Variance (E/C)
DESCRIPTION	2022/23	2022/23		2023/24		2024/25	
TOTAL INCOME	59,691,850	64,110,529	7.4%	69,787,212	16.9%	70,556,329	1.1%
Levies from Petroleum Pipelines Industry	58,408,397	61,613,928	5.5%	67,344,007	15.3%	68,214,207	1.3%
Interest received	1,270,579	2,457,108	93.4%	2,436,241	91.7%	2,334,810	(4.2%,)
Rental Income	12,874	7,769	(39.7%,)	6,964	(45.9%,)	7,312	5.0%
Other Income	-	31,724	0.0%	-	0.0%	-	286.9%
TOTAL OPERATING EXPENDITURE	32,989,843	35,858,459	(8.7%)	37,962,026	15.1%	37,691,181	(0.7%)
National/International/Initiatives	-	-	0.0%	-	0.0%	-	0.0%
Publications and Communications	-	-	0.0%	-	0.0%	-	0.0%
Sponsorships	-	-	0.0%	-	0.0%	-	0.0%
Advertising	1,290,000	1,656,197	(28.4%,)	1,325,000	2.7%	1,950,000	47.2%
Stakeholder Meetings	200,000	-	100.0%	400,000	100.0%	300,000	(25.0%,)
Tribunals and Hearings	90,000	-	100.0%	100,000	11.1%	100,000	0.0%
Advertising, Promotion and Communication	1,580,000	1,656,197	(4.8%)	1,825,000	15.5%	2,350,000	28.8%



Annual Budget for the year 2024/25: Petroleum Pipelines

INCOME AND EXPENDITURE BUDGET FOR 2024/25

	A	B		C	1	D	2
	PETROLEUM PIPELINES REGULATION BUDGET	PETROLEUM PIPELINES REGULATION ACTUAL	% Variance (A/B)	PETROLEUM PIPELINES REGULATION APPROVED BUDGET	% Variance (A/C)	PETROLEUM PIPELINES REGULATION PROPOSED BUDGET	% Variance (E/C)
DESCRIPTION	2022/23	2022/23		2023/24		2024/25	
Gross Salaries	23,392,488	24,138,157	(3.2%,)	26,581,283	13.6%	28,562,630	7.5%
Learnership Allowance	-	-	0.0%	-	0.0%	-	0.0%
Internship Allowance	-	-	0.0%	-	0.0%	-	0.0%
Leave Pay: Staff	-	-46,183	0.0%	-	0.0%	-	0.0%
Leave pay: Regulator Members	-	-	0.0%	-	0.0%	-	0.0%
Performance Bonus: FTRM	-	-	0.0%	-	0.0%	-	0.0%
Performance Bonus: Staff	3,742,797	4,431,972	(18.4%,)	4,253,005	13.6%	4,592,412	8.0%
Cellphone and data allowance Staff	-	425,745	0.0%	431,220	0.0%	431,220	0.0%
Cellphone and data allowance FTRM	-	-	0.0%	-	0.0%	-	0.0%
Remuneration: FTRM	-	-	0.0%	-	0.0%	-	0.0%
Publication Incentives	-	-	0.0%	-	0.0%	-	0.0%
Salaries Temporary Staff	-	-	0.0%	-	0.0%	-	0.0%
Employment cost	27,135,285	28,949,690	(6.7%)	31,265,508	15.2%	33,586,262	7.4%



Annual Budget for the year 2024/25: Petroleum Pipelines Regulation

INCOME AND EXPENDITURE BUDGET FOR 2024/25

	A	B		C	1	D	2
	PETROLEUM PIPELINES REGULATION BUDGET	PETROLEUM PIPELINES REGULATION ACTUAL	% Variance (A/B)	PETROLEUM PIPELINES REGULATION APPROVED BUDGET	% Variance (A/C)	PETROLEUM PIPELINES REGULATION PROPOSED BUDGET	% Variance (E/C)
DESCRIPTION	2022/23	2022/23		2023/24		2024/25	
Maintenance	-	-	0.0%	-	0.0%	-	0.0%
Motor Vehicle Expenses	-	-	0.0%	-	0.0%	-	0.0%
Facility Management Operating expenses	-	-	0.0%	-	0.0%	-	0.0%
Municipal Charges	-	-	0.0%	-	0.0%	-	0.0%
Insurance	-	-	0.0%	-	0.0%	-	0.0%
Facilities Maintenance	-	-	0.0%	-	0.0%	-	0.0%
PPE Tools	50,000	-	100.0%	150,000	200.0%	175,000	16.7%
Office operational expenses-Lease Payments	-	-	0.0%	-	0.0%	-	0.0%
Postage & Courier Services	-	-	0.0%	2,000	0.0%	2,000	0.0%
Information Technology Operations	-	-	0.0%	-	0.0%	-	0.0%
Software License Fees	-	-	0.0%	-	0.0%	-	0.0%
Stationery and Printing	23,750	5,828	75.5%	23,750	0.0%	11,500	(51.6%,)
Organizational Membership Subscriptions	-	-	0.0%	-	0.0%	-	0.0%
Professional Membership Subscriptions	-	-	0.0%	-	0.0%	-	0.0%
Telephone and fax	-	-	0.0%	-	0.0%	-	0.0%
Office Administration	73,750	5,828	92.1%	175,750	138.3%	188,500	7.3%



Annual Budget for the year 2024/25: Petroleum Pipelines Regulation

INCOME AND EXPENDITURE BUDGET FOR 2024/25

	A	B		C	1	D	2
	PETROLEUM PIPELINES REGULATION BUDGET	PETROLEUM PIPELINES REGULATION ACTUAL	% Variance (A/B)	PETROLEUM PIPELINES REGULATION APPROVED BUDGET	% Variance (A/C)	PETROLEUM PIPELINES REGULATION PROPOSED BUDGET	% Variance (E/C)
DESCRIPTION	2022/23	2022/23		2023/24		2024/25	
Consultants' Fees	2,500,000	3,706,979	(48.3%,)	2,500,000	0.0%	-	(100.0%,)
External Auditors	-	-	0.0%	-	0.0%	-	0.0%
Remuneration - PTRM and External Members	-	-	0.0%	-	0.0%	-	0.0%
Recruitment	-	-	0.0%	-	0.0%	-	0.0%
Legal fees	-	-	0.0%	-	0.0%	-	0.0%
Internal Audit	-	-	0.0%	-	0.0%	-	0.0%
Professional fees	2,500,000	3,706,979	(48.3%)	2,500,000	0.0%	-	(100.0%)
Learnership programme	-	-	0.0%	-	0.0%	-	0.0%
Study fees	270,000	294,029	(8.9%,)	220,000	(18.5%,)	220,000	0.0%
External Bursaries	-	-	0.0%	-	0.0%	-	0.0%
Training WEGE	-	-	0.0%	-	0.0%	-	0.0%
Train. & Dev. Full Time Regulator Members	-	-	0.0%	-	0.0%	-	0.0%
Train. & Dev. Part Time Regulator Members	-	-	0.0%	-	0.0%	-	0.0%
Train. & Dev. Staff	350,888	430,252	(22.6%,)	350,888	0.0%	428,439	22.1%
Travel Costs Regulator Members	-	-	0.0%	-	0.0%	-	0.0%
Travel Costs Staff	1,079,920	804,516	25.5%	1,619,880	50.0%	900,480	(44.4%,)
Travel, Accommodation and Training	1,700,808	1,528,797	10.1%	2,190,768	28.8%	1,548,919	(29.3%)



Annual Budget for the year 2024/25: Petroleum Pipelines Regulation

INCOME AND EXPENDITURE BUDGET FOR 2024/25

	A	B		C	1	D	2
	PETROLEUM PIPELINES REGULATION BUDGET	PETROLEUM PIPELINES REGULATION ACTUAL	% Variance (A/B)	PETROLEUM PIPELINES REGULATION APPROVED BUDGET	% Variance (A/C)	PETROLEUM PIPELINES REGULATION PROPOSED BUDGET	% Variance (E/C)
DESCRIPTION	2022/23	2022/23		2023/24		2024/25	
Bank Charges	-	-	0.0%	-	0.0%	-	0.0%
Catering & Entertain	-	10,968	0.0%	5,000	0.0%	17,500	250.0%
Employees Wellness	-	-	0.0%	-	0.0%	-	0.0%
Health and Safety	-	-	0.0%	-	0.0%	-	0.0%
Loss on Disposal of assets	-	-	0.0%	-	0.0%	-	0.0%
Knowledge Centre	-	-	0.0%	-	0.0%	-	0.0%
Other Expenses	-	10,968	0.0%	5,000	0.0%	17,500	250.0%
NET SURPLUS/ (DEFICIT) before Depreciation	26,702,007	28,252,070	(5.8%)	31,825,186	19.2%	32,865,148	3.3%
Support Service	46,521,832	42,815,136	8.0%	50,319,597	8.2%	54,989,161	9.3%
Depreciation		1,124,425	0.0%		0.0%		
NET SURPLUS/ (DEFICIT) for the period	(19,819,825)	(15,687,490)	20.8%	(18,494,411)	0.0%	(22,124,013)	19.6%
TOTAL CAPITAL EXPENDITURE	2,862,794	2,118,022	26.0%	1,512,420	(47.2%)	3,591,000	137.4%
Motor vehicles	388,500	395,361	(1.8%,)	168,000	(56.8%,)	-	(100.0%,)
Computer software	399,000	395,619	0.8%	420,000	5.3%	2,730,000	550.0%
Office furniture and equipment	437,430	24,533	94.4%	268,800	(38.6%,)	126,000	(53.1%,)
Building improvements	146,864	-	100.0%	235,620	60.4%	315,000	33.7%
Computer hardware	1,491,000	1,302,509	12.6%	420,000	(71.8%,)	420,000	0.0%



Annual Budget for the year 2024/25: Support Services

INCOME AND EXPENDITURE BUDGET FOR 2023/24

	A	B		C	1	D	2
	SUPPORT SERVICE BUDGET	SUPPORT SERVICE ACTUAL	% Variance (A/B)	SUPPORT SERVICE APPROVED BUDGET	% Variance (A/C)	SUPPORT SERVICE BUDGET	% Variance (E/C)
DESCRIPTION	2022/23	2022/23		2023/24		2024/25	
TOTAL OPERATING EXPENDITURE	212 870 973	188 266 727	11.6%	221 532 538	4.1%	239 617 127	8.2%
National/International/Initiatives	-	-	0.0%	70,000	0.0%	-	(100.0%)
Publications and Communications	2,520,000	2,430,450	3.6%	2,635,000	4.6%	200,000	(31.7%)
Sponsorships	100,000	23,349	76.7%	100,000	0.0%	865,000	100.0%
Advertising	1,665,000	1,200,592	27.9%	890,000	(46.5%,)	100,000	(2.8%)
Stakeholder Meetings	100,000	48,698	51.3%	-	(100.0%,)	-	0.0%
Tribunals and Hearings	15,000	214,112	(1327.4%,)	30,000	100.0%	50,000	66.7%
Advertising, Promotion and Communication	4,400,000	3,917,201	11.0%	3,725,000	(15.3%)	3,015,000	(19.1%)
Gross Salaries	102,238,846	101,160,314	1.1%	114,023,581	11.5%	121,781,174	6.8%
Learnership Allowance	1,639,241	499,841	69.5%	1,638,636	(0.0%,)	1,638,636	0.0%
Internship Allowance	1,639,241	602,144	63.3%	1,638,636	(0.0%,)	1,638,636	0.0%
Leave Pay: Staff	811,277	(122,138)	115.1%	811,277	0.0%	811,277	0.0%
Leave pay: Regulator Members	140,087	1,432,205	(112.5%,)	140,087	0.0%	140,087	0.0%
Performance Bonus: FTRM	1,255,180	17,146,061	(14.1%,)	1,343,883	7.1%	1,299,113	(3.3%)
Performance Bonus: Staff	16,358,213	1,706,905	(4.8%,)	18,243,773	11.5%	19,622,722	7.6%
Cellphone and data allowance Staff	-	260,400	0.0%	1,953,480	0.0%	1,953,480	100.0%
Cellphone and data allowance FTRM	-	9,432,228	(5.2%)	247,200	0.0%	247,200	100.0%



Annual Budget for the year 2024/25: Support Services

EXPENDITURE BUDGET FOR 2024/25

DESCRIPTION	A	B		C	1	D	2
	SUPPORT SERVICE BUDGET	SUPPORT SERVICE ACTUAL	% Variance (A/B)	SUPPORT SERVICE APPROVED BUDGET	% Variance (A/C)	SUPPORT SERVICE BUDGET	% Variance (E/C)
	2022/23	2022/23		2023/24		2024/25	
Remuneration: FTRM	8,965,576	9,432,228	(5.2%)	9,599,166	7.1%	10,238,910	6.7%
Publication Incentives	75,000	-	100.0%	75,000	0.0%	75,000	0.0%
Salaries Temporary Staff	2,900,000	820,093	71.7%	1,300,000	(55.2%,)	4,100,000	215.4%
Employment cost	136,022,661	133,235,672	2.0%	151,014,719	11.0%	163,546,234	8.3%
Maintenance	3,800,000	2,003,743	47.3%	6,294,926	65.7%	3,800,000	(39.6%)
Motor Vehicle Expenses	400,000	459,445	(14.9%)	180,000	(55.0%,)	450,000	150.0%
Facility Management Operating expenses	3,150,000	2,047,934	35.0%	1,022,729	(67.5%,)	2,090,310	104.4%
Municipal Charges	2,750,000	2,699,196	1.8%	2,500,000	(9.1%,)	2,820,000	12.8%
Insurance	1,059,104	974,969	7.9%	1,400,000	32.2%	1,400,000	0.0%
Facilities Maintenance	11,159,104	8,185,287	26.6%	11,397,655	2.1%	10,560,310	(7.3%)
PPE Tools	-	-	0.0%	-	0.0%	-	0.0%
Office operational expenses-Lease Payments	1,003,385	608,111	39.4%	1,053,554	5.0%	1,159,723	10.1%
Postage & Courier Services	-	923,836	0.0%	-	0.0%	600,000	0.0%
Office Rental	41,151	46,102	(12.0%)	32,500	(21.0%,)	58,500	80.0%
Information Technology Operations	8,250,000	4,476,627	45.7%	8,058,000	(2.3%,)	13,383,576	66.1%
Software License Fees	6,475,245	6,653,232	(2.7%)	6,789,597	4.9%	7,189,385	5.9%
Stationery and Printing	672,000	205,284	69.5%	680,500	1.3%	242,500	(64.4%)
Organizational Membership Subscriptions	1,659,244	1,463,327	11.8%	1,674,244	0.9%	1,884,274	12.5%



Annual Budget for the year 2024/25: Support Services

EXPENDITURE BUDGET FOR 2023/24

	A	B		C	1	D	2
	SUPPORT SERVICE BUDGET	SUPPORT SERVICE ACTUAL	% Variance (A/B)	SUPPORT SERVICE APPROVED BUDGET	% Variance (A/C)	SUPPORT SERVICE BUDGET	% Variance (E/C)
DESCRIPTION	2022/23	2022/23		2023/24		2024/25	
Professional Membership Subscriptions	103,485	58,866	43.1%	1,674,244	0.9%	85,527	(15.7%)
Telephone and fax	767,408	596,351	22.3%	101,475	(1.9%,)	760,000	(1.3%)
Office Administration	18,971,918	15,031,736	20.8%	19,159,779	1.0%	25,363,485	32.4%
Consultants' Fees	8,408,742	7,155,329	14.9%	9,357,414	11.3%	8,820,358	(5.7%)
External Auditors	2,520,308	2,301,394	8.7%	3,209,390	27.3%	3,369,860	5.0%
Remuneration - PTRM and Ext. Members	2,599,507	2,446,476	5.9%	3,957,749	52.3%	3,957,749	0.0%
Recruitment	800,000	487,344	39.1%	600,000	(25.0%,)	600,000	0.0%
Legal fees	19,500,000	20,190,046	(3.5%)	18,000,000	(7.7%,)	18,000,000	0.0%
Internal Audit	2,200,000	2,126,825	3.3%	2,200,000	0.0%	2,200,000	0.0%
Professional fees	36,028,557	34,707,413	3.7%	37,324,554	3.6%	36,947,967	(1.0%)
Learnership programme	950,760	338,500	64.4%	1,250,760	4.2%	1,250,760	0.0%
Study fees	632,000	524,638	17.0%	632,000	(36.2%)	677,000	0.0%
External Busaries	525,000	423,555	19.3%	525,000	0.0%	551,000	0.0%
Training WEGE	-	193,000	0.0%	-	0.0%	400,000	60.0%
Train. & Dev. Full Time Regulator Members	194,484	178,864	8.0%	134,484	(75.1%)	400,000	0.0%
Train. & Dev. Part Time Regulator Members	57,519	24,777	56.9%	157,519	0.0%	500,000	0.0%
Train. & Dev. Staff	2,923,583	3,027,812	(3.6%)	1,533,583	(53.2%)	1,826,718	19.1%
Travel Costs Regulator Members	1,132,396	1,023,371	9.6%	500,000	(86.5%)	1,932,182	7.3%
Travel Costs Staff	2,558,718	2,059,125	19.5%	2,158,718	(50.0%)	2,342,490	(27.7%)



Annual Budget for the year 2024/25: Support Services

EXPENDITURE BUDGET FOR 2023/24

DESCRIPTION	A	B		C	1	D	2
	SUPPORT SERVICE BUDGET	SUPPORT SERVICE ACTUAL	% Variance (A/B)	SUPPORT SERVICE APPROVED BUDGET	% Variance (A/C)	SUPPORT SERVICE BUDGET	% Variance (E/C)
	2022/23	2022/23		2023/24		2024/25	
Travel, Accommodation and Training	8,974,460	7,793,642	13.2%	10,200,420	13.7%	9,880,150	(3.1%)
Bank Charges	65,000	72,320	(11.3%,)	65,000	0.0%	80,000	23.1%
Catering & Entertain	340,000	146,534	56.9%	130,000	(61.8%,)	260,000	100.0%
Employees Wellness	400,000	243,362	39.2%	800,000	100.0%	500,000	(37.5%)
Health and Safety	900,000	877,822	2.5%	800,000	(11.1%,)	700,000	(12.5%)
Loss on Disposal of assets	-	355,804	0.0%	-	0.0%	-	0.0%
Knowledge Centre	4,270,838	4,485,772	(5.0%)	5,000,000	17.1%	,11,000,000	120.0%
Other Expenses	5,975,838	6,181,614	(3.4%)	6,795,000	13.7%	12,540,000	84.5%
Depreciation - Office Equipment	-	1,369,128	0.0%	-	0.0%	-	0.0%
Depreciation - Software	-	451,467	0.0%	-	0.0%	-	0.0%
Depreciation - Motor Vehicles	-	104,153	0.0%	-	0.0%	-	0.0%
Depreciation - Hardware	-	2,106,743	0.0%	-	0.0%	-	0.0%
Depreciation - Building	-	1,322,978	0.0%	-	0.0%	-	0.0%
Depreciation	-	5,354,469	0.0%	-	0.0%	-	0.0%
TOTAL CAPITAL EXPENDITURE	13,632,353	10,085,820	26.0%	7,202,000	(47.2%)	17,100,000	137.4%



Annual Budget for the year 2024/25: Support Services

EXPENDITURE BUDGET FOR 2023/24

	A	B		C	1	D	2
	SUPPORT SERVICE BUDGET	SUPPORT SERVICE ACTUAL	% Variance (A/B)	SUPPORT SERVICE APPROVED BUDGET	% Variance (A/C)	SUPPORT SERVICE BUDGET	% Variance (E/C)
DESCRIPTION	2022/23	2022/23		2023/24		2024/25	
Motor vehicles	1,850,000	1,882,673	(1.8%,)	800,000	(56.8%,)	-	(100.0%)
Computer software	1,900,000	1,883,901	0.8%	2,000,000	5.3%	13,000,000	550.0%
Office furniture and equipment	2,083,000	116,823	94.4%	1,280,000	(38.6%,)	600,000	(53.1%,)
Building improvements	699,353	-	100.0%	1,122,000	60.4%	1,500,000	33.7%
Computer hardware	7,100,000	6,202,422	12.6%	2,000,000	(71.8%,)	2,000,000	0.0%



Annual Budget for the year 2024/25: License Fees and Levy Rate Calculation

	ELECTRICITY	PIPED-GAS	PETROLEUM PIPELINES	TOTAL
Operating Expenditure - Regulated Industries	122,007,597	46,602,535	37,691,181	206,301,313
Expenditure - Support services allocated	151,874,825	54,989,161	54,989,161	261,853,147
Operating Expenditure	273,882,422	101,591,696	92,680,342	468,154,460
Add: Capital Expenditure	9,918,000	3,591,000	3,591,000	17,100,000
Total Expenditure	283,800,422	105,182,696	96,271,342	485,254,460
Less: Interest Received	(6,448,524)	(2,334,810)	(2,334,810)	(11,118,145)
Less: Rental Income	(20,195)	(7,312)	(7,312)	(34,818)
Less: Registration fees	(70,000)	-	-	(70,000)
Less: Refund of prior year surplus funds	(46,500,770)	(14,953,482)	(25,715,013)	(87,169,265)
Leviable amount	230,760,934	87,887,092	68,214,207	386,862,232



Annual Budget for the year 2024/25: License Fees and Levy Rate Calculation

	ELECTRICITY	PIPED-GAS	PETROLEUM PIPELINES
Projected Volumes (from Industry)	200 235 750 MWh	181 931 198 Gj	15 647 000 KI
Electricity license fee (c/kWh)	0.11524		
Piped-Gas levy rate (c/Gj)		48 308	
Petroleum Pipeline levy rate (c/litre)			0.43596
Levy Rate Increase / (Decrease)	10.00%	5.50%	5.50%
2024/25 - Levy Increase / (Decrease) due to change in:			
Volumes	5.07%	(8.86%)	4.21%
Operating Expenditure	2.73%	1.76%	(0.40%)
Support Service Allocation	5.86%	6.08%	6.93%
Capital Expenditure	2.61%	2.70%	3.09%
Interest received and Other Income	0.13%	0.13%	0.15%
Refund of surplus funds	(6.40%)	3.69%	(8.48%)
Total change	10.00%	5.50%	5.50%
Licence fee/Levy Rate 2023/24 FY	0.10477	45.789	0.41323



Annual Budget for the year 2024/25: Programme Budget

	PROGRAMME 1:		PROGRAMME 2:		PROGRAMME 3:		PROGRAMME 4:		PROGRAMME 5:		TOTAL	
	REGULATORY SERVICE DELIVERY		ADVOCACY AND ENGAGEMENT		INNOVATION		OPERATIONAL EFFICIENCY AND QUALITY MANAGEMENT		PEOPLE AND ORGANISATIONAL CULTURE			
Electricity												
Electricity Regulation	8,309,442	80%	2,077,360	20%	-	0%	-	0%	-	0%	10,386,802	100%
Electricity Pricing and Tariffs	35,558,031	80%	8,889,508	20%	-	0%	-	0%	-	0%	44,447,538	100%
Electricity licencing, Compliance, and Dispute Resolution	35,428,225	80%	8,857,056	20%	-	0%	-	0%	-	0%	44,285,281	100%
Electricity Infrastructure Planning	18,310,380	80%	4,577,595	20%	-	0%	-	0%	-	0%	22,887,975	100%
Piped-Gas												
Piped Gas Regulation	11,299,487	90%	1,255,499	10%	-	0%	-	0%	-	0%	12,554,986	100%
Gas Pricing and Tariffs	11,901,478	95%	626,394	5%	-	0%	-	0%	-	0%	12,527,872	100%
Gas Licencing, Compliance and Dispute Resolution	15,050,769	100%	-	0%	-	0%	-	0%	-	0%	15,050,769	100%
Gas Competition and Markets	6,145,463	95%	323,445	5%	-	0%	-	0%	-	0%	6,468,909	100%
Petroleum Pipelines												
Petroleum Pipelines Regulation	6,564,116	50%	6,564,116	50%	-	0%	-	0%	-	0%	13,128,232	100%
Petroleum Pipeline Tariffs	9,436,270	80%	2,359,068	20%	-	0%	-	0%	-	0%	11,795,338	100%
Petroleum Licensing, Compliance and Dispute Resolution	10,214,089	80%	2,553,522	20%	-	0%	-	0%	-	0%	12,767,611	100%



Annual Budget for the year 2024/25: Programme Budget

	PROGRAMME 1:		PROGRAMME 2:		PROGRAMME 3:		PROGRAMME 4:		PROGRAMME 5:			
	REGULATORY SERVICE DELIVERY		ADVOCACY AND ENGAGEMENT		INNOVATION		OPERATIONAL EFFICIENCY AND QUALITY MANAGEMENT		PEOPLE AND ORGANISATIONAL CULTURE		TOTAL	
Corporate Service												
Executive Manager: Corporate services	-	-	302,854	5%	-	0%	5,754,224	95%	-	0%	6,057,077	100%
Legal Advisory Services	-	-	-	0%	-	0%	28,639,141	100%	-	0%	28,639,141	100%
Information Resource Management	-	-	-	0%	1,064,220	5%	20,220,184	95%	-	0%	21,284,404	100%
Communication and Stakeholder Management	-	-	14,294,837	100%	-	0%	-	0%	-	0%	14,294,837	100%
International Co-ordination and Partnership	-	-	6,681,370	100%	-	0%	-	0%	-	0%	6,681,370	100%
Organizational Units												
Chief Operating Officer	-	-	-	0%	-	0%	9,118,776	100%	-	0%	9,118,776	100%
Strategic planning and Monitoring	-	-	-	0%	-	0%	7,289,533	100%	-	0%	7,289,533	100%
Regulatory Support Unit	-	-	-	0%	-	0%	31,819,797	100%	-	0%	31,819,797	100%
Information Communication and Technology	-	-	-	0%	7,051,002	20%	28,204,006	80%	-	0%	35,255,008	100%
Internal Audit Unit	-	-	-	0%	-	0%	13,148,662	100%	-	0%	13,148,662	100%
Regulatory Analysis and Research	-	-	-	0%	-	0%	11,227,031	100%	-	0%	11,227,031	100%
Finance and Administration												
Chief Financial Officer	-	-	-	0%	-	0%	8,074,724	100%	-	0%	8,074,724	100%
Financial Management and Governance	-	-	-	0%	-	0%	12,554,981	100%	-	0%	12,554,981	100%
Supply Chain Management/Facilities/ Projects	-	-	-	0%	-	0%	23,494,223	100%	-	0%	23,494,223	100%



Annual Budget for the year 2024/25: Programme Budget

	PROGRAMME 1:		PROGRAMME 2:		PROGRAMME 3:		PROGRAMME 4:		PROGRAMME 5:		TOTAL		
	REGULATORY SERVICE DELIVERY		ADVOCACY AND ENGAGEMENT		INNOVATION		OPERATIONAL EFFICIENCY AND QUALITY MANAGEMENT		PEOPLE AND ORGANISATIONAL CULTURE				
QUALITY MANAGEMENT													
Human Resources													
Chief Human Capital Officer	-	-	-	0%	-	0%	-	0%	4,205,177	100%	4,205,177	100%	
Human Resources Value Creation	-	-	-	0%	-	0%	-	0%	23,927,318	100%	23,927,318	100%	
Human Resources Transactions	-	-	-	0%	-	0%	-	0%	4,781,085	100%	4,781,085	100%	
	168,217,750	35.93	59,362,624	12.68	8,115,222	1.73	199,545,284	42.62	32,913,581	7.03	468,154,460	100	



Annual Budget for the year 2024/25: Budget Timelines

#	Item	Date
1	Approval of Budget Assumptions and Parameters By Energy Regulator	07 August 2023 Round Robin Approval
2	Proposed budget tabled at NERSA Executive Committee	11 August 2023
3	Proposed budget tabled at NERSA Regulator Executive Committee	14 August 2023
6	Submission of Draft Budget 2024/25 to CEO and RSU	11 August 2023
7	Proposed budget tabled at Finance Committee	17 August 2023
8	Energy Regulator approves the proposed budget	30 August 2023
9	Submission to the Department of Mineral Resources and Energy	31 August 20223
10	Submit approved APP to Minister	31 October 2023
12	Obtain CEO approval to Gazette 2023/24 levies (Petroleum Pipelines and Piped Gas) and invite representation	7 November 2023
13	Closing date for representations	06 December 2023
14	Resolve representations if any, and obtain approval from Energy Regulator for changes to budget by round-robin	06 December 2023
15	Letter to Minister of Mineral Resources and Energy to approve levies in concurrence of the Minister of Finance	11 December 2023
16	Follow-up on Ministerial Approval	11 Dec 2023to 31 Mar 2024
17	Gazette approved levies for 2023/24 (Gas & Petroleum)	After approval by the Minister (approval by 31 March 2024 latest)

NOTES



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